computer system

- Hardware
  - Main board (mother board)
  - Memory (RAM, ROM, cache memory)
  - storage units (hard disks, flash memories)
  - input / output → I/O (keyboard, mouse, speackers, scanner)
  - Cpu
- software
  - OS (operating systems)
  - Application
  - languages (لغات البرمجة)
    - high level
    - low level

*ICs:

OS: windows

*signals: → Analoge تناظرية
→ Digital رقمي

→ numbering systems: أنظمة العد

1) Decimal system النظام العشري

( ) Base 10

digits

bits $(0, 1, 2, \ldots, 9)_{10}$

الأرقام المسموح استخدامها داخل النظام العشري

باقي القسمة $(\%10)_{10}$

المسموح هنا $(0, \ldots, 9)$ → صوته بتعوض أي رقم وبتطلع باقي القسمة عليه دائماً حيطلع هنا من (0 - 9) وماشي قيمة أعلى.

12

2 digits

2 bits

2 + 10

1 2

بالوزن

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weight الوزن

10^4 الوف 10^3 10^2 مئات 10^1 عشرات 10^0 → احاد

Base Base

→ $(1\ 2)_{10}$

10^1 10^0

$\therefore (1*10) + (2*1)$

$= 12$

→ $(5\ 9\ 3)_{10}$

10^2 10^1 10^0

100 10 1

$(5*10^2) + (9*10^1) + (3*10^0)$

$= (5*100) + (9*10) + (3*1) \Rightarrow 593$

2) Binary system : النظام الثنائي

$(\%\ 2)_2$

المسموح هنا → باقي قسمة أي رقم على 2 لازم تكون يا 0 أو 1

(0 و 1) اذا كانت غير معناها فات الـ 2 بطلع باقي القسمة على 2 واذا لا بطلع 1

$(1\ 0\ 1\ 1\ 0)_2$

2^4 2^3 2^2 2^1 2^0

16 8 4 2 1

3) Octal system النظام الثماني

$(\%\ 8)_8$

المسموح هنا → باقي قسمة أي رقم على 8 ممكن تكون أحد الأرقام (0-7)

$(0, 1, 2, \ldots, 7)_8$

$(3\ 5\ 1\ 6)_8$

8^3 8^2 8^1 8^0

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4) Hexadecimal system: النظام السادس عشر

$(\%\ 16)_{16}$

المسموح هنا

$(0, 1, \dots, 15)_{16}$ → Hexa

$(1\ 5\ 9)_{16}$

$16^2\ \ 16^1\ \ 16^0$

ex) $(135)_{16}$ 2digits or 3digits ??

$(1\ 3\ 5)$ or $(13\ 5)$

$16^2\ 16^1\ 16^0$ $\quad$ $16^1\ 16^0$

← صوبه صار في عنا مشكلة لانه النظام الوحيد اللي بقبل خانتين و ممكن تحل احقانيه حالما الرقميه وقع بعض زو كل خانة لحال عشانه نحل هاي المشكلة عالنا هاد الاسئ ←

لحل هذه المشكلة 16 %

15 و 14 و 13 و 12 و 11 و 10 و 9 و ... و 1 و 0

10 → A

11 → B

12 → C

13 → D

14 → E

15 → F

$(1\ A\ 9)_{16}$: $(1\ 10\ 9)$

$16^2\ 16^1\ 16^0$ $\quad$ $16^3\ 16^2\ 16^1\ 16^0$

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تحويل من أي نظام الى نظام عشري

* system conversion from any system to decimal:: ↑

→ multiplying the digit by the corresponding weight and find the summation.

* Example #1:

* Binary to decimal:: ثنائي الى عشري

→ $(1\ 0\ 1\ 1)_2 \equiv (11)_{10}$

$2^3\ 2^2\ 2^1\ 2^0$

$8\ 4\ 2\ 1$

$\Rightarrow (1*8) + (0*4) + (1*2) + (1*1)$

$= 8 + 0 + 2 + 1 = 11 \Rightarrow (11)_{10}$

→ $(1\ 0\ 1\ 1\ 0\ 1\ 0)_2 \Rightarrow (90)_{10}$

$64\ 32\ 16\ 8\ 4\ 2\ 1$

$\Rightarrow (64) + 0 + 16 + 8 + 0 + 2 + 0$

$= (90)_{10}$

* Octal to decimal:: ثماني الى عشري

→ $(1\ 7)_8 \approx (15)_{10}$

$8^1\ 8^0 \Rightarrow (8*1) + (7*1)$

$= 8 + 7 = (15)_{10}$

→ $(1\ 0\ 1\ 6)_8 \approx (526)_{10}$

$8^3\ 8^2\ 8^1\ 8^0 \Rightarrow (1*8^3) + (0*8^2) + (1*8) + (6*1)$

$= (1*512) + 0 + 8 + 6 = (526)_{10}$

* Hexa decimal to decimal:: سادس عشر الى عشري

→ $(1\ 5\ 9)_{16} \equiv (345)_{10}$

$1\ 5\ 9$

$16^2\ 16^1\ 16^0 \Rightarrow (1*256) + (5*16) + (9*1) = 256 + 80 + 9 = (345)_{10}$

→ $(3\ F)_{16} \equiv (63)_{10}$

$3\ F \Rightarrow (3*16) + (F*1) = (48) + (15*1) \Rightarrow (63)_{10}$

$1_{16}\ 16^0$

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تحويل من عشري الى اي نظام

*Decimal to any system:

dividing the decimal # by the system I want to convert then the remainder is the answer as the following:

$10 \rightarrow 2$ بقسم على ٢

$10 \rightarrow 8$ بقسم على ٨

$10 \rightarrow 16$ بقسم على ١٦

*From decimal to Binary: من عشري الى ثنائي

Example #1:

$\rightarrow (13)_{10} \equiv (1101)_2$

2	13	remainder
2	6	1
2	3	0
2	1	1
	0	1

$(1101)_2$

to make sure:

$1\ 1\ 0\ 1 \overset{?}{=} 13$

$8 + 4 + 0 + 1 = \boxed{13}$ ✓

Example #2:

$\rightarrow (37)_{10} \equiv (100101)_2$

2	37	1
2	18	0
2	9	1
2	4	0
2	2	0
2	1	1
	0	

$(100101)_2$

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* From decimal to octal: من عشري لثماني

→ $(17)_{10} \Rightarrow (21)_8$

8	17	Rem
8	2	1
	0	2

$(21)_8$

$8^1 \quad 1 = 16 + 1 = \boxed{17}$ ✓

→ $(67)_{10} \equiv (103)_8$

8	67	3
8	8	0
8	1	1
	0	

$(1\ 0\ 3)_8$

$64 \quad 8 \quad 1 = 64 + 0 + 3 = \boxed{67}$ ✓

* From decimal to hexadecimal: من عشري الى سادس عشر

Example #1:

→ $(15)_{10} \equiv (F)_{16}$

16	15	Rem
	0	15

$15 \rightarrow (F)_{16}$

→ $(91)_{10} = (5B)_{16}$

16	91	Rem
16	5	11 → B
	0	5

$(5B)_{16}$

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* lecture (2) :: * اذا بدي احول الكسور العشرية الى اي نظام فبقوم بالضرب بقيمة ذلك النظام *

Fractions :: الكسور

* بضرب فقط الكسر (على نظام الطلبة)

→ Example: convert $(0.5)_{10}$ to Binary.

multiply the # by 2

العدد الصحيح ← العدد الكسري

$0.5 * 2$

* على الحاسوب بدي اضرب اثنا فالي بالعدد الصحيح بضرب العدد الكسري بس *

$1\overline{1}.0 \quad .0 * 2 = 0$ finish

$(0.1)_2$

→ Example: $(0.25)_{10} \equiv (0.01)_2$ * بضل أكرر العملية لحد ما يصير بتكرر الناتج وهي (يصير دوري) *

$.25 * 2$

$= 0.50$

$.50 * 2 = 1.00 \quad .00 * 2$ finish $(0.01)_2$

→ Example: $(0.4)_{10} \equiv (0.0110)_2$

$0.4 * 2 = \boxed{0}.8 * 2$

$= \boxed{1}.6 * 2$

$= \boxed{1}.2 * 2$

$= \boxed{0}.4 \rightarrow$ رجعنا لبداية المسألة

$= (0.0110)_2$

→ Example: $(0.375)_{10} \equiv (0.011)_2$

$0.375 * 2$

$\boxed{0}.750 * 2$

$\boxed{1}.500 * 2$

$\boxed{1}.00 \rightarrow$ Finish

$(0.011)_2$

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* Example: Convert $(0.5)_{10}$ to octal. ⟹ *8

0.5 * 8

= 4.00 Finish

$\therefore (0.4)_8$

* Example: $(0.3)_{10} \equiv (0.23146)_8$

0.3 * 8 =

[2].4 * 8

[3].2 * 8

[1].6 * 8

[4].8 * 8

[6].4 * 8 .4 * 8 Finish

$\therefore (0.23146)_8$

* Example: Convert $(0.5)_{10}$ to Hexadecimal ⟹ *16

0.5 * 16

= 8.00 00 * 16 Finish

$(0.8)_{16}$

* Example: $(0.3)_{10} \equiv (0.4C)_{16}$

0.3 * 16

4.8 * 16

12.8 ⟶ Finish, Ci

$(0.4C)$

~~* Example: (illegible)~~

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* تحويل كسري من النظام الثنائي الى النظام العشري:

Example: $(0.1)_2 \equiv (\ 0.5\ )_{10}$

int. Fraction

بالكسور ← O . X X X X X ← بالاعداد العادية

$2^3\ 2^2\ 2^1\ 2^0$ | $2^{-1}\ 2^{-2}\ 2^{-3}\ 2^{-4}\ 2^{-5}$ ...

$\frac{1}{2}\ \frac{1}{4}\ \frac{1}{8}\ \frac{1}{16}\ \frac{1}{32}$ ...

0.1 weight

$2^{-1} \Rightarrow 1 * 2^{-1} \Rightarrow 1 * 0.5$

$\Rightarrow 0.5 \quad (0.5)_{10}$

* Example: $(0.011)_2 \equiv (0.375)_{10}$

0 . 011

$2^{-1}\ 2^{-2}\ 2^{-3}$

$\frac{1}{2}\ \frac{1}{4}\ \frac{1}{8}$

0.5 0.25 0.125

$\Rightarrow$

$(0 * \frac{1}{2}) + (1 * \frac{1}{4}) + (1 * \frac{1}{8})$

$= 0 + 0.25 + 0.125 = (0.375)_{10}$

* تحويل كسري النظام الثماني الى النظام العشري:

Octal to decimal

$(0.4)_8 \equiv (\ 0.5\ )_{10}$

. X X X X

$8^{-1}\ 8^{-2}\ 8^{-3}\ 8^{-4}$

$\frac{1}{8}\ \frac{1}{64}\ \frac{1}{512}$

$\Rightarrow$ 0 . 4 $\quad 4 * \frac{1}{8}$

$\frac{1}{8} \nearrow = \frac{4}{8} = 0.5$

$(0.5)_{10}$

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Example: $(0.74)_8 \equiv (0.\quad)_{10}$

0.74
$\downarrow \quad \downarrow$
$8^{-1} \quad 8^{-2}$

$= (7 * \frac{1}{8}) + (\frac{4}{64})$

$= \frac{7*8}{8*8} + \frac{4}{64} = \frac{60}{64} = (\dots)_{10}$

← تحويل كسر في النظام الثماني عشري الى النظام العشري

$(0.8)_{16} \equiv (0.5)_{10}$

$. \quad X \quad X \quad X \quad X \quad \dots\dots$
$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
$16^{-1} \quad 16^{-2} \quad 16^{-3} \quad 16^{-4}$

$\rightarrow 0.8 \Rightarrow 8 * \frac{1}{16}$
$\downarrow$
$16^{-1} = \frac{8}{16} = 0.5$

$\Rightarrow (0.5)_{10}$

Example: $(0.4C)_{16} \equiv (0.3)_{10}$

$0 . 4 \quad C$
$\downarrow \quad \downarrow$
$16^{-1} \quad 16^{-2}$

$\rightarrow \frac{4}{16} + \frac{12}{256} = 0.25 + 0.046 \approx (0.3)_{10}$

* تحويل الكسور من ثنائي الى ثماني :

Example: $(0.1)_2 \equiv (0.4\quad)_8$

I] Binary → Decimal → octal

$(0.1)_2 \rightarrow (0.5)_{10}$
$\downarrow 2^{-1} \rightarrow \frac{1}{2}$

$(0.5)_{10} \rightarrow (0.4)_8$
$\downarrow$
$*8$

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Subject :

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* Relation Between Binary and octal:

→ each 3 Binary digits (bits) ≡ 1 octal digit (bit)

$2^{③} = 8$ ⟶ $B^{[xxx]} = oct^{X}$

* Relation Between Binary and Hexadecimal.

→ each 4 Binary bits ≡ 1 hexadic Bit.

$2^{④} = 16$ $B^{[xxxx]} \equiv Hex^{(x)}$

* Relation Between octal and Hex: No relation

$8^{?} = 16$ X

{→ oct → Bin → Hex}

* Example: $(1011011)_2 = (133)_8$ لما بدي اجي اقسم لـ bits بالأعداد الصحيحة باخذ بهذا الاتجاه ←

[001][011][011]

001 → 421 → 1

011 → 421* → 3

011 → 421* → 3

$\Rightarrow (133)_8$

$(1\ 011\ 011)_2$

64 | 32 16 8 | 4 2 1

* (بطريقة وحدة اكثر) *

64 + 16 + 8 + 2 + 1

$(91)_{10} \Rightarrow$

8	91	Rem
8	11	3
8	1	3
	0	1

$\Rightarrow (133)_8$

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ثنائي لثماني

* Example: $(011\,110\,111\,011\,01)_2 = (16735)_8$

01 110 111 011 101
21 421 421 421 421

$(1\ 6\ 7\ 3\ 5)_8$

ثنائي لسادس عشر

* Example: $(1011\,1011\,0111)_2 = (BB7)_{16}$

1011 1011 0111
8421 8421 8421
(11) (11) 7
B B

(BB7)

ثنائي لسادس عشر

* $(133)_8 = (5B)_{16}$

$(1\ 33)_8 \Rightarrow$ Binary

XXX XXX XXX
421 421 421
001 011 011

$(01\,01\,1\,011)_2 \Rightarrow (\ )_{16}$
5 B (5B)

سادس عشر لثماني

* $(5B)_{16} \Rightarrow (\ 8\ )_8$

$(5\ B)_{16}$

XXXX XXXX
8421 8421
0101 1011

$(01\,01\,1011)_2$

$(1\ 3\ 3)_2$

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* Example: $(3CBFA)_{16} \equiv (0011\ 1100\ 1011\ 1111\ 1010)_2$

سادس عشر لثنائي

$3^{12}\ C^{11}\ B^{15}\ F\ A^{10}$

XXXX XXXX XXXX XXXX XXXX
8421 8421 8421 8421 8421
0011 1100 1011 1111 1010

ثنائي لثماني

000 111 100 101 111 111 010
421 421 421 421 421 421 421
7 4 5 7 7 2

$(745772)_8$

$\therefore (3CBFA)_{16} \equiv (745772)_8$

سادس عشر لثماني

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* $(0.1)_2 \Rightarrow (0.4)_8$

* الكسور

0.100
421

$\Rightarrow (0.4)_8$

* $(0.1)_2 \Rightarrow (0.8)_{16}$

0.1000
8421

$\Rightarrow (0.8)_{16}$

* Example: $(1001101 . 11011010)_2 \equiv$ octal

هاد العدد صحيح فبدي ←

001001101
421 421 421
1 1 5

هاد العدد كسري فبدي →

110110100
421 421 421
6 6 4

$\Rightarrow (115.664)_8$

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Subject Date No. 14

*Binary addition:

	sum	carry
0 + 0 =	0	0
0 + 1 =	1	0
1 + 0 =	1	0
1 + 1 =	0	1

*Example:

$61 + 23 \Rightarrow \equiv (1010100)_2$

1 1 1 1 1 1
1 1 1 1 0 1

1 0 1 1 1
1 0 1 0 1 0 0
64 32 16 8 4 2 1

*octal addition:

* $(37)_8 + (106)_8 = (145)_8$ *في النظام الثماني وعاي اجمع ارقام لحد 7 بدونما ما طلع فوق 7 وجرد ما تعدى الـ 7 بصير في عندي carry

1 1
1 0 6
3 7 +
1 4 5

وابدأ من اليمين 1 بس قيمته الاصلية 8 هاد العدد اكبر من 8 فبطرح منه 8 والي ظل بكونه هو الجواب

$6 + 7 = 13$

$13 - 8 = 5$

* $(36)_8 + (62)_8 = (120)_8$

1 1
6 2 +
3 6
1 2 0

$6 + 2 = 8$

$8 - 8 = 0$

$7 + 3 = 10$

$10 - 8 = 2$

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* $(76)_8 + (71)_8 = (167)_8$

1 7 6 $\quad 7 + 7 = 14$

7 1 + $\quad \frac{8-}{6}$

(1 6 7)

* $(145)_8 + (566)_8 = 733$

1 1

5 6 6 $\quad 6 + 5 = 11$

1 4 5 + $\quad \frac{8-}{3}$

7 3 3

* Hexadecimal Addition:: * اذا الرقم تجاوز عدد 15 فبنقص 16 ونحط الناتج

→ $(40C)_{16} + (97)_{16} = (4A3)_{16}$ ونحط 1 carry وقيمته بتكون 16

1

4 0 C $\quad$ C

$\quad$ 12 $+ 7 = 19 - 16 = 3$

9 7 +

4 A 3

10

→ $(BFC)_{16} + (8AB)_{16} \rightarrow (14A7)_{16}$

1 1

1 B F C $\quad$ C B

$\quad$ 12 $+ 11 = 23 - 16 = 7$

8 A B + $\quad$ F A

$\quad$ 15 $+ 10 + 1 = 26 - 16 = 10 \rightarrow A$

1 4 A 7 $\quad$ B

$\quad$ 11 $+ 9 = 20 - 16 = 4$

→ $(EF4)_{16} + (DC)_{16} \rightarrow (FD0)_{16}$

1 1

E F 4 $\quad 4 + 12 = 16 - 16 = 0$

D C + $\quad 15 + 13 + 1 = 29 - 16 = 13 \rightarrow D$

F D 0 $\quad 14 + 1 = 15 \rightarrow F$

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* Binary subtraction:

* عملية الطرح بالنظام الثنائي

	sum	Borrow
0 - 0 =	0	0
0 - 1 =	1	1
1 - 0 =	1	0
1 - 1 =	0	0

ماي قيمتها 2

$$\begin{array}{r} {}^{0}\not{1}\,{}^{10}\not{0} \\ 1 \\ \hline 1 \end{array}$$

$$\rightarrow \begin{array}{r} {}^{0}\not{1}\,{}^{10}\not{0}\,{}^{1}\not{1}\,{}^{10}\not{0}\,{}^{1}\not{1}\,{}^{10}\not{0} \\ 1\;0\;1\;0\;1\;1 \\ \hline (1\;1\;1\;1\;0\;1)_2 \end{array}$$

$$\rightarrow \begin{array}{r} {}^{0}\not{1}\,{}^{10}\not{0}\,{}^{0}\not{1}\,{}^{10}\not{0} \\ 1\;1\;0\;1\;1 \\ \hline (1\;0\;0\;1\;1)_2 \end{array}$$

* octal subtraction:

* عملية الطرح بالنظام الثماني

← هون لما بدي استعير من الخانة اللي جنبها بستعير 1 بتكون قيمتها 8

$$\rightarrow (733 - 145)_8$$

$$\begin{array}{r} {}^{6}\not{7}\,{}^{12}\not{3}\,{}^{13}\not{3} \\ 1\;4\;5 \\ \hline (5\;6\;6)_8 \end{array}$$

← ماي جنبي 13 ماي عبارة عن 8 و 3 يعني 8+3 = 11

بالنظام العشري فقط بدي اطرح 13-5 حتى 11-5

$$\begin{array}{r} {}^{8}\not{1}\,{}^{10}\not{2} \\ 4 \\ \hline 6 \end{array}$$

$$\rightarrow (120 - 36)_8$$

$$\begin{array}{r} {}^{0}\not{1}\,{}^{11}\not{2}\,{}^{10}\not{0} \\ 3\;6 \\ \hline (6\;2)_8 \end{array}$$

$^{8}\not{1}\,{}^{10}0 \rightarrow 8 - 6$

$9 - 3 = 6$

* Hexadicimal subtraction:-

← هون لما بدي استعير من الخانة اللي جنبها بستعير 1 بتكون قيمتها 16

$$\rightarrow (FD0 - DC)_{16}$$

$$\begin{array}{r} {}^{E}\not{F}\,{}^{1C}\not{D}\,{}^{10}\not{0} \\ D\;C \\ \hline E\;F\;4 \end{array}$$

$^{16}\not{1}\,{}^{1}0 \rightarrow 16 - 12 = 4$

$^{16}\not{1}\,{}^{1}C \rightarrow 12 + 16 = 28 - 13 = \boxed{15} \rightarrow F$

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Example: 16¹ 16⁰
13 19 17

X Y A 7 → 23 - 12 = 1111 → B

B F C → 25 - 15 = 1010 → A

$(8\ A\ B)_{16}$ → 19 - 11 = 8

---

* BCD: Binary coded decimal

$(17)_{10}$

BCD

XXXX → (0001 0111)
8 4 2 1

* دائماً الـ BCD يمثل بـ 4 خانات

باخد كل 4 خانات سوا

0001 0111

← هاد العدد اكبر من ١٠ فبالتالي فابقدر اعثله.

→ Decimal	BCD (8 4 2 1)
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001

مسموح لحد ٩ بس

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هدول عبارة عن 2 digits فبمثل كل وحدة لحال (3 7)

xxxx xxxx

0011 0111

* في عملية الجمع اذا طلع عندي الناتج بيع الجمع اكبر من 9 أو طلع في عندي carry فبيكون باخر العملية فاذا داخل بيجمع الجواب رقم 6 ← 0110 عشان أو صل لـ 15 وأقدر اضيف خانة جديدة

(1 3 5)

0001 0011 0101

* BCD Addition :

ex): 4 + 5 ⇒ 9

0100 0101 = 1001

0101

+ 0100

1001 → صار الرقم 9

example: 4 + 8 ⇒ 12

0100

+ 1000

1100

+ 0110

10010 → 2 ، 1

(12)

* example: 8 + 9 ⇒ 17

1000

+ 1000

10001

+ 0110

10111 → 7 ، 1

* الطريقة الأسهل:

8 + 9 = 17

1000 0111

∴ الجواب (00010111)

صاي الطريقة لما يطلب مني أحل بالقانون صاي أو بالطريقة هاي اطالما طلب بقدر استخدم الطريقة الأسهل

example: 184 + 576 → 760 → (0111 0110 0000)

0100 0111 0110

+ 0001 1000 0100

0110 1111 1010 → باخذ كل وحدة لحالها

لـ صاي طابعتلج — اكبر من 9 — اكبر من 9

+ 0110 + 0110

اضيفلها 6

0111 0110 0000

7 6 0

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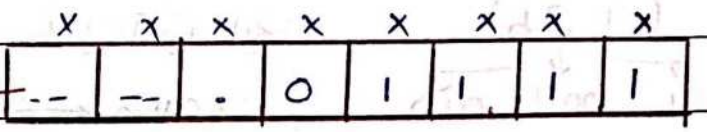
* Signed numbers:

\+ −

+15 −15

→ 1 byte = 8 bit



Register 8 bit

+15

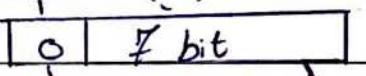
خانة الاشارة

1) 0 1111 ⇒ +15

2) 0 000 1111 ⇒ +15 (8bit)

* اذا كانت خانة الاشارة فيها 0 معناها موجب

واذا كانت خانة الاشارة فيها 1 معناها سالب



0 ← هدول للإشارة

7 bit ← هدول الرقم

Sign ← سالب

3) 1 1111 ⇒ −15

1 000 1111 ⇒ −15 (8bit)

* Signed numbers can be represented by:

1) ones complement (1st complement).

2) Two's complement (2nd complement).

* 1's complement: بتم عن طريقه عكس اسها toggle والي هي عكس قلب الـ 1 لـ 0 والـ 0 لـ 1

example:

$(-4)_{10}$ ⇒ 0000 0100

↓ Toggle

negative (1111 1011) ← 1st comp

(0000 0100)

هي النتيجة

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Subject Date No. 20

* 2's complement:

$(-13)_{10} \Rightarrow$ 00001101

(1) 1's comp

11110010 1st comp

1 +

(1) 11110011 → 2nd comp negative

(2) 00001101

11110011

* الطريقة الثانية هي اني ابحث عن أول 1 بلا فيها بالرقم الاصلي وبيم ما كانت اليم بتكون أول وحدة للباقي من اليسار لليمين بعديها بنزلها كما هي بعديم بقلب الارقام الي بعدها.

* الطريقة الاولى هي اني اجيب الـ 1st comp العدد وبعديم اضيف 1

* Example:: Find the 1st complement and the 2nd complement for the # -25

00011001

سالب (1) 11100110 1st comp

1+

11100111 2nd comp

11100111 دايركت

* Gray code:

* الـ Gray code علاقة دائماً مع الـ Binary

$(10110)_2 \equiv (11101)_{Gr}$

* لما بدي اعمل Binary لـ Gray code:

1. بنزل اول 1 محاضر

2. ببلش اجمع كل بت مع الي جنبه وبنزل الجواب واذا طلع في عندي carry بهمله

10110

(11101)Gray

$(11000110)_2 \Rightarrow (10100101)_{Gr}$

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*convert Gray to Binary:

→ $(1\ 1\ 1\ 0\ 1)_{Gr} \Rightarrow (1\ 0\ 1\ 1\ 0)_2$

$(1\ 0\ 1\ 1\ 0)_2$

→ $(1\ 0\ 1\ 0\ 0\ 1\ 0\ 1)_{Gr} \Rightarrow (1\ 1\ 0\ 0\ 0\ 1\ 1\ 0)_2$

$(1\ 1\ 0\ 0\ 0\ 1\ 1\ 0)_2$

*التحويل من Gray code لـ Binary:

١. بنزل أول رقم كما هو

٢. بجمع الرقم اللي بالجواب للعدد اللي بعده وبحط الناتج

*Excess -3 code:

→ Example:

*$(21)_{10} \equiv (01010100)_{Ex-3}$

2 → 3+ → 0101

1 → 3+ → 0100

*$(11101)_2 \equiv (01011100)_{Ex-3}$

16 8 4 2 1

11101 $\Rightarrow (29)_{10}$

$(29)_{10} \Rightarrow (01011100)_{Ex-3}$

2 → 3+ → 0101

9 → 3+ → 1100

*Relationship with decimal by adding 3 For each digit.

هو برضه حينفع نحول على طول لانه كلنا انه

صار الكود علاقة مع الـ decimal بس

فبحول من Binary لـ decimal وبعدين لكل

حل

*9's complement:

→ Example: Find the 9's comp fo 1234.

9 9 9 9

1 2 3 4 -

8 7 6 5

* لما بدي احل على الـ 9's comp باخذ مقابل كل digit رقم 9 وبقوم بعملية طرح السطات

هذا الرقم اللي بالسؤال * علاقة مع الـ decimal فقط *

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* 10's complement:: لحابي احول على الـ 10's comp بعمل نفس خطوات الـ 9's comp وبالآخر بضيف 1 للناتج

Example::

Find the 10's complement for 8634

↓ 9's complement +1

9999
8634 -
1365
1 +
1366 → 10's complement

* Example:: Find $(96-23)_{10}$ using::

1) 9's complement:: بالمثال هاد الرقم اللي بدي اقوم بطرحه هو اللي بدي اجيبله الـ 9's comp واللي هو هون عنا اياه 23

9's complement
99 -
23
76

96
76 +
172
overflow ← (1)72 ← هيك بستخدم الـ overflow لما اعمل عملية بينهم

1172
↳1 +
173 ← بكون في عندي overflow وبجمع العدد

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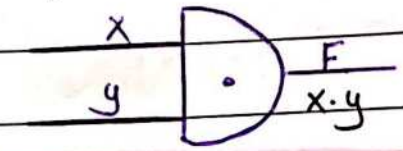
* Logical gates:

* AND: بوابة و

2 inputs and تستقبل مدخلين



تمثل بعملية الضرب

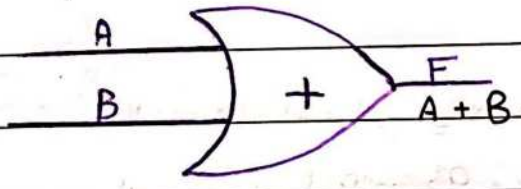
*The only time the output is positive is when all the inputs are positive.

Truth table:

A	B	F (A.B)
0	0	0
0	1	0
1	0	0
1	1	1

* OR:

2 inputs OR



تمثل بعملية الجمع

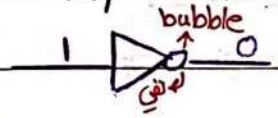
Truth table:

A	B	F(A+B)
0	0	0
0	1	1
1	0	1
1	1	1

* The output will be positive when any one or all inputs are positive.

* Not:

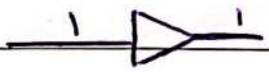
1 input not



$A' / \bar{A}$

Truth table:

A	Y($\bar{A}$)
0	1
1	0



buffer

هي بستخدمها لأكون بين أجزاء العملية لسبب ما.

*The output is the opposite of the input

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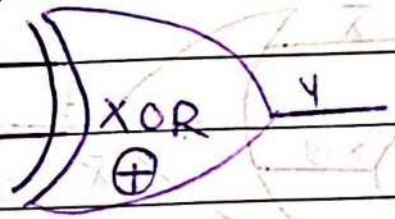
Subject

Date

* XOR:

Exclusive OR (خاصة OR)



A	B	y(A⊕B)
0	0	0
0	1	1
1	0	1
1	1	0

اذا المدخلين متشابهين الناتج ← 0

اذا المدخلين مختلفين الناتج ← 1

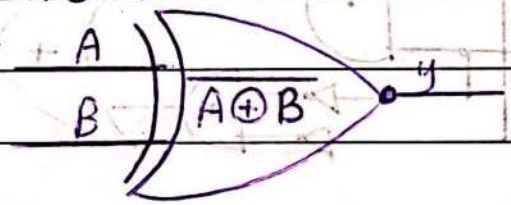
XOR تكافئ $\bar{A}B + A\bar{B} = A \oplus B$

*The only time the output is positive is when the inputs are not the same

* XNOR:

Exclusive NOR (عبارة عن XOR ونفيه)

$\overline{A \oplus B}$



A	B	$\overline{A \oplus B}$
0	0	1
0	1	0
1	0	0
1	1	1

متشابهين ← 1

مختلفين ← 0

$A \oplus A = 0$

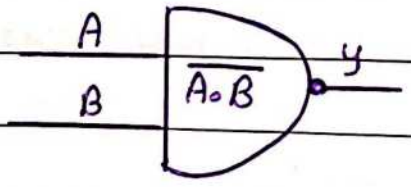
$A \oplus 0 = A$

$A \oplus 1 = \bar{A}$

*The only time the output is positive is when all the inputs are negative

* NAND:

(AND + Not)



A	B	$\overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

*The output is positive provided all the inputs are not positive

NOTEBOOK

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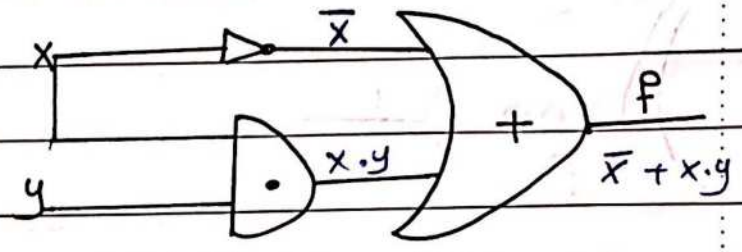
لازم تحتوي على جميع المتغيرات x و y

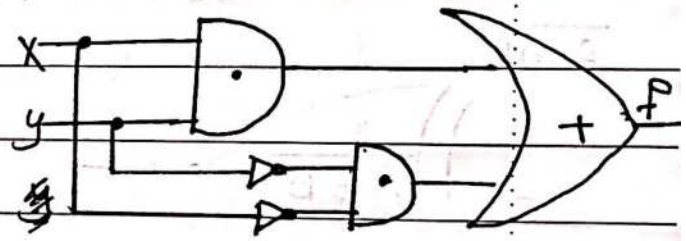
* Example:: $F(x,y) = \bar{x} + xy$ Truth table

x	y	$\bar{x}$	$x \cdot y$	$\bar{x} + x \cdot y$
0	0	1	0	1
0	1	1	0	1
1	0	0	0	0
1	1	0	1	1



* Example:: $F(x,y) = xy + \bar{x}\bar{y}$

x	y	$\bar{x}$	$\bar{y}$	$x \cdot y$	$\bar{x} \cdot \bar{y}$	$x \cdot y + \bar{x} \cdot \bar{y}$
0	0	1	1	0	1	1
0	1	1	0	0	0	0
1	0	0	1	0	0	0
1	1	0	0	1	0	1



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Boolean Algebra:

* الجبر البولي *

- has 2 values:: الجبر البولي هو الذي يهتم بالحياة بـ 0 أو 1

False (0)

True (1)

* Laws::

1) Commutativity of addition and multiplication $(A+B)=(B+A)$

$AB=BA$

2) Associativity of addition and multiplication $(A+(B+C))=((A+B)+C)$

$(A(BC))=((AB)C)$

3) Distributivity of multiplication over addition $(A(B+C))=AB+AC$

$(A+B)(C+D)=AC+AD+BC+BD$

← قواعد إضافية:

A ← وإنما القيمة بـ 0 أو 1

$A.0=0$

$A.1=A$ ← $0.1=0$, $1.1=1$

$A+0=A$ ← $0+0=0$, $1+0=1$

$A+1=1$

$A.A=A$

$A+A=A$

4) Absorption $A(A+B)=A$

$A+AB=A$

$A+BC=AB+AC$

$A.\bar{A}=0$

$A+\bar{A}=1$

$\bar{\bar{A}}=A$

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$\bar{+} \rightarrow \cdot$

$\bar{\cdot} \rightarrow +$

* Demorgan's Law:

$\overline{(AB)} = \bar{A} + \bar{B}$

$\overline{(A+B)} = \bar{A}\,\bar{B}$

$(\overline{ABC}) = \bar{A} + \bar{B} + \bar{C}$

$(\overline{AB}) = (\bar{A} + \bar{B})$

A	B	$\overline{AB}$	$\bar{A}+\bar{B}$
0	0	1	1
0	1	1	1
1	0	1	1
1	1	0	0

$\overline{(A+B)} = (\bar{A} \cdot \bar{B})$

A	B	$\overline{A+B}$	$\bar{A}\cdot\bar{B}$
0	0	1	1
0	1	0	0
1	0	0	0
1	1	0	0

* $(A+B+C)' = \bar{A} \cdot \bar{B} \cdot \bar{C}$

$\overline{(A+B+C)}$ تكافئ ←

* simplify:

$AB + A(B+C) + B(B+C)$ بنوزع الـ A والـ B ع الأقواس

$AB + AB + AC + BB + BC$ بنجمع المتشابه

$AB + AC + B + BC$ بناخد B عامل مشترك

$AB + AC + B(1 + C)$

$AB + AC + B$ بناخد B عامل مشترك

$B(A + 1) + AC$

$\boxed{B + AC}$

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*simplify:

$(A+C)(AD+A\bar{D})+AC+C$

$(A+C)A(D+\bar{D})+AC+C$ بناخد A عامل مشترك

$(A+C)A+C(A+1)$ بناخد C عامل مشترك

$(A+C)A+C$ بنوزع A ع القوس

$AA+AC+C$

$A+AC+C$

$A(1+C)+C$ بناخد A عامل مشترك

$\boxed{A+C}$

* simplify:

$(A+B)(A+C)$ بنفك الأقواس

$AA+AC+BA+BC$

$A+AC+BA+BC$ بناخد A عامل مشترك

$A(1+C+B)+BC$

$\boxed{A+BC}$

* simplify:

$(\bar{A}+B)(A+B)$

$\bar{A}A+\bar{A}B+BA+BB$

$0+\bar{A}B+BA+B$

$B(\bar{A}+A+1)$

$\boxed{B}$

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*simplify::

$$(A+\bar{B}) \cdot (C+\bar{D})$$

① ↓ ②

$$\overline{(A+\bar{B})} \cdot \overline{(C+\bar{D})}$$

$$(\bar{A} \cdot \bar{\bar{B}}) + (\bar{C} \cdot \bar{\bar{D}})$$

$$(\bar{A} \cdot B) + (\bar{C} \cdot D)$$

$$\boxed{\bar{A}B + \bar{C}D}$$

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* Conanical and standard forms:-

→ Conanical:
Each term has all the Literals (المتغيرات) *يحتوي على جميع المتغيرات بكل اشكالهم سواء بالإثبات او بالنفي*
Ex→ F(AB) ⇒ AB + $\bar{A}\bar{B}$

→ Standard:
Each term doesn't have all literals *يحتوي على أحد المتغيرات والمتغير الآخر يكون مفقود بعض سواء بالإثبات أو بالنفي*
Ex→ F(AB) = A + $\bar{A}$B
B missed

* minterms:
The literals expressed as product of literals. *كل المتغيرات الموجودة عندي بالضرب*
في انه تكون العلاقة بسبب أي متغير التي هي ضرب (•)
⇒ xyz minterm
⇒ $\bar{x}y\bar{z}$ minterm
*There when A=0 it expressed as $\bar{A}$ and when A=1 it expressed as A.

So. A B C ⇒ minterm? *ليش الـ ٣ متغيرات اخدوا؟ لانهم الـ ٣ مثبتات*
1 1 1
2^2 2^1 2^0
4 + 2 + 1
= |7| ⇒ minterm 7
m7

→ Example: f(A, B, C) = $\bar{A}\bar{B}\bar{C}$ + AB$\bar{C}$ write it as minterms.
F(A, B, C)

$\bar{A}\bar{B}\bar{C}$ AB$\bar{C}$
0 0 0 1 1 0
2^2 2^1 2^0 4 2 1
m_0 m_6

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Maxterms: the literals are expressed as sum of literals.

⇒ $(X + y + z)$ maxterm

⇒ $(\bar{X} + \bar{y} + \bar{z})$ maxterm

مثل المتغيرات الموجودة حتى بالجمع، يعني انه تكون العلاقة بين أي متغير والتالي (+)

→ Here when A = 0 it is expressed as A and when A = 1 it's expressed as $\bar{A}$

→ $A + \bar{B} + C$

0 ↑ 1 ↑ 0

M_2 → Maxterm2

→ $\bar{A} + B + \bar{C}$

1 0 1

M_5 ⟶ Maxterm5

* The logical function can be expressed by 2 ways:

1) Sum of product: (SOP) جمع المضاربين.

$Xyz + \bar{x}\bar{y}\bar{z} + \text{-----}$

≡ Sum of minterms $\Sigma m(*,*,*,*)$

2) product of sum: (POS) ضرب المجاميع

$(\bar{X} + \bar{y} + z)(X + y + \bar{z}) \ldots$

≡ Product of sum $\pi M(*,*,*)$

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$F(x,y,z) \Rightarrow$ Truth table

	x	y	z	minterm	Maxterm
0	0	0	0	$\bar{x}\cdot\bar{y}\cdot\bar{z}$	$x+y+z$
1	0	0	1	$\bar{x}\cdot\bar{y}\cdot z$	$x+y+\bar{z}$
2	0	1	0	$\bar{x}\cdot y\cdot\bar{z}$	$x+\bar{y}+z$
3	0	1	1	$\bar{x}\cdot y\cdot z$	$x+\bar{y}+\bar{z}$
4	1	0	0	$x\cdot\bar{y}\cdot\bar{z}$	$\bar{x}+y+z$
5	1	0	1	$x\cdot\bar{y}\cdot z$	$\bar{x}+y+\bar{z}$
6	1	1	0	$x\cdot y\cdot\bar{z}$	$\bar{x}+\bar{y}+z$
7	1	1	1	$x\cdot y\cdot z$	$\bar{x}+\bar{y}+\bar{z}$

demorgan law

$\bar{x}\cdot\bar{y}\cdot\bar{z} \Rightarrow \overline{(x\cdot y\cdot z)}$

$\bar{\bar{x}}+\bar{\bar{y}}+\bar{\bar{z}}$

$x+y+z$

0 0 0

m_0

minterm ← m_0

maxterm ← M_0

x	y	z	minterm	
0	0	0	$\bar{x}\cdot\bar{y}\cdot\bar{z}$	$\rightarrow m_0$
0	0	1	$\bar{x}\cdot\bar{y}\cdot z$	$\rightarrow m_1$
0	1	0	$\bar{x}\cdot y\cdot\bar{z}$	$\rightarrow m_2$
0	1	1	$\bar{x}\cdot y\cdot z$	$\rightarrow m_3$
1	0	0	$x\cdot\bar{y}\cdot\bar{z}$	$\rightarrow m_4$
1	0	1	$x\cdot\bar{y}\cdot z$	$\rightarrow m_5$
1	1	0	$x\cdot y\cdot\bar{z}$	$\rightarrow m_6$
1	1	1	$x\cdot y\cdot z$	$\rightarrow m_7$

x	y	z	maxterm	
0	0	0	$x+y+z$	$\rightarrow M_0$
0	0	1	$x+y+\bar{z}$	$\rightarrow M_1$
0	1	0	$x+\bar{y}+z$	$\rightarrow M_2$
0	1	1	$x+\bar{y}+\bar{z}$	$\rightarrow M_3$
1	0	0	$\bar{x}+y+z$	$\rightarrow M_4$
1	0	1	$\bar{x}+y+\bar{z}$	$\rightarrow M_5$
1	1	0	$\bar{x}+\bar{y}+z$	$\rightarrow M_6$
1	1	1	$\bar{x}+\bar{y}+\bar{z}$	$\rightarrow M_7$

* Example:: Express the following boolean function using sum of minterms.

1) conanical (canonical) $F(A,B) = \bar{A} + AB$

$\bar{A}(\bar{B}+B) + AB \Rightarrow \bar{A}B + \bar{A}\bar{B} + AB$

0 1 0 0 1 1

m_1 m_0 m_3

$F(A,B) = \Sigma m(m_0, m_1, m_3)$

* جدول minterm لازم الاقوس يكون عشان كل بيعة

* conanical ال

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* Example: $F(A,B,C) = A\bar{B} + \bar{C}$

$A\bar{B} \leq \bar{C}$

$\bar{C} \rightarrow \bar{A}\bar{B}, \bar{A}B, A\bar{B}$

$A\bar{B}\bar{C} + A\bar{B}C + \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + AB\bar{C}$

$F(A,B,C) = \Sigma m(4,5,0,2,6)$

* Example: $F(w,x,y,z) = \bar{w}y$

ordering

ordinary الترتيب

$w\ x\ y\ z$

$\bar{w}\ \bar{x}\ y\ \bar{z} \Rightarrow (0010) \Rightarrow m2$

$\bar{w}\ \bar{x}\ yz \rightarrow 0011 \Rightarrow m3$

$\bar{w}\ x\ y\bar{z} \rightarrow 0110 \Rightarrow m6$

$\bar{w}\ x\ yz \rightarrow 0111 \Rightarrow m7$

$F(w,x,y,z) = \Sigma m(2,3,6,7)$

* $F(x,y,z) = \Sigma m(1,2,3)$ Algebric Function

$m_1 + m_2 + m_3$

$\bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz$

NOTE B

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* Express the following function using Π of maxterm

$F(A,B) = (A+B)(A+\bar{B})$

$A+B$ ← $A+\bar{B}$ ↙

$0 \quad 0$ $\quad$ $0 \quad 1$

M_0 $\quad$ M_1

$F(A,B) = \Pi M(0,1)$

* $F(x,y,z) = x+y$ → هون ما في z بالحالتين بتعوضها عشان يتحول لـ canonical صورة ما في standard فبالتالي لازم نضيف canonical حتى

$x+y+z\bar{z}$

$(x+y+z).(x+y+\bar{z})$

$0\ 0\ 0 \quad 0\ 0\ 1$

$M_0 \quad M_1$

$F(x,y,z) = \Pi M(0,1)$

→ Karnaugh maps: simplification "minimization"

# of squares (هي عبارة عن عدد المربعات)

it depends on the # of variables (عدد المربعات يعتمد على عدد المتغيرات)

↳ How ?!

# of k map squares = 2^n , since n is the number of variables

→ 2 variables ⇒ k map will be $2^2 = \boxed{4}$ squares.

→ 3 variables ⇒ k map will be $2^3 = \boxed{8}$ squares.

→ 4 variables ⇒ k map will be $2^4 = \boxed{16}$ squares.

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Subject Date No.

* 2 variables → F(x,y) $\boxed{n=2}$

→ K map squares here will be $2^2 = \boxed{4}$

x \ y	$\bar{y}$ 0	y 1
$\bar{x}$ 0	$\bar{x}\bar{y}$ 00 m_0	$\bar{x}y$ 01 m_1
x 1	$x\bar{y}$ 10 m_2	xy 11 m_3

⟹

x \ y	0	1
0	m_0	m_1
1	m_2	m_3

* 3 variables → F(x,y,z) $\boxed{n=3}$

→ K map squares here will be $2^3 = \boxed{8}$

x \ yz	$\bar{y}\bar{z}$ 00	$\bar{y}z$ 01	yz 11	$y\bar{z}$ 10
$\bar{x}$ 0	$\bar{x}\bar{y}\bar{z}$ 000 m_0	$\bar{x}\bar{y}z$ 001 m_1	$\bar{x}yz$ 011 m_3	$\bar{x}y\bar{z}$ 010 m_2
x 1	$x\bar{y}\bar{z}$ 100 m_4	$x\bar{y}z$ 101 m_5	xyz 111 m_7	$xy\bar{z}$ 110 m_6

→

x \ yz	00	01	11	10
0	m_0	m_1	m_3	m_2
1	m_4	m_5	m_7	m_6

* 4 variables → F(w,x,y,z)

→ K map squares here will be $2^4 = \boxed{16}$

wx \ yz	00	01	11	10
00	m_0	m_1	m_3	m_2
01	m_4	m_5	m_7	m_6
11	m_{12}	m_{13}	m_{15}	m_{14}
10	m_8	m_9	m_{11}	m_{10}

* Filling k map:

minterms → SOP $\boxed{1}$

Maxterms → POS $\boxed{0}$

NOTEBOOK

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Subject Date No. 36

*Example #1: $F(x,y) = \bar{x} + xy$ minterms

$\bar{x} \begin{matrix} \nearrow y \\ \searrow \bar{y} \end{matrix} \Rightarrow \bar{x}y + \bar{x}\bar{y} + x\bar{y}$

m_1 m_0 m_3

01 00 11

$F(x,y) = \Sigma m(0,1,3)$

n=2

x \ y	0	1
0	m_0 1	m_1 1
1	m_2 0	m_3 1

*Example #2: $F(x,y) = \bar{y}$

minterms → $\bar{y} \begin{matrix} \nearrow x \\ \searrow \bar{x} \end{matrix} \Rightarrow \bar{x}\bar{y} + x\bar{y}$

m_0 m_2

00 10

$F(x,y) = \Sigma m(0,2)$

x \ y	0	1
0	m_0 1	m_1 0
1	m_2 1	m_3 0

*Example #3: $F(x,y,z) = \bar{x}y$

minterms → $\bar{x}y \begin{matrix} \nearrow z \\ \searrow \bar{z} \end{matrix} \Rightarrow \bar{x}y\bar{z} + \bar{x}yz$

m_2 m_3

010 011

$F(x,y,z) = \Sigma m(2,3)$

x \ yz	00	01	11	10
0	m_0 0	m_1 0	m_3 1	m_2 1
1	m_4 0	m_5 0	m_7 0	m_6 0

*Example #4: $F(x,y,z) = \Sigma m(1,2,5,7)$

x \ yz	00	01	11	10
0	m_0 0	m_1 1	m_3 0	m_2 1
1	m_4 0	m_5 1	m_7 1	m_6 0

*Example #5: $F(w,x,y,z) = \Sigma m(1,2,5,7,9,11,12)$

wx \ yz	00	01	11	10
00	0 0	1 1	3 0	2 1
01	4 0	5 1	7 1	6 0
11	12 1	13 0	15 0	14 0
10	8 0	9 1	11 1	10 0

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* How to minimize:

→ gathering the adjacent (1's) or (0's)

→ 2 1's together
→ 4 1's together
→ 8 1's together
→ 16 1's together
} → SOP

→ 2 0's together
→ 4 0's together
→ 8 0's together
→ 16 0's together
} → POS

SOP ↑
* اذا بدي اشتغل بدلالة الـ minterm بدي اعمل grouping لـ (1's)

POS ←
* اذا بدي اشتغل بدلالة الـ Maxterm بدي اعمل grouping لـ 0's

* Ex:

x \ y	$\bar{y}$ (0)	y (1)
$\bar{x}$ (0)	1 (m_0)	1 (m_1)
x (1)	0 (m_2)	0 (m_3)

→ horizontally its similar → as $\bar{x}$
→ vertically → no similarity
→ $\bar{x}$

اثبات انو الحل صح:
$m_0 + m_1 \rightarrow \bar{x}\bar{y} + \bar{x}y$
$\bar{x}(\bar{y}+y) = \bar{x}$

← هيك بدي اعمل grouping واخلص بدي اتطلع على المتغيرات وشو فيه اللي ما اختلفت قيمته القيمة اللي ما فضلت ثابتة في جميع المربعات اللي عملنا لها grouping بقوم بالغائها *

x \ y	$\bar{y}$ (0)	y (1)
$\bar{x}$ (0)	1 (m_0)	1 (m_1)
x (1)	0 (m_2)	1 (m_3)

→ $F(x,y) = \bar{x}\bar{y} + \bar{x}y + xy$
$F(x,y) = \Sigma m(0,1,3)$ *
→ $\bar{x} + y$

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x \ yz	$\bar{y}\bar{z}$ 00	$\bar{y}z$ 01	yz 11	$y\bar{z}$ 10
$\bar{x}$ 0	m_0 1	m_1 1	m_3 0	m_2 1
x 1	m_4 1	m_5 1	m_7 0	m_6 0

$F(x,y,z) = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + x\bar{y}z$

$F(x,y,z) = \sum m(0,1,2,4,5)$

$\bar{y}$, $\bar{x}\bar{z}$

$\rightarrow \bar{y} + \bar{x}\bar{z}$

x \ yz	$\bar{y}\bar{z}$ 00	$\bar{y}z$ 01	yz 11	$y\bar{z}$ 10
$\bar{x}$ 0	1	1	1	1
x 1	m_4 0	1	1	1

$F(x,y,z) = \sum m(0,1,2,3,5,6,7)$

$\bar{x}$, z , y

$\rightarrow \bar{x} + z + y$

x \ y	$\bar{y}$ 0	y 1
$\bar{x}$ 0	0	(1)
x 1	(1)	0

$F(x,y) = \sum m(1,2)$

لا اختصار

$\rightarrow \bar{x}y + x\bar{y}$

x \ yz	$\bar{y}\bar{z}$ 00	$\bar{y}z$ 01	yz 11	$y\bar{z}$ 10
$\bar{x}$ 0	1	0	0	1
x 1	1	0	0	1

$F(x,y,z) = \sum m(0,2,4,6)$

$\bar{z}$

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$f(x,y,z) = \Sigma m(0,2,3,4,5,7)$

x \ yz	$\bar{y}\bar{z}$ 00	$\bar{y}z$ 01	yz 11	$y\bar{z}$ 10
$\bar{x}$ 0	1	0	1	1
x 1	1	1	1	0

$\bar{y}\bar{z}$ xz $\bar{x}y$

$F(x,y,z) \rightarrow \bar{y}\bar{z} + xz + \bar{x}y$

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	1	1	1	1
$\bar{A}B$ 01	0	1	1	1
AB 11	0	1	1	1
$A\bar{B}$ 10	0	1	1	0

$\bar{A}\bar{B}$ D BC

$F(A,B,C,D) = \bar{A}\bar{B} + D + BC$

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	1	1	1	1
$\bar{A}B$ 01	0	0	0	1 → $BC\bar{D}$
AB 11	1	1	0	1
$A\bar{B}$ 10	0	1	1	0

$\bar{A}\bar{B}$ $AB\bar{C}$ $\bar{B}D$

$F(A,B,C,D) = \bar{A}\bar{B} + AB\bar{C} + BC\bar{D} + \bar{B}D$

x \ yz	$\bar{y}\bar{z}$ 00	$\bar{y}z$ 01	yz 11	$y\bar{z}$ 10
$\bar{x}$ 0	0	1	0	1 → $\bar{x}y\bar{z}$
x 1	1	1	0	0

$x\bar{y}$ $\bar{y}z$

∴ $F(x,y,z) = x\bar{y} + \bar{y}z + \bar{x}y\bar{z}$

NOTEBOOK

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← هون لاني بدي اشتغل pos بكونه الـ 0 مثبت والـ 1 نفي وبكونه بدي اعمل grouping للـ 0s

x \ yz	yz 00	yz̄ 01	ȳz̄ 11	ȳz 10
x 0	1	0	0	1
x̄ 1	1	0	0	1

z̄ → pos

x \ yz	yz 00	yz̄ 01	ȳz̄ 11	ȳz 10
x 0	0	1	1	1
x̄ 1	0	0	1	0

(y+z) (x̄+y) (x̄+z)

$$F(x,y,z) = (y+z)(\bar{x}+y)(\bar{x}+z)$$

---

* Don't Care :

الـ Don't care ممكن عادة يرمز X أو يرمز له والمقصود به انه هذه الخانة ممكن انه يتحول ويكون اما 1 أو 0 حسب الظرف والحاجة.

x \ yz	ȳz̄ 00	ȳz 01	yz 11	yz̄ 10
x̄ 0	0	X	X	X
x 1	0	0	1	1

مع استخدام الـ dont care بكونه

الناتج ← y

بدون استخدام الـ dont care بكونه

الناتج ← xy

فبالتالي استخدام الـ dont care وفر علينا استخدام الكيت مع بوابة وهاد هو الاختصار الأفضل.

wx \ yz	ȳz̄ 00	ȳz 01	yz 11	yz̄ 10
w̄x̄ 00	0	0	0	0
w̄x 01	0	0	1	1
wx 11	X	X	X	X
wx̄ 10	1	1	X	X

w ، xy

$$F(w,x,y,z) = w + xy$$

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wx \ yz	$\bar{y}\bar{z}$ 00	$\bar{y}z$ 01	yz 11	$y\bar{z}$ 10
$\bar{w}\bar{x}$ 00	X	1	1	X
$\bar{w}x$ 01	0	X	1	0
wx 11	0	0	1	0
$w\bar{x}$ 10	0	0	1	0

$\bar{w}\bar{x}$ yz

$\therefore F(wx,y,z) = yz + \bar{w}\bar{x}$

* Example: $F(wxy) = \Sigma m(1,3,7) + \Sigma d(2,5)$

هي وضعناها عشان انوفي عندي don't care بالمربعين (2,5)

w \ xy	$\bar{x}\bar{y}$ 00	$\bar{x}y$ 01	xy 11	$x\bar{y}$ 10
$\bar{w}$ 0	0	1	1	X
w 1	0	X	1	0

y

$\therefore F(wxy) = y$

AB \ CD	$\bar{C}\bar{D}$ 00	$\bar{C}D$ 01	CD 11	$C\bar{D}$ 10
$\bar{A}\bar{B}$ 00	0	1	0	0
$\bar{A}B$ 01	X	1	0	1
AB 11	1	1	0	1
$A\bar{B}$ 10	0	0	0	0

$B\bar{C}$ $\bar{A}\bar{C}D$ $B\bar{D}$

$\therefore F(ABCD) = B\bar{C} + B\bar{D} + \bar{A}\bar{C}D$

NOTE

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*combinational circuits: (الدارة التجميعية)
combine 2 gates or more to construct a curcuit that helps.

→ mechanism:
1) Scenario Analysis * تحديد الـinputs والـoutputs وشو الهدف من الدارة كلها
2) Building the T.T * بناء الـtruth table وبحدد الـmaxterms والـminterms
3) Finding the function by K map. * بقوم باختصار للمعادلة بأقل عدد ممكن باستخدام

① Adders:
Half adder
Full adder
Half subtractor
Full subtractor
4 Bit Full adder
Adder / subtractor

② multiplexer.
③ De - multiplexer.
④ Decoder.
⑤ Encoder.
⑥ Comparator.

*Example: Design a circuit that produces "1" as output if the number is odd otherwise produces "0" assume that the number is 2 bit. * لازم يحددلي بالسؤال كم bit بده عشان اعرف اختار الـinputs

→ solution: ① X = X_1 X_0
Bit ↵ ↳ Bit

0 → 3 : الأرقام اللي بدكم تمثيلها باستخدام 2 bits

③ X_1	X_0	F → هاد اللي بقوره حسب السؤال	② X_1	X_0	
0	0	0 even	0	0	0 ⇒ even
0	1	1 odd	0	1	1 ⇒ odd
1	0	0 even	1	0	2 ⇒ even
1	1	1 odd	1	1	3 ⇒ odd

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$F(x_1, x_0) = \Sigma m(1, 3)$

x_1 \ x_0	x_0 = 0	x_0 = 1
$\bar{x}_1$ = 0	0	1
x_1 = 1	0	1

$\overline{F = x_0}$

→ Example #2:

Design a circuit that produces "1" if the number is prime. otherwise produces "0". الدائرة هاي بتنتج الرقم 1 اذا كان الرقم أولي و بتنتج 0 اذا كان عن أولي

Hint: the number is 3 bits.

$X = x_2\, x_1\, x_0$

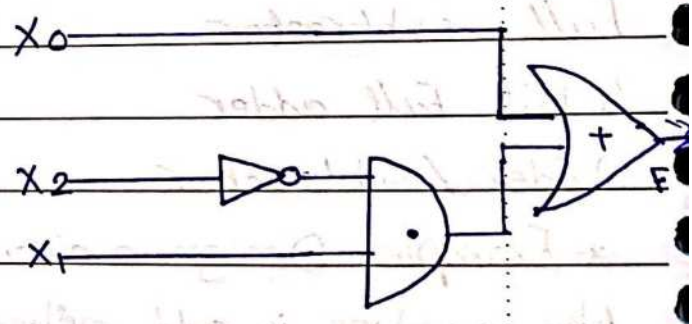
	x_2	x_1	x_0	F	
0 ←	0	0	0	0	
1 ←	0	0	1	1	prime
2 ←	0	1	0	1	prime
3 ←	0	1	1	1	prime
4 ←	1	0	0	0	
5 ←	1	0	1	1	prime
6 ←	1	1	0	0	
7 ←	1	1	1	1	prime

$F = \Sigma m(1, 2, 3, 5, 7)$

x_2 \ $x_1 x_0$	$\bar{x}_1 \bar{x}_0$ = 00	$\bar{x}_1 x_0$ = 01	$x_1 x_0$ = 11	$x_1 \bar{x}_0$ = 10
$\bar{x}_2$ = 0	0	1	1	1
x_2 = 1	0	1	1	0

$F = x_0 + \bar{x}_2 x_1$

$x_0 + \overline{x_2}\, x_1$



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→ Half adder :- «نصف الجامع»

A	B	Sum	carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

بالـ half adder بطلع الـ minterm
الـ function الأول 2 functions
الـ sum والتاني الـ function
carry

① Sum = $\Sigma m(1,2)$

A \ B	$\bar{B}$ 0	B 1
$\bar{A}$ 0	0	(1)
A 1	(1)	0

$F = \bar{A}B + A\bar{B} \Rightarrow A \oplus B$

هاد الاقتران ← نفسه بيكافئ اقتران الـ XOR

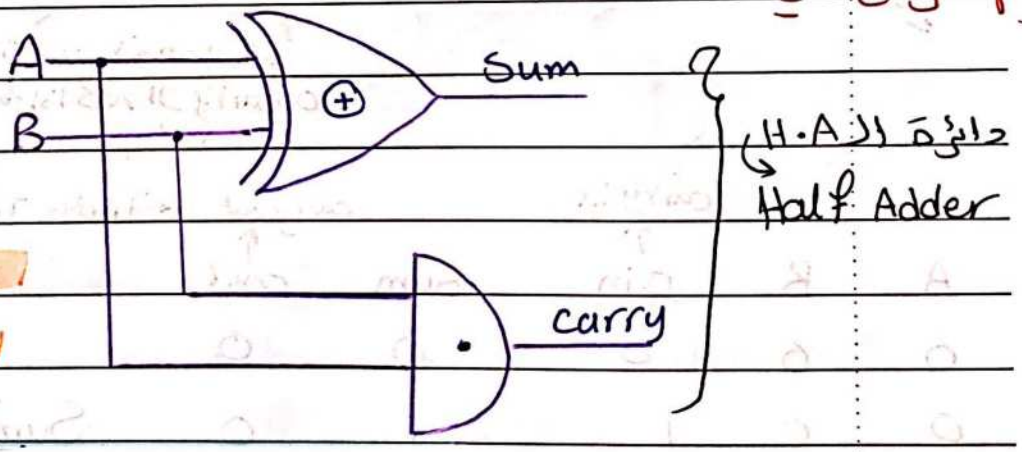
② carry بيدا عمل minimisation

A \ B	$\bar{B}$ 0	B 1
$\bar{A}$ 0	0	0
A 1	0	(1)

Carry = $\Sigma m(3)$

هو بنادي لو عملتها K map لأنها قيمة واحدة فقط فش وحال اعملها grouping مع اشي.

$F = AB$



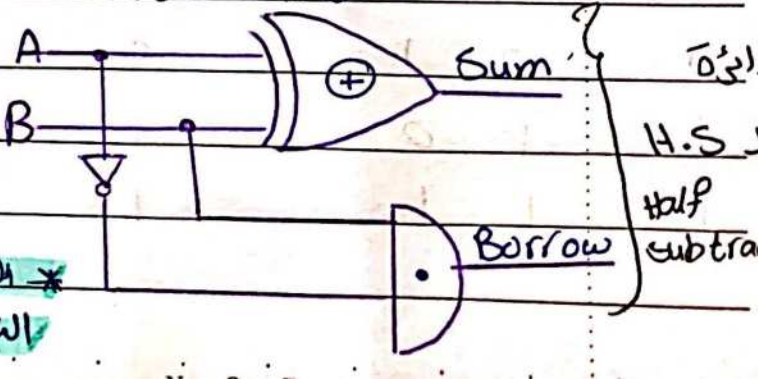
* لفوق بسيرة H.A والـ H.S فقط انه طبقت inverter ع A*

→ Half subtractor «نصف الطارح» (A-B)

A	B	sum	borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Sum = $A \oplus B$

borrow = $\bar{A}B$



* الـ sum الناتج من الـ H.S هو نفس الـ sum الناتج من H.A لكنه انا بجمع بطرح بجي

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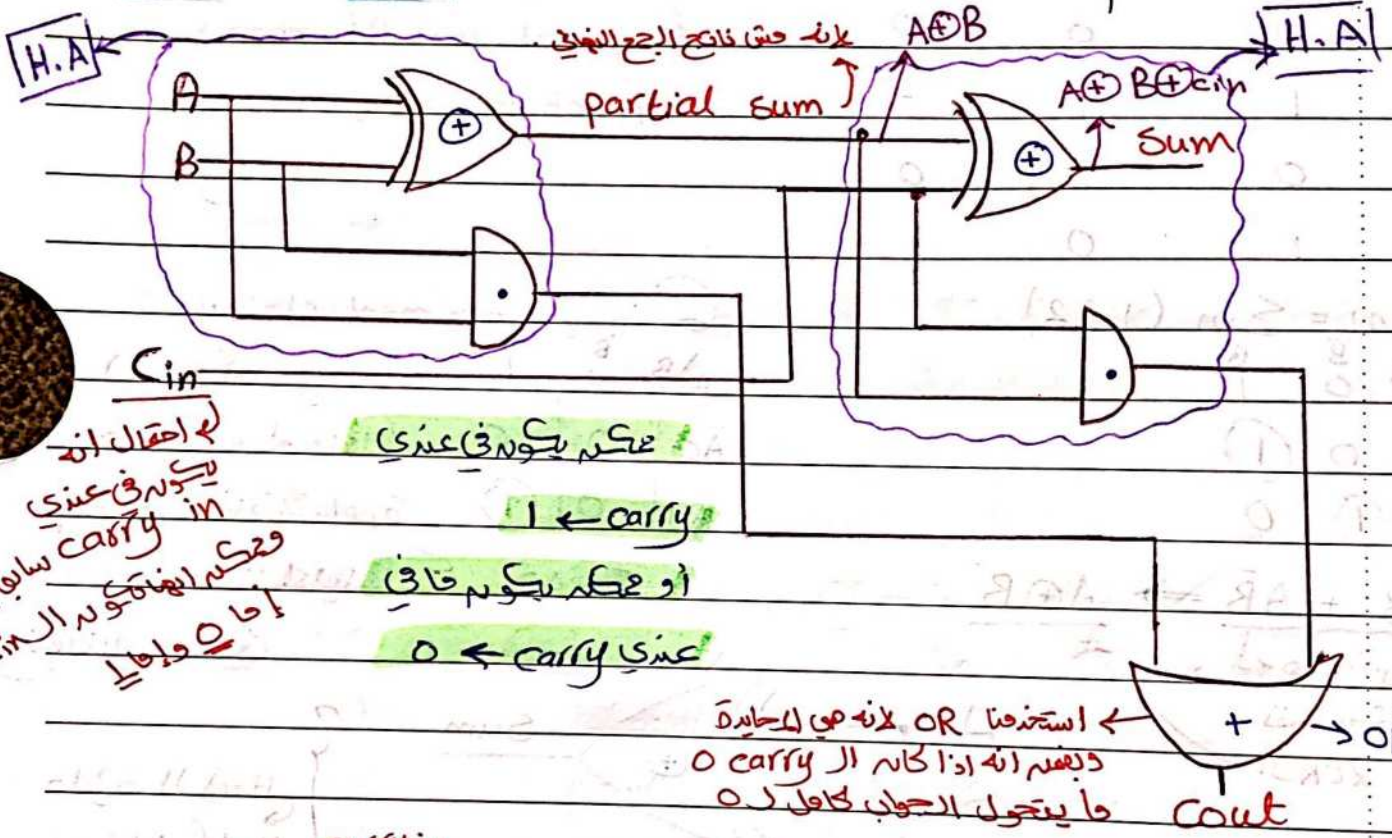
* Full adder :-

H.A + H.A = F.A

صمم بالرسم



احتمال انه يكون في عندي in carry سابع وعكس انها تكون السنت إما 0 واما 1

عكس يكون في عندي

1 ← carry

او عكس يكون فاتي

عندي carry ← 0

استخدمنا OR لانه هي الوحيدة
ونفهم انه اذا كان ال carry 0
فا يتحول الجواب كامل لـ 0

carry in ↑ Cin

carry out ↑ Cout

على شكل Truth Table :

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$Sum = \Sigma m(1,2,4,7)$

$Cout = \Sigma m(3,5,6,7)$

تابع

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Sum →

A \ B_{cin}	$\bar{B}\bar{C}$ 00	$\bar{B}C$ 01	BC 11	$B\bar{C}$ 10
$\bar{A}$ 0	0	1	0	1
A 1	1	0	1	0

$Sum = \bar{A}B\bar{cin} + \bar{A}B\bar{cin} + A\bar{B}\bar{cin} + ABcin$

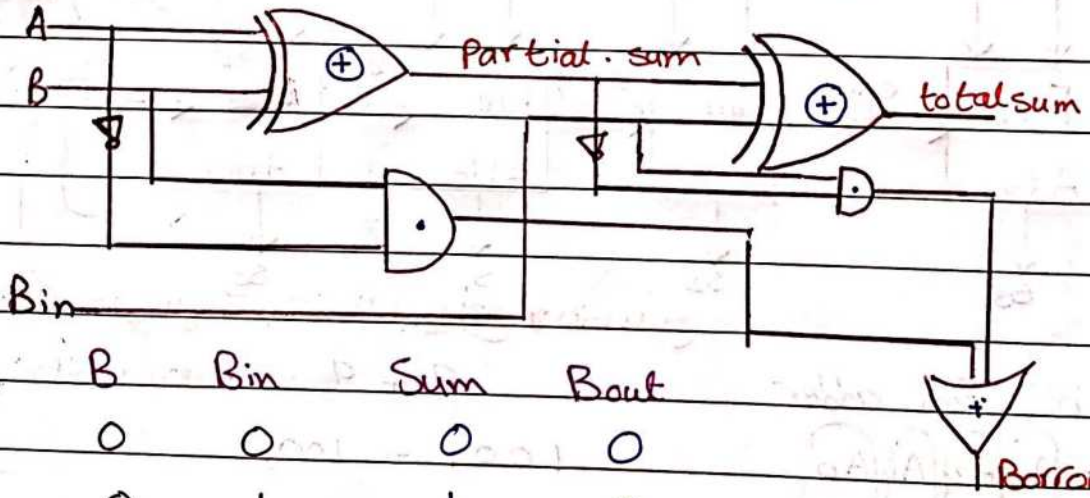
Cout →

A \ B_{cin}	$\bar{B}\bar{cin}$ 00	$\bar{B}cin$ 01	$Bcin$ 11	$B\bar{cin}$ 10
$\bar{A}$ 0	0	0	1	0
A 1	0	1	1	1

$Cout = Acin + Bcin + AB$

* Full subtractor:

H.S + H.S = F.S



A	B	Bin	Sum	Bout
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$Sum = \Sigma m(1,2,4,7)$

$Bout = \Sigma m(1,2,3,7)$

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→ Sum →

A \ B_{in}	$\bar{B}\bar{B}_{in}$ 00	$\bar{B}B_{in}$ 01	BB_{in} 11	$B\bar{B}_{in}$ 10
$\bar{A}$ 0	0	1	0	1
A 1	1	0	1	0

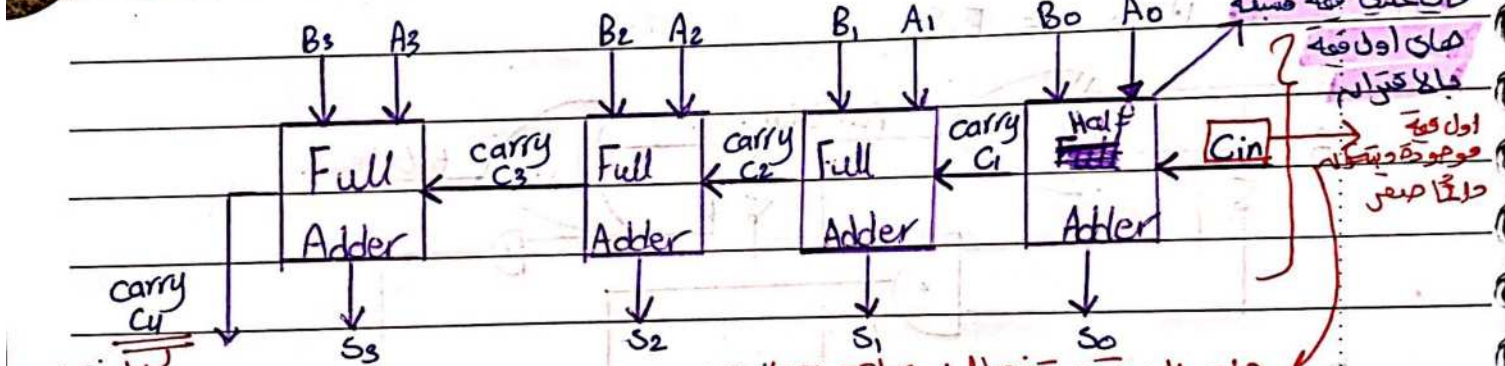
$Sum \rightarrow \bar{A}\bar{B}B_{in} + \bar{A}B\bar{B}_{in} + A\bar{B}\bar{B}_{in} + ABB_{in}$

* Bout →

A \ B_{in}	$\bar{B}\bar{B}_{in}$ 00	$\bar{B}B_{in}$ 01	BB_{in} 11	$B\bar{B}_{in}$ 10
$\bar{A}$ 0	0	1	1	1
A 1	0	0	1	0

$Bout \rightarrow \bar{A}B_{in} + \bar{A}B + BB_{in}$

هاي هو اللي فيها A و B لكنه فاتي عندي قيمة سابقة هان اول قيمة فالاكترلم



اول قيمة موجودة وبتكون دائما صفر

اخر قيمة بتكون موجودة

هاي الرومة بستخدمها لما بدي اجمع 4bits مع 4bits وبس

4 bit Full adder

$A = A_3 A_2 A_1 A_0$

$B = B_3 B_2 B_1 B_0$

$Cout = \Sigma_3 \Sigma_2 \Sigma_1 \Sigma_0$

9 + 9 ∴ 5bits هتل

1001 + 1001

Full adder

يعني كل bit بدي اجمعه بحتاج لـ Full adder

فبشكل اختصر على حالي وضع بستخدم قطعة اسمها 4 bit Full adder واللي هي موجودة بالرومة اللي فوق

ال carry الناتج من العملية فابظهر مع النواتج ولكنه بكون ضمنياً بالحل فيعني الناتج اللي بجده بكون ناتج جمع A + B + carry من العملية اللي قبل

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9 + 10 ⇒

9 → 1001 (A)

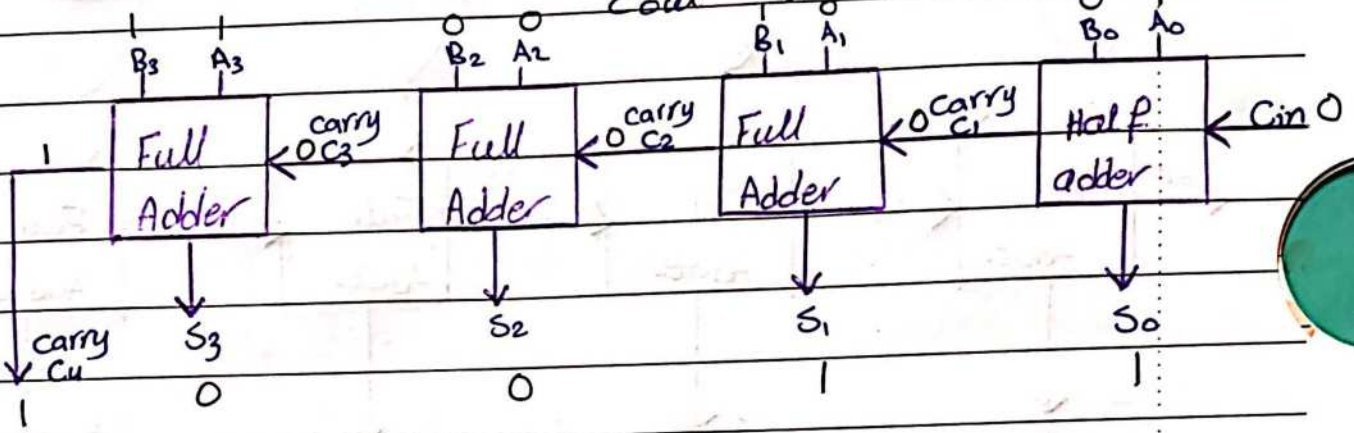
10 → 1010 (B)

F.A F.A F.A F.A

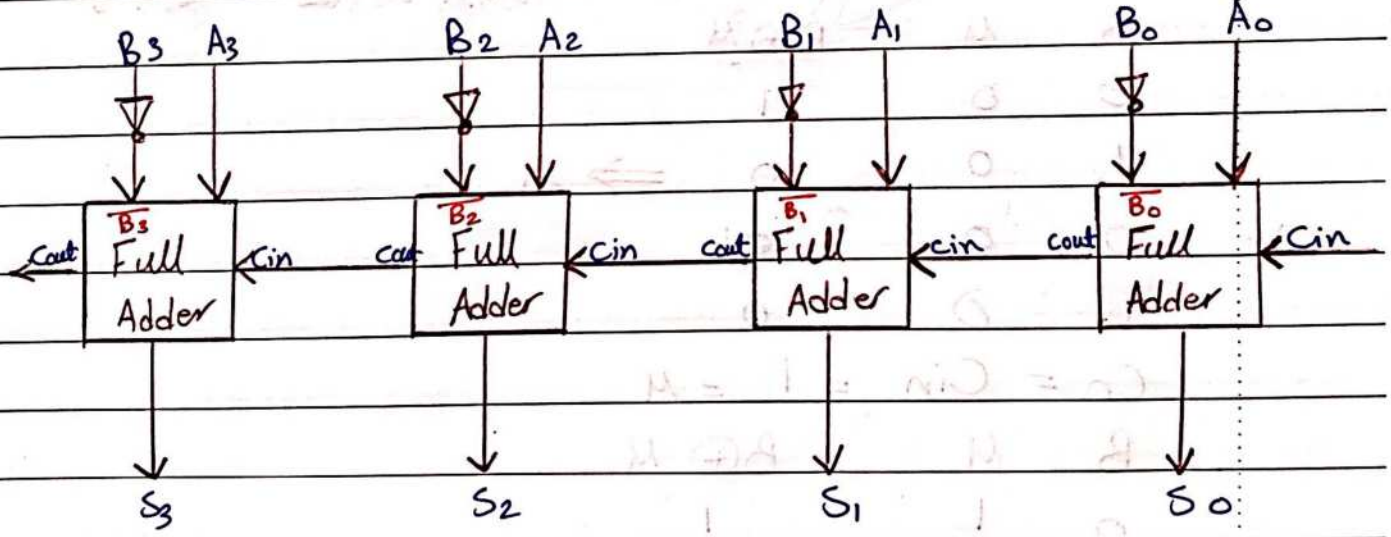
1 0 1 0 +

1 0 0 1 1 → Σ_0

Σ_1 Σ_2 Σ_3 Cout



1 0 0 1 1 → 19



A − B ⇒ (A + (−B))

1's complement + 1 ← 2nd complement

انا بدي اعمل subtracter من خلال الـ Adder

كيف؟ عن طريق اني اضيف inverter

هنالك بتتحول الـ 1st comp على اساس الـ 2nd

وبتصير طرح.

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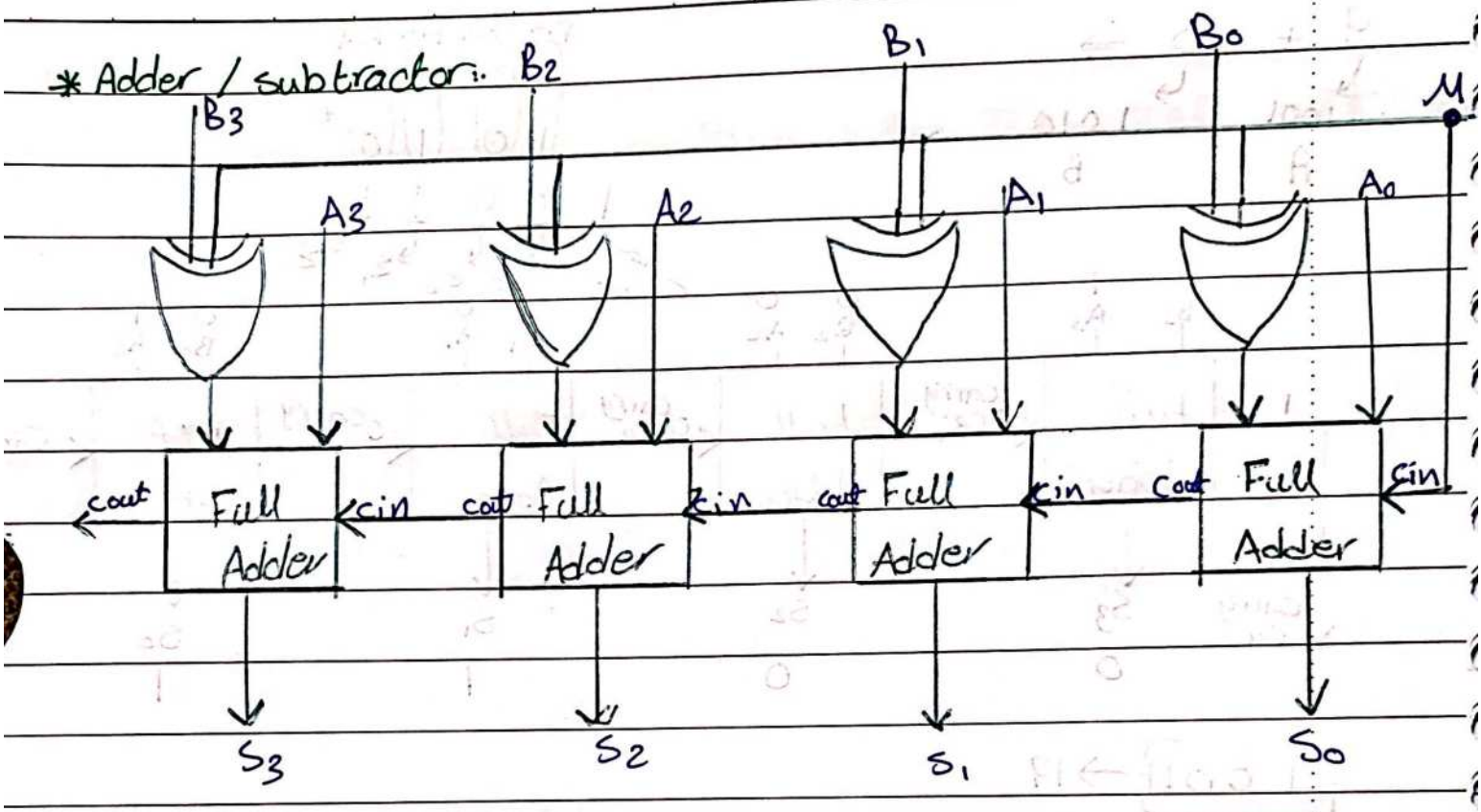
* Adder / subtractor:



$C_0 = M \Longrightarrow M \oplus B$ حرة بثبت M ⊥ وحرة بثبتها 0

B	M	B⊕M
0	0	1
1	0	0 ⟹ B
0	0	1
1	0	0

$C_0 = C_{in} = 1 = M$

B	M	B⊕M
0	1	1
1	1	0 ⟹ $\bar{B}$ طرح
0	1	1
1	1	0

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$$(10275.253)_{10}$$ (4 3 2 1 0 -1 -2 -3) * decimal Numbers

$$= 5*10^0 + 7*10^1 + 2*10^2 + 0*10^3 + 1*10^4 + 2*10^{-1}$$
$$+ 5*10^{-2} + 3*10^{-3}$$

$$(325)_8$$ (2 1 0) * Octal Numbers.

$$5*8^0 + 2*8^1 + 3*8^2 \rightarrow$$ decimal $\rightarrow$ (ZWI)

$$(11010101)_2$$ (7 6 5 4 3 2 1 0) * Binary Numbers.

$$1*2^0 + 0*2^1 + 1*2^2 + 0*2^3 + 1*2^4 + 0*2^5 + 1*2^6 + 1*2^7$$
$$1 + 0 + 4 + 0 + 16 + 0 + 64 + 128 = (\quad)_{10}$$
decimal

Hexa

- computers $\rightarrow$ 64 bit $\rightarrow$ binary + decimal
- 8 bits $\rightarrow$ 1 byte

Five Apple

* Convert to decimal

• ex: convert $(1011.11)_2 = (\quad)_{10}$

$1*2^0 + 1*2^1 + 0*2^2 + 1*2^3 + 1*2^{-1} + 1*2^{-2}$

$= 1 + 2 + 0 + 8 + 0.5 + 0.25$

$= (11.75)_{10}$

• ex: $(127.4)_8 = (\quad)_{10}$

$= 7*8^0 + 2*8^1 + 1*8^2 + 4*8^{-1}$

$= 7 + 16 + 64 + 0.5 = (87.5)_{10}$

• ex: $(10B.A)_{16} = (267.625)_{10}$

$= 11*16^0 + 0*16^1 + 1*16^2 + 10*16^{-1}$

$= 11 + 0 + 256 + 0.625$

$= (267.625)_{10}$

2^{10} → kilo

2^{20} → mega ÷ 1024

2^{30} → giga

2^{40} → Tera

0	
1	
2	
3	
4	
5	
6	
7	
8	
9	
A	10
B	11
C	12
D	13
E	14
F	15

* Binary Addition :

```
  1        1        1
  0 +      1 +      1 +
  1        10       11
```

* Convert from decimal to Numbering systems :

ex : $(41)_{10} = (\quad)_2$

$= (1*2^5) + (0*2^4) + (1*2^3) + (0*2^2) + (0*2^1) + (1*2^0)$

$= 32 + 0 + 8 + 0 + 0 + 1$

	41	
	20	1
2÷	10	0
	5	0
	2	1
	1	0
	0	1

$= (101\ 001)_2$

ex : $(41)_{10} = (\quad)_8$

	41	
8÷	5	1
	0	5

$= (51)_8$

• ex : $(41)_{10} = (\quad)_{16}$

$= (29)_{16}$

16÷

41	
2	9
0	2

ex : $(0.625)_{10} = (\quad)_2$

$0.625 * 2 = 1.25$

$0.25 * 2 = 0.5$

$0.5 * 2 = 1.0$

$= (0.101)_2$

- ex : $(12.125)_{10} = (\quad)_2$

12	
6	0
3	0
1	1
0	1

$= (1100.\quad)_2$

$0.125 * 2 = 0.25$

$0.25 * 2 = 0.5$

$0.5 * 2 = 1$

$= (1100.001)_2$

* Octal ⇄ Binary

0	0 0 0
1	0 0 1
2	0 1 0
3	0 1 1
4	1 0 0
5	1 0 1
6	1 1 0
7	1 1 1

° ex: $(5631)_8 = (\quad)_2$

$= (101\ 110\ 011\ 001)_2$

° ex: $(736)_8 = (\quad)_2$

$= (111\ 011\ 110)_2$

° ex: $(011\ 010\ 111\ 101)_2 = (\quad)_8$

$= (3265)_8$

° ex: $(01\ 010\ 111\ 011)_2 = (\quad)_8$

$(1273)_8$

0	0	0	0	0	
1	0	0	0	1	
2	0	0	1	0	
3	0	0	1	1	→ octal
4	0	1	0	0	
5	0	1	0	1	
6	0	1	1	0	
7	0	1	1	1	
8	1	0	0	0	
9	1	0	0	1	
A	1	0	1	0	
B	1	0	1	1	
C	1	1	0.	0	
D	1	1	0	1	
E	1	1	1	0	
F	1	1	1	1	

- ex : $(A3C1)_{16} = (1010\,0011\,1100\,0001)_2$

- ex : $(FE3)_{16} = (1111\ 1110\ 0011)_2$

- ex : $(1011\ 0101\ 1010\ 1101\ 0110)_2 = (B5AD6)_{16}$

- ex : $(A3172)_{16} = (\quad)_8$

↳ $(A3172)_{16} = (\quad)_2 = (\quad)_8$

$= (1010\,0011\,0111\,0010)_2 = (2430562)_8$

- ex : $(7653\,1207)_8 = (\quad)_{16}$

$(111\,110\,101\,011\,001\,010\,000\,111)_2 = (FAB2\,87)_{16}$

- ex : $(45.12)_8 = (\quad)_{16}$

$= (\underline{001}\ \underline{001}\ \underline{01}.\ \underline{001}\ \underline{010}\underline{00})_2 = (25.28)_{16}$

- ex : $(72.5A)_{16} = (\quad)_8$

$\underline{011}\ \underline{111}\ \underline{001}\ \underline{0}.\ \underline{010}\ \underline{110}\ \underline{100} = (162.264)_8$

** Arithmatic Operations :

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0\ 1 \\ +\ 1\ 0\ 0\ 1\ 1\ 1 \\ \hline 1\ 0\ 1\ 0\ 1\ 0\ 0 \end{array} \qquad \begin{array}{r} 0\ 1^{0}\ 1^{0} \\ 1\ 0\ \not{1}\not{X}\ 0\ 1 \\ -\ 1\ 0\ 0\ 1\ 1\ 1 \\ \hline 0\ 0\ 0\ 1\ 1\ 0 \end{array}$$

$$\begin{array}{r} 1\ 0\ 1\ 1 \\ *\quad 1\ 0\ 1 \\ \hline 1\ 0\ 1\ 1 \\ t\ 0\ 0\ 0\ 0 \\ 1\ 0\ 1\ 1 \\ \hline 1\ 1\ 0\ 1\ 1\ 1 \end{array}$$

* Decimal Arithmatic :

```
  7 5 2
+   5 8 6
-------
1 3 3 8
```

* Octal Arithmatic :

```
  7 6 3
+   2 5 +
-------
1 1 1 0
```

* Hexadecimal Arithmatic :

```
  A 3 1 4
+ A 9 6 7
---------
1 4 C 7 B
```

A + A = 20

20	
1	4
0	1

16=

0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
A	10
B	11
C	12
D	13
E	14
F	15

lecture (2)

** Binary Codes :

o BCD : Binary Coded Decimal = 1, 2, 4, 8

0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0
5	0 1 0 1
6	0 1 1 0
7	0 1 1 1
8	1 0 0 0
9	1 0 0 1

used in calculators

benefits : less extensive in potential power.

*ex : $(567)_{10}$ = (0101 0110 0111) BCD

* BCD Addition :

• 6 + 2 = 8

0110 +
0010
1000

• 5 + 5 = 10

0101 +
0101
1010
0110 +
00010000
1 0

** لما يطلع الرقم اكبر من 9
بزيد 6

• 7 + 9 = 16

1 1 1
0111
1001 +
10000 → 10 > 9
0110 + +6
00010110
1 6

• 184 + 476 = 660

0001 1000 0100
0100 0111 0110 +
0101 1111 1010 >9
0110 + +6
01100000 0000
0000 0110 0000 + +6
0110 0110 0000
6 6 0

• 73 + 62 = 135

0110 0010 +
0111 0011
(13) 1101 0100 (5) 13 > 9
0110 +
0001 0011 0101
1 3 5

مضيف لا يمكن الله
اكبر من 9

Five Apple

** Binary to Gray code :

* ساعات بالسيد .

1 0 1 1 0 → 1 1 1 0 1

1 1 0 0 1 → 1 0 0 0 1

** Gray to Binary code :

1 1 0 1 1 → 1 0 0 1 0

** ASCII Code نوع من انواع Binary code

** Binary Subtraction :

$(01101)_2 - (00111)_2$

1 1 0 1 0

0 0 1 1 1

0 1 1 1 1

• 1's complement :

$\bar{1} \rightarrow 0$

$\bar{0} \rightarrow 1$

$(\overline{11010}) \xrightarrow{1's} 00101$

المكمل لحتى يوصل لعشرة $(\bar{5})_{10} : 5$

• 2's complement :- ⇝ 1's complement + 1

$(11010) \xrightarrow{2's} 00101 + 1 = 00110$

او اول صفر خليه بعض لحد اول واحد بصلوفه اجليه بعدين بعمله 1's complement

↑ ex = 101 0001 000 $\xrightarrow{2's}$

0 1011 1000

نخلي اصفار لحد ما اصادف اول 1 بعدين نقلب الباقي.

*ex : 101 000 1 001 $\xrightarrow{2's}$

0 10 111 0 111

** one's complement subtraction :

• 7 - 3 = 4

0 1 1 1 - (0011)

$\xrightarrow{1's}$

0 1 1 1 .

1 1 0 0 +

carry (1) 0 0 1 1

نقلب الرقم السالب

اجمعين بجمع.

~~الباقي من الـ carry~~

~~اسقطه~~

* بجمع الـ carry للناتج.

1 +

0 1 0 0

• $(1011)_2 - (0110)$

نقلب

وجمع.

1 0 1 1

1 0 0 1

(1) 0 1 0 0

1 +

0 1 0 1 5

Five Apple

* 2's complement Subtraction :

- 1011 − 0110

1011 — لعمله 2's لازم

1010 + — لجمع

(1)0101

Ignore

in 1's complement ⊕ the carrier

in 2's complement (ignore) the carrier

- 4 − 9 =

100 − 1001

0100 − 1001 صفر

00100 − 01001 صفر

01001 → 2's

00100 — رقم موجب

10111 + — رقم سالب

(1)10111 ← سالب — لانه بتكون عندنا زيادة خانات لما ضفت واحد عرفت انه سالب.

00101 ← 2's

(-5) ← لما يطلع سالب بجمع بعمله 2's

1000 → most significant bit

0001 → less significant bit

- 7 − 14 =

111 − 1110

00111 − 01110

01110 → 2's

00111

(1)0010 +

(1)1001 ← سالب.

00111 ← سالب — 2's

(-7)

0 -5 -8

-(0101) -(1000)

↳ -(00101) -(01000)

↳ نعمل لـ ② داخ

11011 +

11000

ignore ← (1)10011

2's ← 01101

-13

** Signed Binary numbers :

① Signed magnitude

② 1's complement.

③ 2's complement.

(-9) represent this number :

1 1001 : signed magnitude

01001 : 10110 → 1's complement

9 -9

01001 : 10111 → 2's complement

9 -9

1111 $\xrightarrow{2's}$ -0001 (-1)

lecture (3)

** Signed numbers :

- The possible 3-bits signed numbers :

Binary	signed mag	1's comp	2's comp
0 0 0	+0	0	0
0 0 1	+1	+1	+1
0 1 0	+2	+2	+2
0 1 1	+3	+3	+3
1 0 0	-0	-3	-4
1 0 1	-1	-2	-3
1 1 0	-2	-1	-2
1 1 1	-3	-0	-1

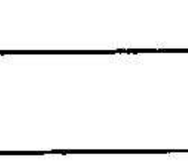
* we used 2's complement (is best) because it has one zero (not positive or negative)

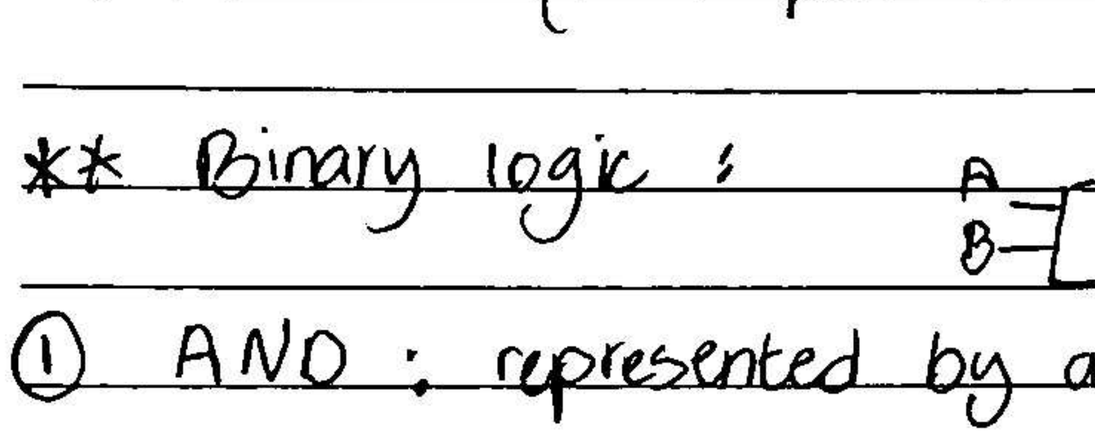
** Binary logic :

$Y = A \cdot B$

① AND : represented by a dot or abscence of operator.

② Inverter : $Y = \bar{A}$

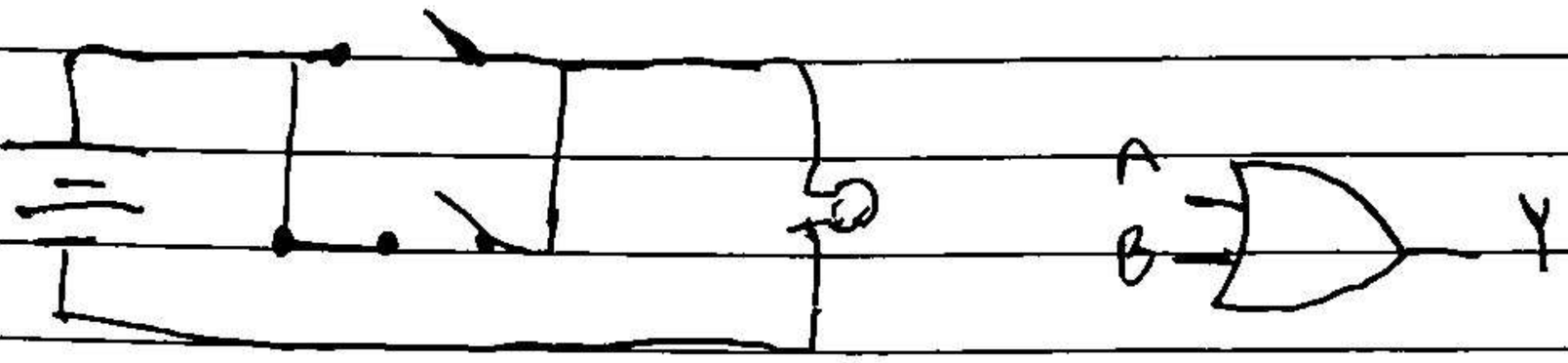




AND

Two switches on series

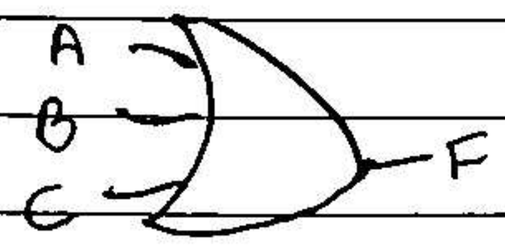
③ OR : $Y = A + B$



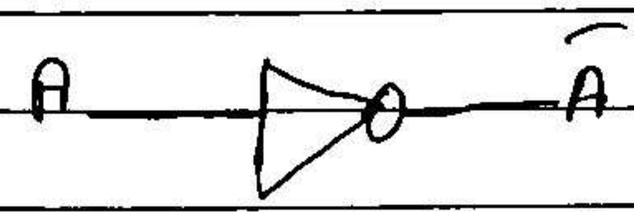
OR gate → Two switches in parallel

* wave forms (on slides).

x	0	1	1	0	0	
y	0	0	1	1	0	
and	0	0	1	0	0	
OR	0	1	1	1	0	
Not	1	0	0	1	1	



OR



inverter.

* three input gates : 2^n

A	B	C	$F = A + B + C$
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

3 - input OR gate.

A	B	C	F = ABC
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

3 input AND gate.

** Boolean Algebra ;

00 Axiomatic Definitions ;

① Identity element :

$x + 0 = x$

$x \cdot 1 = x$

A	B	A ⊕ B (OR)	A · B
0	0	0	0
1	0	1	0
0	1	1	0
1	1	1	1

② commutative law :

$x + y = y + x$

$x \cdot y = y \cdot x$

③ Distributive law :

$x \cdot (y + z) = (x \cdot y) + (x \cdot z)$

$x + (y \cdot z) = (x + y) \cdot (x + z)$

④ complement ;

$x + \bar{x} = 1$

$0 + 1 = 1$

$1 + 0 = 1$

$x \cdot \bar{x} = 0$

⑤ Associative law :

$x + (y + z) = (x + y) + z$

$(x \cdot y) \cdot z = x (yz)$

⑥

$x + 1 = 1$

$x + 0 = x$

$x \cdot 1 = x$

$x \cdot 0 = 0$

$\bar{\bar{x}} = x$

* Find the output : (Simplify)

① $xy + x\bar{y} =$

$= x(y + \bar{y})$

$= x \cdot 1 = x$

② $x + xy$

$= x(1 + y)$

$= x \cdot 1 = x$

③ $(x + \bar{y})\, y$

$= xy + \bar{y}y$

$= xy + 0 = xy$

④ $(x + y)(x + \bar{y})$

$= xx + x\bar{y} + yx + y\bar{y}$

$= x + x\bar{y} + xy + 0$

$= x + x(\bar{y} + y)$

$= x + x(1) = x$

⑤ $x(x + y)$

$= xx + xy$

$= x + xy$

$= x(1 + y)$

$= x(1) = x.$

⑥ $(x\bar{y}) + y$

$= (x + y)(\bar{y} + y)$

$= (x + y)\, 1$

$= x + y$

⑦ $A + \bar{A}B$

$= (A + \bar{A})(A + B)$

$= A + B$

** Demorgan's theorem :

$(x + y) = \bar{x} \cdot \bar{y}$

$(x \cdot y) = \bar{x} + \bar{y}$

x	y	$x+y$	$\overline{x+y}$	$\bar{x}\bar{y}$	$\bar{x} \cdot \bar{y}$
0	0	0	1	1	1
0	1	1	0	0	0
1	0	1	0	0	0
1	1	1	0	0	0

** Boolean Functions :

$F_1 = x + \bar{y}z$

يوصف متى الـ Function بمستوى واحد

$F_2 = \bar{x}z + xy$

x	Y	x	F_1
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

x	Y	x	$\bar{y}$	$\bar{y}z$	F_1
0	0	0	1	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	0	0	1
1	0	0	1	1	1
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

lecture (4)

* Simplify the following :

Duality

- $F = AB + \bar{A}C + BC$

$= AB + \bar{A}C + BC(A + \bar{A})$

$= AB + \bar{A}C + ABC + \bar{A}BC$

$= AB(1 + C) + \bar{A}C(1 + B)$

$= AB + \bar{A}C$

- $F = (A+B) \cdot (\bar{A}+C) \cdot (B+C)$

$= (A+B) \cdot (\bar{A}+C) \cdot (B+C+A\bar{A})$

$= (A+B) \cdot (\bar{A}+C) \cdot (A+B+C) \cdot (\bar{A}+B+C)$

$= (A+B) \cdot (\bar{A}+C)$

Assume $A+B = x$

$x(x+C)$

$x + xC$

$x(1+C) = x$

assume

$\bar{A}+C = y$

$= y$.

$(\bar{A}+C)(\bar{A}+B+C)$

$= y \cdot (y+B)$

$= y + yB$

$= y(1+B)$

$= y$

$= (\bar{A}+C)$

- $F = C + (B\bar{C})$

$= C + \bar{B} + C$

$= C + C = C$

$= C + \bar{B}$

$C + \bar{B}C$

$C(1 + \bar{B})$

- $F = \overline{C + (B\bar{C})}$

$F = \bar{C} + \overline{(B\bar{C})}$

$F = \bar{C} + \bar{B} + \bar{\bar{C}}$ $\quad C + \bar{C} = 1$

$= 1 + \bar{B}$

$= 1$

- $F = \overline{(AB)}\,\overline{(\bar{A} \cdot B)} \cdot \overline{(B + \bar{B})}$

$= (\bar{A} + \bar{B})(\bar{A} + B) \cdot (B + \bar{B})'$

$= \bar{A}\bar{A} + \bar{A}B + \bar{B}\bar{A} + B\bar{B}$

$= \bar{A} + \bar{A}B + \bar{A}\bar{B} + 0$

$= \bar{A}(1 + B + \bar{B})$

$= \bar{A}$

- $F = xy\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + \bar{x}\bar{y}\bar{z}$

$= y\bar{z}(x + \bar{x}) + \bar{y}\bar{z}(x + \bar{x})$

$= y\bar{z} + \bar{y}\bar{z}$

$= \bar{z}(y + \bar{y}) = \bar{z}$

** Table 2.1 in Book
Page 43 important

** proof :-

- $F = (A+C)(AD + A\bar{D}) + AC + C = A+C$

$= (A+C)(A(D+\bar{D})) + AC + C$

$= A(A+C) + AC + C$

$= (A + AC) + AC + C$

$= A(1+C) + AC + C$

$= A + AC + C$

$= A(1+C) + C$

$= A + C$ #

** Complement of a function :

The complement of a function 'F' is '$\bar{F}$' and is obtained from interchanging 0's and 1's, and 1's and 0's in the value of F

$F_1 = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z$

$\bar{F}_1 = \bar{x}y\bar{z} + \bar{x}\bar{y}z$

$= (x + \bar{y} + z)(x + y + \bar{z})$

** Canonical and Standard forms :

- Minterms and Maxterms :

- a binary variable may appear either in normal form (x) or it's complement ($\bar{x}$).

* consider x and y two binary variables with an AND operator.

xy, $\bar{x}y$, $x\bar{y}$, $\bar{x}\bar{y}$

- for a 3- variables the minterms and maxterms are as follows :

	Minterms		Maxterms.	
x y z	term	designation	term	designation
0 0 0	$\bar{x}\bar{y}\bar{z}$	m_0	$x+y+z$	M_0
0 0 1	$\bar{x}\bar{y}z$	m_1	$x+y+\bar{z}$	M_1
0 1 0	$\bar{x}y\bar{z}$	m_2	$x+\bar{y}+z$	M_2
0 1 1	$\bar{x}yz$	m_3	$x+\bar{y}+\bar{z}$	M_3
1 0 0	$x\bar{y}\bar{z}$	m_4	$\bar{x}+y+z$	M_4
1 0 1	$x\bar{y}z$	m_5	$\bar{x}+y+\bar{z}$	M_5
1 1 0	$\bar{x}\bar{y}z$	m_6	$\bar{x}+\bar{y}+z$	M_6
1 1 1	xyz	m_7	$\bar{x}+\bar{y}+\bar{z}$	M_7

↳ the zero is the complement (صفرية عن الواحد)

↳ the one is the complement. (صفرية عن الصفر)

** A Boolean Function can be expressed Algebrically from a truth table by :

① Sum of Minterms (Sum of product) (best)

② product of Maxterms (product of Sum)

ex :

x	y	z	F_1	F_2
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$\rightarrow F_1$

$F_1 = \bar{x}\bar{y}z + x\bar{y}\bar{z} + xyz$

$= \Sigma(m_1, m_4 + m_7)$

$= \Sigma m(1, 4$

$F_1 = (x+y+z)(x+\bar{y}+z)(x+\bar{y}+\bar{z})$

$(x+y+\bar{z})\ (\bar{x}+\bar{y}+z)$

$F_1 = M_0 M_2 M_3 M_5 M_6$

$F_1 = \pi(0, 2, 3, 5, 6)$

↳ F_1 complement

** Conversion between Canonical forms :

- The complement of a function expressed as the sum of Minterms equals to the sum of Minterms missing from the original form :

$F_1 = \Sigma(0, 2, 3, 5, 6)$ sum of the missing values

$= \overline{m_0 + m_2 + m_3 + m_5 + m_6}$ complement.

$= M_0 \cdot M_2 \, M_3 \, M_5 \, M_6$

$= \Pi(0, 2, 3, 5, 6)$

⇝ complement of minterms = maxterms

lecture (5)

function of sum of product

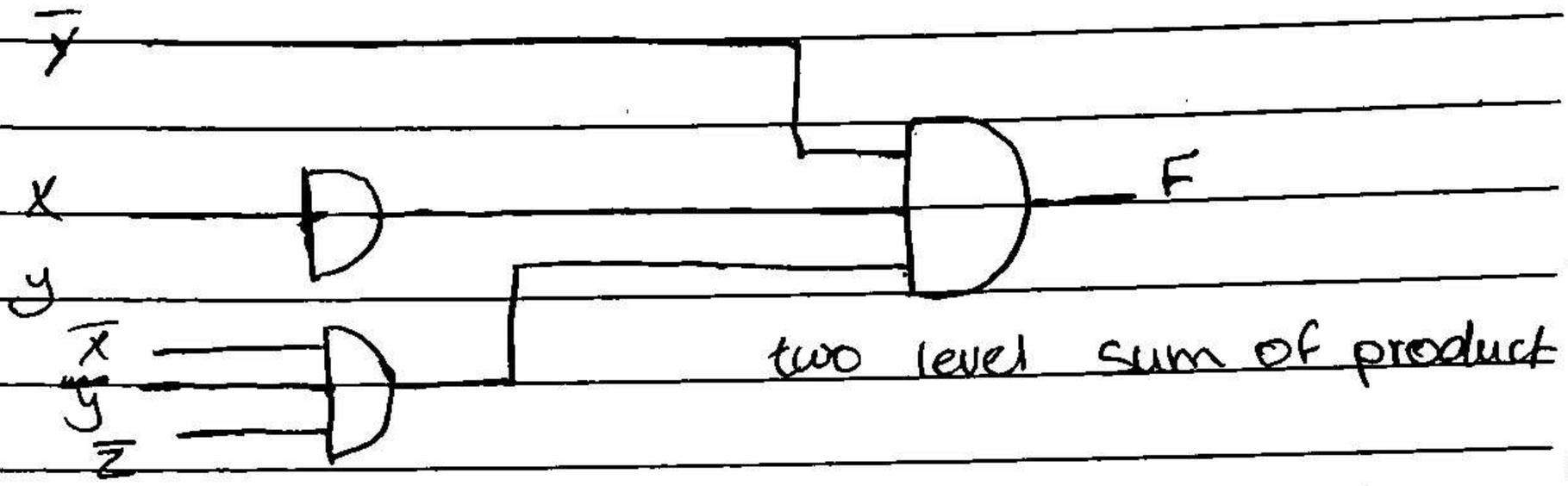
** Standard forms :

- each minterm or maxterm must contain by defention all the variables either complementer or uncomplemented.
- The standard forms are :

① Sum of product SoP

② product of Sum PoS

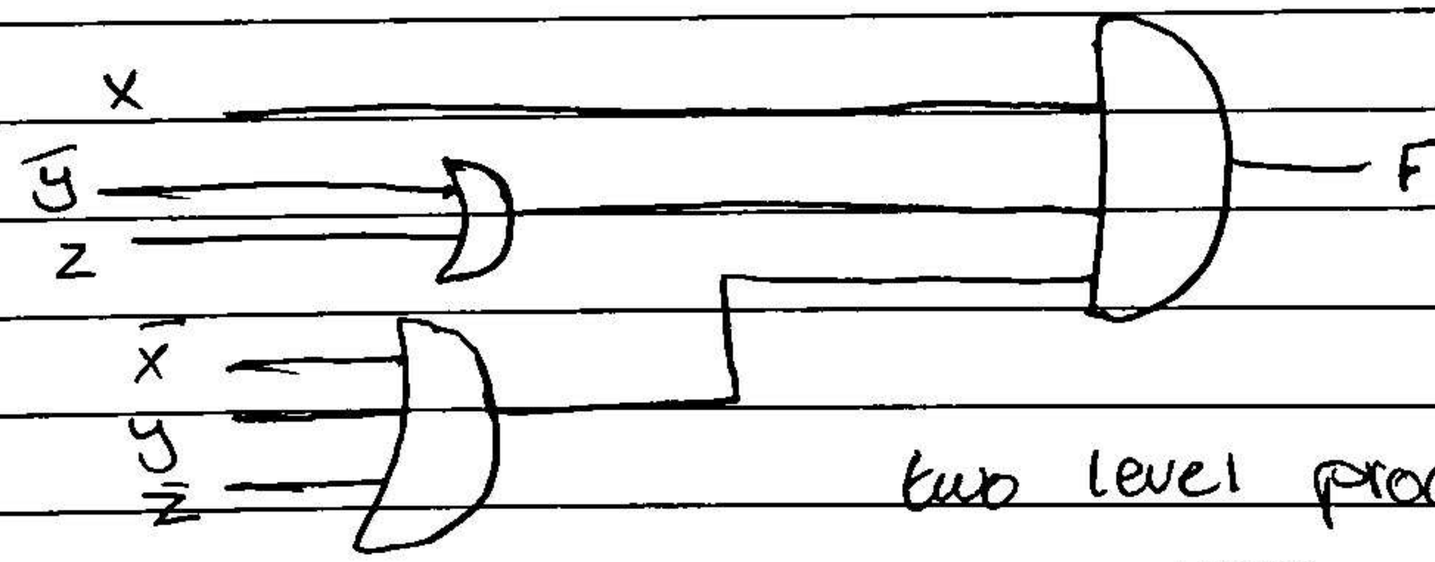
* $F = \bar{x} + xy + \bar{x}y\bar{z}$ (this is not a standard form)



two level sum of product

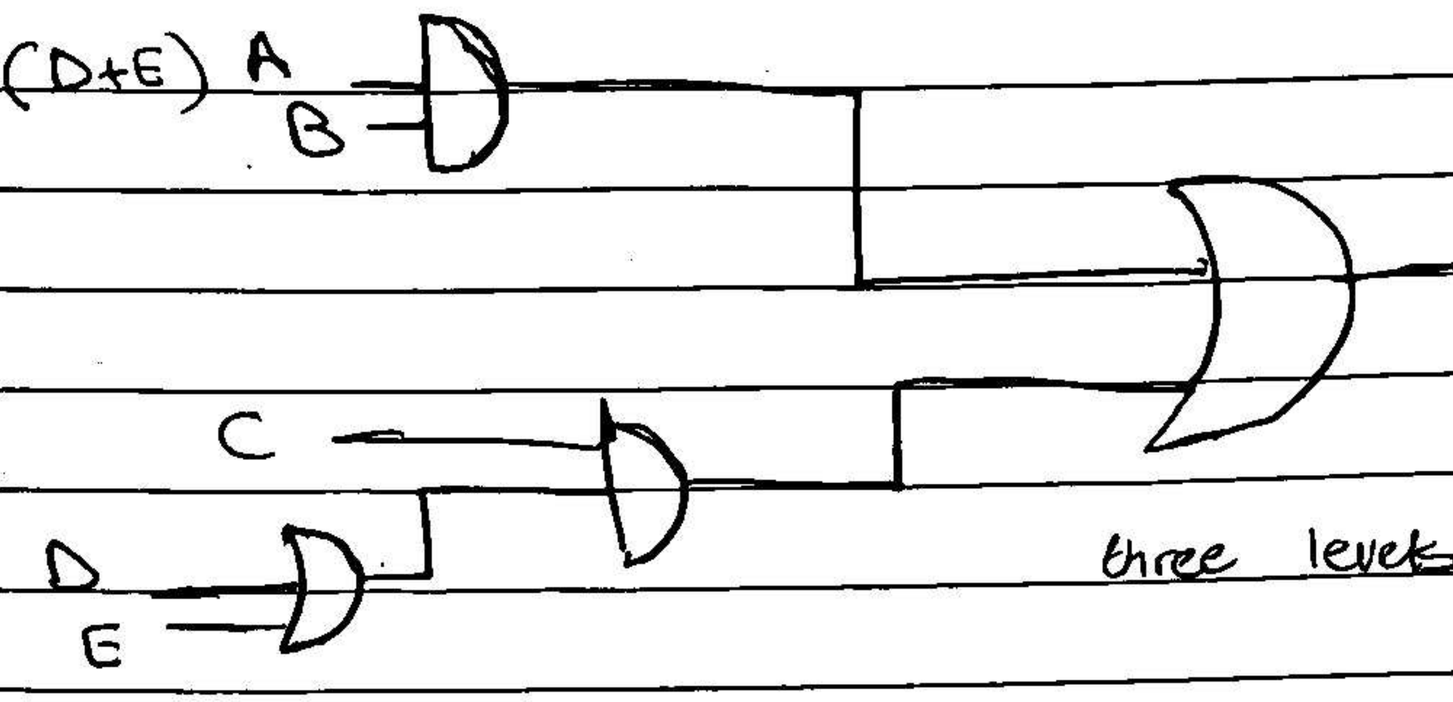
- function of sum of product

* $F = x(\bar{y}+z)(\bar{x}+y+\bar{z})$



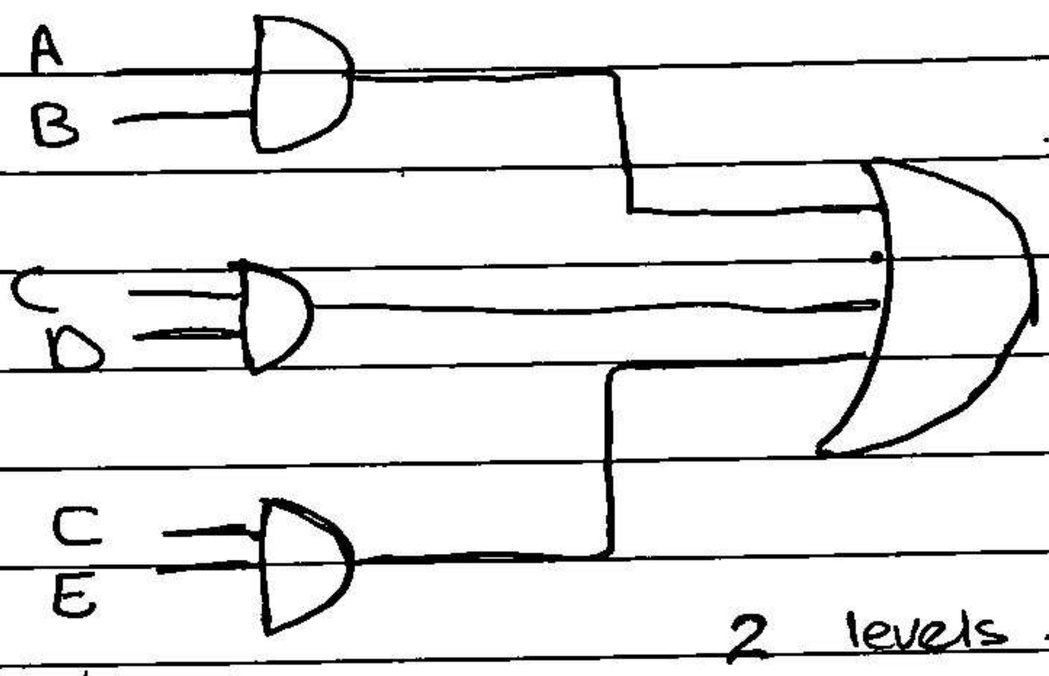
two level product of sum.

* $F = AB + C(D+E)$



three levels.

* $F = AB + CD + CE$



* two levels better.

2 levels.

** Convert from Standard form to Canonical form:

• ex: express the Boolean function $F = A + B\bar{C}$ as a sum of minterms

$= \Sigma(1, 4, 5, 6, 7)$

$= A(\bar{B}+B) + B\bar{C}(A+\bar{A})$

$= \cancel{A(\bar{B}+B)+}$

$= A\bar{B} + AB + ABC + \bar{A}BC$

$= A\bar{B}(C+\bar{C}) + AB(C+\bar{C}) + ABC + \bar{A}BC$

$= A\bar{B}C + A\bar{B}\bar{C} + ABC + AB\bar{C} + AB\bar{C} + \bar{A}BC$

$= m_5 + m_4 + m_7 + m_6 + m_5 + m_1$

or

- From the truth table : → the best.

$F = A + \bar{B}C$

A B C	$\bar{B}$	$\bar{B}C$	$F = A + \bar{B}C$	
0 0 0	1	0	0	m_0
0 0 1	1	1	1	m_1
0 1 0	0	0	0	m_2
0 1 1	0	0	0	m_3
1 0 0	1	0	1	m_4
1 0 1	1	1	1	m_5
1 1 0	0	0	1	m_6
1 1 1	0	0	1	m_7

$F = \Sigma (1, 4, 5, 6, 7)$

* Express the Boolean function :

$F = xy + \bar{x}z$ using PoS

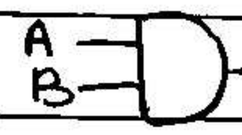
x y z	$\bar{x}$	$\bar{x}z$	xy	$F = xy + \bar{x}z$	
0 0 0	1	0	0	0	m_0
0 0 1	1	1	0	1	m_1
0 1 0	1	0	0	0	m_2
0 1 1	1	1	0	1	m_3
1 0 0	0	0	0	0	m_4
1 0 1	0	0	0	0	m_5
1 1 0	0	0	1	1	m_6
1 1 1	0	0	1	1	m_7

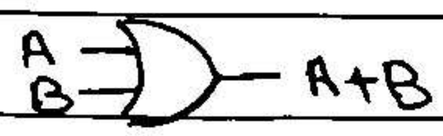
$\Sigma (1, 3, 6, 7) = \Pi (0, 2, 4, 5) = F$

$F = (x+y+z)(x+\bar{y}+z)(\bar{x}+\bar{y}+\bar{z})(\bar{x}+\bar{y}+\bar{z})$

** Digital logic circuits :

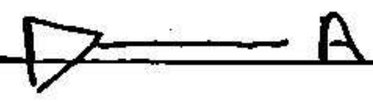
Name Symbol

① AND 

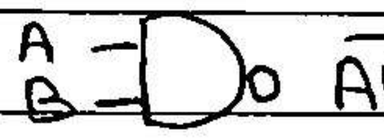
② OR 

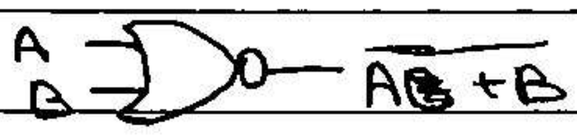
③ NOT

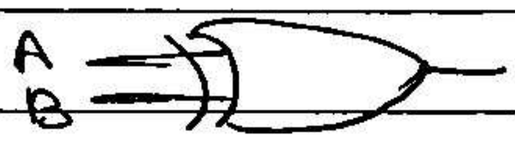
(inverter)

④ Buffer 

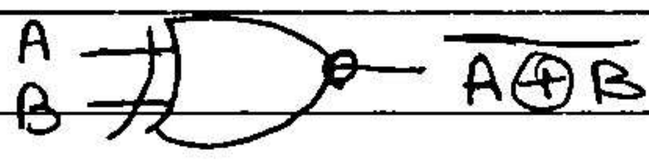
Restore the logic levels

⑤ NAND 

⑥ NOR 

⑦ XOR 

ADDer / Substarcter circuits.
error checking

⑧ XNOR 

A	B	AB	A+B	$\bar{A}$	A	$\overline{AB}$	$\overline{A+B}$	A⊕B	$\overline{A \oplus B}$
0	0	0	0	1	0	1	1	0	1
0	1	0	1	1	0	1	0	1	0
1	0	0	1	0	1	1	0	1	0
1	1	1	1	0	1	0	0	0	1

NAND, NOR → Most important in design.

then XOR

** Gate level minimization :

* (k arnaugh map or k-map)

- 2 variable map :

	0 $\bar{y}$	1 y
0 $\bar{x}$	m_0 00	m_1 01
1 x	m_2 10	m_3 11

*o minimize the following functions using k-map :

o $F_z \Sigma(0,1)$

	$\bar{y}$	y
$\bar{x}$	1 0	1
x	0 2	3 0

$F = \bar{x}$

o $F = \Sigma(0,2$

	$\bar{y}$	y
$\bar{x}$	1 0	1 0
x	1 2	3 0

$F = \bar{y}$

o $F = \Sigma(1,2)$

	$\bar{y}$	y
$\bar{x}$	0 0	1
x	1 2	3 0

دايما بتزل مكان الاصفار فاضي

$F = x\bar{y} + \bar{x}y$

o $F = \Sigma(0,1,2,3)$

	$\bar{y}$	y
$\bar{x}$	1 0	1
x	1 2	3 1

$F = 1$

o الواحدات باخدهم يا اما صفر يا اما 2 يا اما 4

** Simplify using k-maps :

$F(x,y) = \bar{x}y + x\bar{y} + xy$

	$\bar{y}$	y
$\bar{x}$	0	1 1
x	1 2	3 1

$F(x,y) = y + x$

** three variable map

	00 $\bar{y}$	01	11 y	10
$\bar{x}$	m_0 0	m_1 1	m_3 3	m_2 2
x	m_4 4	m_5 5	m_7 7	m_6 6
	$\bar{z}$	z		$\bar{z}$

lecture (6)

** Simplify $F(x,y,z) = \Sigma(2,3,4,5)$

	$\bar{y}$		y	
$\bar{x}$	0	1	1 3	1 2
x	1 4	1 5	7	6
	$\bar{z}$	z	$\bar{z}$	

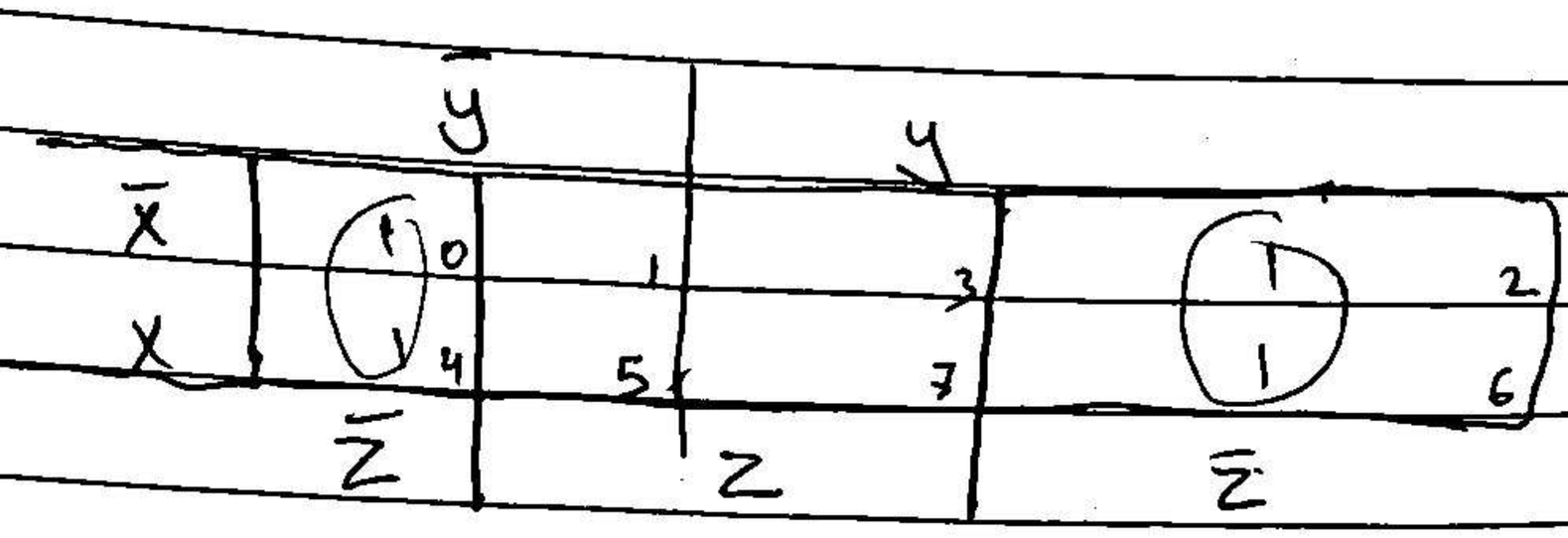
x is most significant

$\hookrightarrow F = \bar{x}y + x\bar{y}$

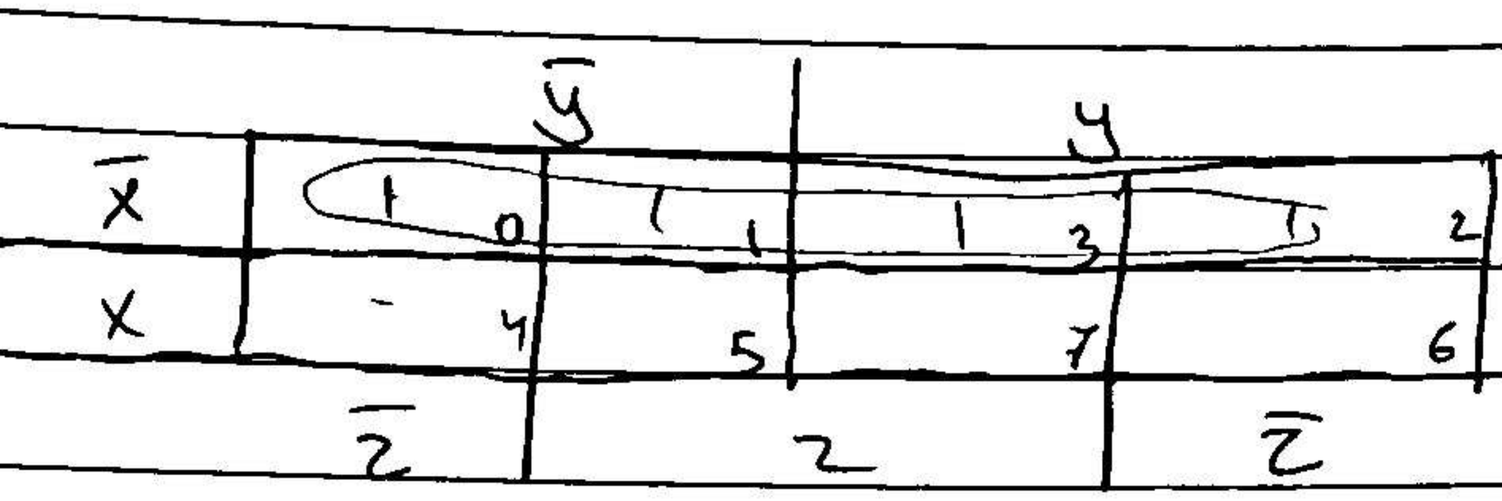
** if you group :

- one variable you tem will have x, y, z
- two variables only two (x, y / y, z / x, z)
- 4 variables only one (x / y or z)

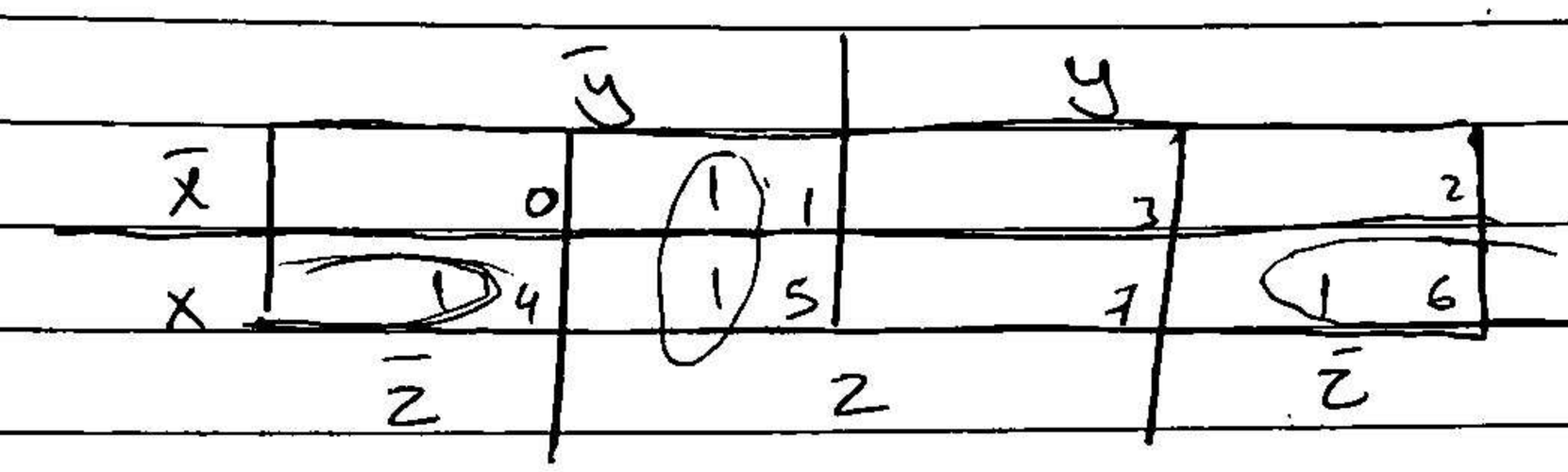
o $F(x, y, z) = \Sigma(0, 2, 4, 6)$



o $F(x, y, z) = \Sigma(0, 1, 2, 3)$



$F(x, y, z) = \Sigma(1, 6, 4, 5)$



$F = x\bar{z} + \bar{y}z$

* Let $F = \bar{A}C + \bar{A}B + A\bar{B}C + BC$

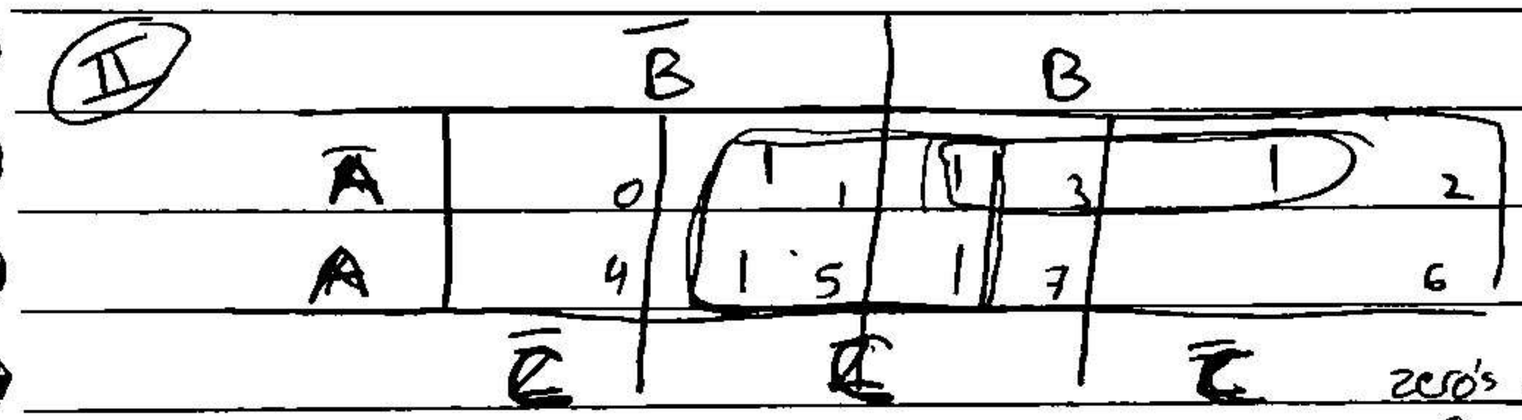
① Express F as the sum of minterms.
② Find the minimal SoP

(I)

A	B	C	$\bar{A}$	$\bar{A}C$	$\bar{A}B$	$A\bar{B}C$	BC	F
0	0	0	1	0	0	0	0	0
0	0	1	1	1	0	0	0	1
0	1	0	1	0	1	0	0	1
0	1	1	1	1	1	0	1	1
1	0	0	0	0	0	0	0	0
1	0	1	0	0	0	1	0	1
1	1	0	0	0	0	0	0	0
1	1	1	0	0	0	0	1	1

$F = \Sigma\ (1, 2, 3, 4, 7)$

(II)

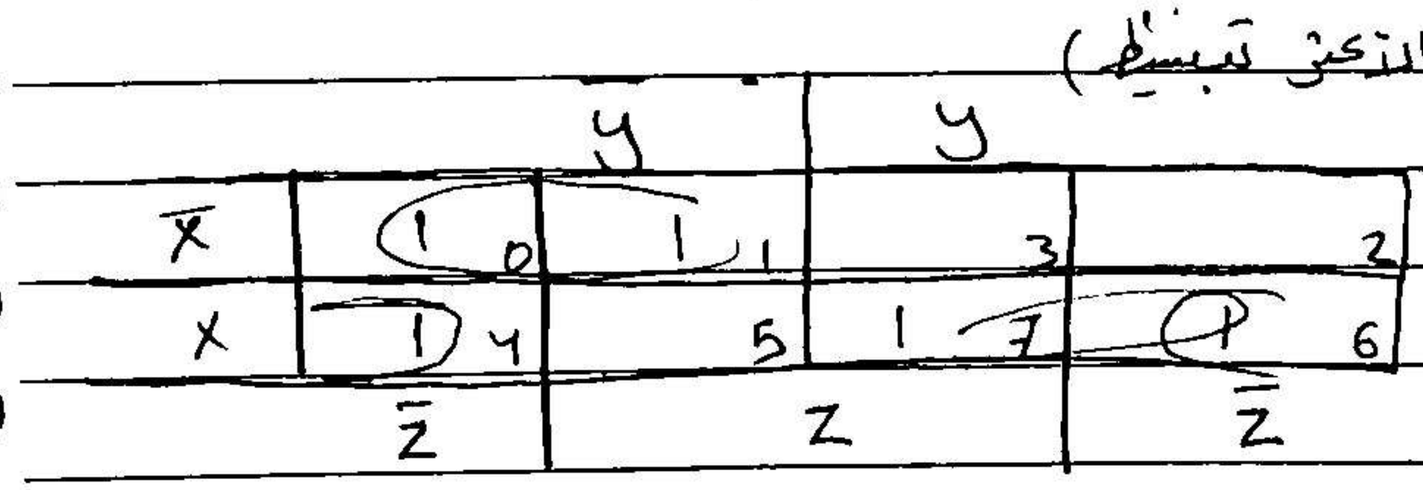


$F = C + \bar{A}B$

عادة منجمع الـ ones على الـ zero's ← minterms

* $F = \pi\ (2, 3, 5) \rightarrow = \Sigma\ (0, 1, 4, 6, 7)$ ← max terms ← (x, y, z)

(الطريقة العادية للسؤال على الذكر تبسيطه)



6 and 4 = $x\bar{z}$
0 and 1 = $\bar{x}\bar{y}$
7 and 6 = xy

$F = x\bar{z} + \bar{x}\bar{y} + xy$

or

$F = \bar{y}\bar{z} + \bar{x}\bar{y} + xy$

$F = \bar{x}\bar{y} + x\bar{z} + xyz$ ← X false.

ما ننفع آخذ إلـ 7 لحال.

** Four variable K-map :

w is most significant then x then y then z

	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$	
$\bar{w}$	m_0 0	m_1 1	m_3 3	m_2 2	$\bar{x}$
	m_4 4	m_5 5	m_7 7	m_6 6	x
w	m_{12} 12	m_{13} 13	m_{15} 15	m_{14} 14	
	m_8 8	m_9 9	m_{11} 11	m_{10} 10	$\bar{x}$

• $F(w,x,y,z) = \Sigma(0,1,2,4,5,6,8,9,12,13,14)$

	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$	
$\bar{w}$	1 0	1 1	3	1 2	$\bar{x}$
	1 4	1 5	7	1 6	x
w	1 12	1 13	15	1 14	
	1 8	1 9	11	10	$\bar{x}$

$\bar{z}$ | z | $\bar{z}$

0, 4, 12, 8 → $\bar{y}$

6, 14, 4, 12 → $x\bar{z}$

0, 4, 2, 6 → $\bar{w}\bar{z}$

$F = \bar{y} + x\bar{z} + \bar{w}\bar{z}$

• $F(w,x,y,z) = \Sigma(0,2,8,12,14,10,15)$

	$\bar{y}$		y		
$\bar{w}$	1 0	1	3	1 2	$\bar{x}$
	4	5	7	6	x
w	$\bar{1}$ 12	13	1 15	1 14	
	1 8	9	11	1 10	$\bar{x}$

$\bar{z}$ | z | $\bar{z}$

0, 2, 8, 10 → $\bar{x}\bar{z}$

14, 10, 12, 8 → $w\bar{z}$

14, 15 → wxy

$F = \bar{x}\bar{z} + w\bar{z} + wxy$

* Simplify :

$F(A,B,C,D) = \bar{A}\bar{B}\bar{C} + \bar{B}C\bar{D} + \bar{A}BC\bar{D} + A\bar{B}\bar{C}$

$\bar{A}$	$\bar{C}\bar{D}$	$\bar{C}D$	$C D$	$C\bar{D}$	$\bar{B}$
	0	1	3	2	
	4	5	7	6	B
A	12	13	15	14	
	8	9	11	10	$\bar{B}$

$0, 2, 8, 10 \longrightarrow \bar{B}\bar{D}$

$0, 1, 8, 9 \longrightarrow \bar{B}\bar{C}$

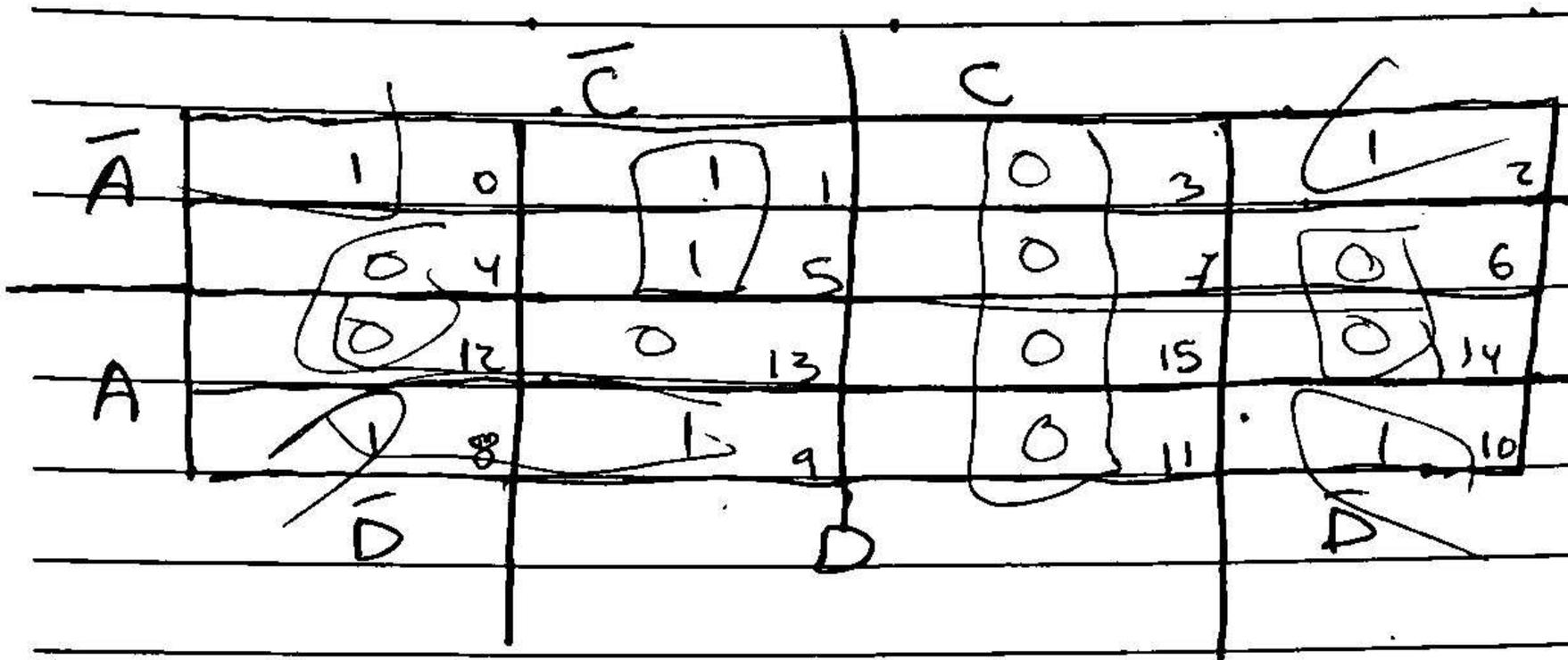
$2, 6 \longrightarrow \bar{A}C\bar{D}$

$\longrightarrow F = \bar{B}\bar{D} + \bar{B}\bar{C} + \bar{A}C\bar{D}$

lecture (7)

* Simplify :

$F(A,B,C,D) = \Sigma(0,1,2,5,8,9,10)$



$F = \bar{B}\bar{D} + \bar{A}CD + A\bar{B}\bar{C}$ SoP

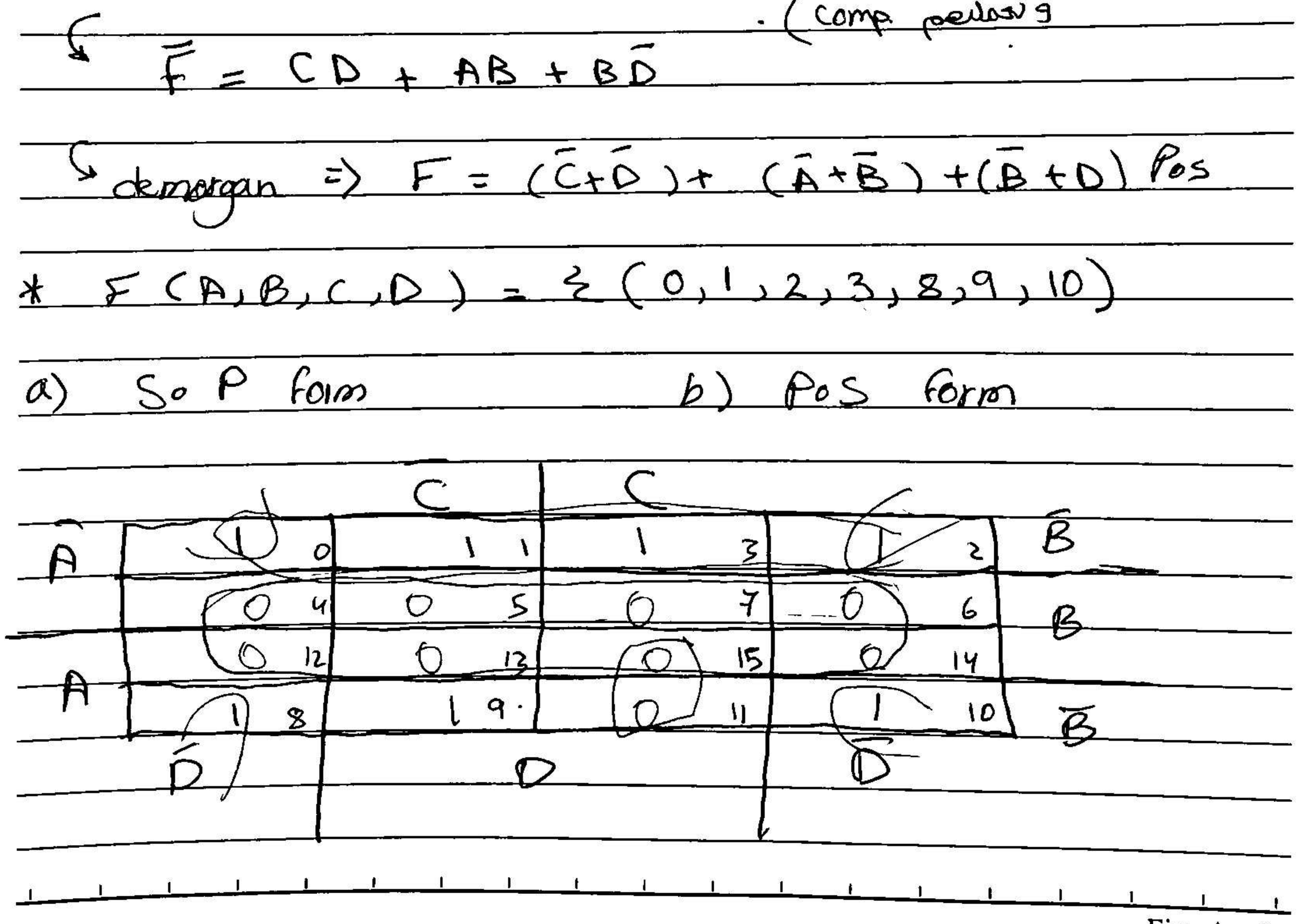
• ↳ Find the product of sum : zeros الـ (يعني)

↳ $\bar{F} = CD + AB + B\bar{D}$. (complement)

↳ demorgan ⇒ $F = (\bar{C}+\bar{D}) + (\bar{A}+\bar{B}) + (\bar{B}+D)$ PoS

* $F(A,B,C,D) = \Sigma(0,1,2,3,8,9,10)$

a) SoP form b) PoS form



* اها يكونا عندي مجال تاخدي 4 وتروحي تاخدي 2 غلط

0, 2, 8, 10 → $\bar{A}\bar{B}$

0, 1, 3, 2 → $\bar{B}\bar{D}$

0, 1, 5, 10

$F = \bar{A}\bar{B} + \bar{B}\bar{D} + \bar{B}\bar{C}$

b) PoS :

4, 5, 7, 6, 12, 13, 15, 14 → B

15, 11 → ACD

$\bar{F} = B + ACD$

$\overline{\bar{F}} = \overline{B + ACD} = (\bar{B})(\bar{A} + \bar{C} + \bar{D})$

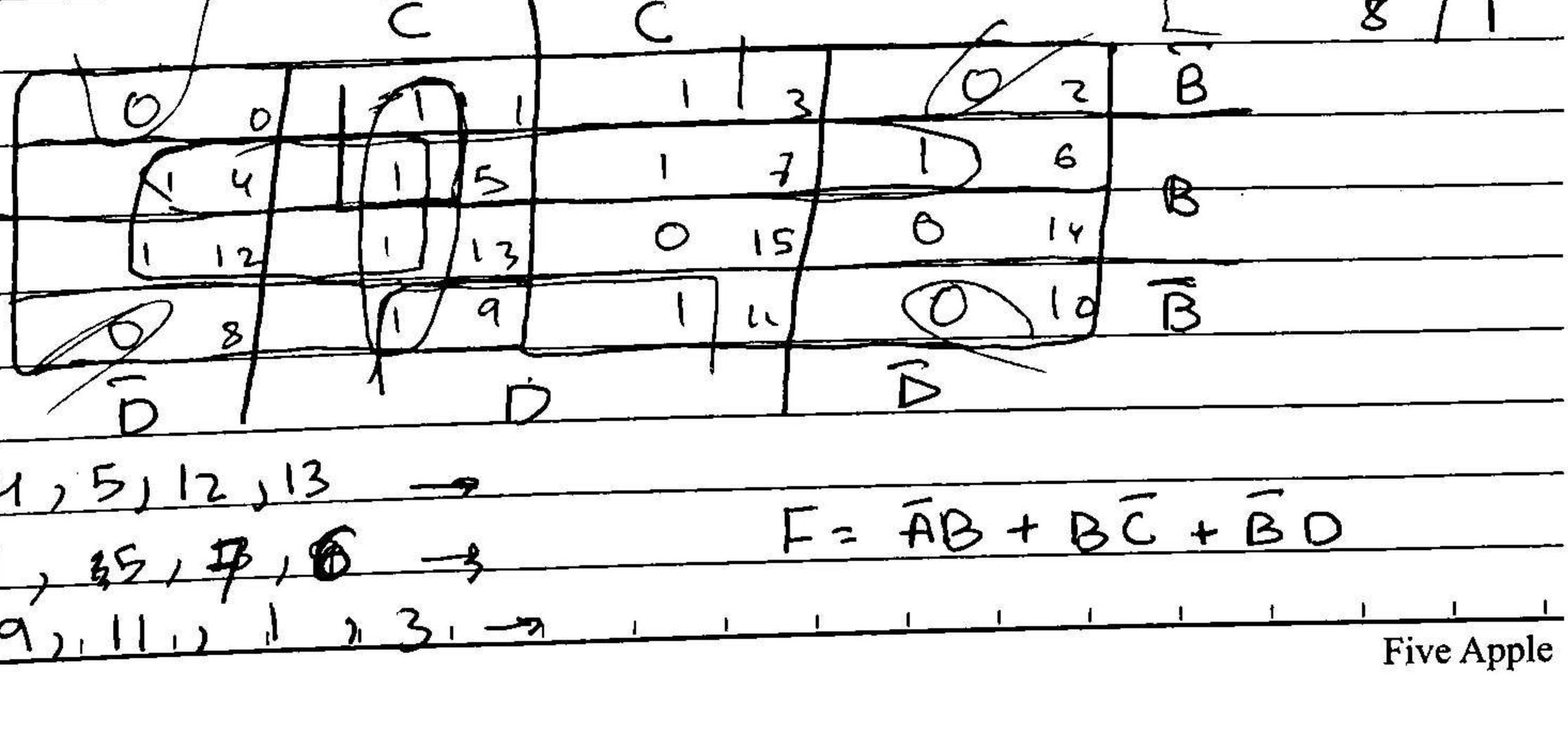
** Enter the following PoS function into the map and find the Sop for it :

$F = (\bar{A} + \bar{B} + \bar{C})(B + D)$

$\bar{F} = \overline{(\bar{A} + \bar{B} + \bar{C})(B + D)}$

$\bar{F} = ABC + \bar{B}\bar{D}$

4 variable map	
# squares	# variable
1	4
2	3
4	2
8	1



4, 5, 12, 13 →

4, 5, 11, 6 →

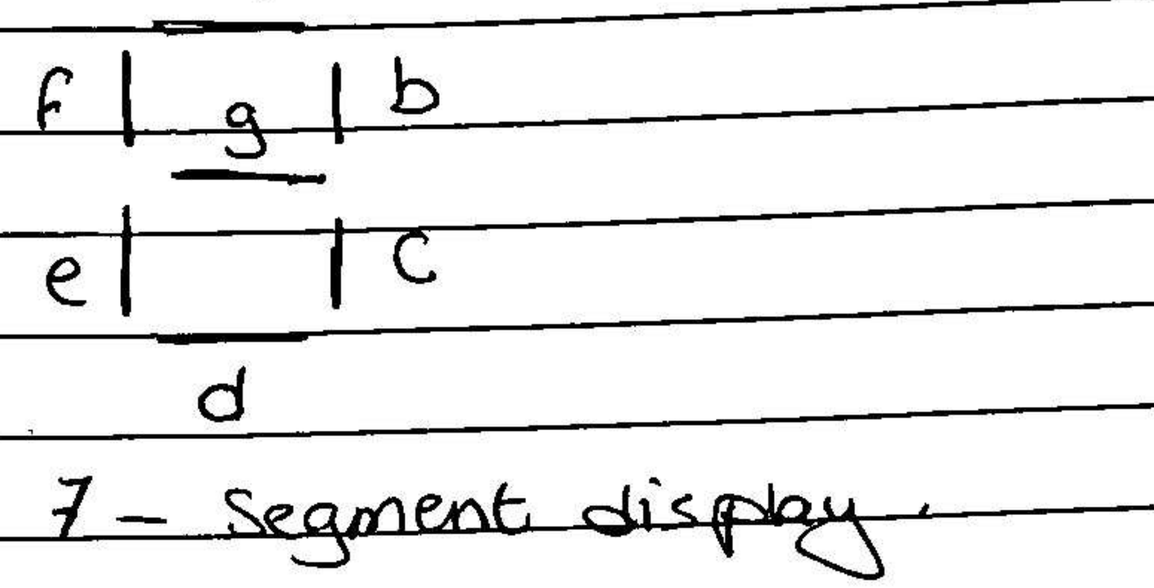
9, 11, 1, 3 →

$F = \bar{A}B + B\bar{C} + \bar{B}D$

** Don't care: بعتبر احلها صفر وبعتبر واحد حسب شو بحتاج.

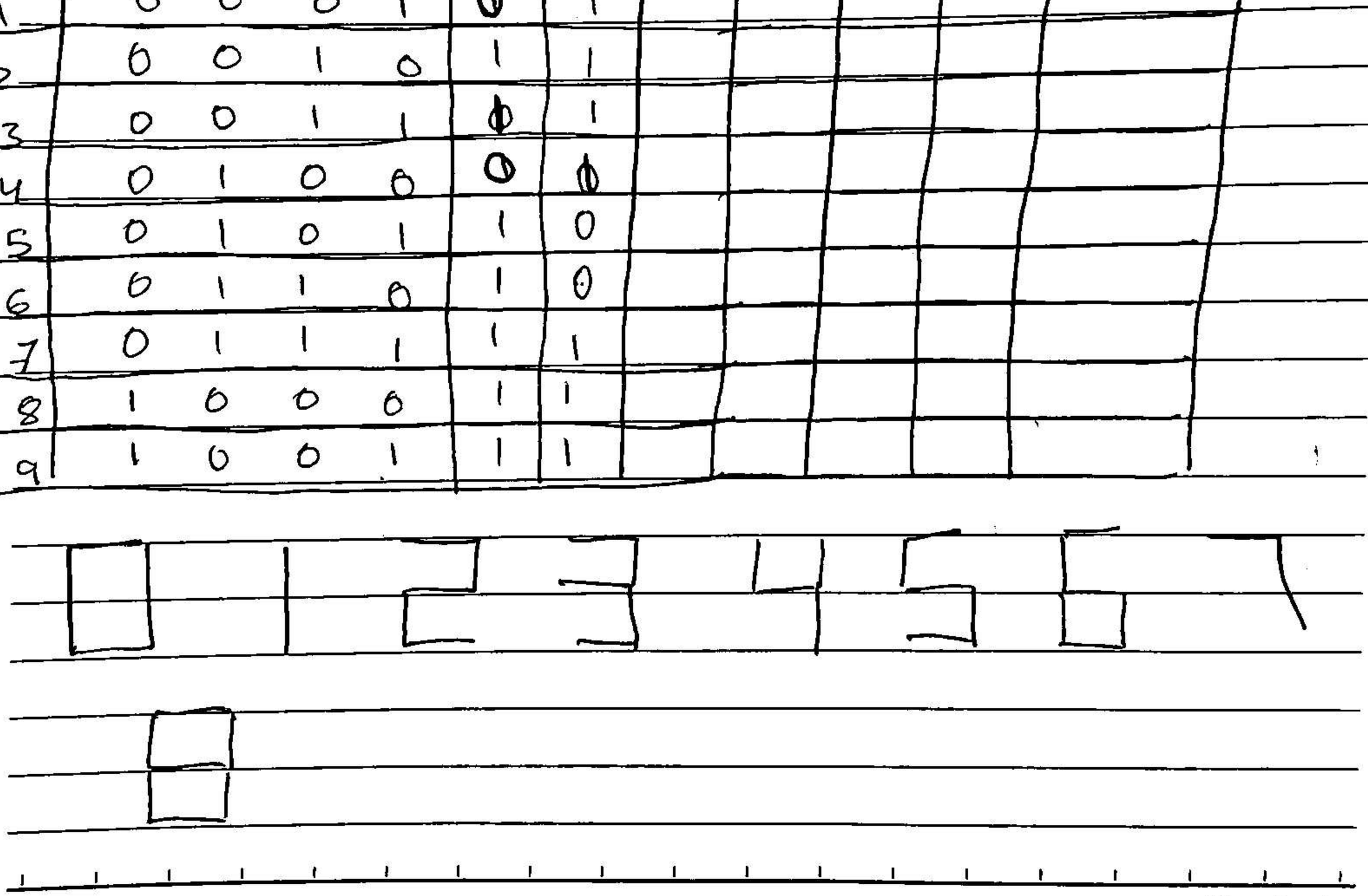
BCD

0	0 0 0 0
1	0 0 0 1
2	0 0 1 0
3	0 0 1 1
4	0 1 0 0
5	0 1 0 1
6	0 1 1 0
7	0 1 1 1
8	1 0 0 0
9	1 0 0 1



7 - Segment display.

		a	b	c	d	e	f	g
0	0 0 0 0	ϕ	1					
1	0 0 0 1	0	1					
2	0 0 1 0	1	1					
3	0 0 1 1	ϕ	1					
4	0 1 0 0	0	ϕ					
5	0 1 0 1	1	0					
6	0 1 1 0	1	0					
7	0 1 1 1	1	1					
8	1 0 0 0	1	1					
9	1 0 0 1	1	1					



Five Apple

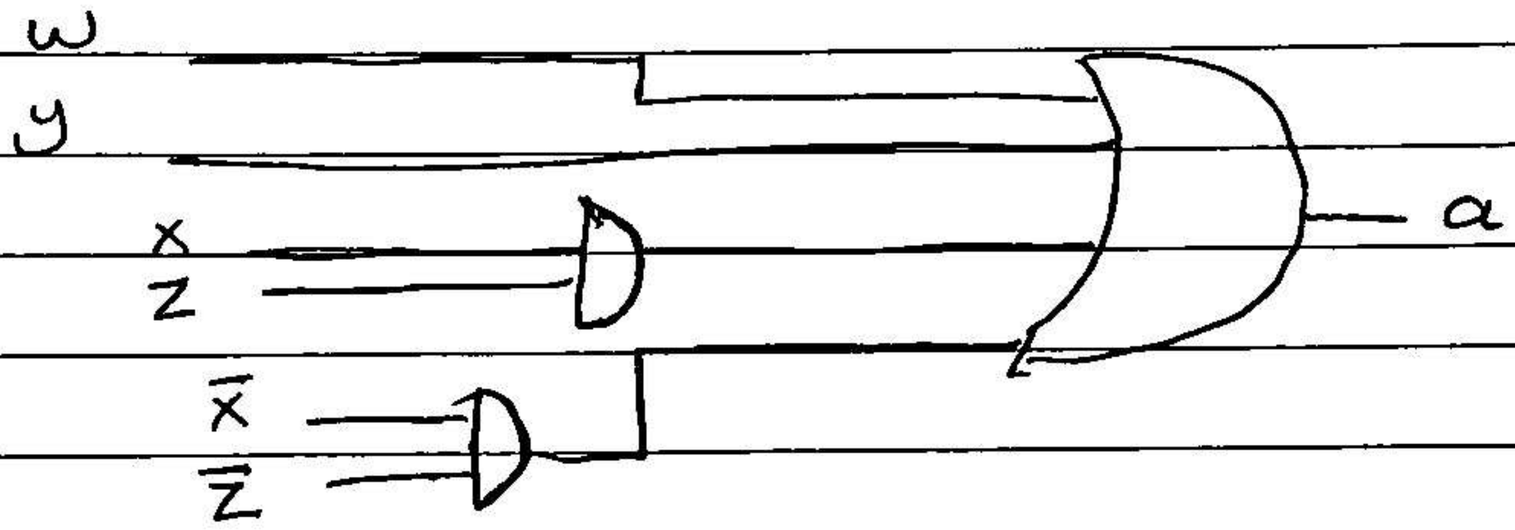
k-map for a

	$\bar{y}$		y		
$\bar{w}$	1 0	0 1	4 3	2	$\bar{x}$
	0 4	* 5	1 7	6	X
w	X 12	X 13	X 15	X 14	
	8	9	X 11	X 10	$\bar{x}$
	z		z	$\bar{z}$	

X ⟶ Don't care

$a = w + y + xz + \bar{x}\bar{z}$

↳ logical circuit



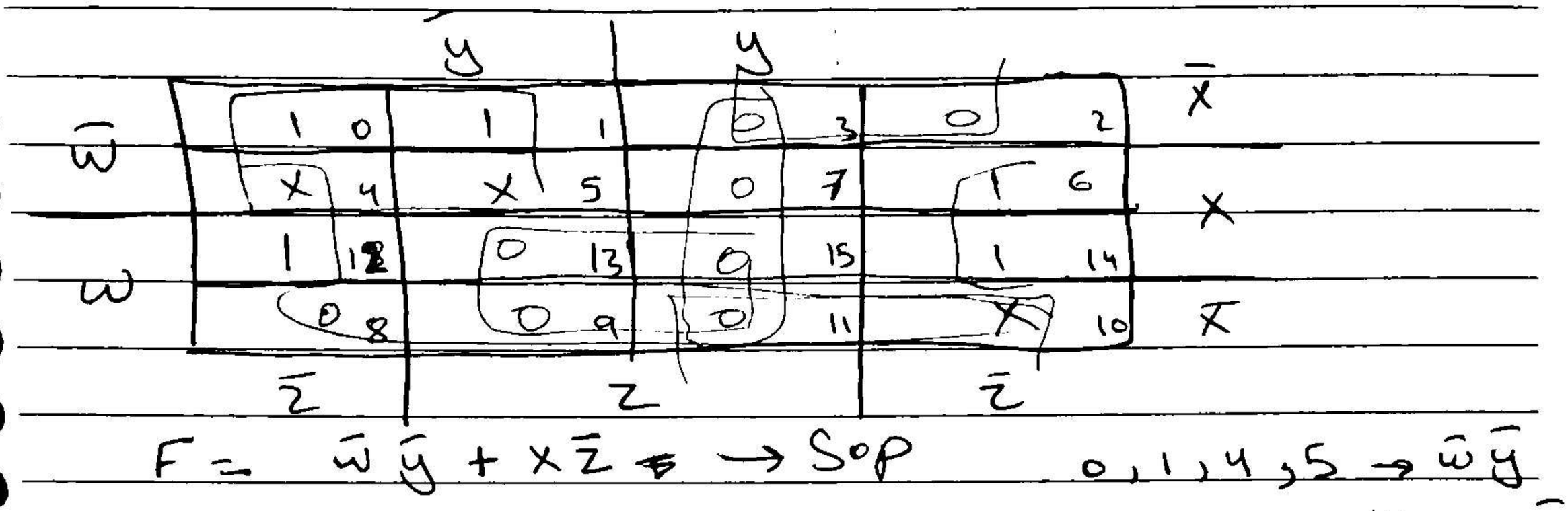
lecture (8)

** Given the following Boolean function :

$$F(w,x,y,z) = xy\bar{z} + \bar{x}\bar{y}\bar{w} + x\bar{y}\bar{z}w$$

$$d(w,x,y,z) = \bar{x}y\bar{z}w + x\bar{y}\bar{w} \rightarrow \text{function of don't care}$$

find ① SoP ② PoS



$F = \bar{w}\bar{y} + x\bar{z}$ → SoP

0, 1, 4, 5 → $\bar{w}\bar{y}$

4, 12, 6, 14 → $x\bar{z}$

3, 7, 15, 11 → yz

13, 15, 9, 11 → zw

8, 9, 11, 10 → $w\bar{x}$

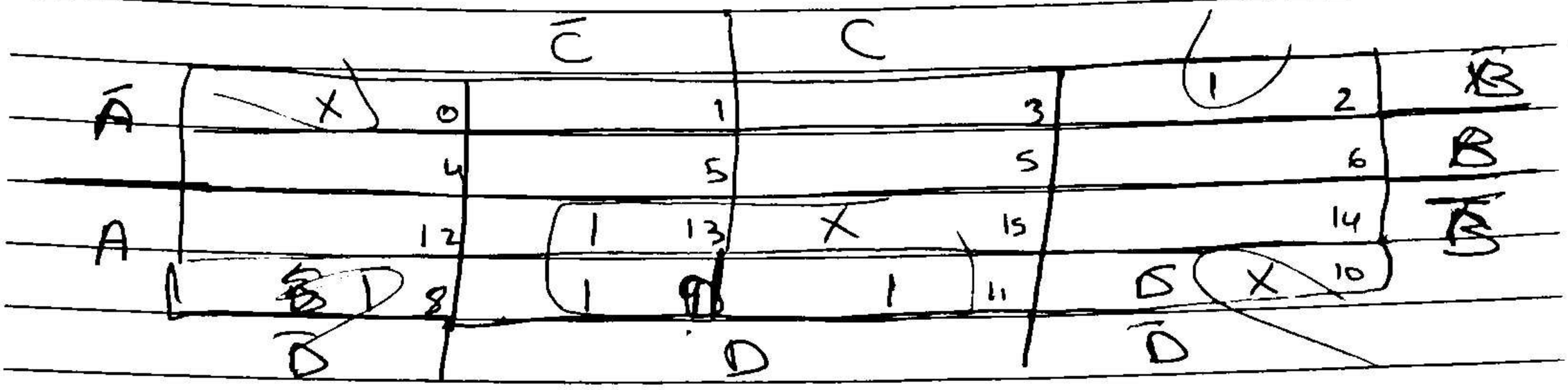
3, 2, 11, 10 → $\bar{x}y$

$\bar{F} = yz + zw + w\bar{x} + \bar{x}y$

$F = \overline{yz + zw + w\bar{x} + \bar{x}y}$

$= (\bar{y}+\bar{z})(\bar{z}+\bar{w})(\bar{w}+x)(x+\bar{y})$ → PoS

** ex: the function $F(A,B,C,D) = \Sigma(2,8,9,11,13)$ is simplified to $F = AD + \bar{B}\bar{D}$, what is the 'don't care' function used ?



$13, 11, 1, 15 \rightarrow AD$

$0, 8, 2, 10 \rightarrow \bar{B}\bar{D}$

$d(A,B,C,D) = \Sigma(0,10,15)$

* Design a digital circuit that converts BCD to excess3 (+3) state form

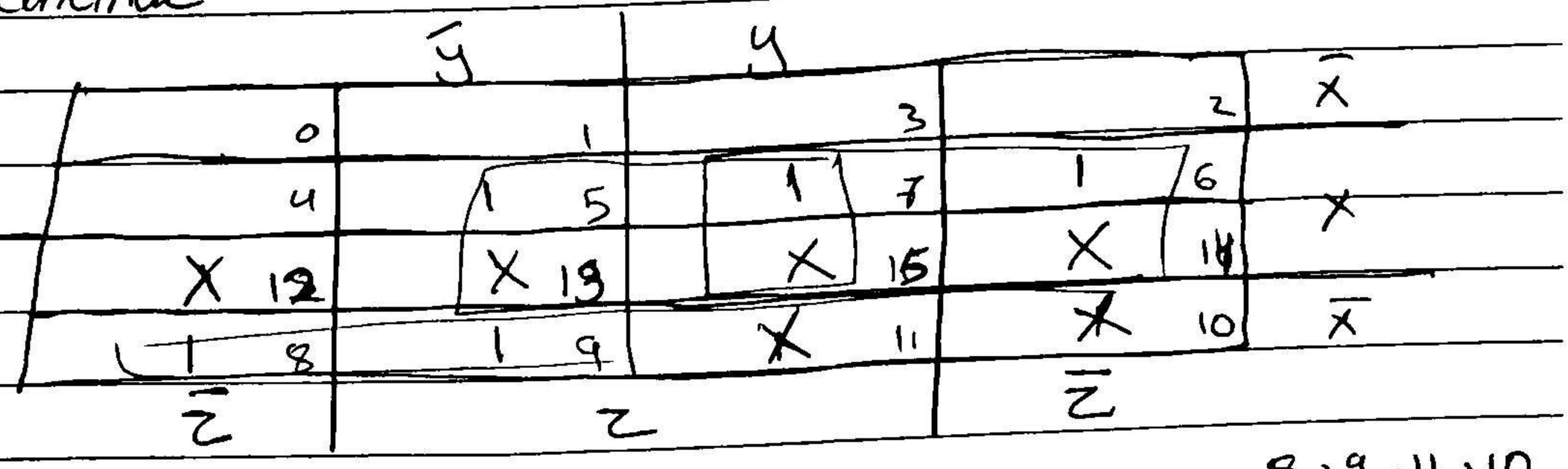
	w	x	y	z	A	B	C	D
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0

$A(w,x,y,z) = \Sigma(5,6,7,8,9)$

$d(w,x,y,z) = \Sigma(10,11,12,13,14,15)$ don't care

الفكرة من السؤال انه يكمل اذا don't care فيه الـ 10 لـ 15

Continue



$A = w + xy + xz$

$w \rightarrow 8, 9, 11, 10$

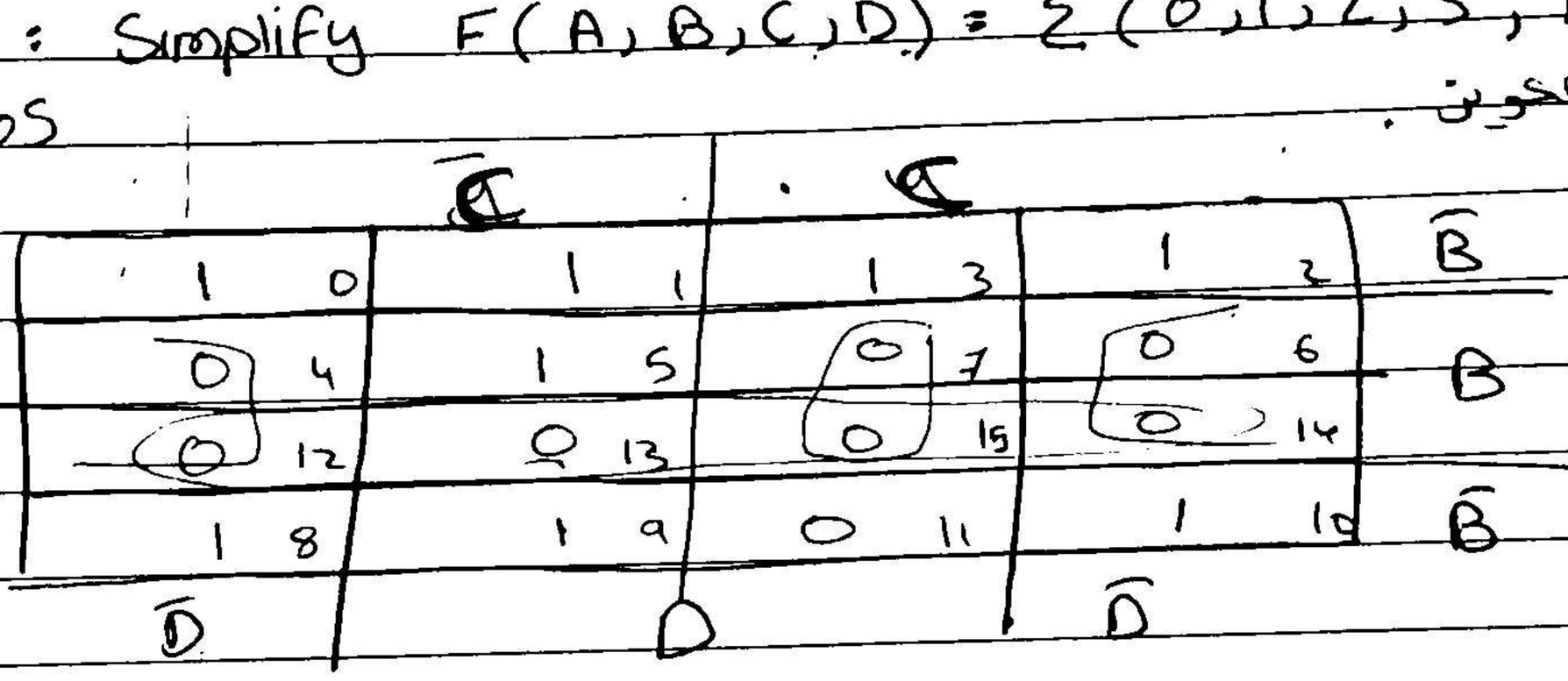
$xy \rightarrow 7, 6, 15, 14$

$xz \rightarrow 5, 7, 13, 15$

~~[redacted]~~

** ex: Simplify $F(A,B,C,D) = \Sigma(0,1,2,3,5,8,9,10)$ in POS

سؤال الكوين.



$\bar{F} = AB + \bar{D}B + BC + ACD$

$= (\bar{A}+\bar{B})(D+\bar{B})(\bar{B}+\bar{C})(\bar{A}+\bar{C}+\bar{D})$

$\bar{D}B \rightarrow 6, 14, 4, 12$

$AB \rightarrow 12, 13, 14, 15$

$BC \rightarrow$

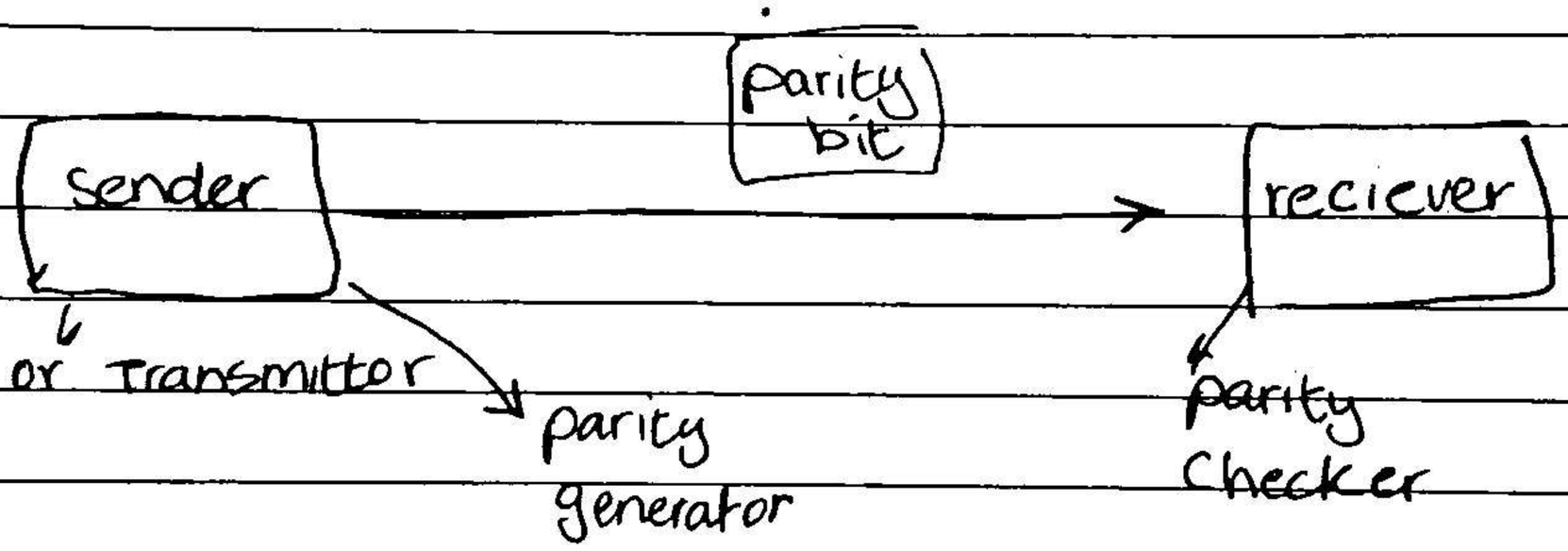
$ACD \rightarrow$

** BCD to Gray

A	B	C	D	w	x	y	z
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1

يعني ايه don't care وبيشتغلي عليه.

* Parity :



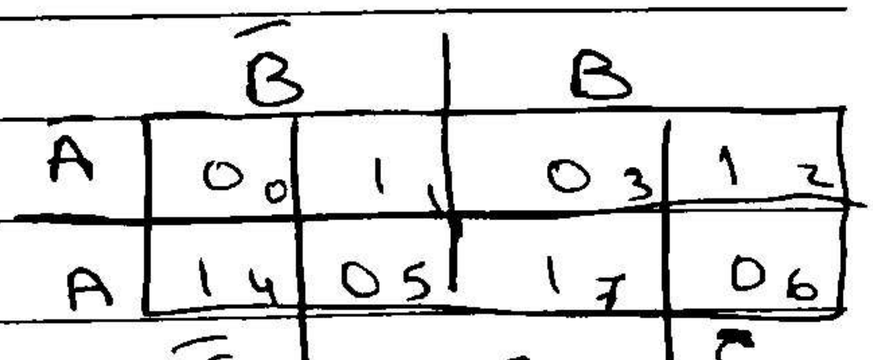
• In even parity, the parity Bit is 0 if the number of 1s is even

• In odd parity, the parity Bit is 1 if the number of 1s is even

** even parity Generator :

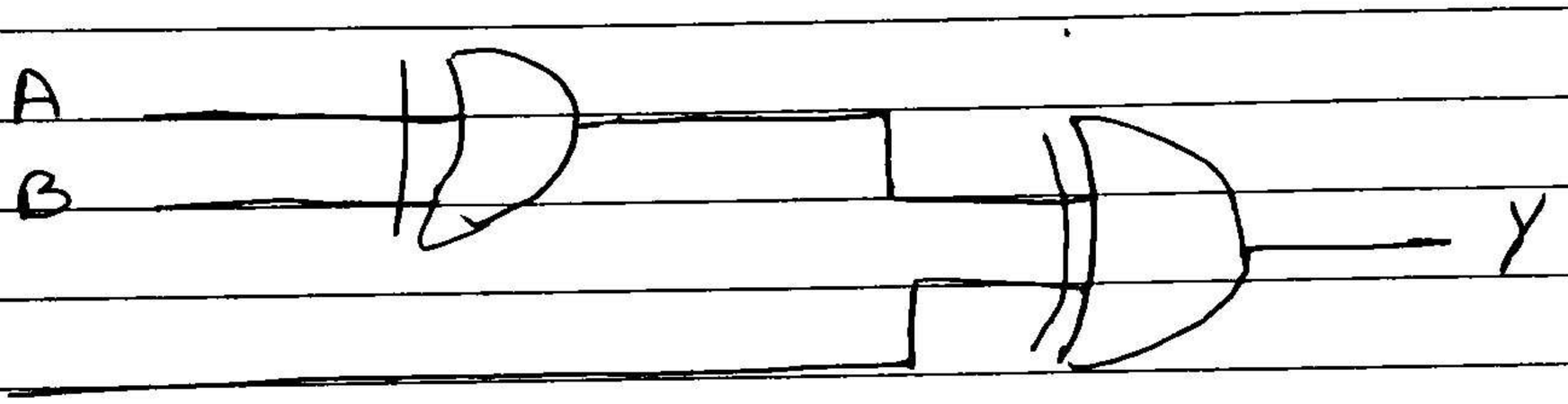
3 - bit message even parity (P)

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1



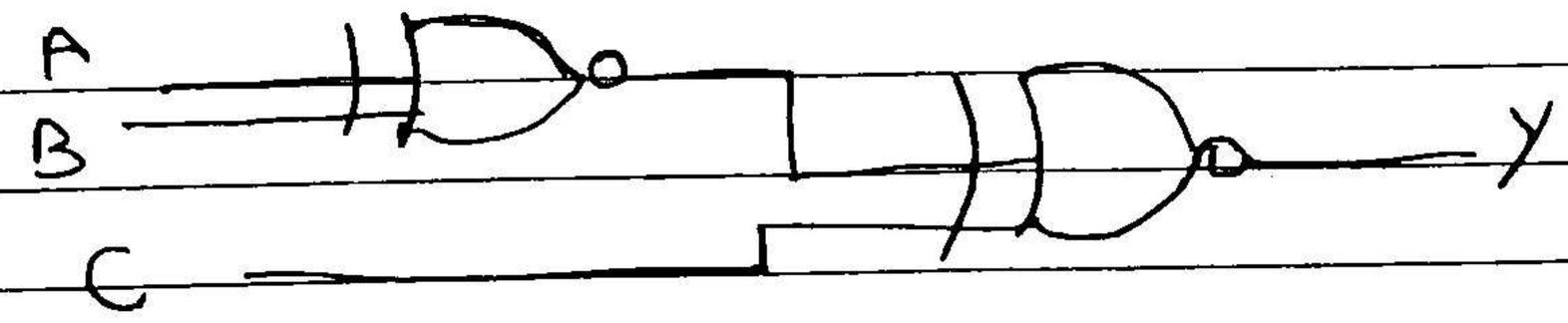
$Y = A \oplus B \oplus C$

ما اله حل



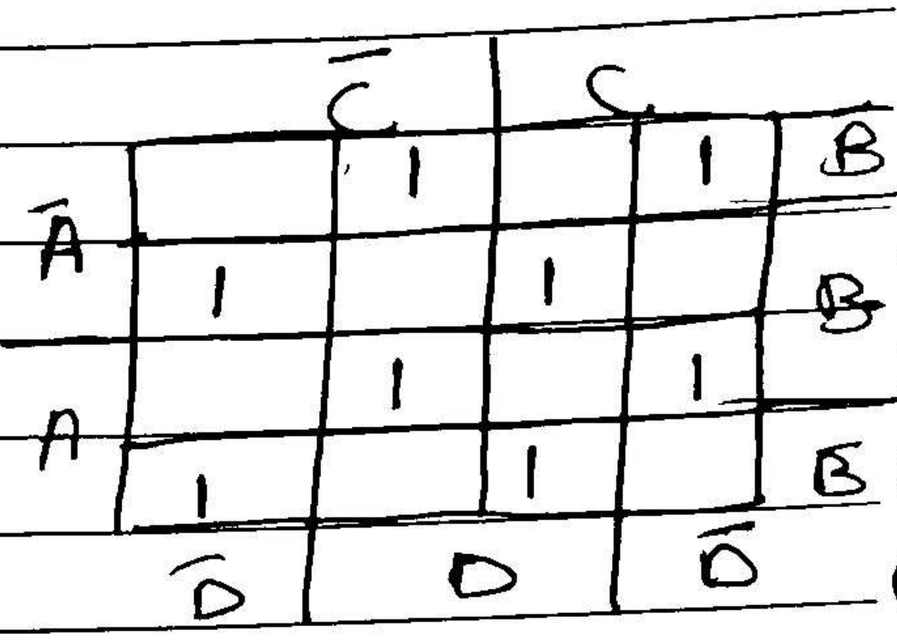
* odd parity

$F = \overline{A \oplus B \oplus C}$

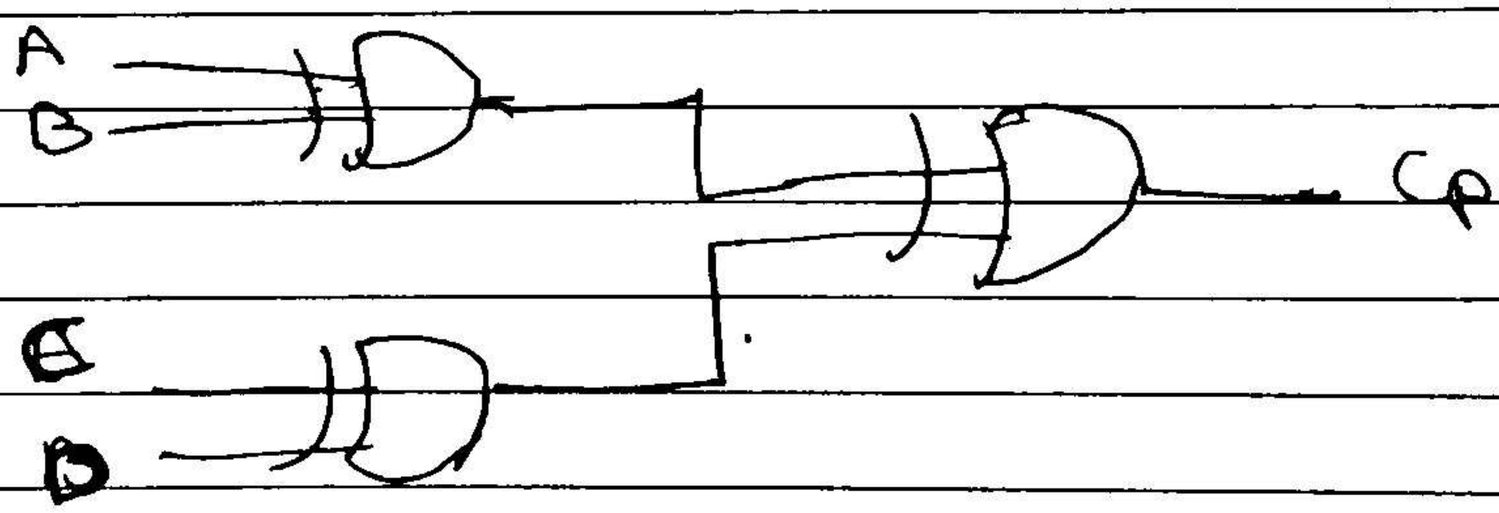


** parity checker

4 bit message				parity error Cp
A	B	C	P	
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1



$C_P = (A \oplus B) \oplus (C \oplus D)$



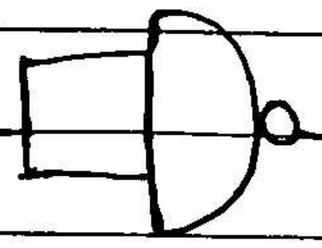
lecture (9)

** Universal Gates :

° NAND Circuits :

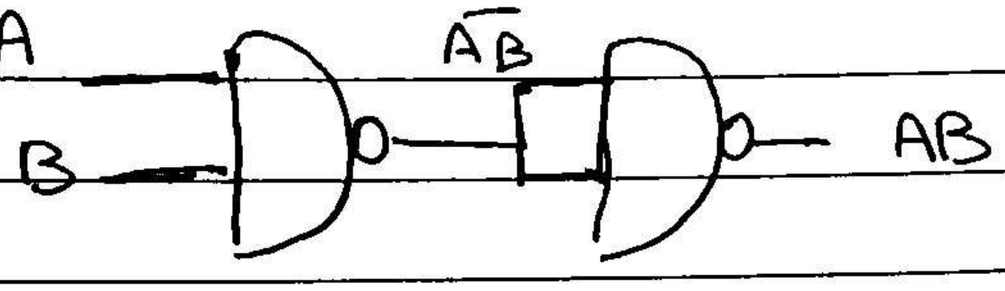
A	B	$\overline{AB}$
0	0	1
0	1	1
1	0	1
1	1	0

[1] NOT : $F = \overline{x}$

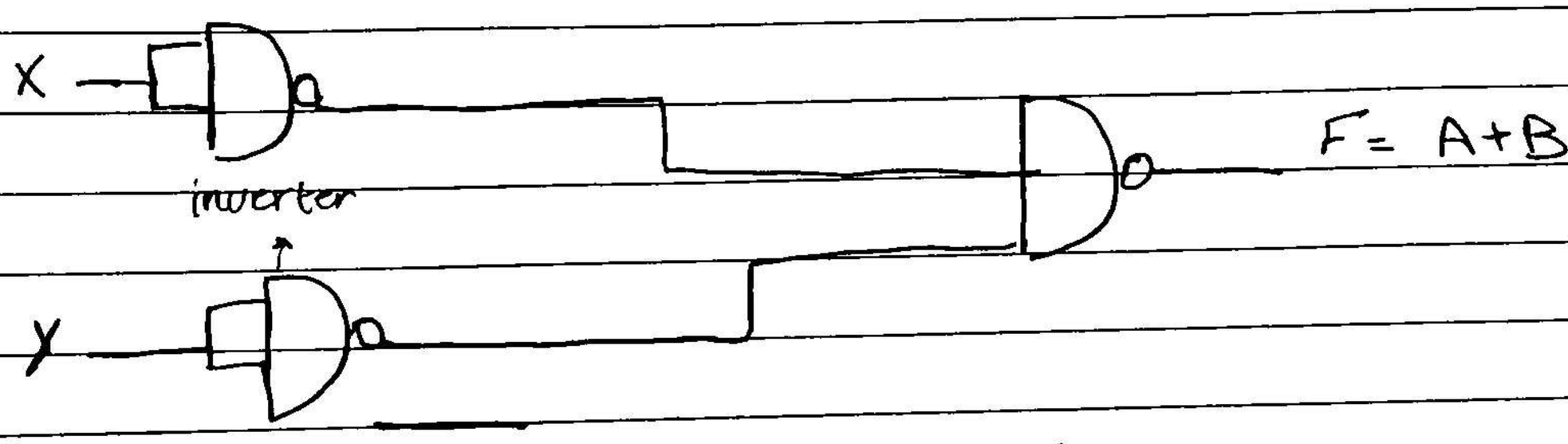


$F = (\overline{xx}) = \overline{x}$

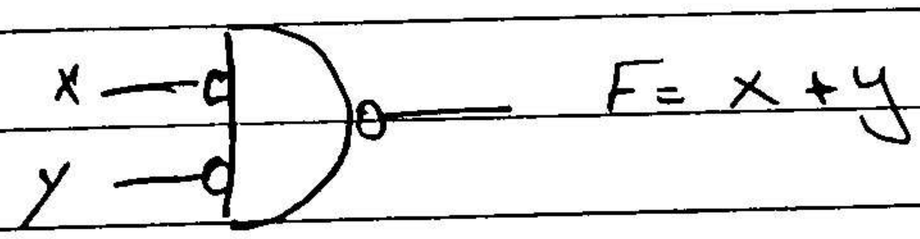
[2] AND : $F = AB$



[3] OR : $F = A + B$



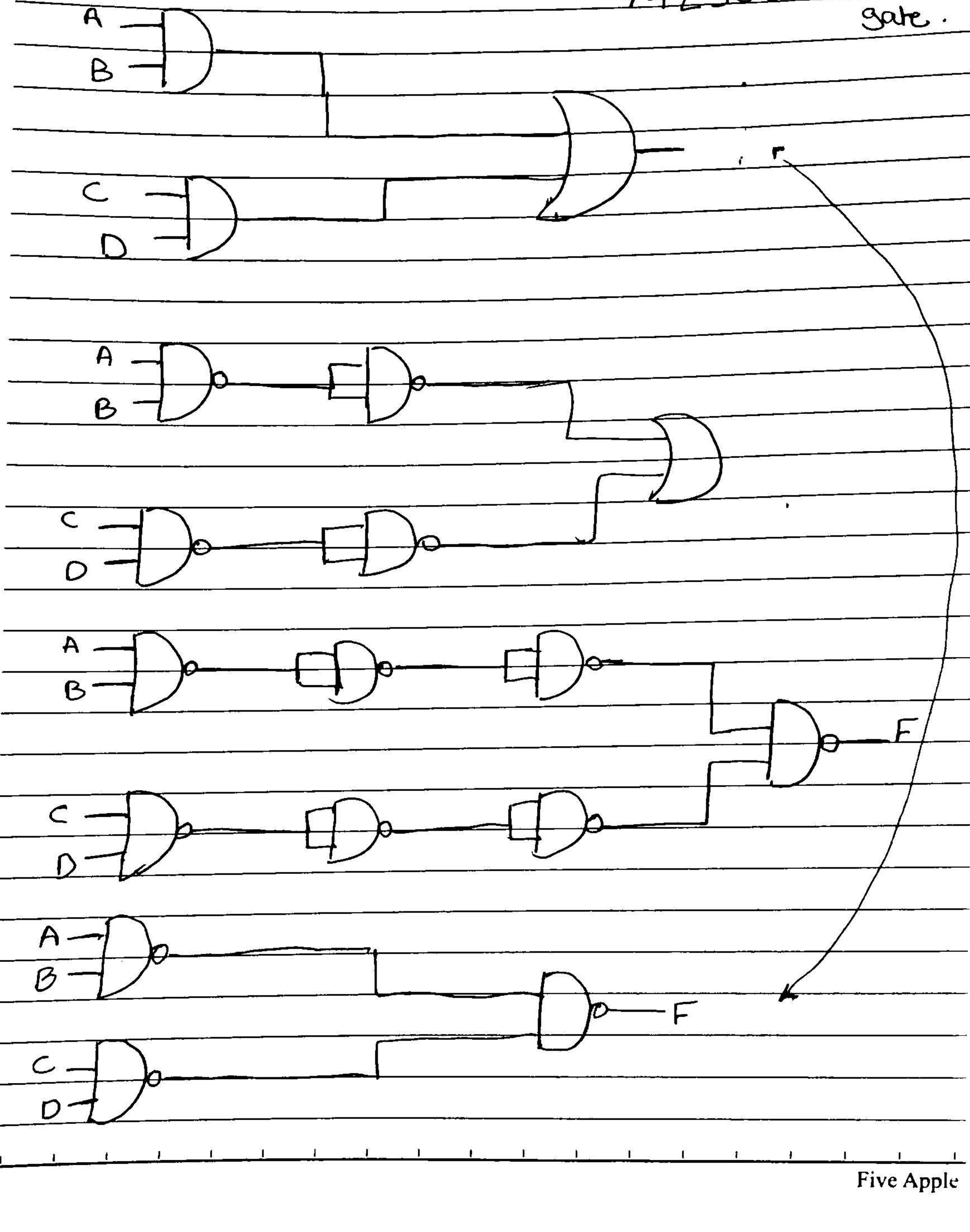
$F = \overline{x}\,\overline{y} = x + y$



* ex: implement the following using NAND Gates:

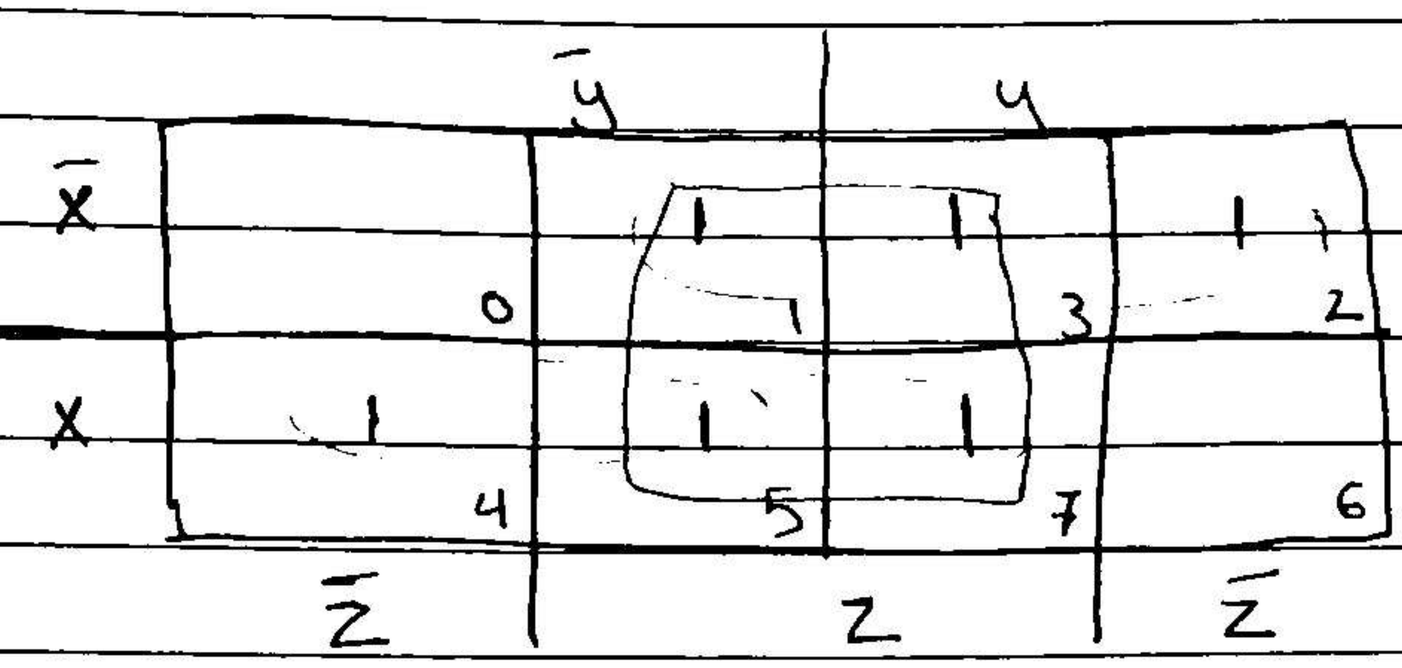
$F = AB + CD$

74LS00 → NAND gate.



*ex: Implement the following Boolean function with NAND gates.

$F(x,y,z) = \Sigma(1,2,3,4,5,7)$

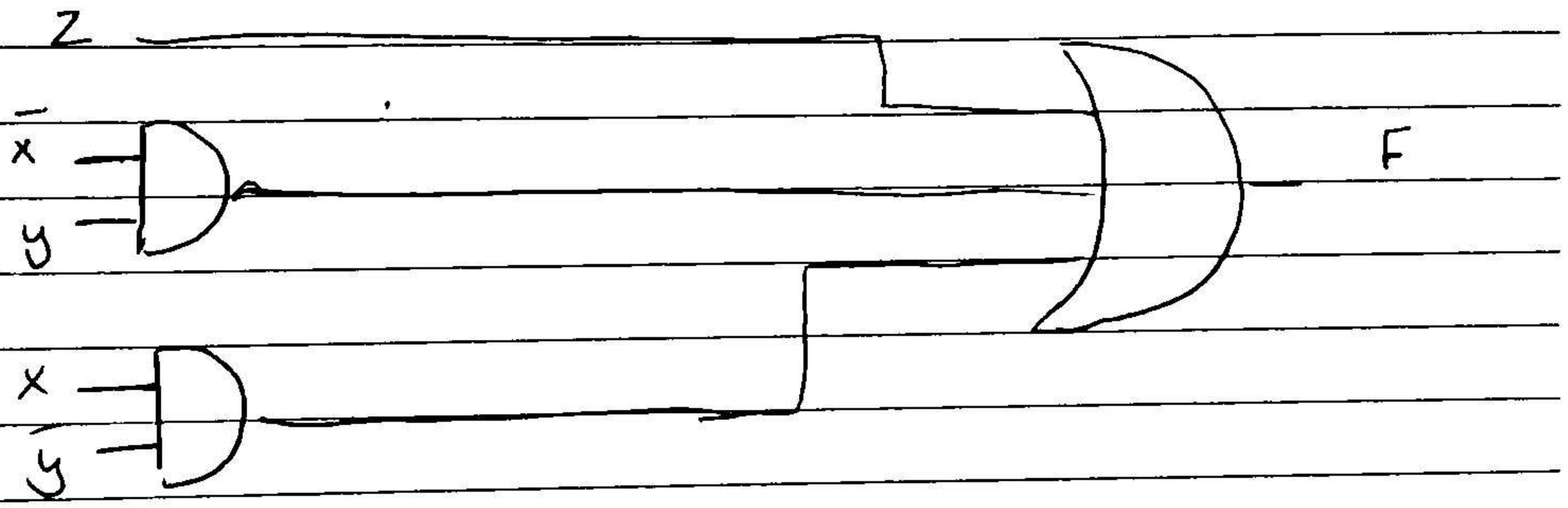


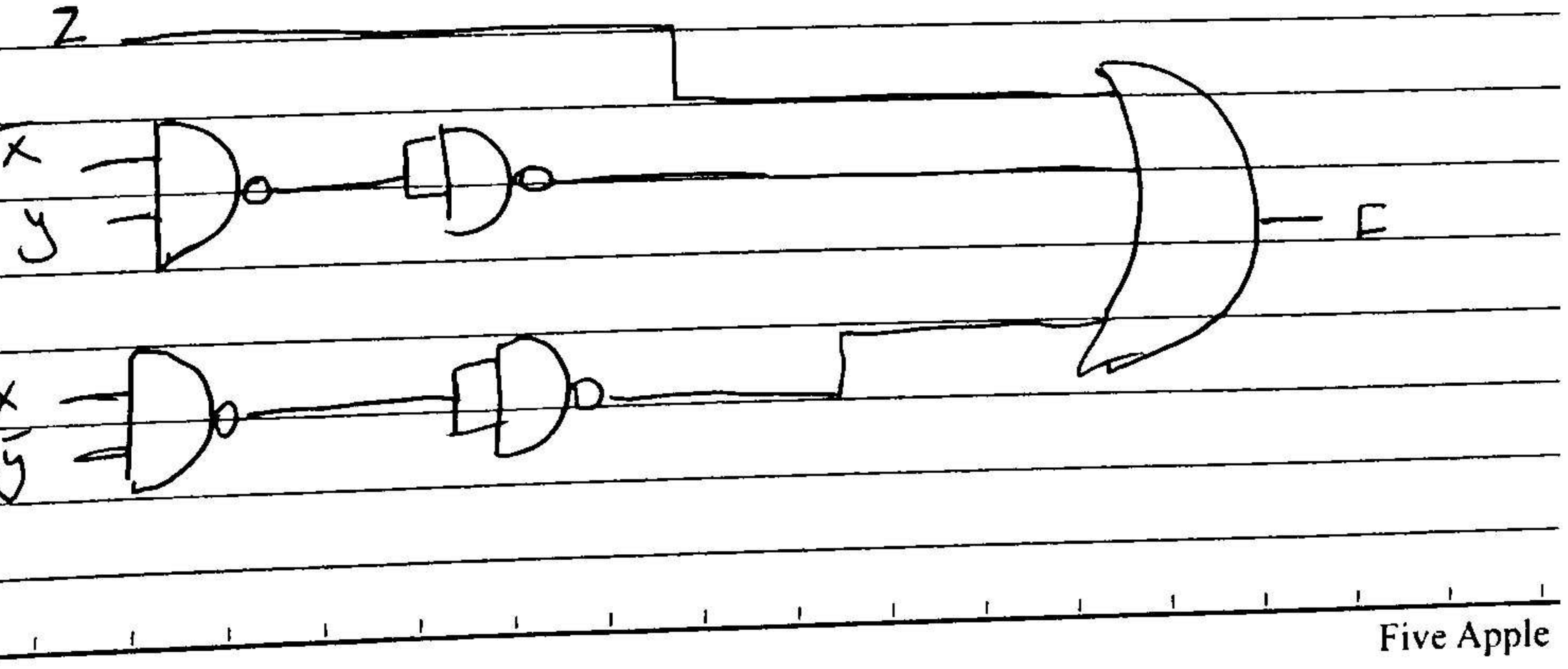
5, 7, ~~1, 3~~ → ~~x̄~~ z

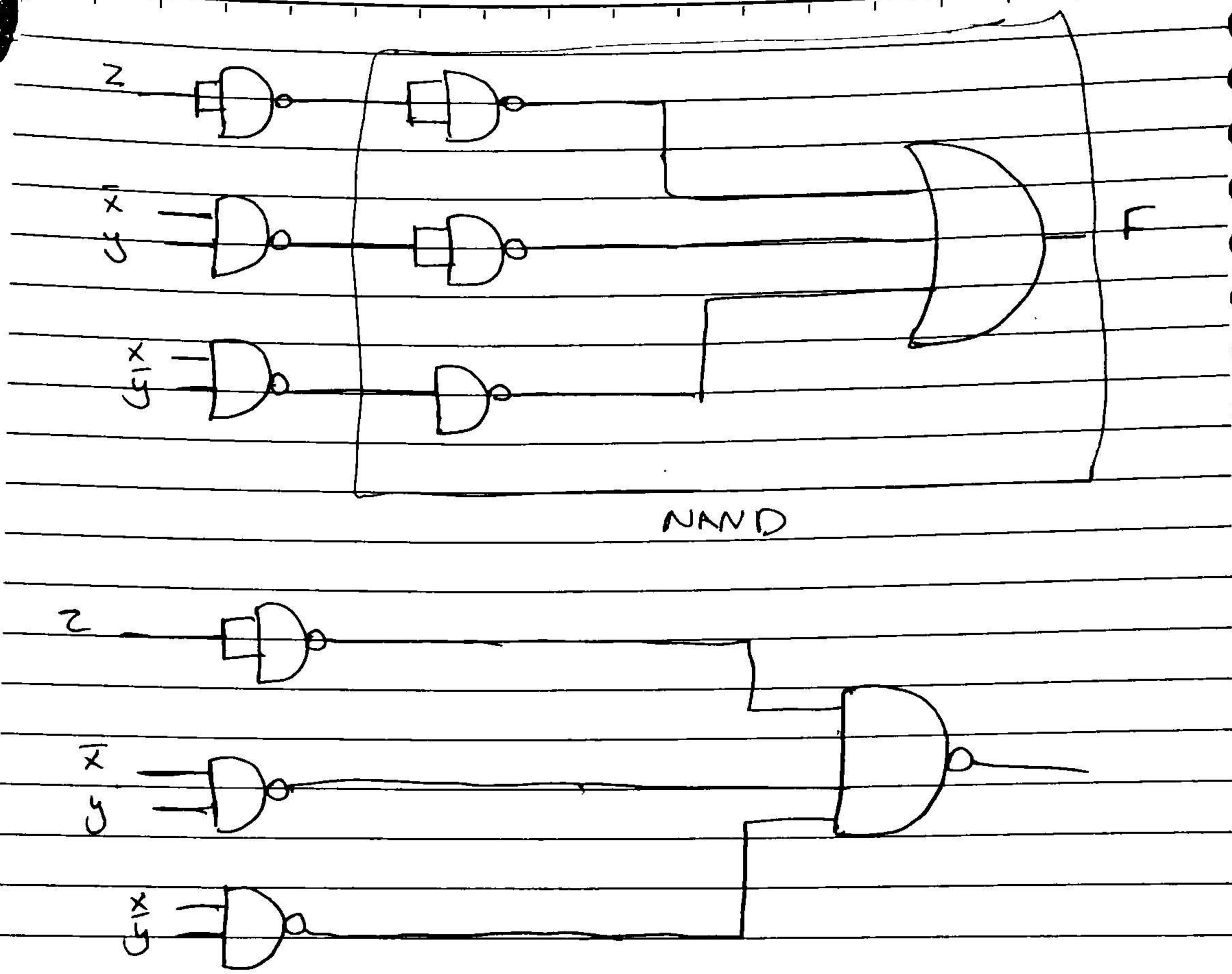
~~3~~, 2 → x̄ ~~y~~

~~y~~, 5 → x ȳ

$F = z + \bar{x}y + x\bar{y}$

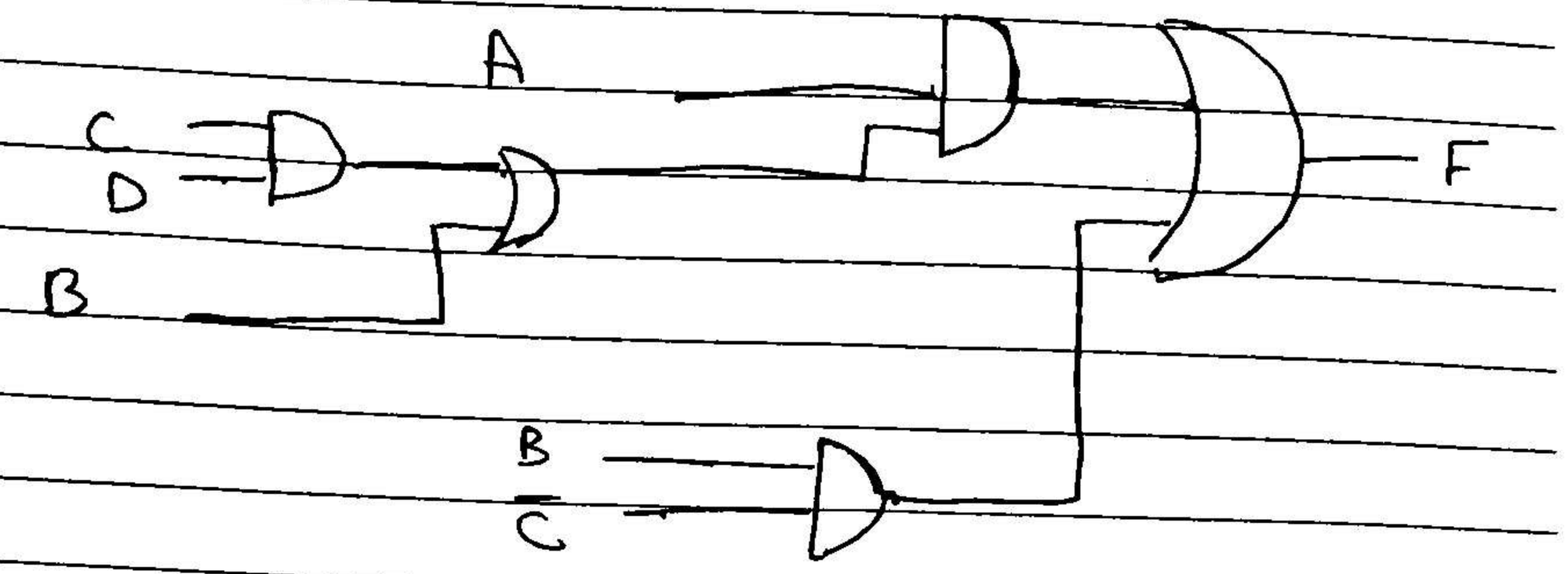




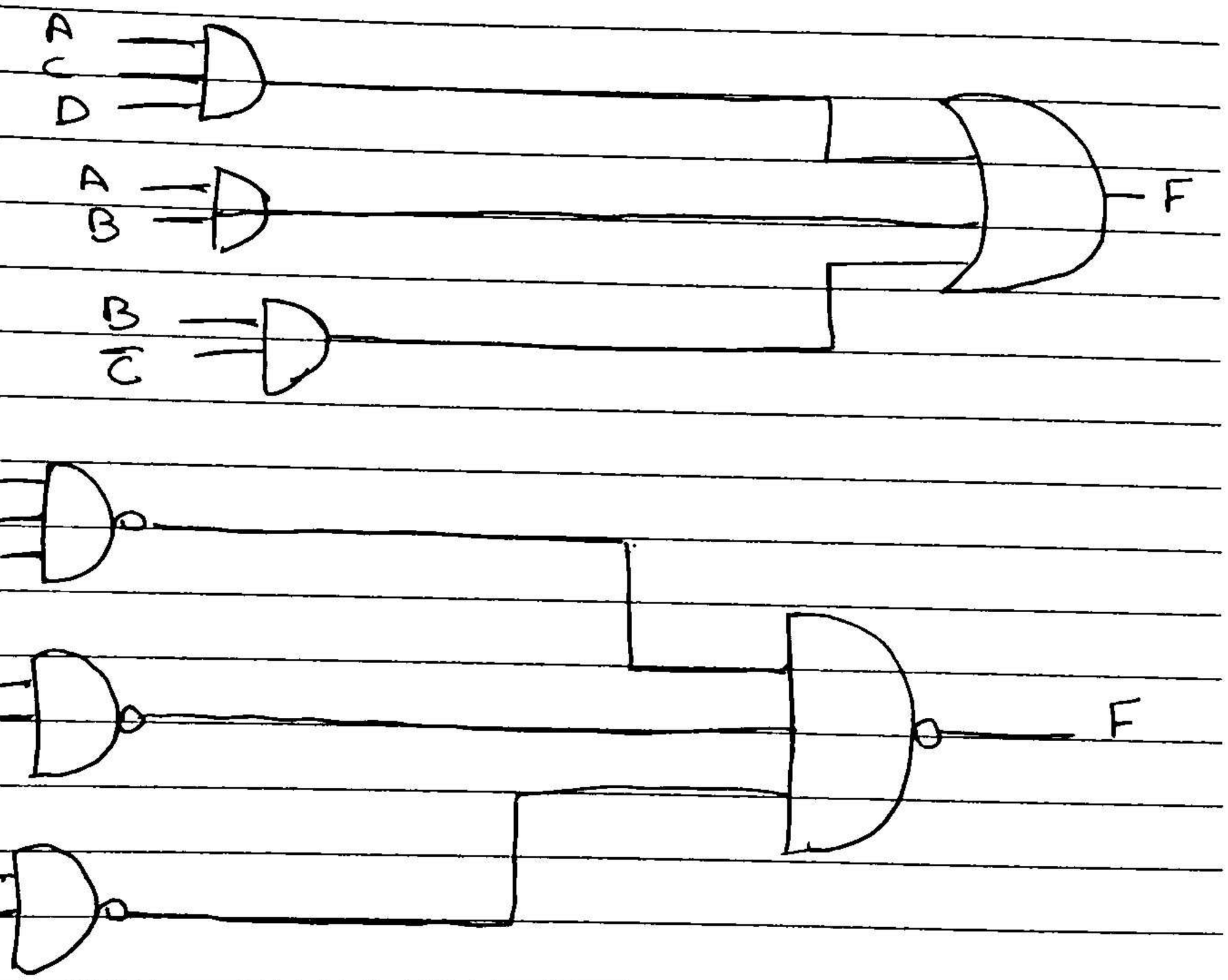

z
x̄
y
F
x̄
y
NAND
z
x̄
y
x
y

*

$$F = A(CD + B) + B\bar{C}$$



or $F = ACD + AB + B\bar{C}$

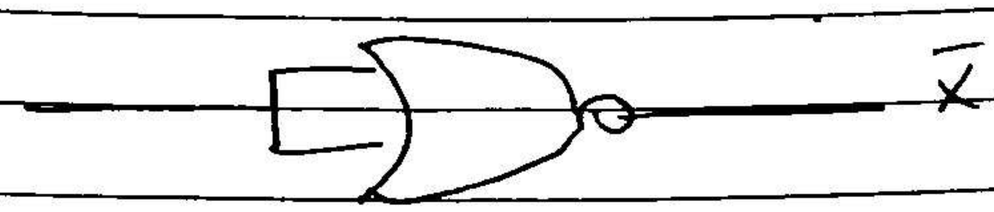


Nor ⟶ product of sum

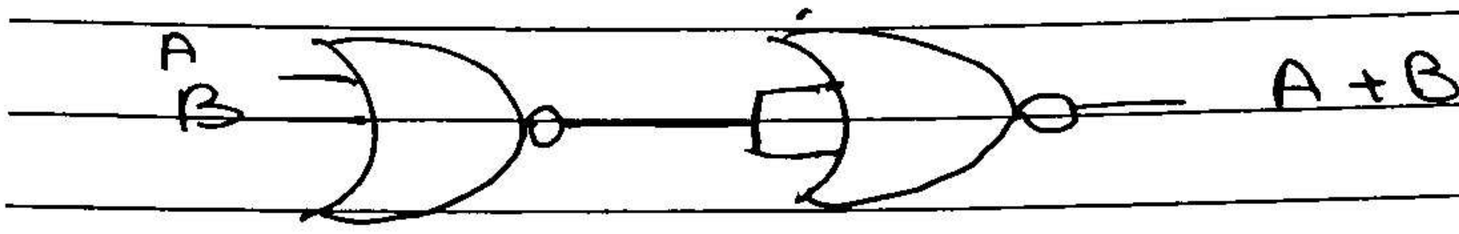
NAND ⟶ Sum of product.

- NOR Implementation :

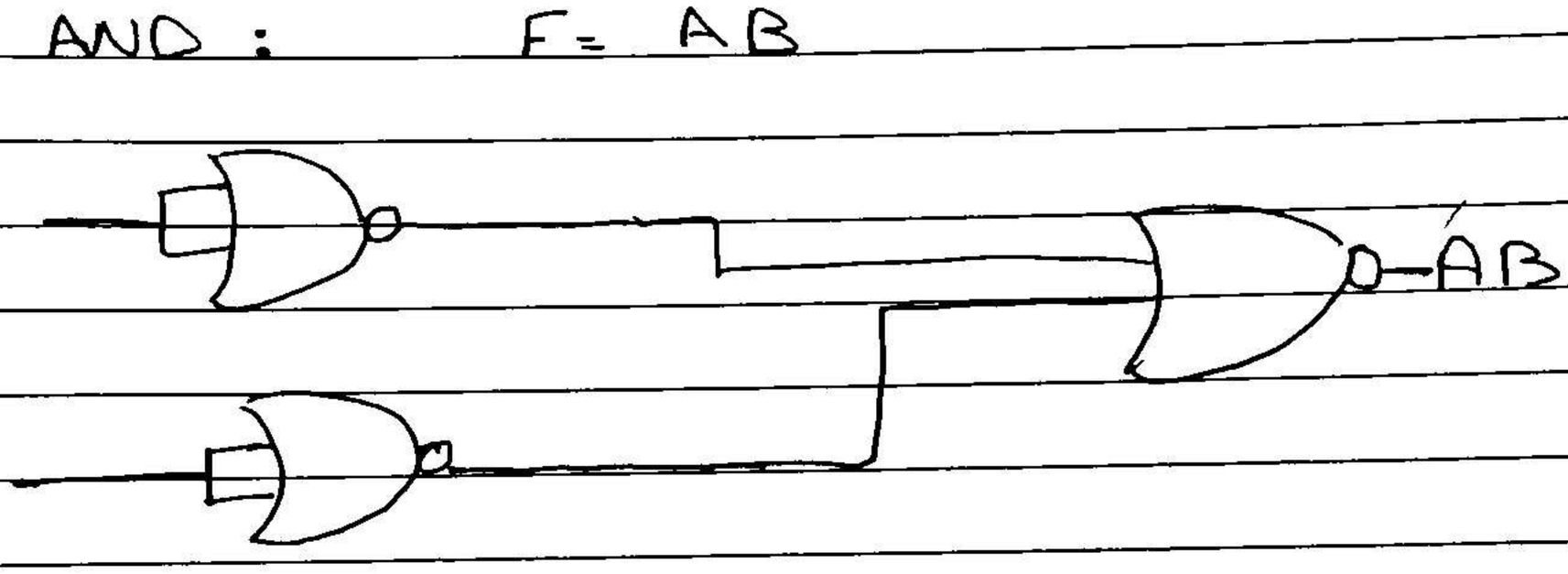
① NOT : $F = \bar{x}$



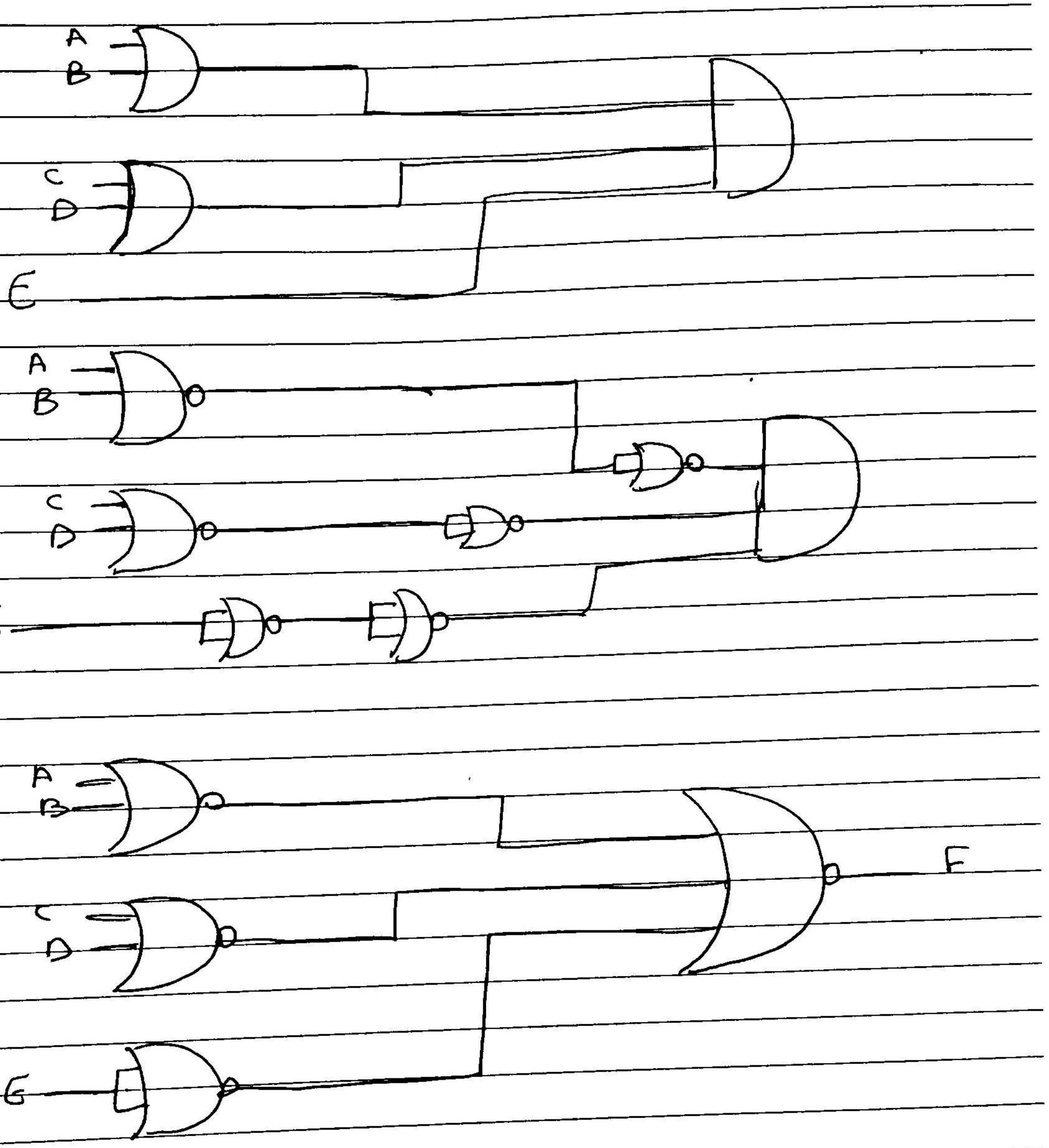
② OR : $F = A + B$



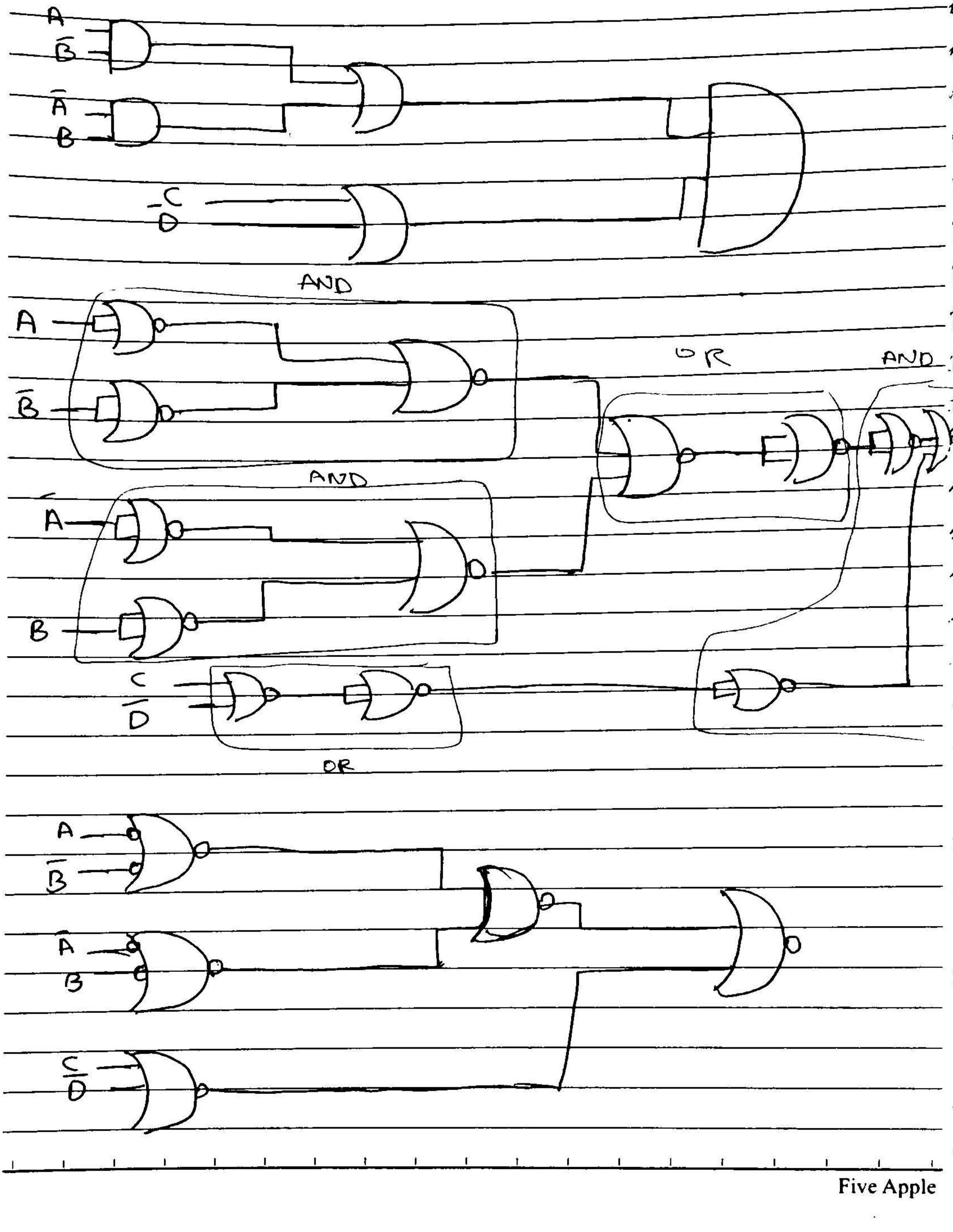
③ AND : $F = AB$



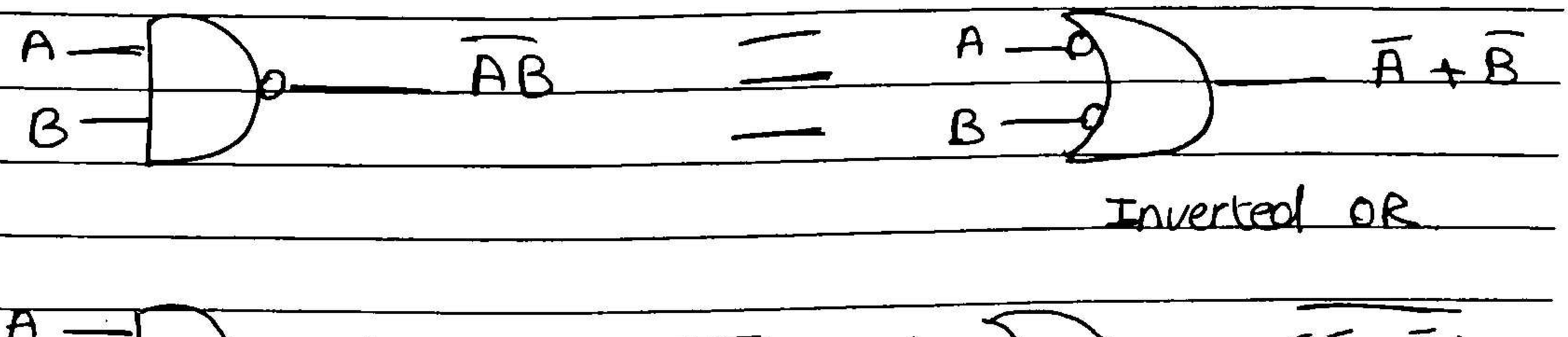
ket : $F = (A+B)(C+D)E$



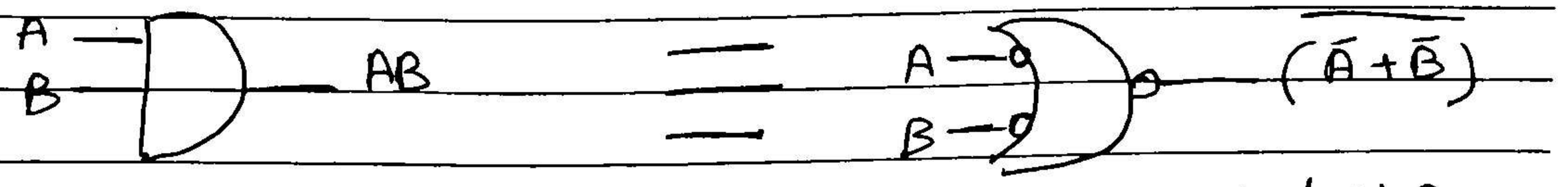
*ex : $F = (A\bar{B} + \bar{A}B)(C + \bar{D})$



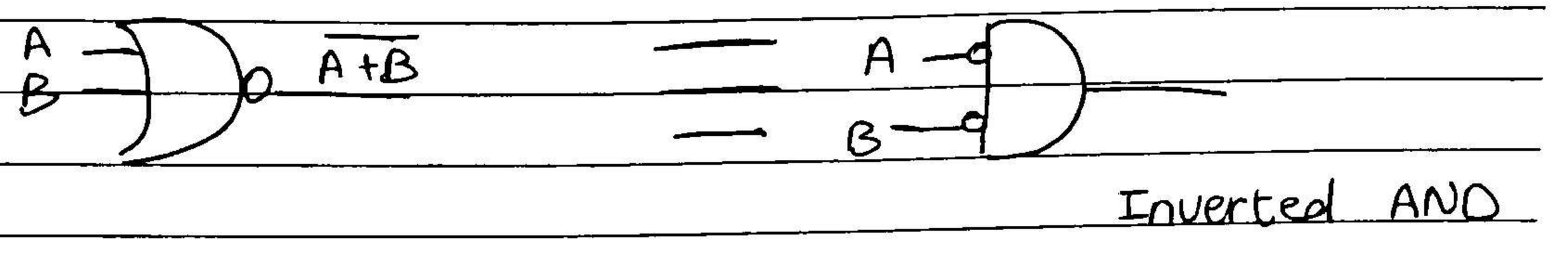
** Equivalent gates :



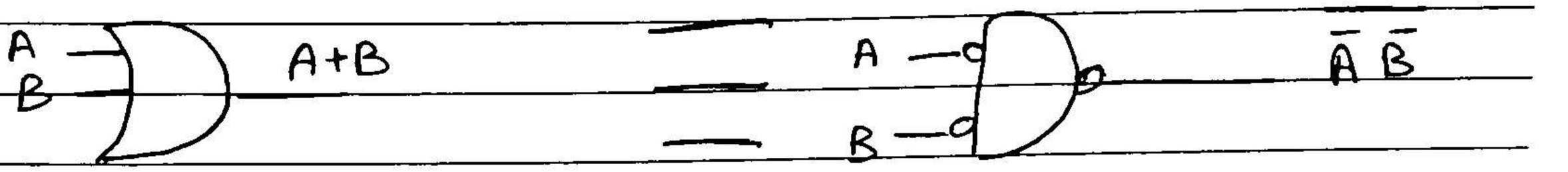
Inverted OR



Inverted NOR



Inverted AND



Inverted NAND

- XOR gate :

$F = x \oplus y = \bar{x}y + x\bar{y}$

- XNOR gate :

$F = \overline{x \oplus y}$

$F = \bar{x}\bar{y} + xy$

x	y	$x \oplus y$
0	0	0
0	1	1
1	0	1
1	1	0

$x \oplus 0 = x$

$x \oplus 1 = \bar{x}$

$x \oplus x = 0$

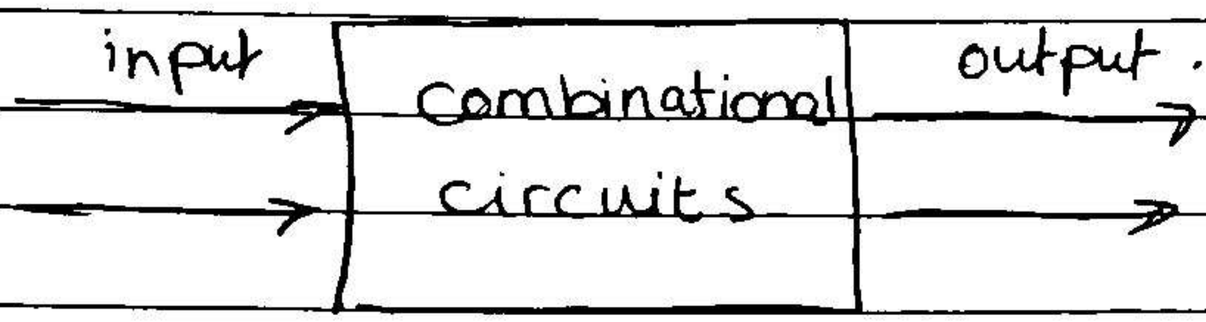
$x \oplus \bar{x} = 1$

lecture (10)

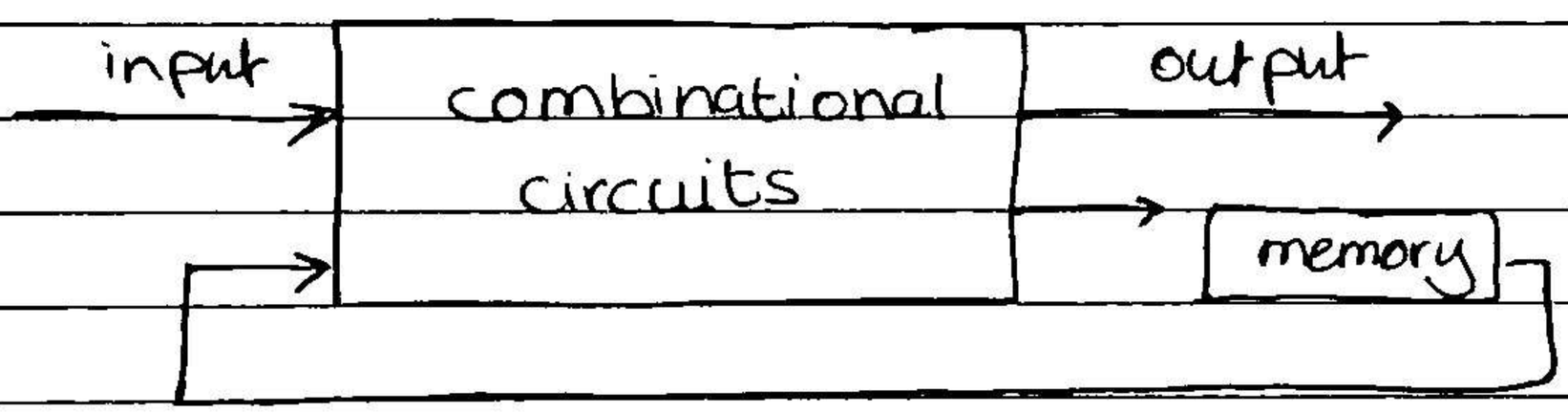
** Combinational Logic :

Logic circuits are divided into :

① Combinational circuits :
consists of logic gates whose output at any time are determined from the input.



② Sequential circuits :
Employ storage elements in addition to logic gates, their outputs are a function of the input and the storage element



** Design procedure :

① Circuit specifications.
② Truth table
③ k-map
④ Logic design.

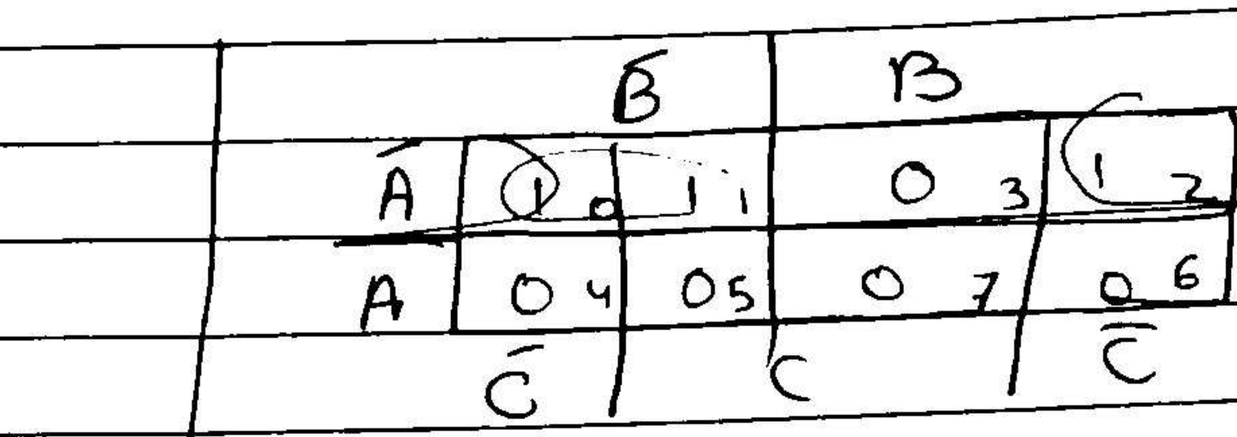
* ex: Design a Combinational circuit which has 3-bit input number and a single output 'F', where

F = 1, if the input is less than 3

F = 0, Other wise

sol:

A	B	C	F
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
0	0	0	0
1	0	1	0
1	1	0	0
1	1	1	0

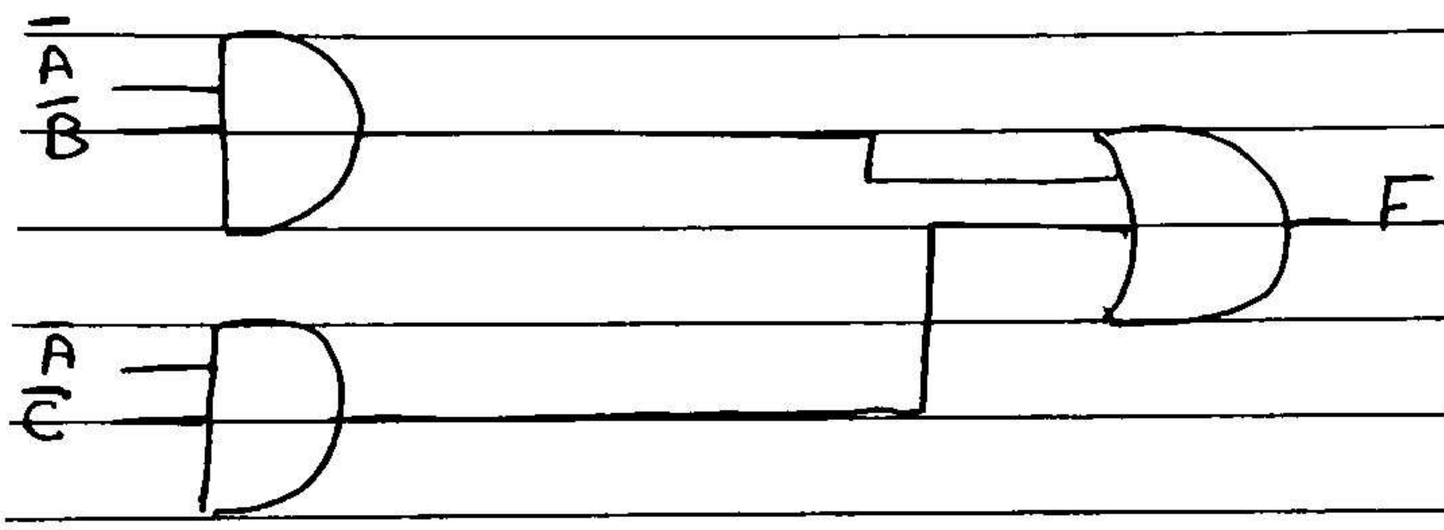


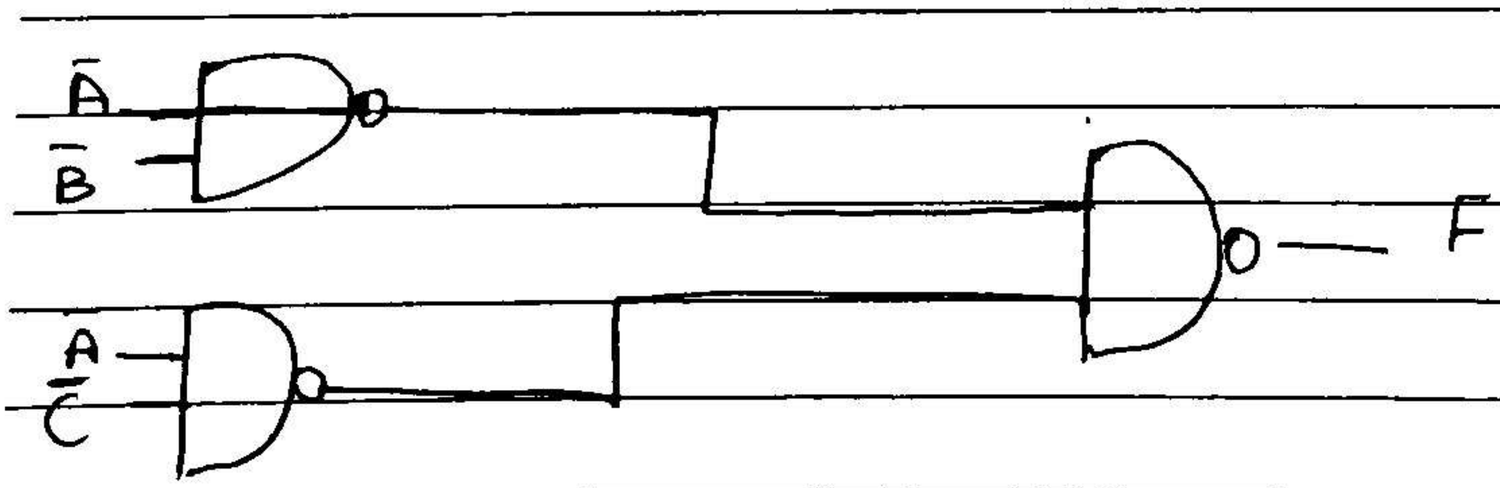
0,1 → $\bar{A}\bar{B}$

0,2 → $\bar{A}\bar{C}$

$F = \bar{A}\bar{B} + \bar{A}\bar{C}$

X (ما باخد الـ $\bar{A}$ عامل مشترك)





* ex: Design a circuit that converts BCD to excess 3 code:

A	B	C	D	W	X	Y	Z
0	0	0	0	0	0	1	1
0	0	0	1	0	1	0	0
0	0	1	0	0	1	0	1
0	0	1	1	0	1	1	0
0	1	0	0	0	1	1	1
0	1	0	1	1	0	0	0
0	1	1	0	1	0	0	1
0	1	1	1	1	0	1	0
1	0	0	0	1	0	1	1
1	0	0	1	1	1	0	0
X	X	X	X	X	X	X	X

: don't care

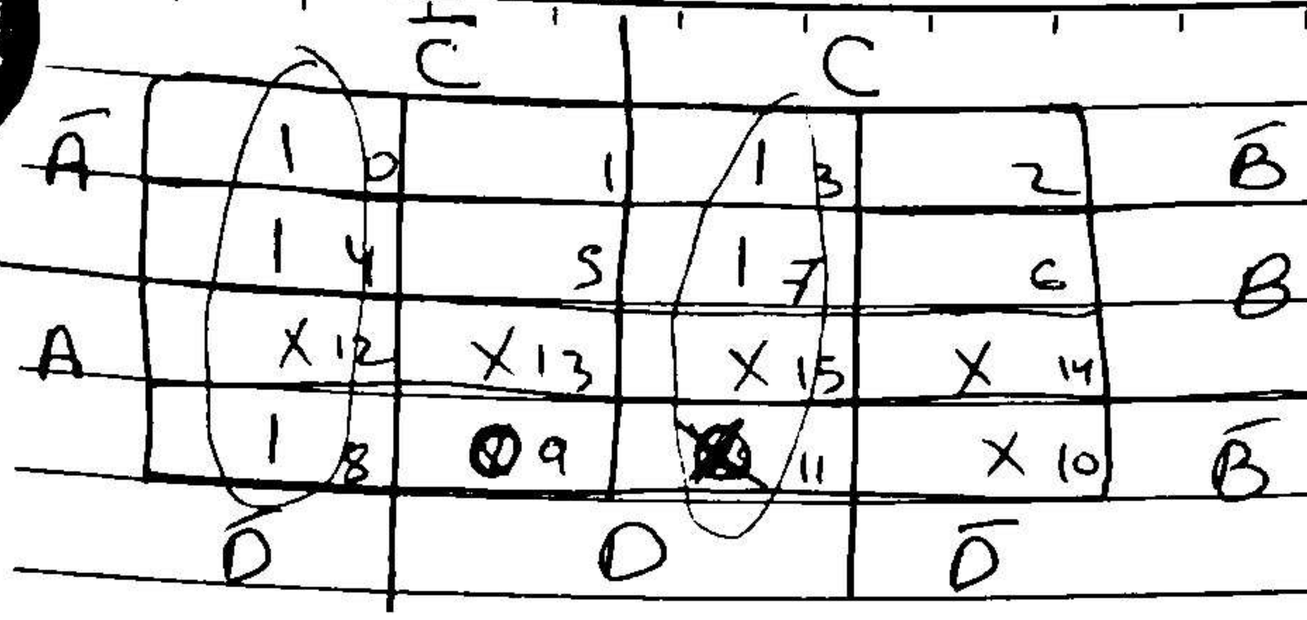
* باخد الـ kmap لـ w, x, y, z
(4 k-maps)

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$	
$\bar{A}$	0	1	3	2	$\bar{B}$
	4	15	17	16	
	X 12	X 13	X 15	X 14	B
A	18	19	X 11	X 10	$\bar{B}$
	$\bar{D}$	D		$\bar{D}$	

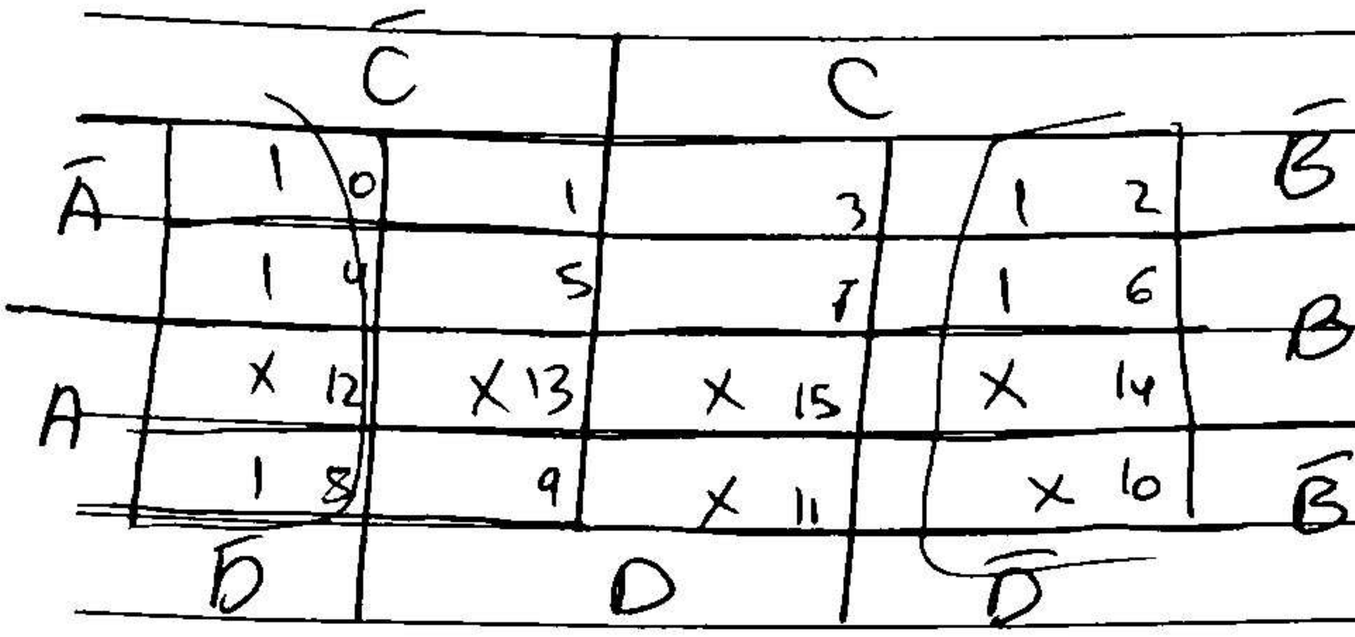
$W = A + BC + BD$

	$\bar{C}\bar{D}$	$\bar{C}D$	CD	$C\bar{D}$	
$\bar{A}$	0	1	3	2	$\bar{B}$
	4	5	7	6	
	X 12	X 13	X 15	X 14	B
A	8	9	X 11	X 10	$\bar{B}$
	$\bar{D}$	D		$\bar{D}$	

$X = \bar{B}D + \bar{B}C + B\bar{C}\bar{D}$

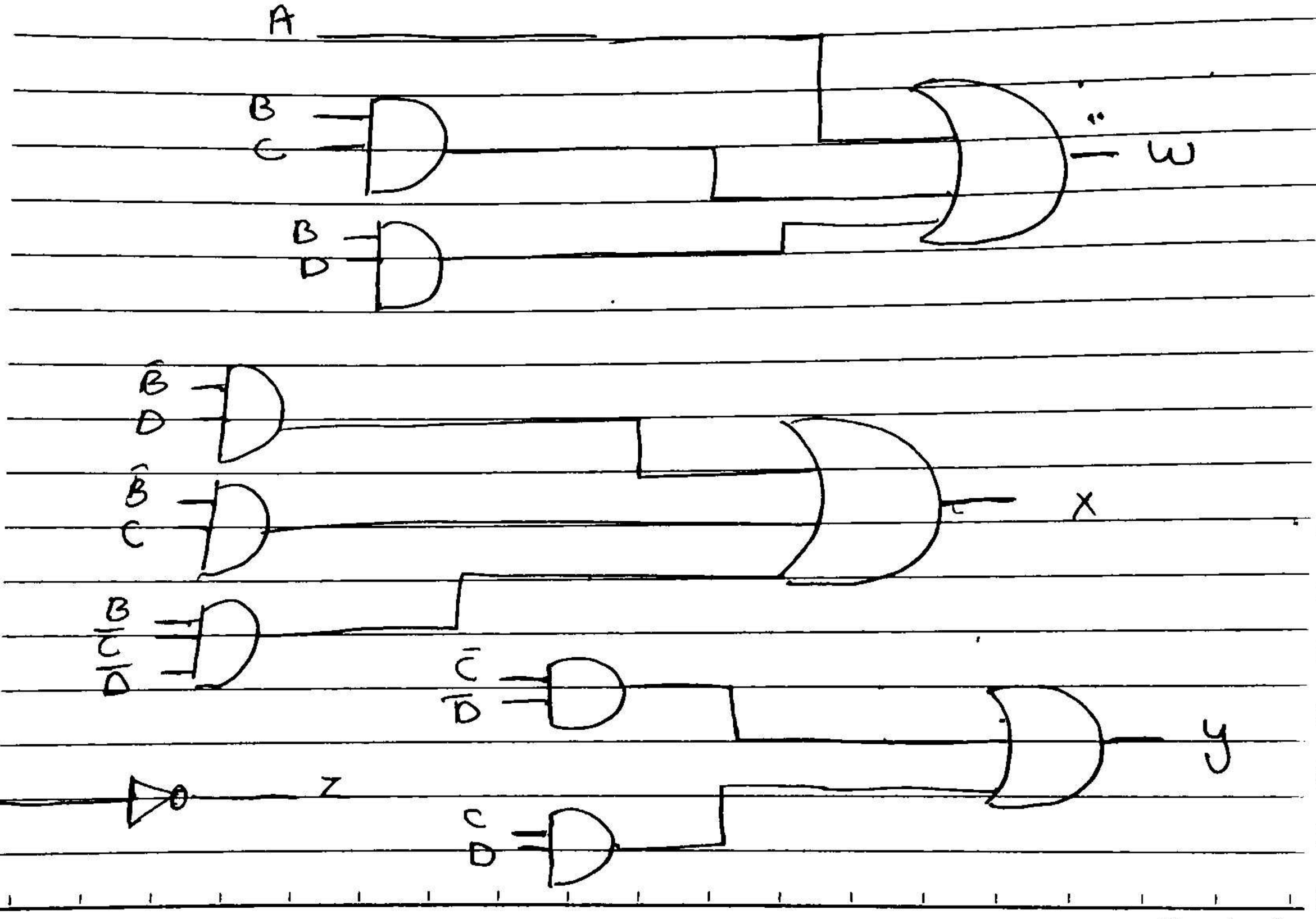


$y = \bar{D}\bar{C} + CD$



$Z = \bar{D}$

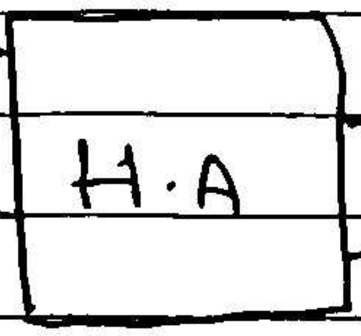
logic design Class, E



lecture (11)

** Binary Adders and Subtractors :

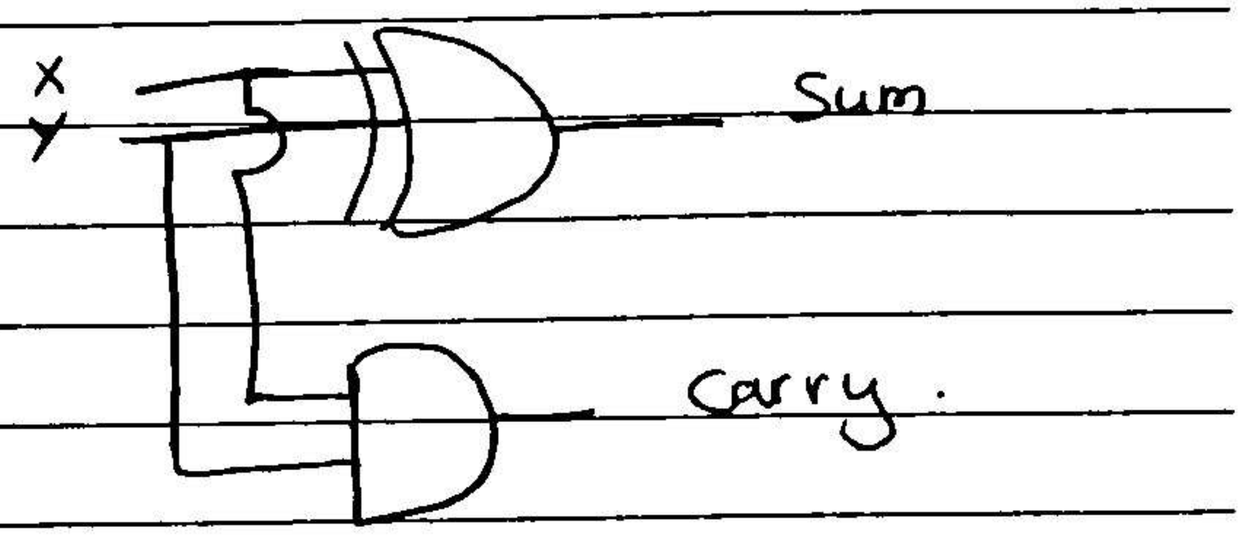
* Half Adders :



x	y	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Sum = $x \oplus y$

Carry = $x \cdot y$



* Full adder :

x	y	z / Cin	Sum	Cout (Carry)
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Sum =

	$\bar{y}$		y	
$\bar{x}$	0 0	(1) 1	0 3	(1) 2
x	(1) 4	0 5	(1) 7	0 6
	$\bar{z}$		z	$\bar{z}$

$Sum = \bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xyz$

$= \bar{x}(\bar{y}z + y\bar{z}) + x(\bar{y}\bar{z} + yz)$

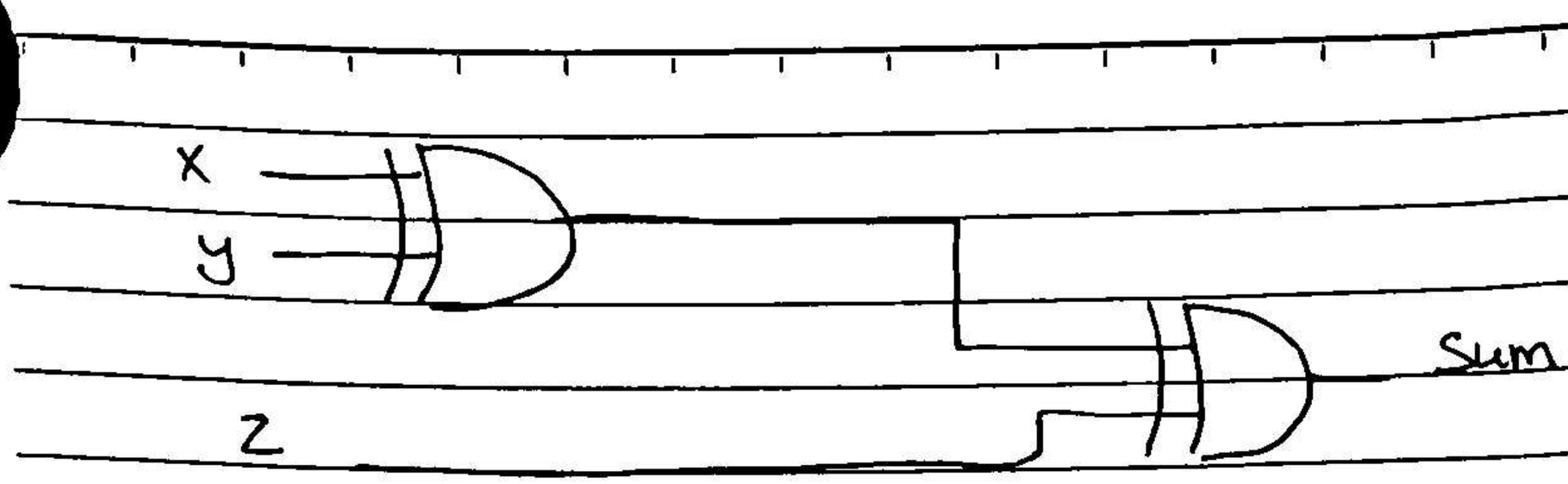
$= \bar{x}(y \oplus z) + x\overline{(y \oplus z)}$

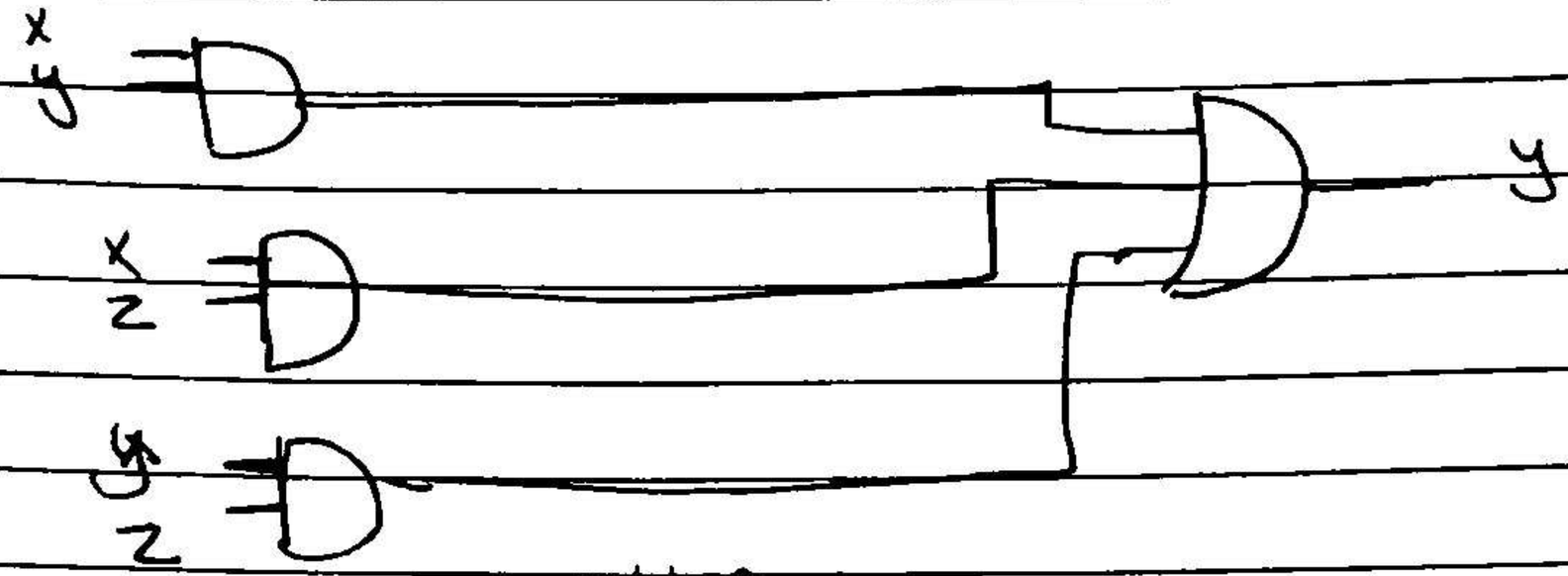
$= x \oplus y \oplus z$

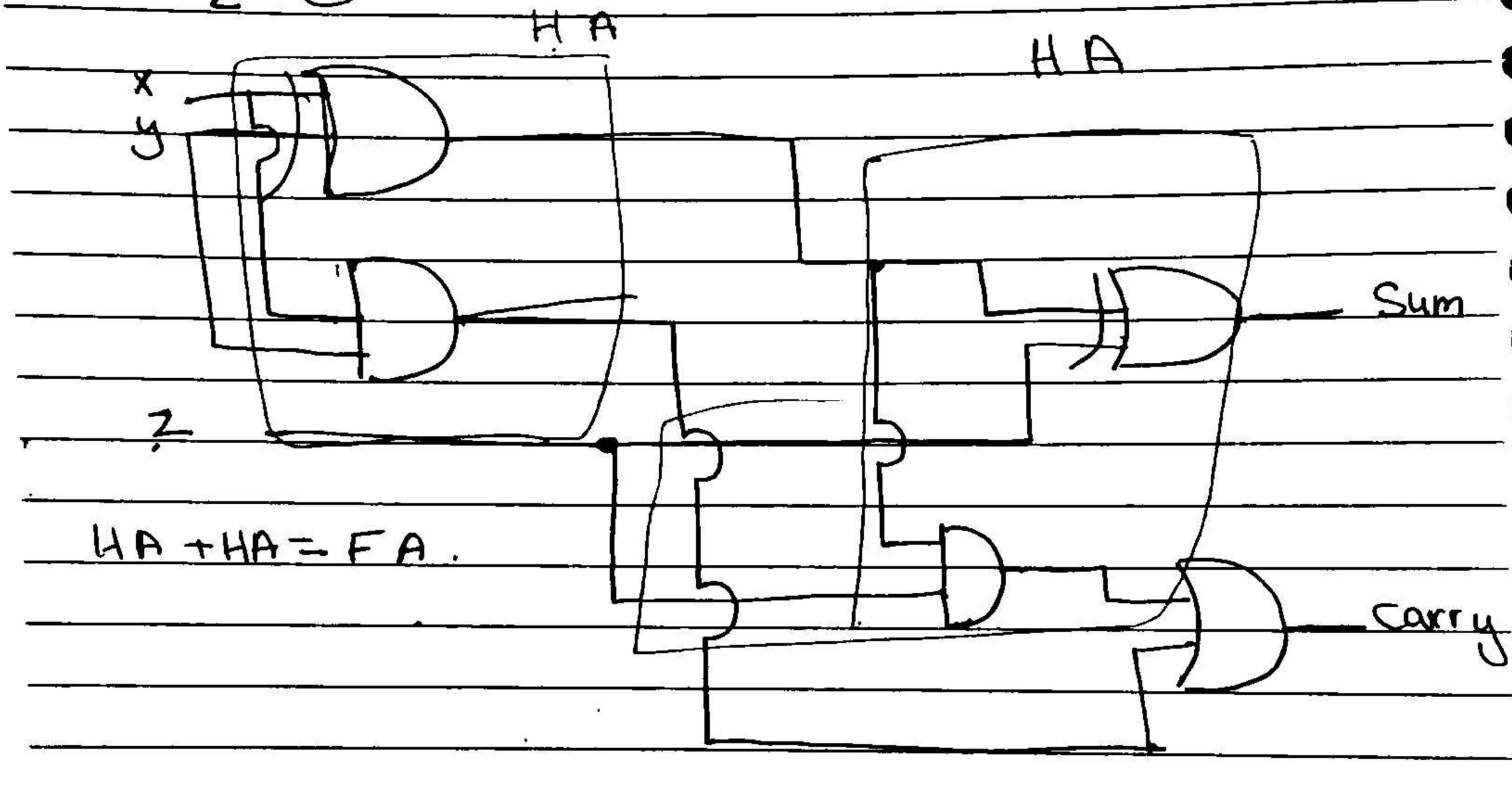
Carry =

	$\bar{y}$		y	
$\bar{x}$	0 0	0 1	(1) 3	0 2
x	0 4	(1 5	(1) 7	1 6
	$\bar{z}$	z		$\bar{z}$

Carry = $xz + xy + yz$

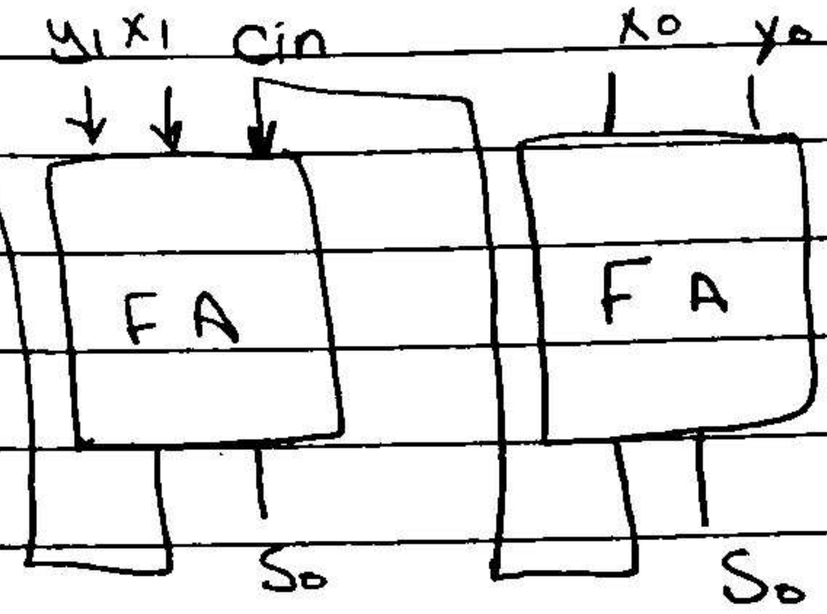






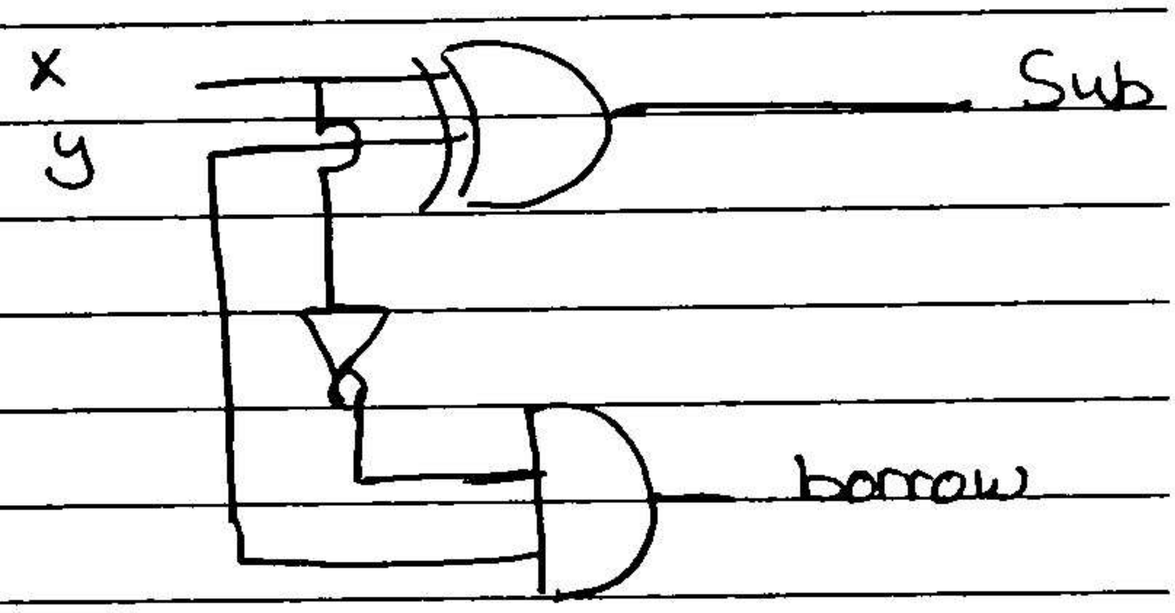
HA + HA = FA.

** Half Subtractor :



x	y	Sub	borrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

H.S.



** Full subtractor : X - y - Bin

X	y	z Bin	Sub	Bout.
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

*10
0
1 ← نقترض من الـ Borrow
1
0

Sub :

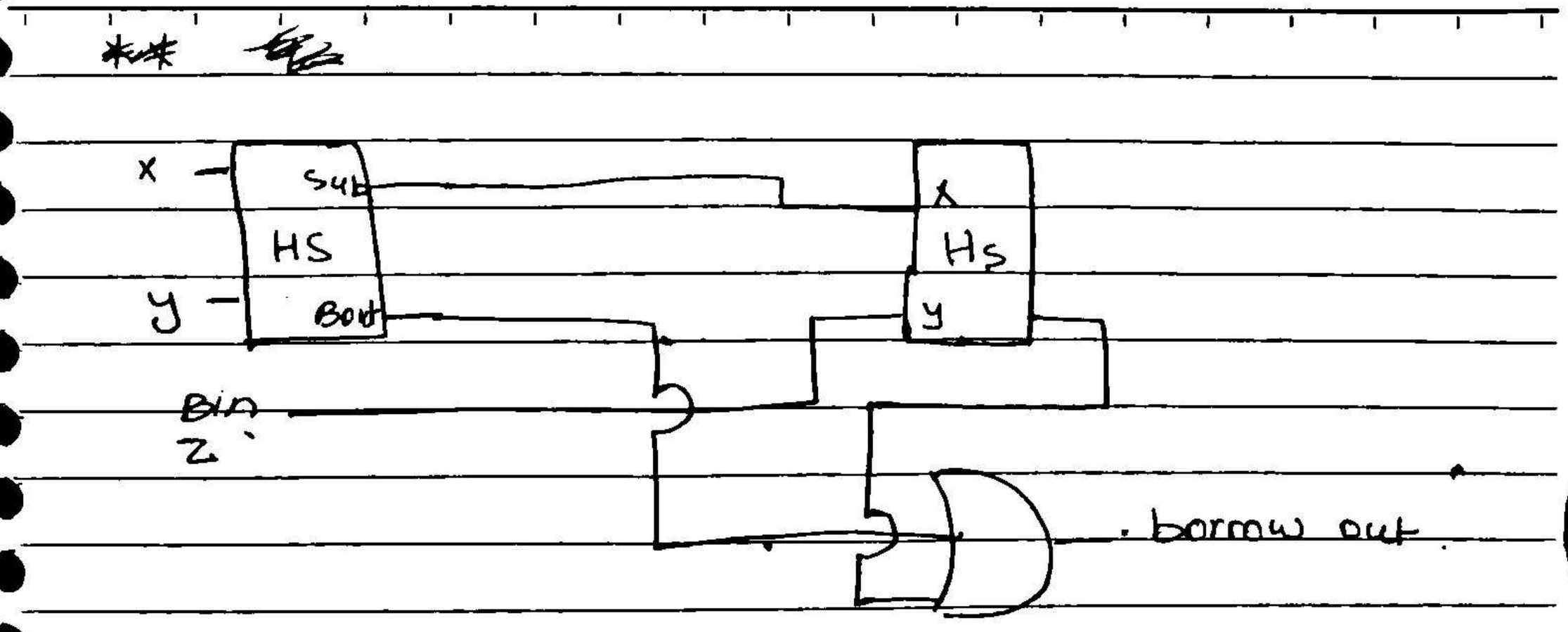
	$\bar{y}$		y	
$\bar{x}$	0 0	1 1	3	1 2
x	1 4	5	1 7	6
	$\bar{B_{in}}$ $\bar{z}$	B_{in} z		z $\bar{B_{in}}$

Sub = $x \oplus y \oplus z = x \oplus y \oplus$ Bin

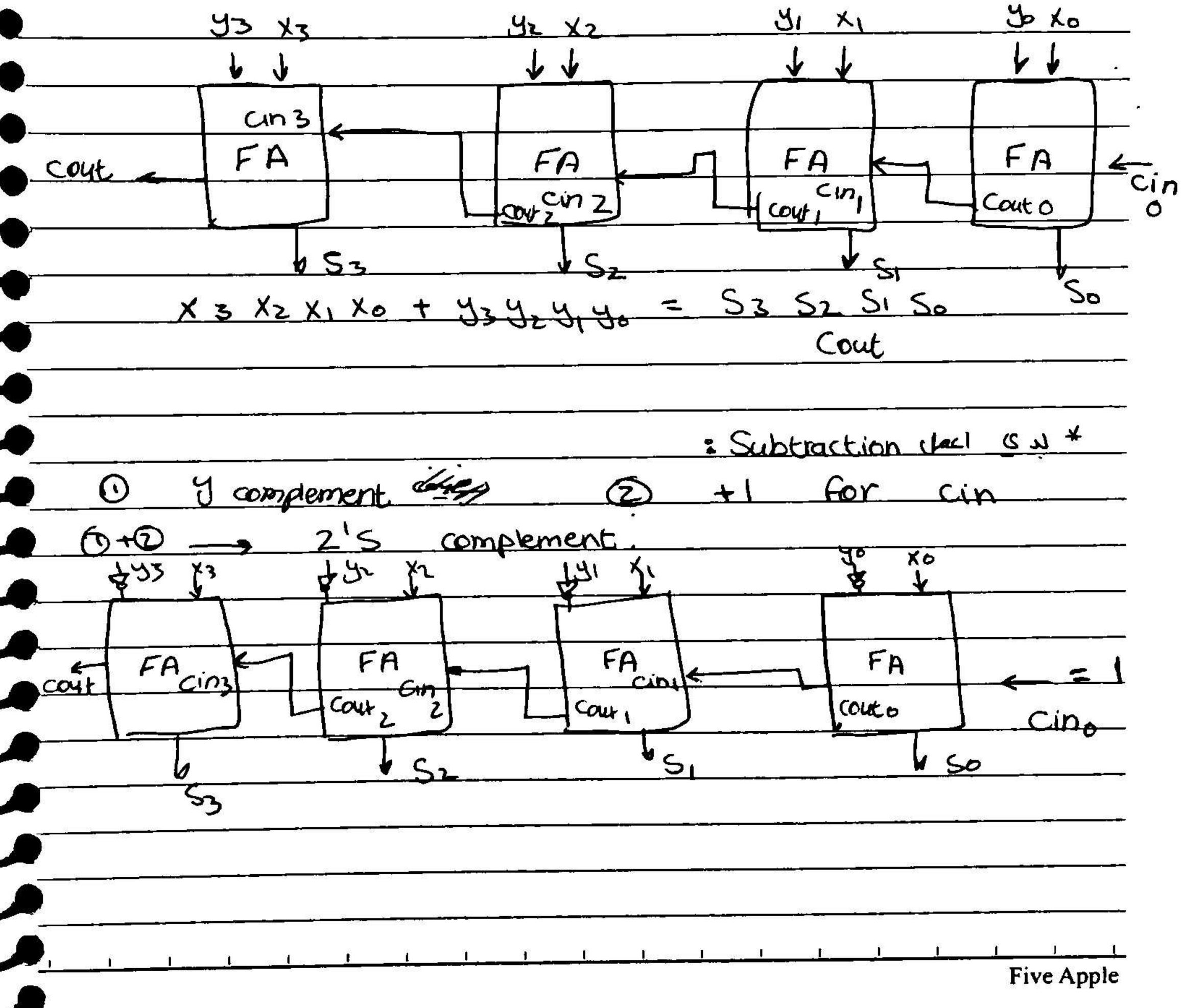
Bout :

	$\bar{y}$		y	
$\bar{x}$	0	1 1	3	1 2
x	4	5	1 7	6
	$\bar{B_{in}}$ $\bar{z}$	B_{in} z		$\bar{z}$ $\bar{B_{in}}$

Bout = $x\bar{y} + \bar{x}z + yz = x\bar{y} + \bar{x}$ Bin + y Bin



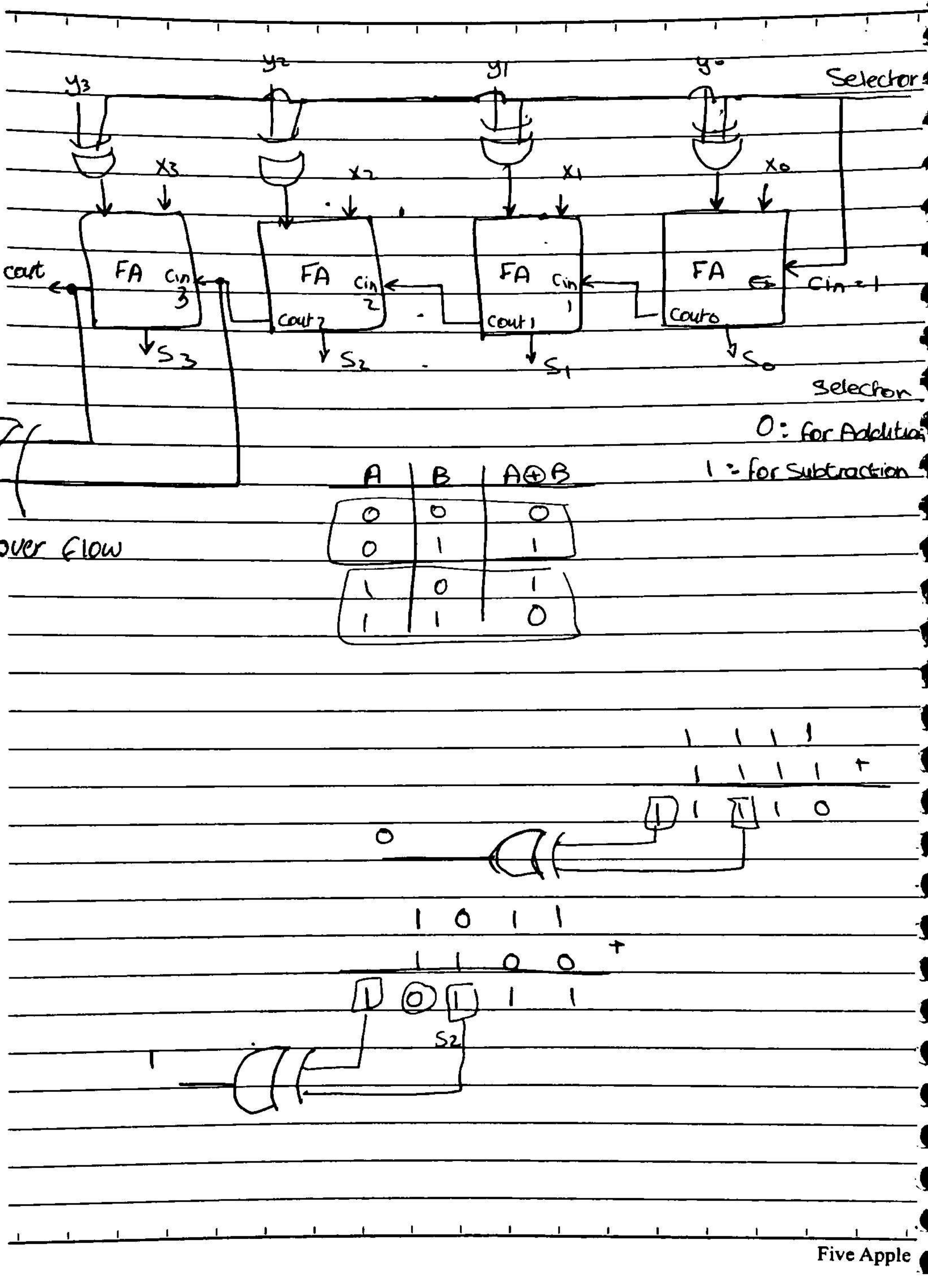
** 4 - bit Adder :



* بدي اعمل Subtraction :

① y complement ② +1 for cin

① + ② → 2's complement.


y3
y2
y1
y0
Selector
x3
x2
x1
x0
cout
FA
Cin 3
Cin 2
Cin 1
Cin = 1
cout3
cout2
cout1
cout0
S3
S2
S1
S0
Selection
0 : for Addition
1 : for Subtraction
over Flow
A
B
A⊕B
0 0 0
0 1 1
1 0 1
1 1 0
1 1 1 1
+
0
1 1 1 1 0
1 0 1 1
+
1 1 0 0
1 0 1 1 1
S2
1

lecture (12)

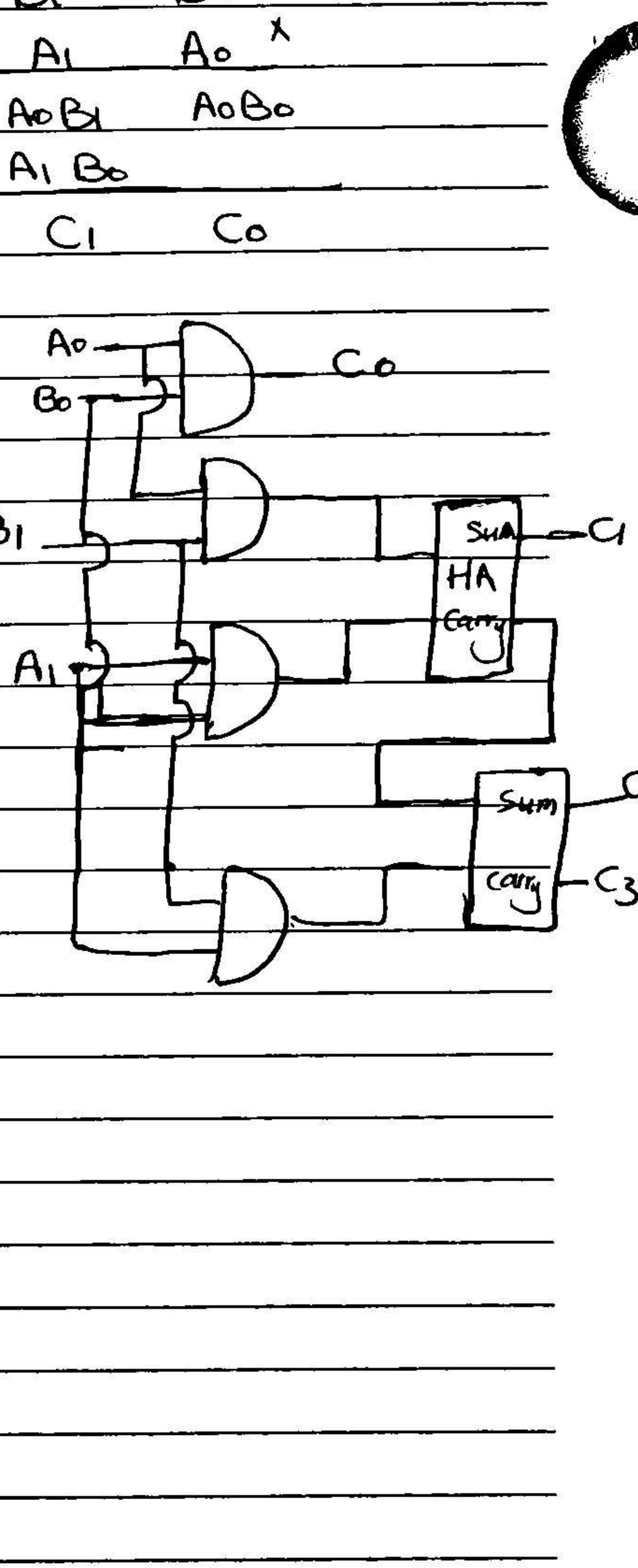
** Binary Multiplier :

ex: Design a Multiplier for two 2-bit numbers : A*B

C = A * B

A = A_1 A_0

B = B_1 B_0



* Design a Combinational circuit that determine the relative magnitude of A * B

$A = A_1 A_0$

$B = B_1 B_0$

Sol:

The two numbers are equal if $A_1 = B_1$ and $A_0 = B_0$

So

The equality is expressed by X nor.

$X_i = A_i B_i + \bar{A}_i \bar{B}_i$

$X_1 = A_1 B_1 + \bar{A}_1 \bar{B}_1$

$X_0 = A_0 B_0 + \bar{A}_0 \bar{B}_0$

* لتحديد من اكبر B_1 or A_1 The most Significant

① لشوف

② اذا متساويين في الـ Significant يروح لـ A_0 & B_0

② $A > B = A_1 \bar{B}_1 + X_1 A_0 \bar{B}_0$

اذا هدول متساويين ؛ بتنقل الي هذا الحد.

اذا هدول متساويين معناها $A > B$

Ex: $A_1 \bar{B}_1 + X_1 A_0 \bar{B}_0$ — 1 0

$1 \ 0 + D \cdot 1 \cdot \bar{D}$ — 0 1

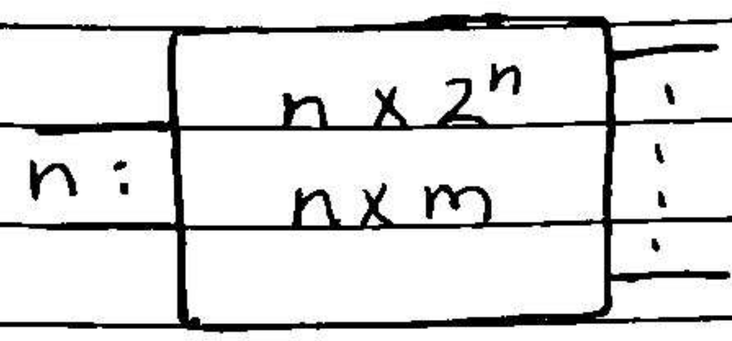
$= 1 \rightarrow A > B$.

③ $A < B = \bar{A}_1 B_1 + X_1 \bar{A}_0 B_0$

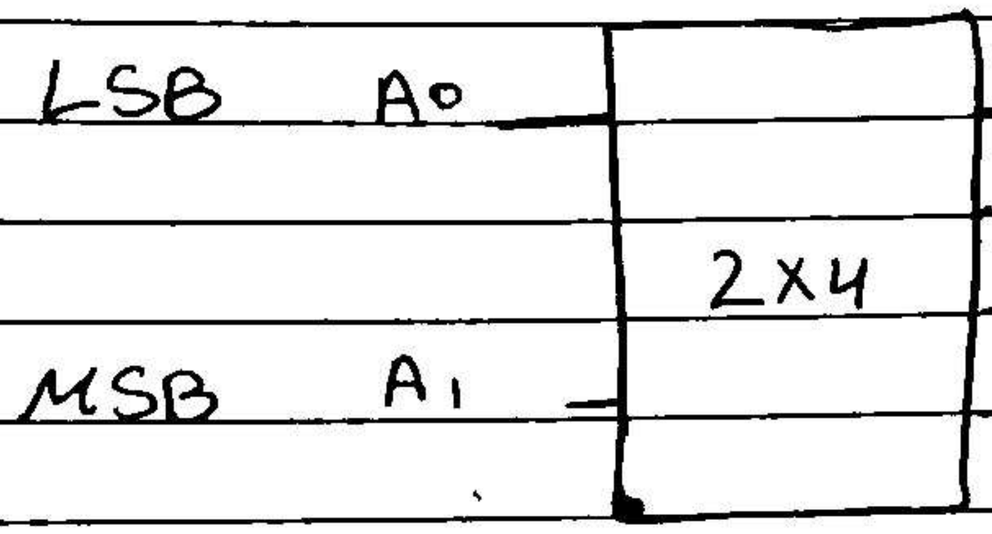
** Decoders :

- A decoder is a combinational circuit that converts binary informations from (n) inputs to 2^n outputs

- The decoders are called n to m decoders

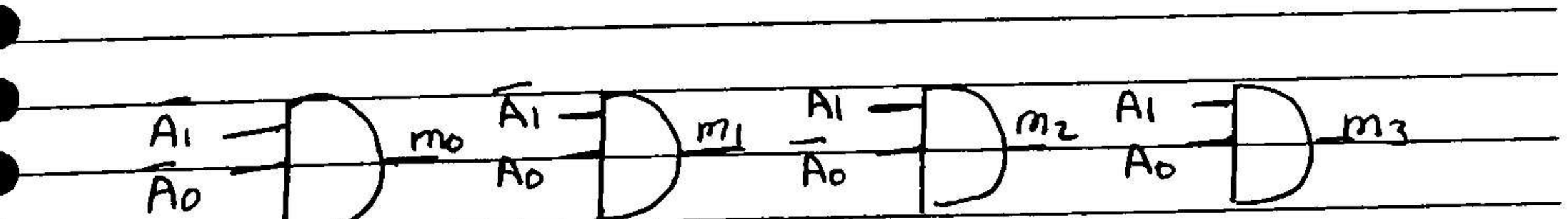


• ex: 2 to 4 decoder :



باخذ Decoder 2 to 4 وطلع 3 inputs & outputs

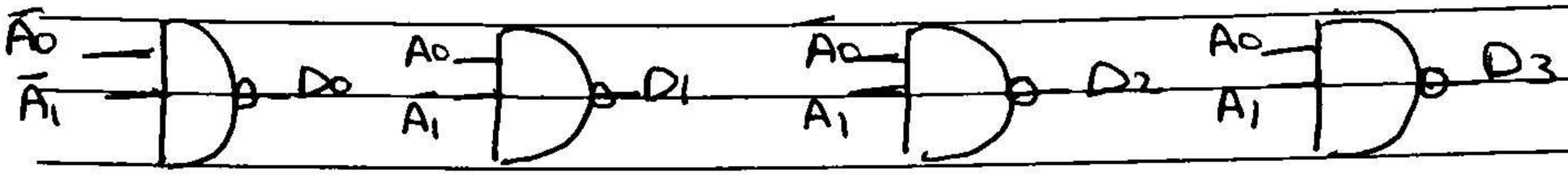
A1	A0	D3	D2	D1	D0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0



page 151 3 by 8 decoders. (on book)

** 2 to 4 decoder (active low) using NAND :

A_1	A_0	D_3	D_2	D_1	D_0
0	0	1	1	1	0
0	1	1	1	0	1
1	0	1	0	1	1
1	1	0	1	1	1



** 2 to 4 decoder with enable : (Active Low)

En	A_1	A_0	D_3	D_2	D_1	D_0
0	X	X	1	1	1	1
1	0	0	1	1	1	0
1	0	1	1	1	0	1
1	1	0	1	0	1	1
1	1	1	0	1	1	1

X → don't care

- Active High :

En	A_1	A_0	D_3	D_2	D_1	D_0
0	X	X	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

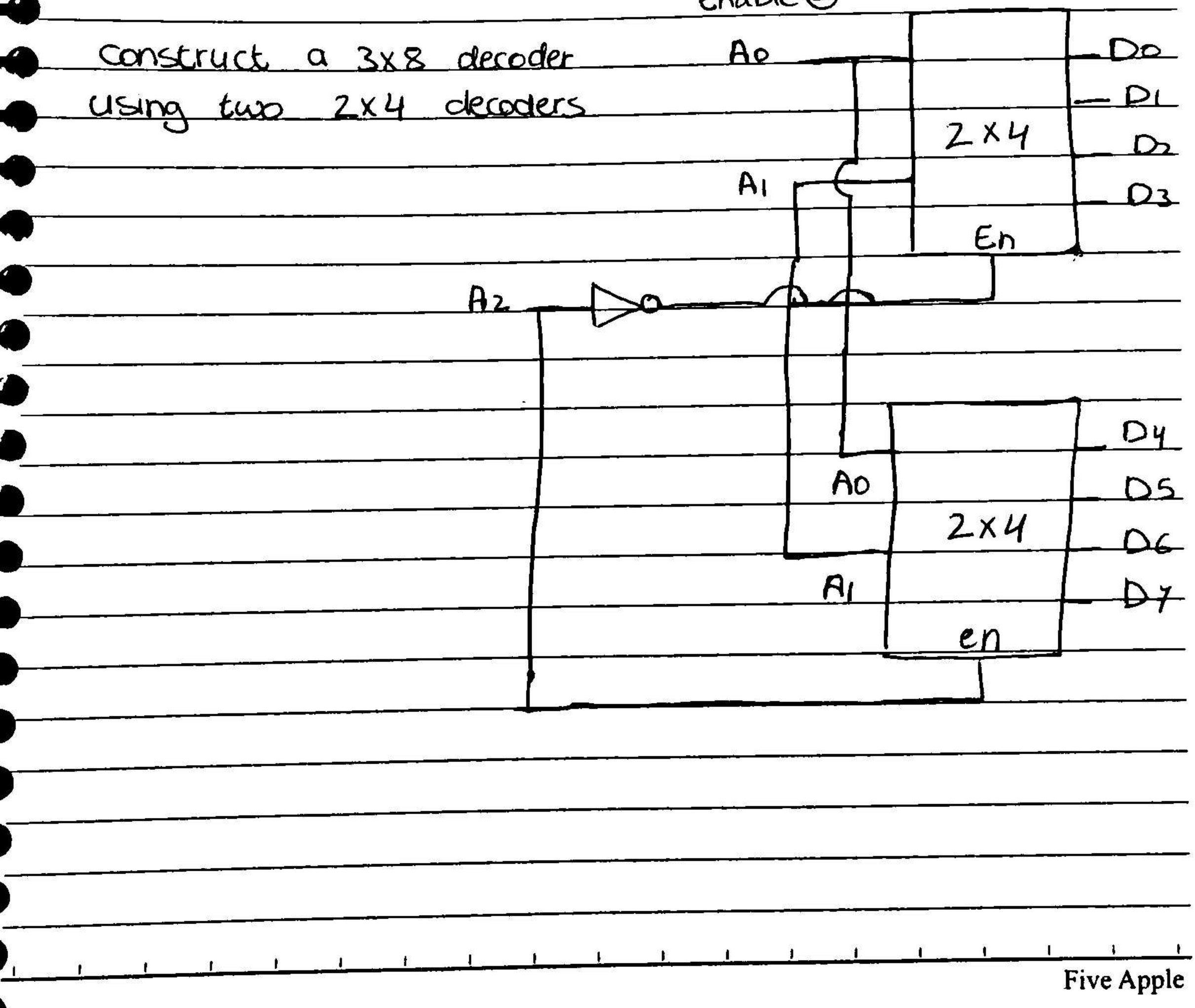
تتحتاج

** 3 to 8 Decoder : 2 decoder Enable (1)

A_2	A_1	A_0	D_0	D_1	D_2	D_3	D_4	D_5	D_6	D_7
0	0	0	1	0	0	0	0	0	0	0
0	0	1	0	1	0	0	0	0	0	0
0	1	0	0	0	1	0	0	0	0	0
0	1	1	0	0	0	1	0	0	0	0
1	0	0	0	0	0	0	1	0	0	0
1	0	1	0	0	0	0	0	1	0	0
1	1	0	0	0	0	0	0	0	1	0
1	1	1	0	0	0	0	0	0	0	1

Enable (2)

Construct a 3x8 decoder using two 2x4 decoders

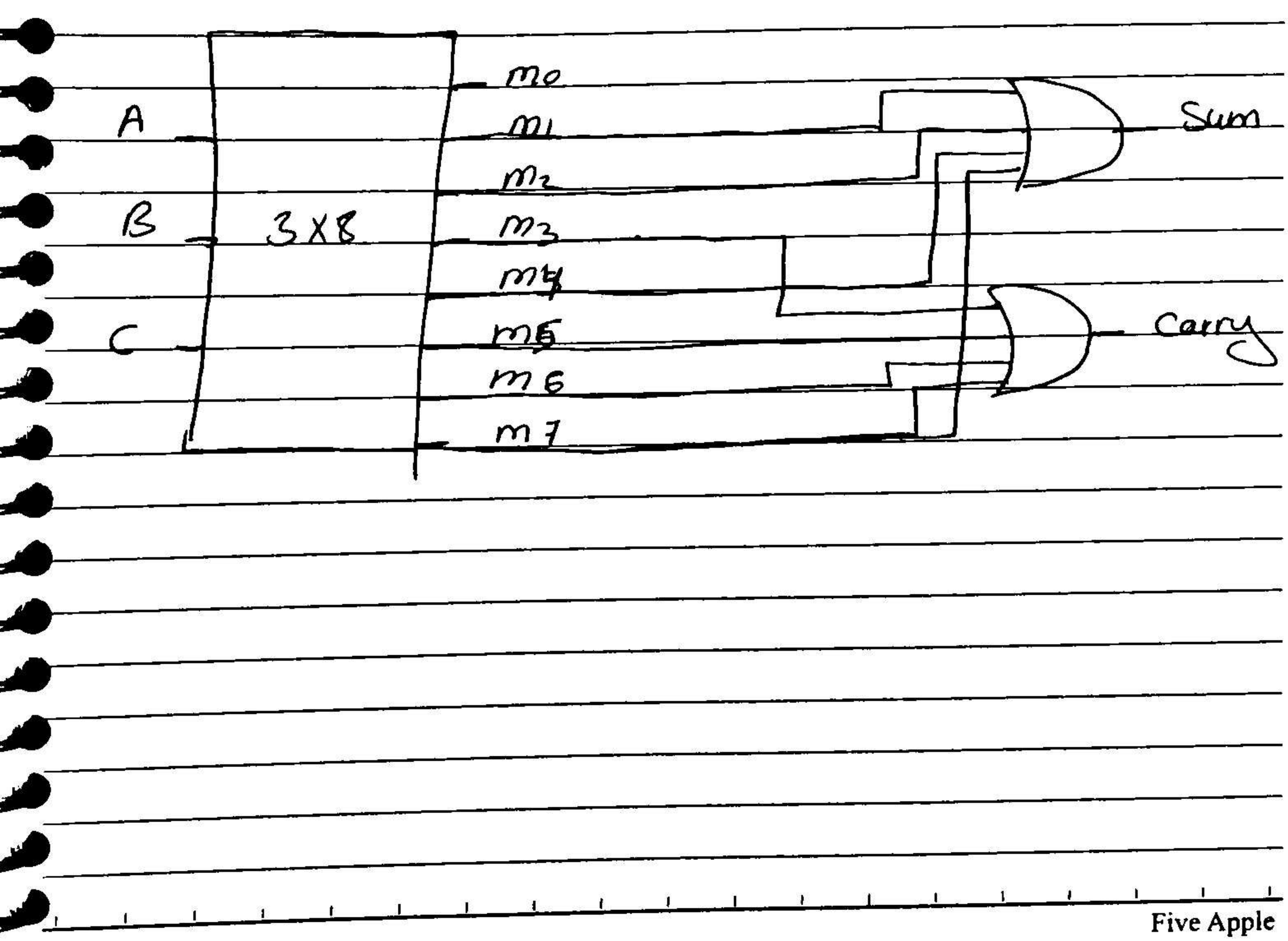


lecture (13)

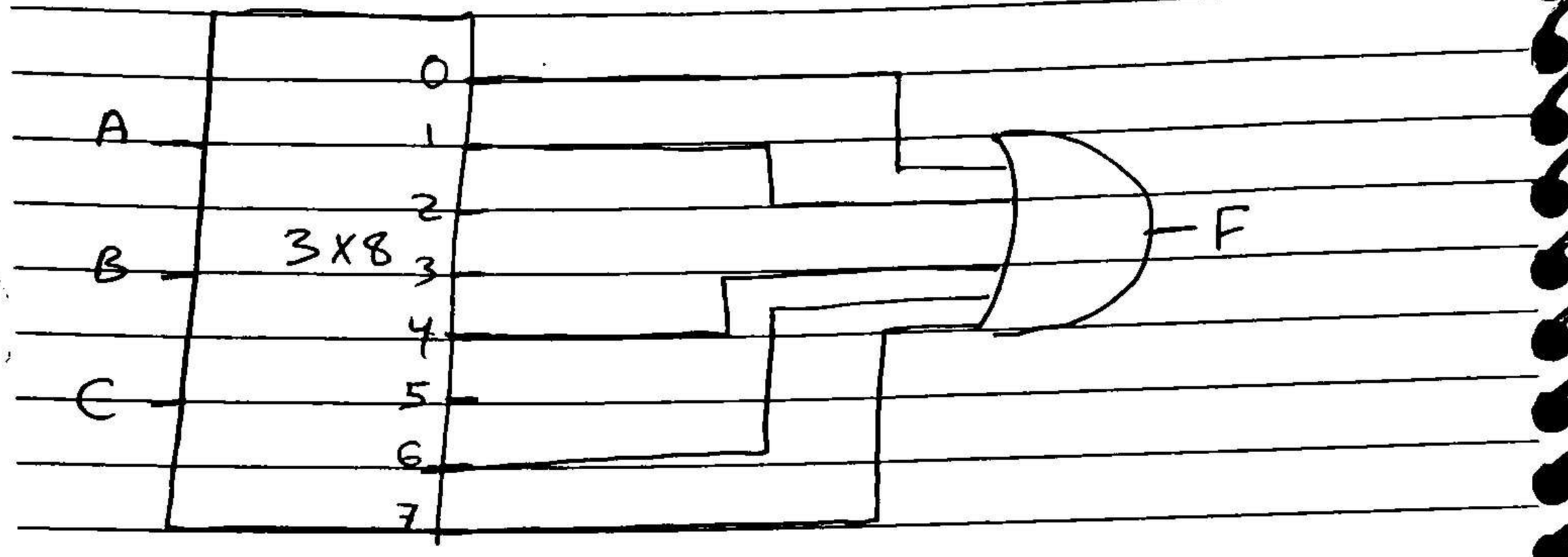
** Combinational Circuit implementation using Decoders :-

implement a Full Adder using a 3x8 decoder :-

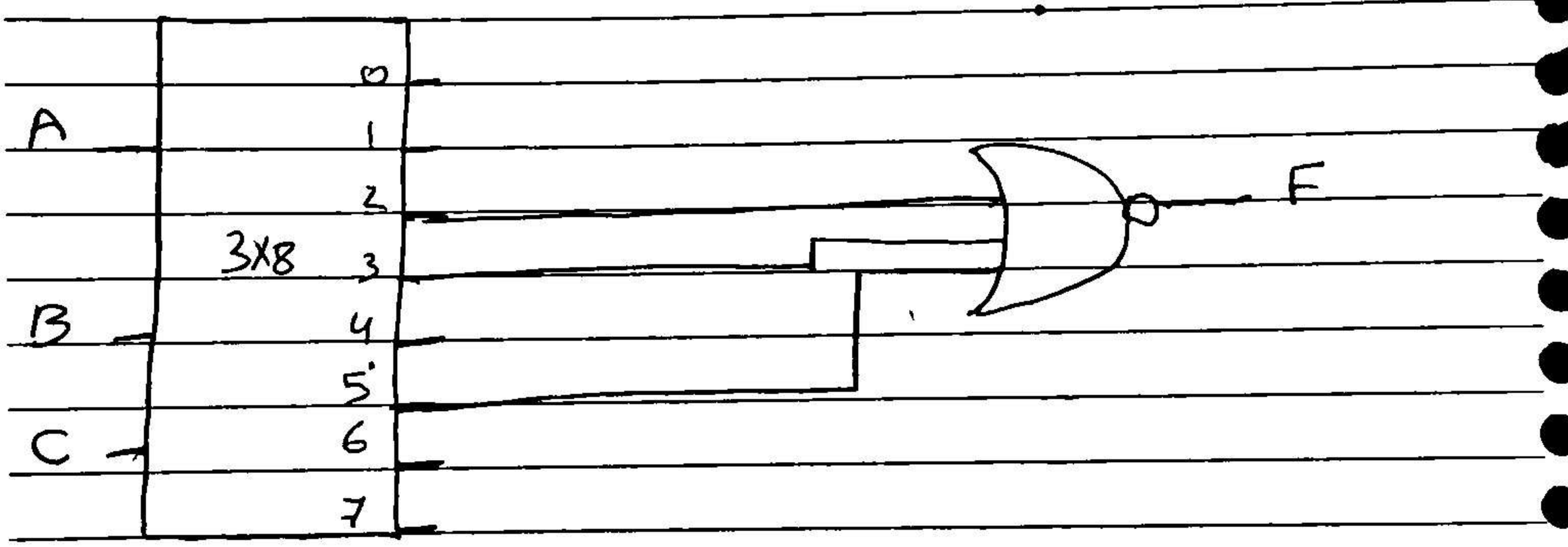
A	B	C	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1



* ex: implement $F(A,B,C) = \Sigma(0,1,4,6,7)$ using 3x8 decoder.



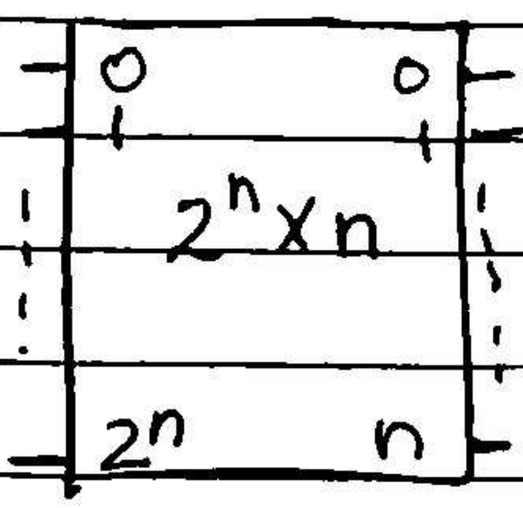
* ex ② $F(A,B,C) = \Sigma(2,3,5)$



** Encoder circuits :

- An Encoder performs the inverse operation of a decoder

- It has 2^n inputs an 'n' outputs



* The 4 x 2 encoder :

input				output	
A_3	A_2	A_1	A_0	S_1	S_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

* The encoder has two major issues :

① if two inputs are '1' at the same time, what should we do ? (piority encoder)

② the output '00' can be obtained :
   ① if all inputs are zero] → valid
   ② if $A_0 = 1$

* priority encoder :

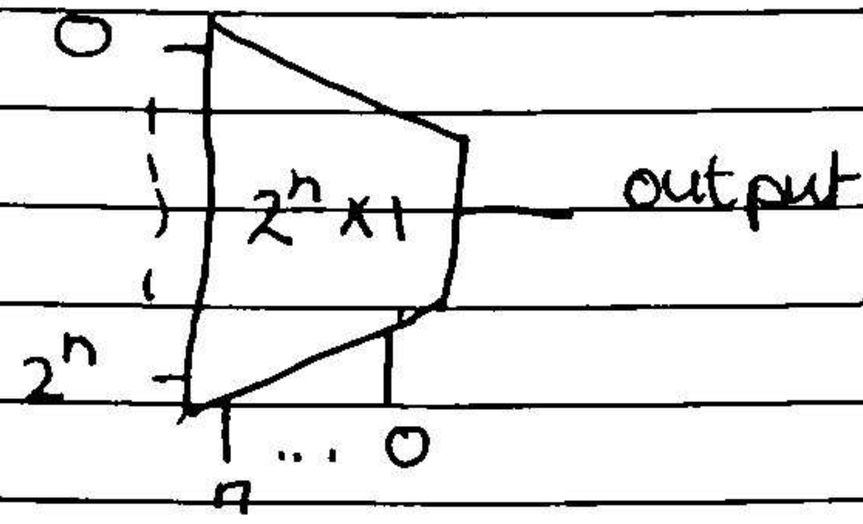
input				output		
A3	A2	A1	A0	S1	S0	valid
0	0	0	1	0	0	1
0	0	1	X	0	1	1
0	1	X	X	1	0	1
1	X	X	X	1	1	1
0	0	0	0	X	X	0

*ex: design a combinational circuit that has an output active if the input is even, the input is 4-bits

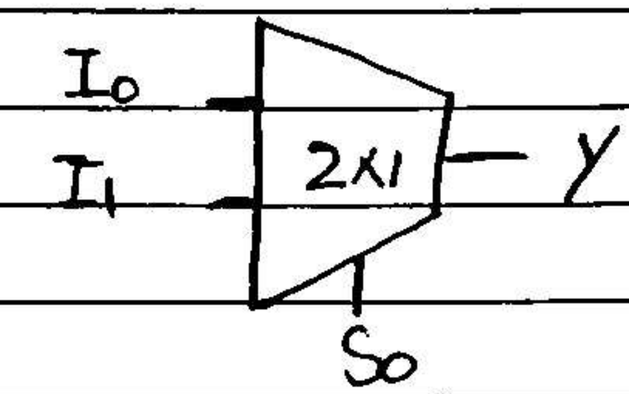
A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0

** Multiplixers or Data Selectors :

= A Multiplixers is a Combinational Circuit that selects one of 2^n inputs to the output



• 2X1 Multiplixer :



S_0	Y.
0	I_0
1	I_1

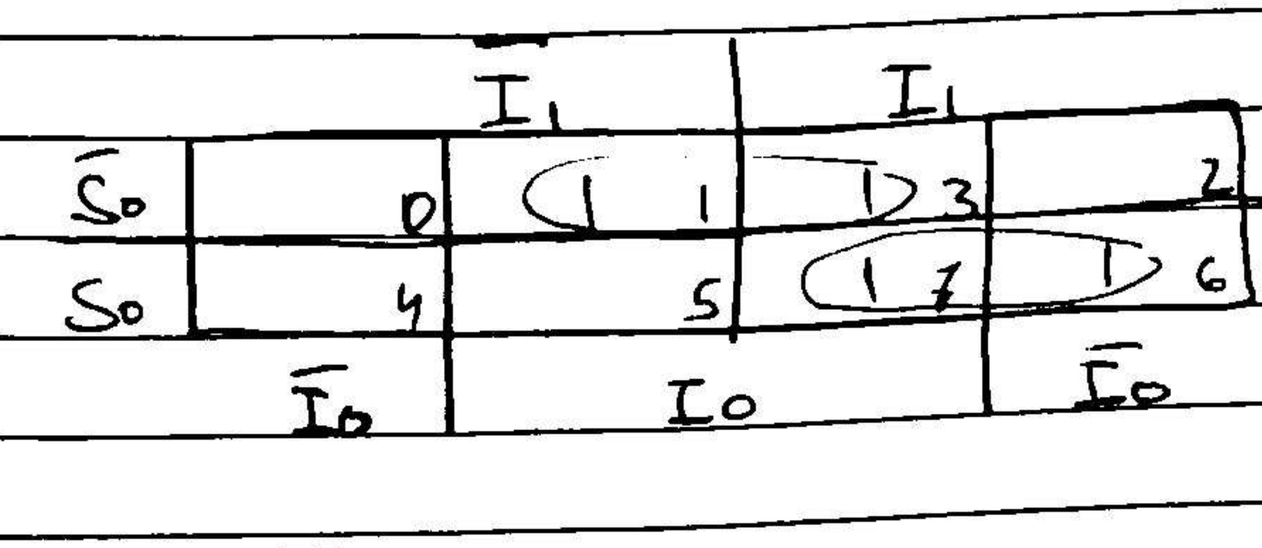
*ex : Design a Combinational logic circuit for the 2X1 mux

S_0	I_1	I_0	Y
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

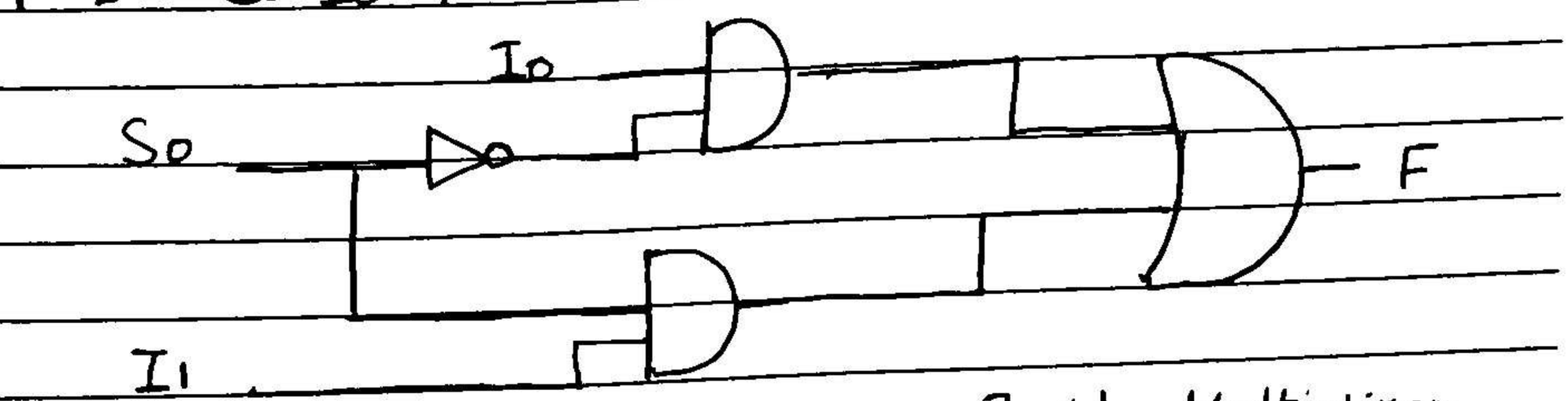
when $S_0 = 0$
I_0 is the output

when $S_0 = 1$
I_1 is the output

*(three inputs (I_0, I_1, S_0))

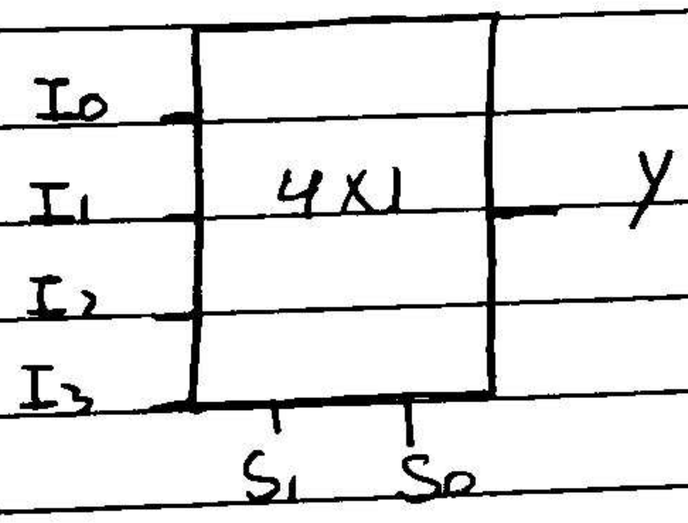


$F = \bar{S_0} I_0 + S_0 I_1$



2 x 1 Multiplixer.

** 4 x 1 Multiplixer :



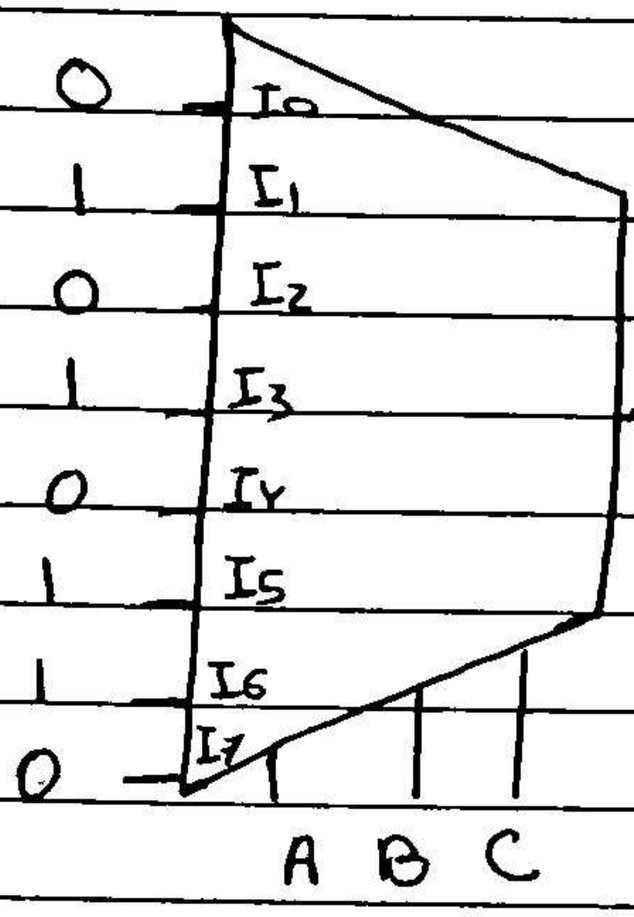
$F = I_0 \bar{S_1} \bar{S_0} + I_1 \bar{S_1} S_0 + I_2 S_1 \bar{S_0} + I_3 S_1 S_0$

Building Combinational Circuits using Mux :-

Given a function of n-variables, Show how to use a mux with :

① n select input (state form)
② n-1 select input imp

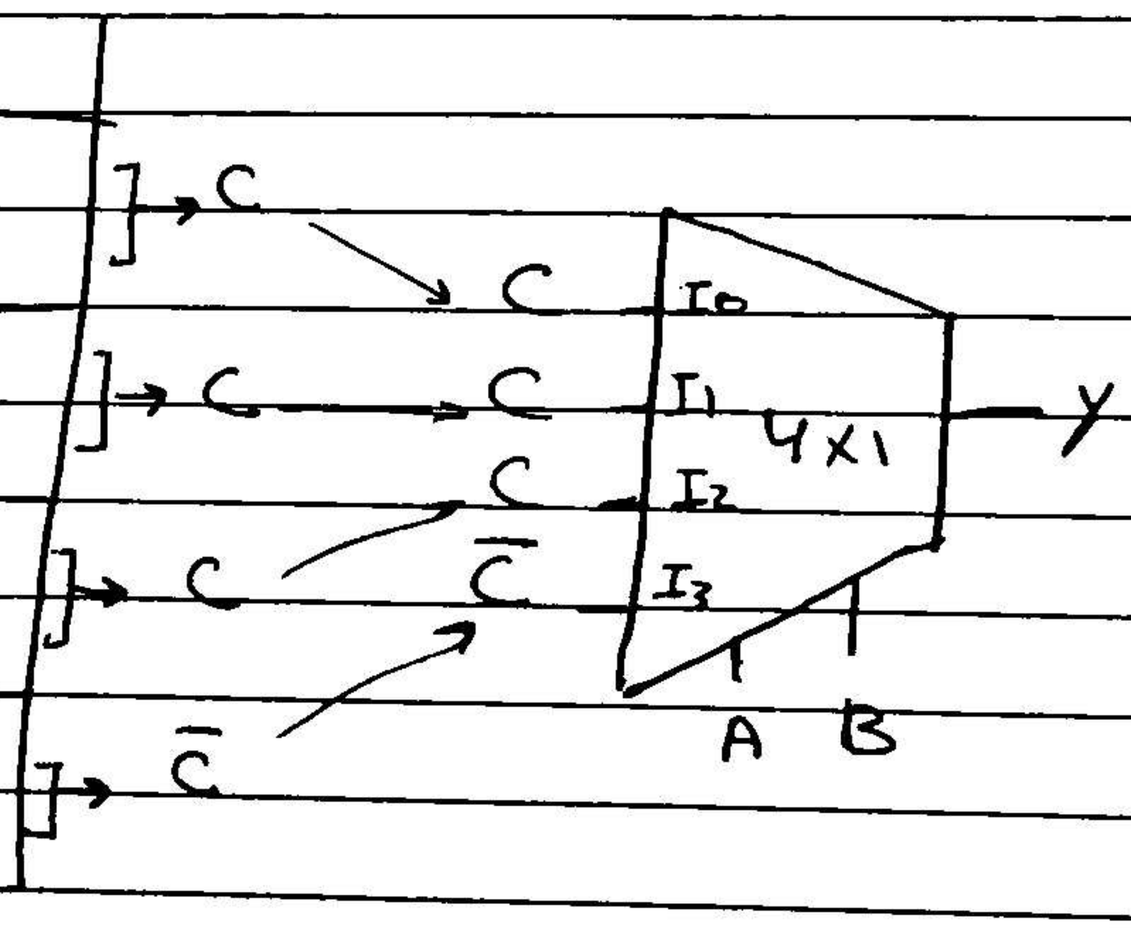
$F(A,B,C) = \Sigma(1,3,5,6)$



- using n-select.

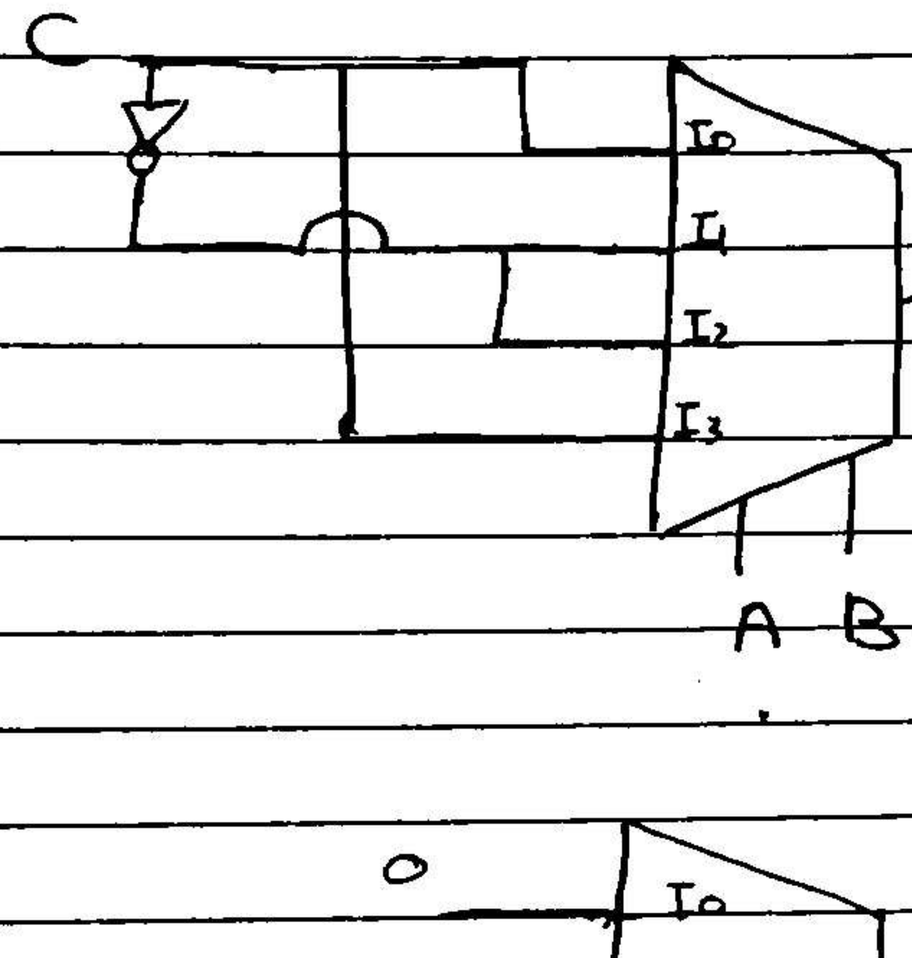
- using n-1 : (4x1) mux

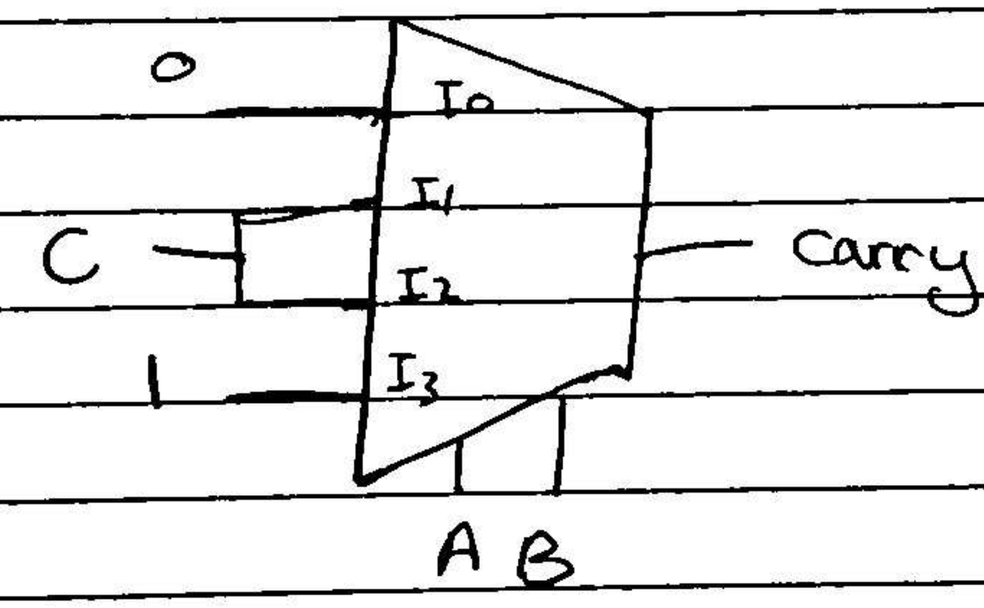
Selector A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0



*ex: Implement the Full Adder circuit using two 4×1 Muxes:

A	B	C	Sum	Carry	
0	0	0	0	0	(compare with C)
0	0	1	1	0	
0	1	0	1	0	
0	1	1	0	1	
1	0	0	1	0	
1	0	1	0	1	
1	1	0	0	1	
1	1	1	1	1	





zero → ground
1 → Vcc.
(power supply of circuit)

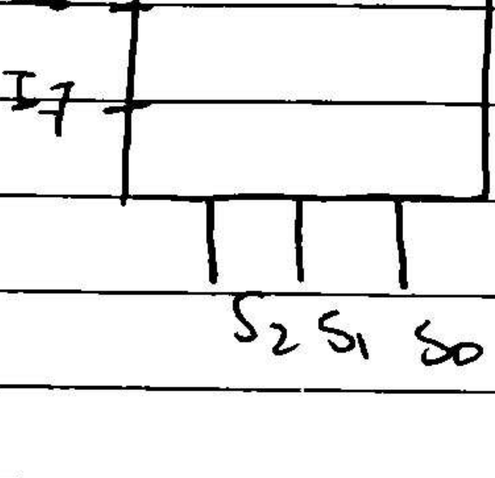
*ex: implement $F(A,B,C,D) = \Sigma(1,3,4,11,12,13,14,15)$ using 8x1 Mux :

Selectors

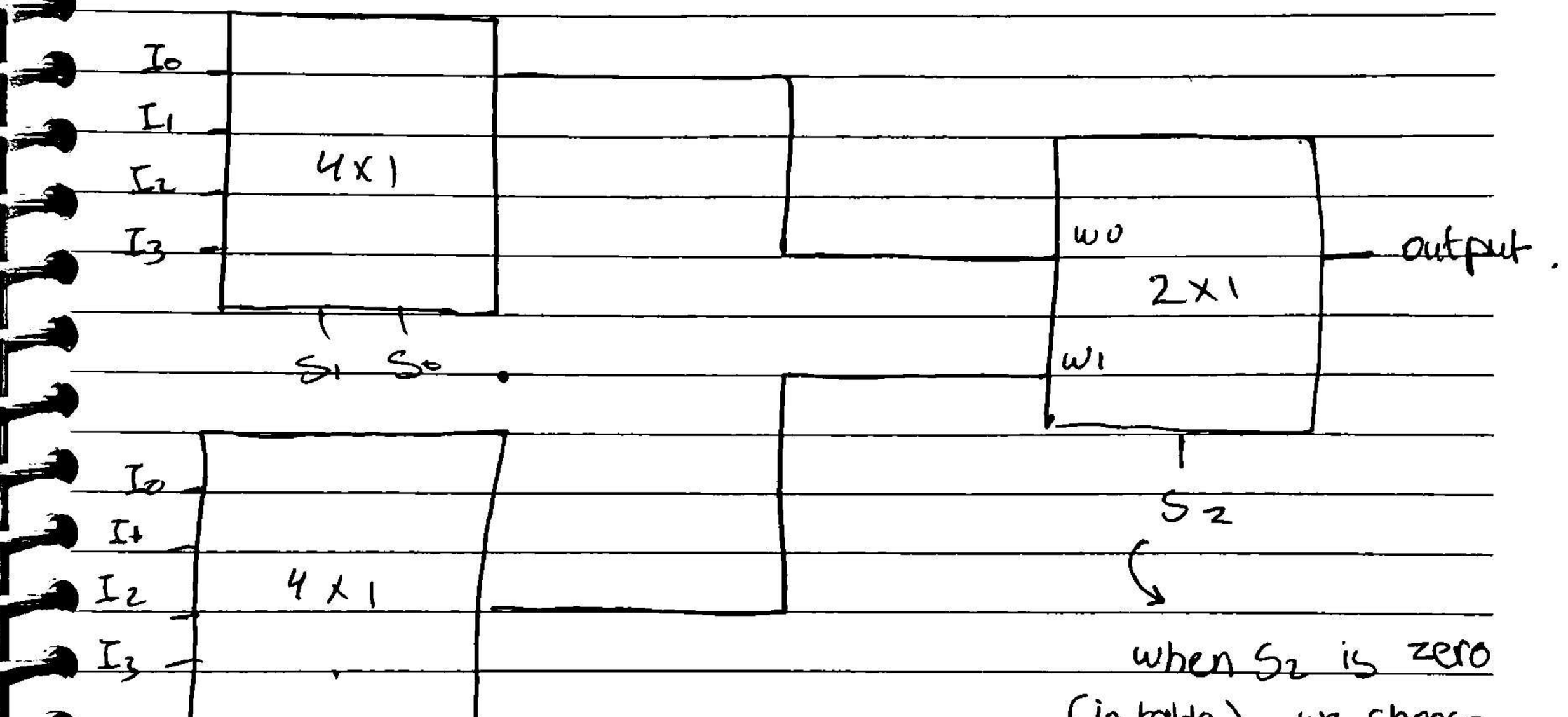
A	B	C	D	F	(compare with D)
0	0	0	0	0	D
0	0	0	1	1	
0	0	1	0	0	D
0	0	1	1	1	
0	1	0	0	1	$\bar{D}$
0	1	0	1	0	
0	1	1	0	0	0
0	1	1	1	0	
1	0	0	0	0	0
1	0	0	1	0	
1	0	1	0	0	D
1	0	1	1	1	
1	1	0	0	1	1
1	1	0	1	1	
1	1	1	0	1	1
1	1	1	1	1	

lecture (14)

** Construct an 8x1 mux using 4x1 and 2x1 muxs



S_2	S_1	S_0	output
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

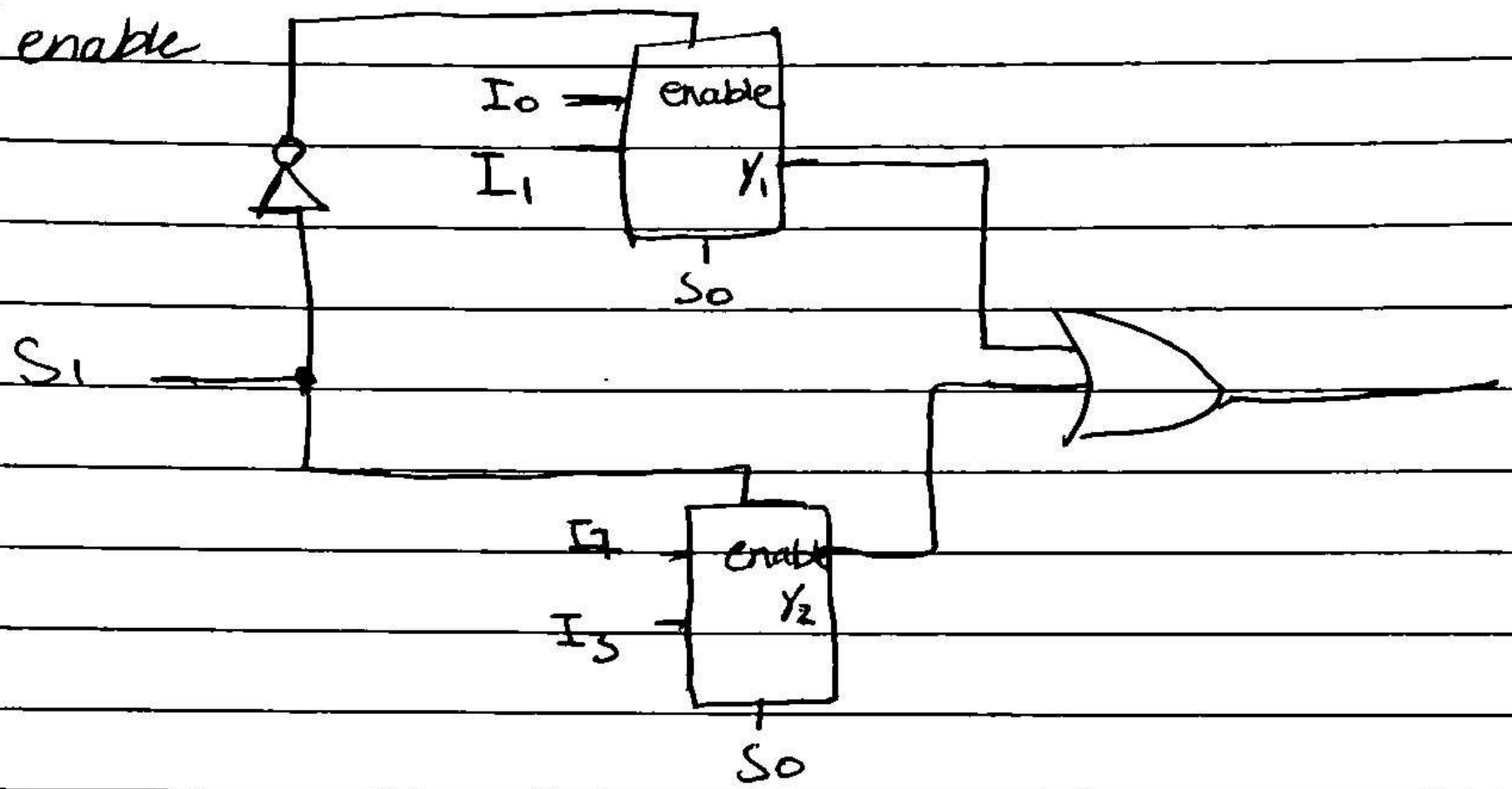


when S_2 is zero (in table) we choose the 1st (4x1) mux

when S_2 is one we choose 2nd (4x1) mux

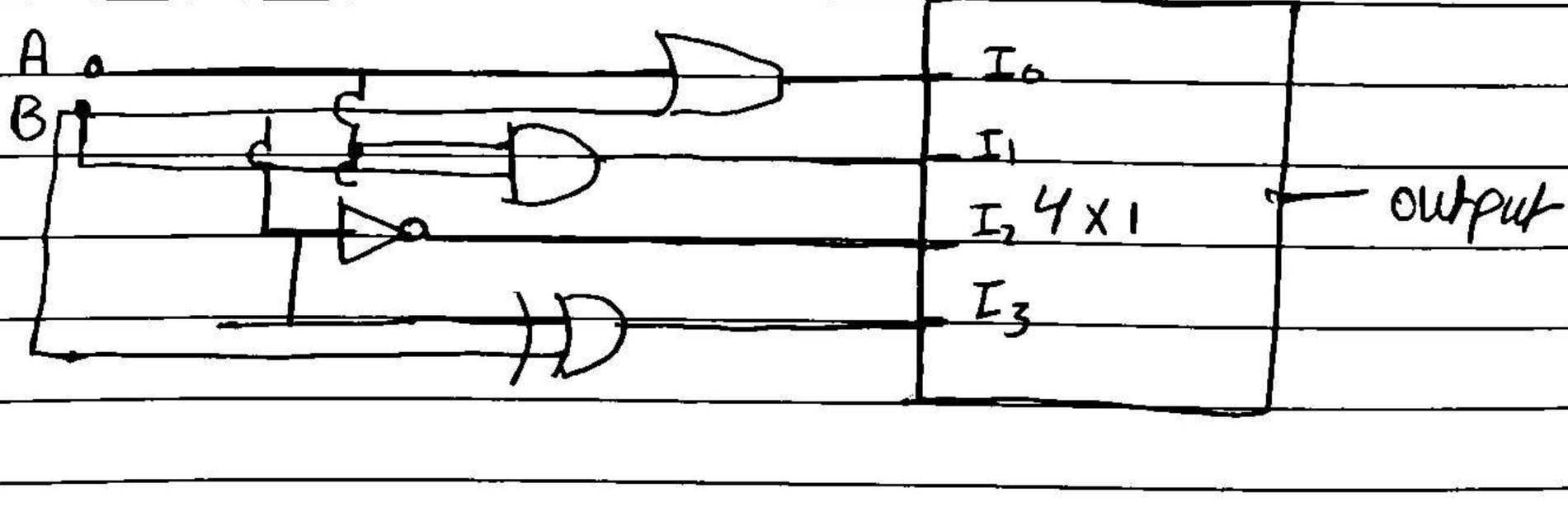
* Construct a 4x1 mux using 2x1 mux with enable :-

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3



* Design a circuit representing the following function :-

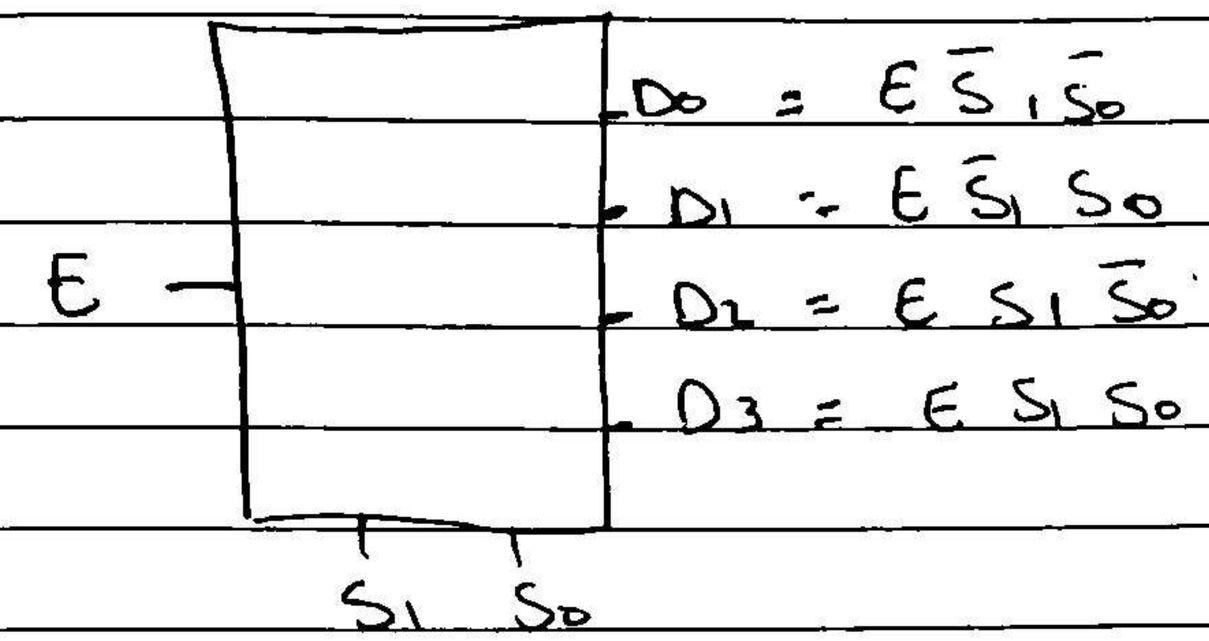
S_1	S_0	output.
0	0	A OR B
0	1	A AND B
1	0	$\bar{A}$
1	1	A XOR B



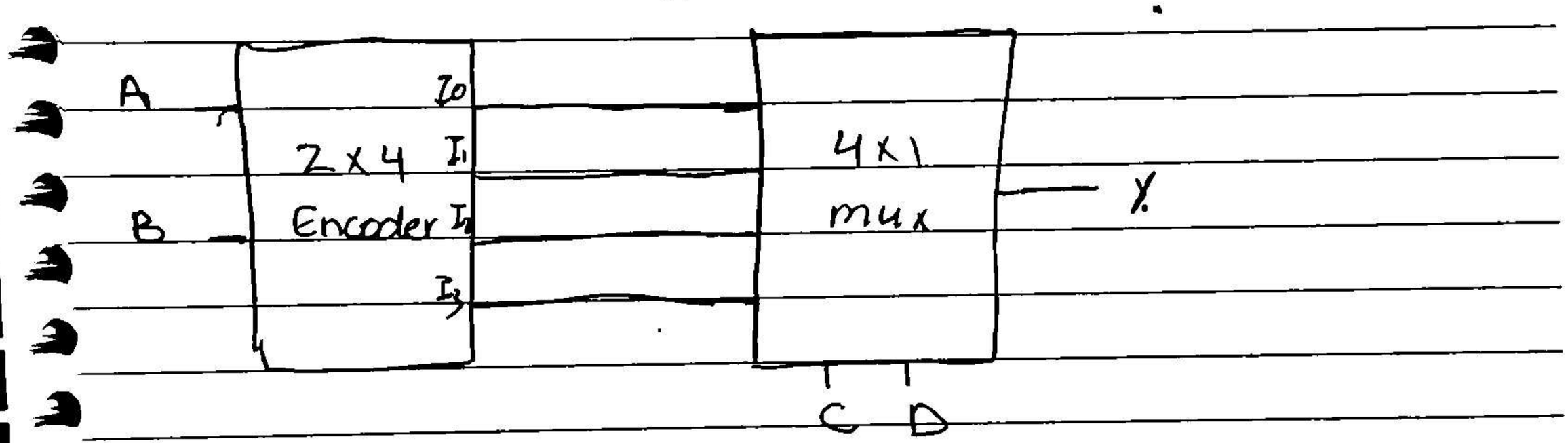
** Demultiplexer :

- performs inverse of multiplexing
- It has one input and 2^{nd} outputs

ex: 1 x 4 Demux



* ex: Fill the following truth table :



B	A	I_3	I_2	I_1	I_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

A	B	C	D	Y
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

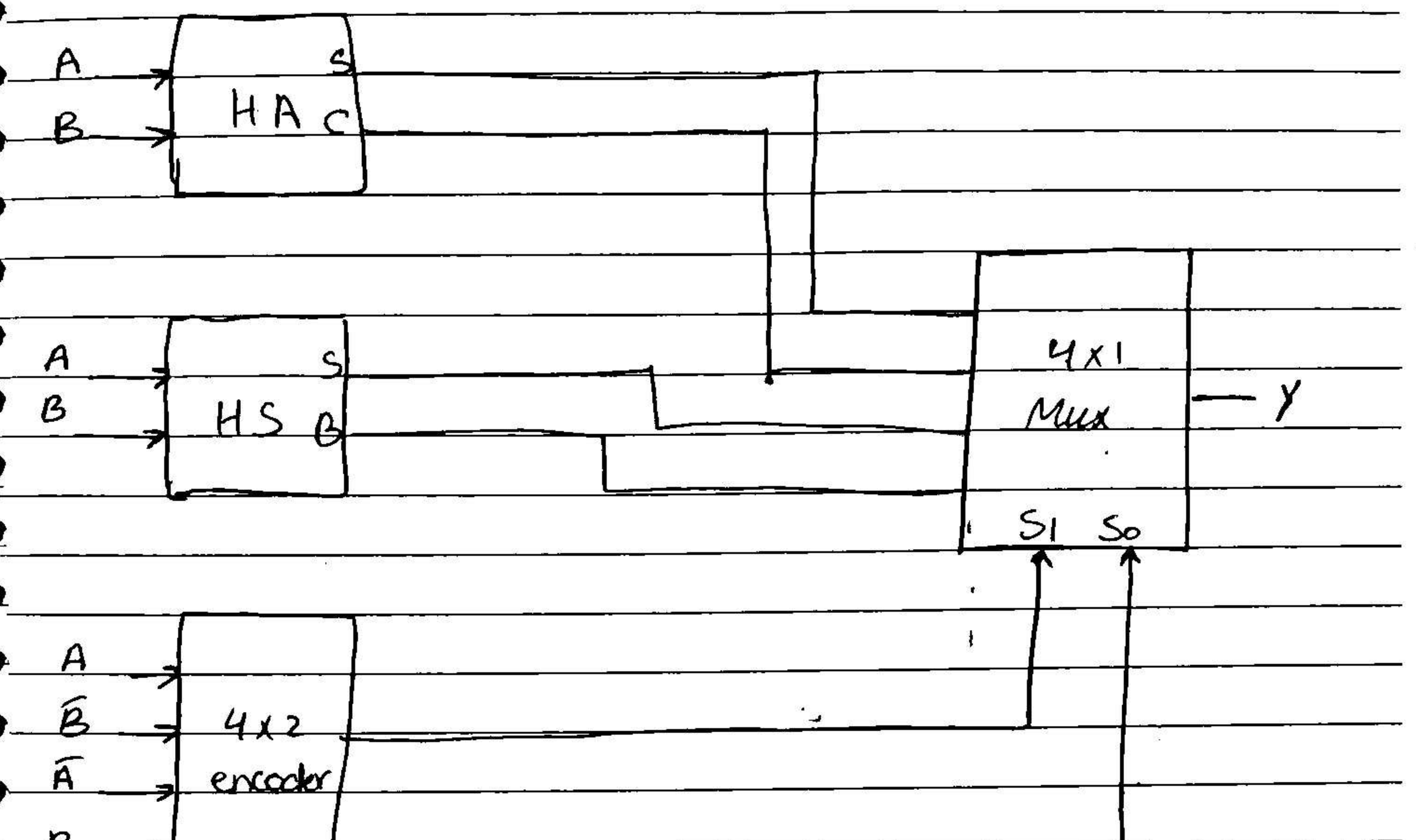
(عدد اول) AB → 00 → 3 is one.

(عدد دوم) AB → 01 → 2 is one.

(عدد سوم) AB → 10 → 1 is one (9)

(عدد چهارم) AB → 11 → zero is one (12)

ex : Fill the following truth table :



→ encoder table

w_3	w_2	w_1	w_0	S_1	S_0	V
0	0	0	1	0	0	1
0	0	1	X	0	1	
0	1	X	X	1	0	
1	X	X	X	1	1	

b' Sub

HA

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

C

HS

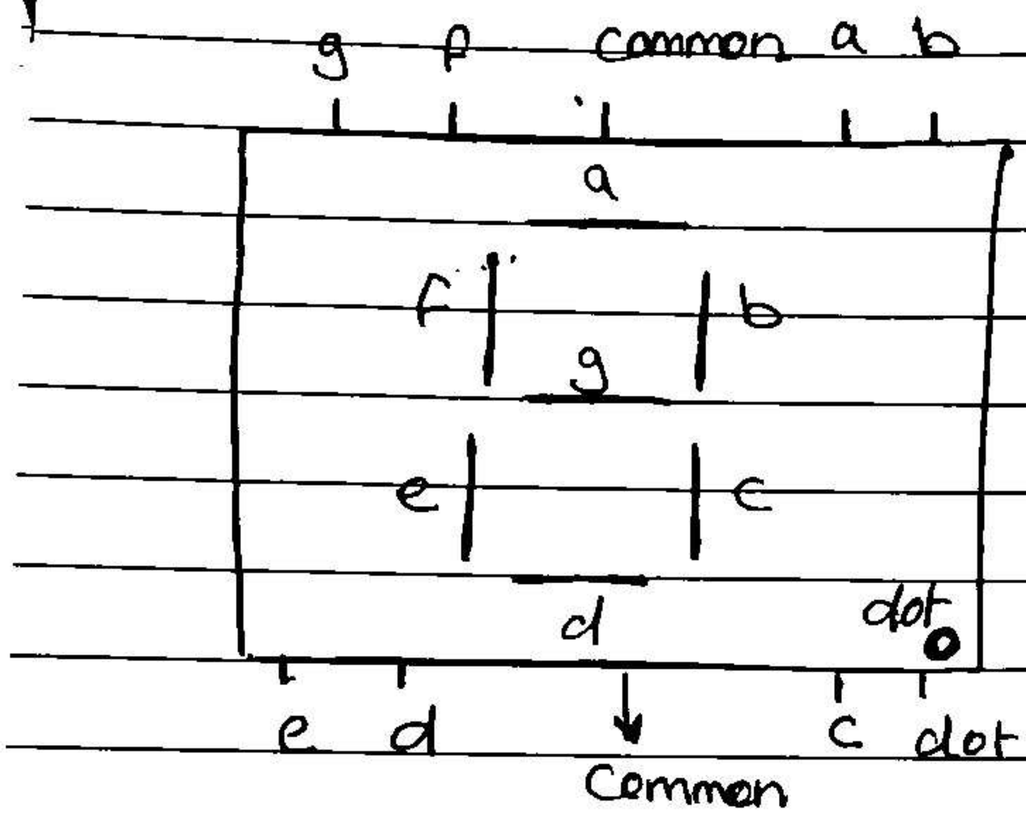
S	b
0	0
1	1
1	0
0	0

Sum

A	B	S_1	S_0	I_0	I_1	I_2	I_3	Y
0	0	1	0	0	0	0	0	0
0	1	0	1	1	1	0	1	1
1	0	1	1	0	1	0	1	1
1	1	1	1	0	0	1	0	0

** BCD to 7-Segment Display decoder :

- The 7-Segment display contains 7-Leds
- There are two types :
  1. Common Anode (Active low)
  2. Common cathod (Active High)



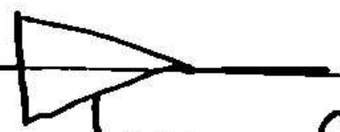
CA = 5V

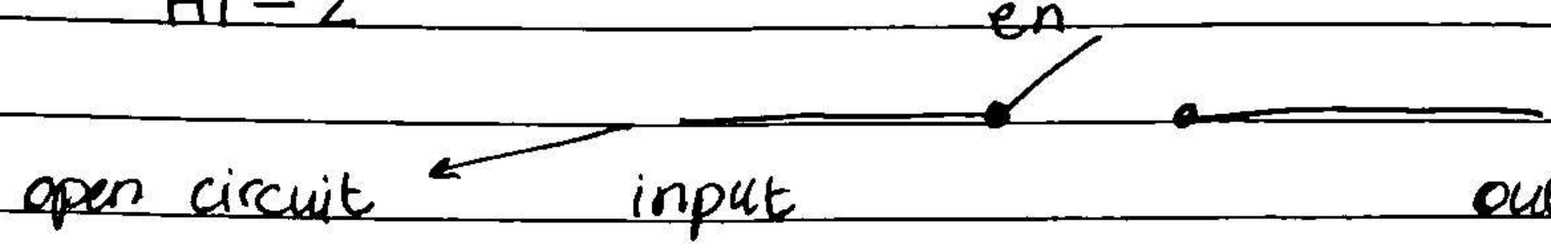
CC = GND

H.W

A	B	C	D	a	b	c	d	e	f	g	display
0	0	0	0	1	1	1	1	1	1	0	
0	0	0	1	0	1	1	0	0	0	0	
0	0	1	0	1	1	0	1	1	0	1	
0	0	1	1								
0	1	0	0								
0	1	0	1								
0	1	1	0								
0	1	1	1								
1	0	0	0								

* Tri - State Buffer : enable → تفعيل الـ output عند السيركت (حيث صفر) open circuit.





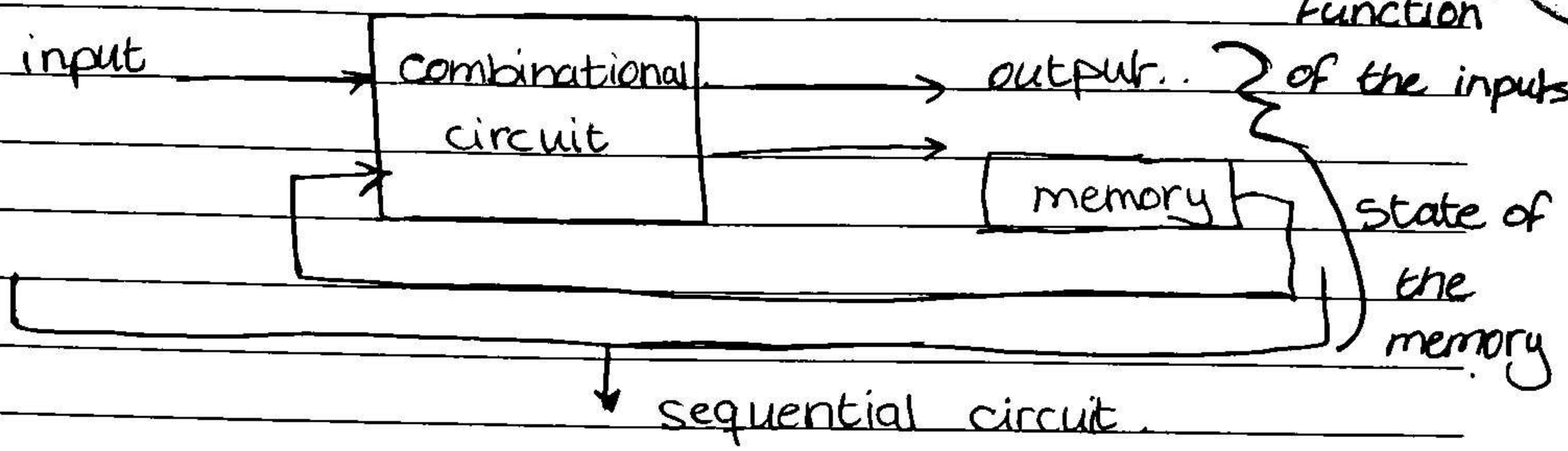
Chapter (5) lecture (15)

** Sequential Circuits :

* Synchronous sequential circuits :

- Unlike combinational circuits, sequential circuits include memory.

sequential circuit = combinational circuit + memory component.



-

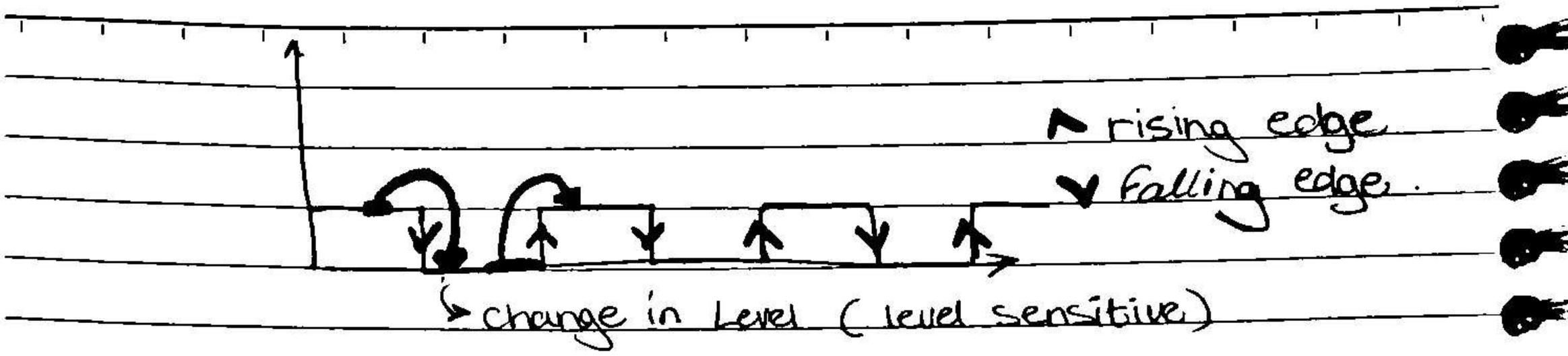
- The outputs in a sequential circuit are :
  ① A function of the inputs.
  ② present state of the storage element

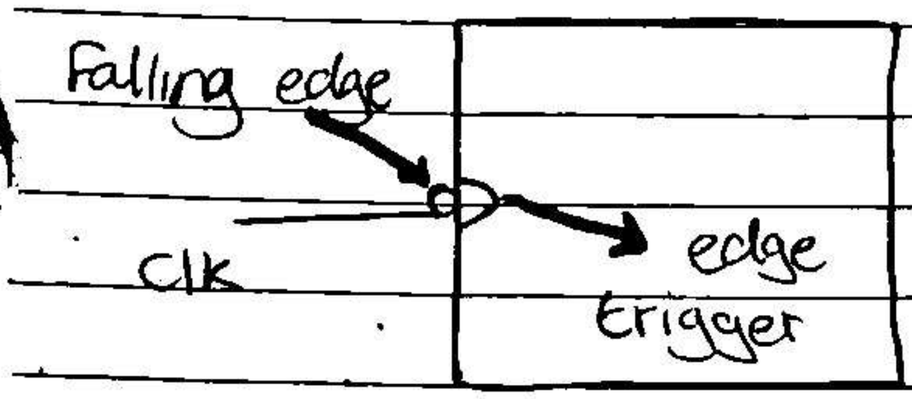
- There are two types of sequential circuits :
  ① Synchronous
  ② Asynchronous

* Synchronous circuits :

- The state is allowed to change at specific instances of time.

- There is a clock (CLK) input





* Asynchronous :

- The stored value is allowed to change at any instance of time.

⇝ There is no clock.

⇝ The design is complex.

⇝ circuit is faster.

** Basic memory elements :

① Latch :

- operate with signal levels (Level sensitive)
- The output can change multiple times during a clock cycle
- The output changes while the clock is active.

② Flip - Flops :

- controlled by clock transition (edge sensitive)
- The output can change only ONCE during a clock cycle.

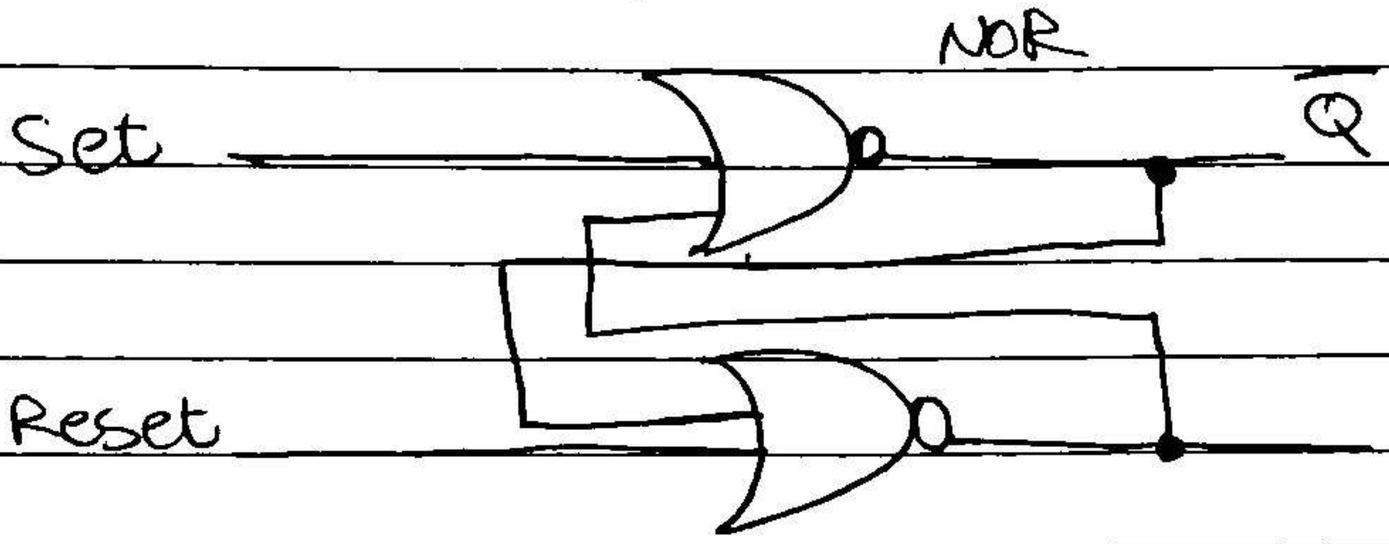
** Latches :

o SR Latch :

- has two inputs :

① Set (S) input.

② Reset (R) input.



- it has three modes of operation :

▲ Set : Latch output set to '1' (Q=1)

↳ S=1 R=0

▲ Reset : Latch output set to '0' (Q=0)

↳ S=0 R=1

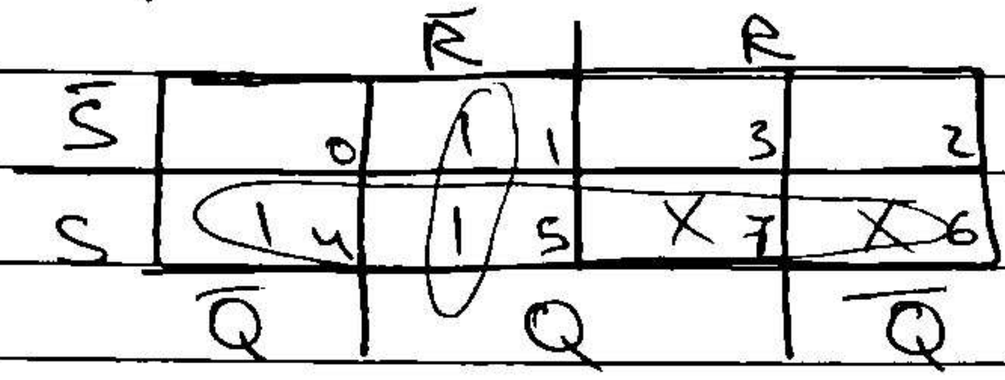
▲ Store : Latch output set to "dont change" ($Q^+=Q$)

↳ S=0 R=0

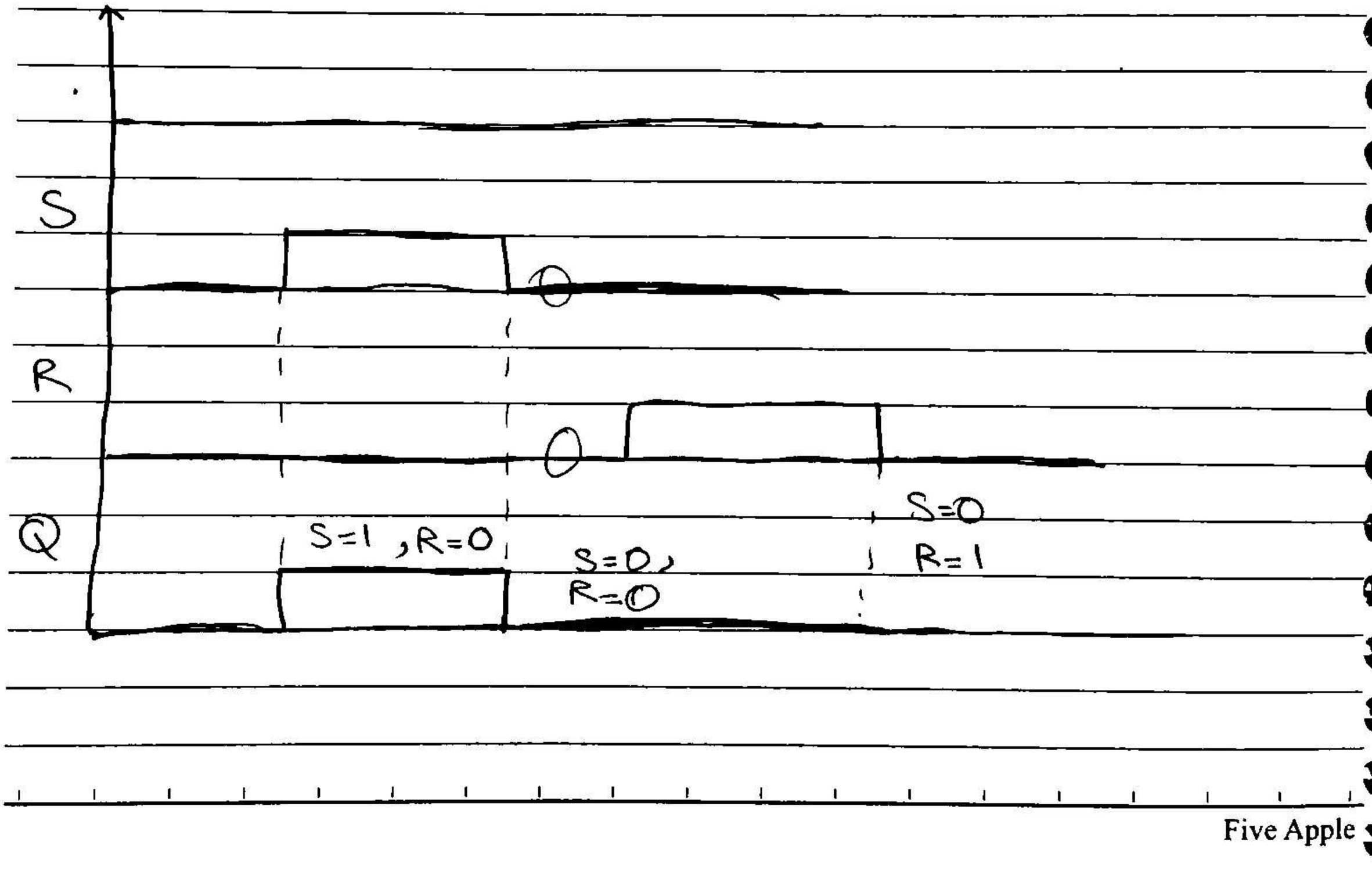
current state			next state	
S	R	Q	Q^+	
0	0	0	0	→ No change
0	0	1	1	
0	1	0	0	→ Reset $Q^+=0$
0	1	1	0	
1	0	0	1	→ Set $Q^+=1$
1	0	1	1	
1	1	0	–	→ Not allowed
1	1	1	–	

(undetermined) Not don't care X

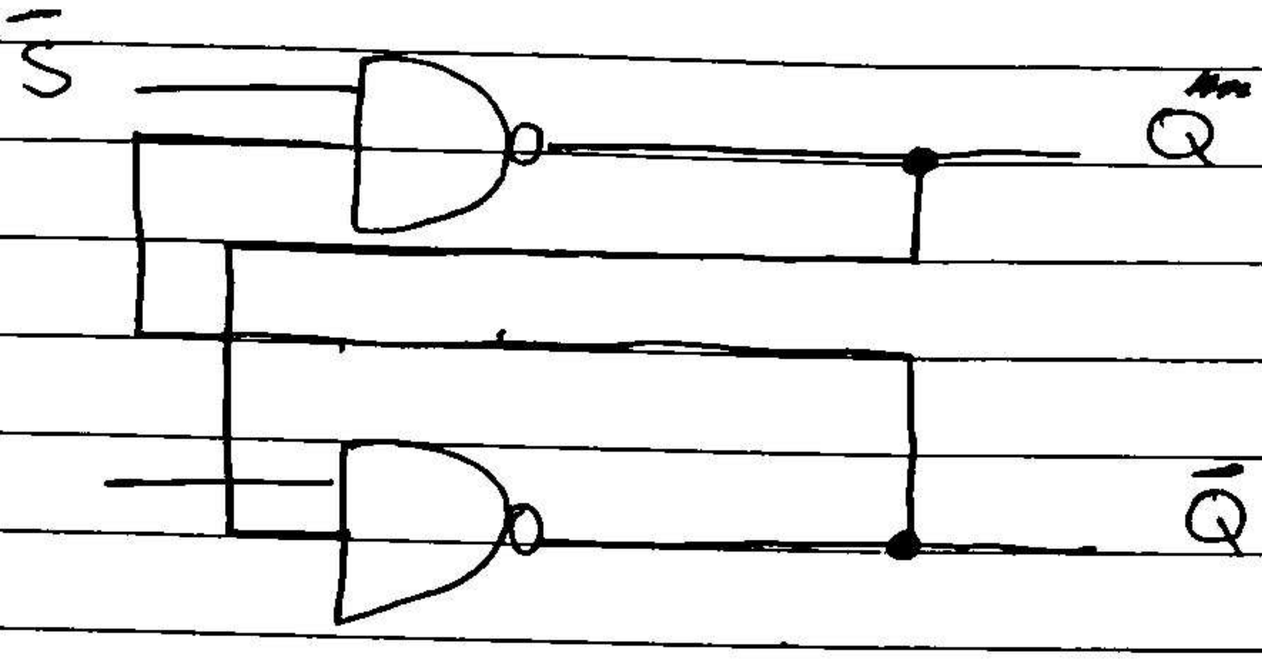
** The SR equation :

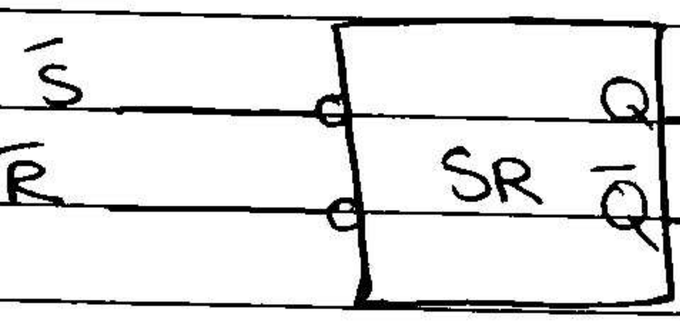


$Q^+ = S + \bar{R}Q$



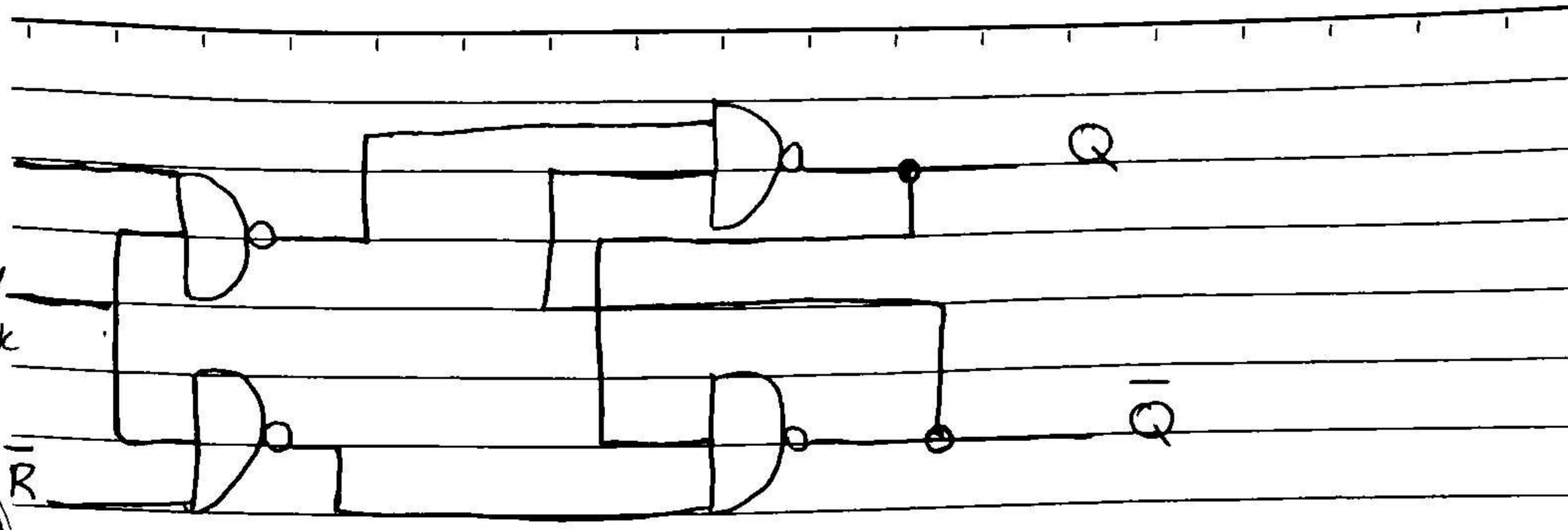
* SR Latch using NAND gates:





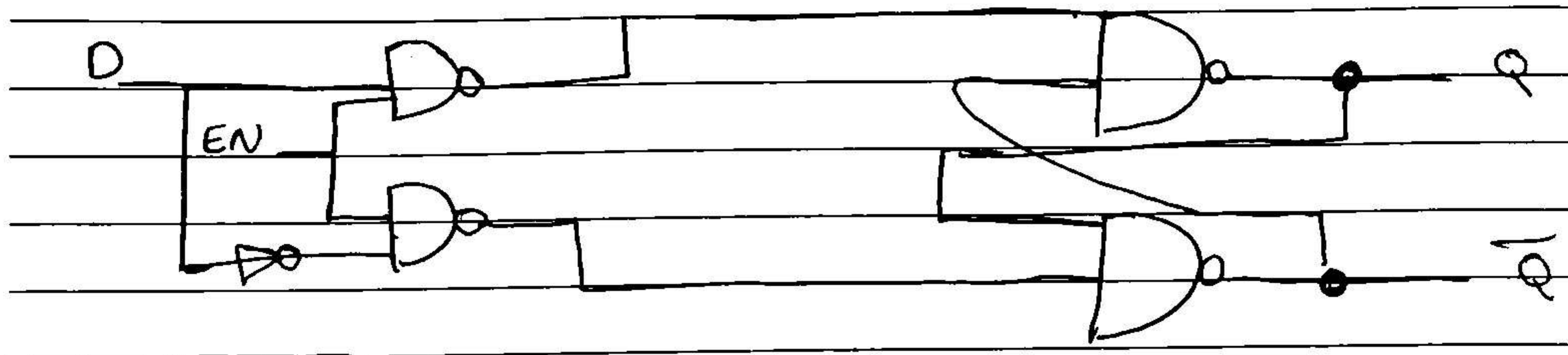
$\bar{S}$	$\bar{R}$	Q	$\bar{Q}$	
0	0	0	1	Not allowed.
0	0	1	1	
0	1	0	1	Set.
0	1	1	1	
1	0	0	0	Reset
1	0	1	0	
1	1	0	0	No change
1	1	1	1	

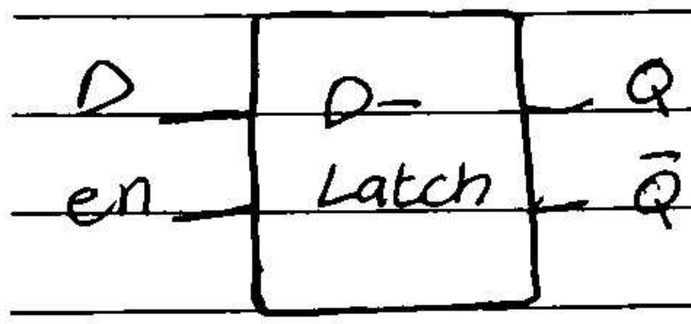
- The operation of the Basic SR-Latch can be modified by providing an additional input signal enable



EN	S	R	Q^+	$\overline{Q}^+$	
0	X	X	No change		
1	0	0	No change		
1	1	0	1	0	
1	0	1	0	1	
1	1	1	Not allowed		

* D- Latch :



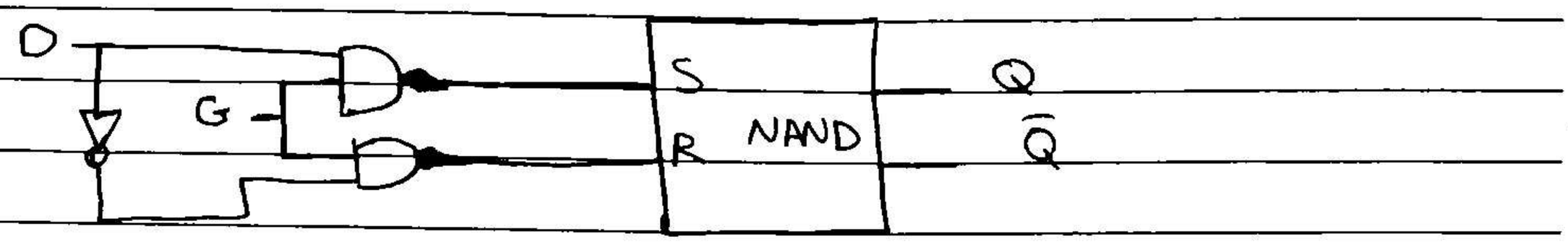


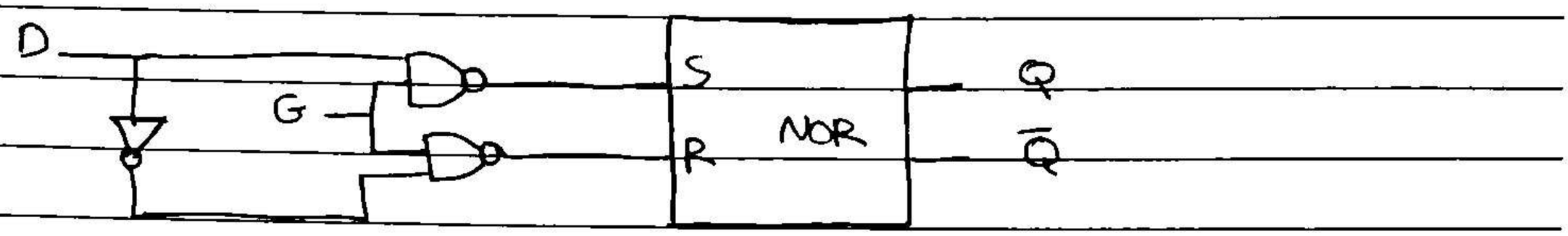
EN	D	Q	$\overline{Q}$	
0	X	No Change		
1	1	1	0	Set
1	0	0	1	Reset

Not allowed ← تخلصنا من حالة ال
D-Latch من طرق ال

* Gated D- Latch (transparent Latch) :

- it has two inputs :
  1. Gate input (G)
  2. Data input (D)





° Gated D - Latch :

G	D	Q	Q^+	
0	0	0	0	No change
0	0	1	1	No change
0	1	0	0	No change
0	1	1	1	No change
1	0	0	0	
1	0	1	0	
1	1	0	1	
1	1	1	1	

when $\boxed{G=0}$ Q is the output

when $\boxed{G=1}$ D is the output

	G		G	
$\bar{Q}$	0	1	3	2
Q	4	5	7	6
	$\bar{D}$		D	$\bar{D}$

$$Q^+ = \bar{G}Q + GD$$

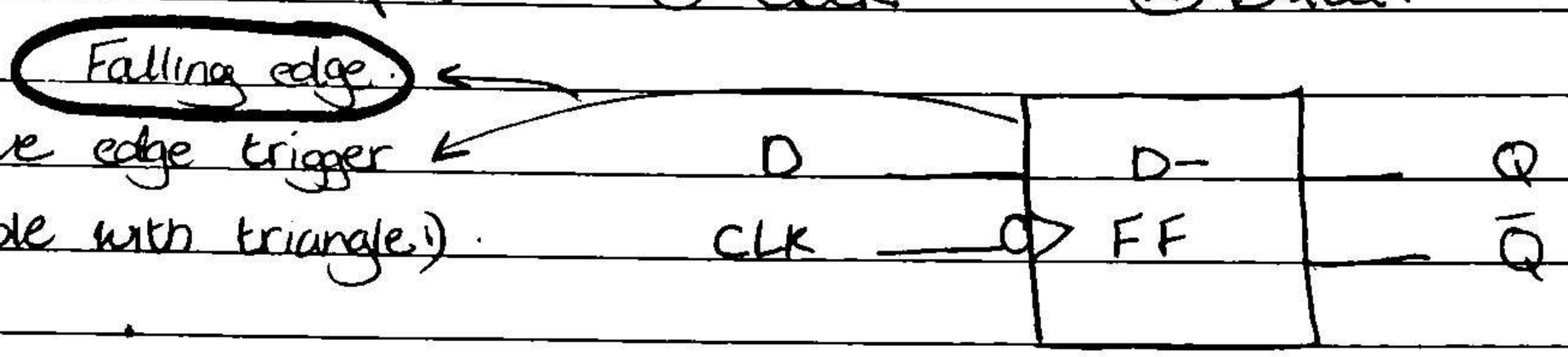
lecture (16)

** Flip - Flops :

- The D-Flip-Flop is the most commonly used F-F.
- other types exist : ' SR , Jk , T
- Three operations can be performed with F-F
  ① Set to '1'
  ② Reset to '0'
  ③ Complement Q

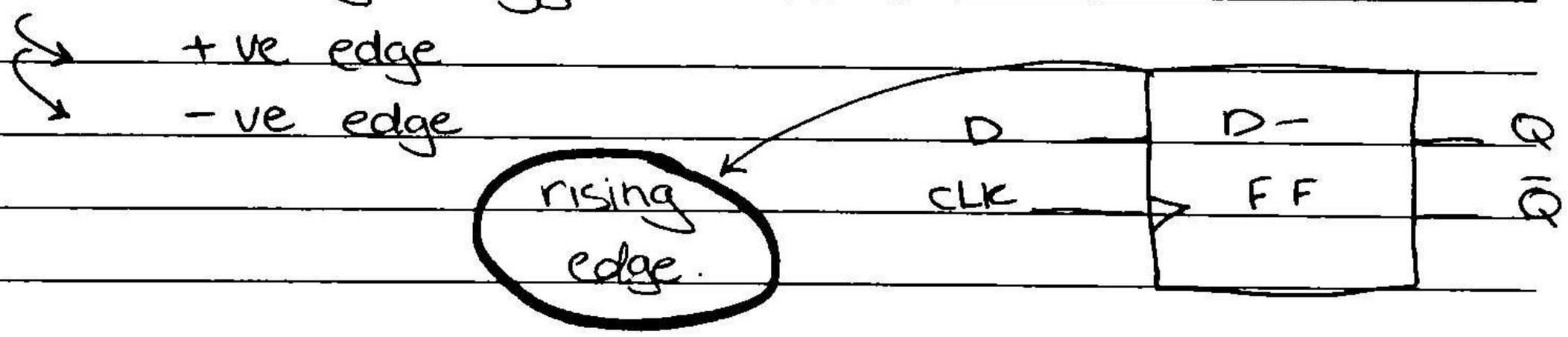
* D- Flip-Flop :

- it has two inputs : ① clock ② Data.



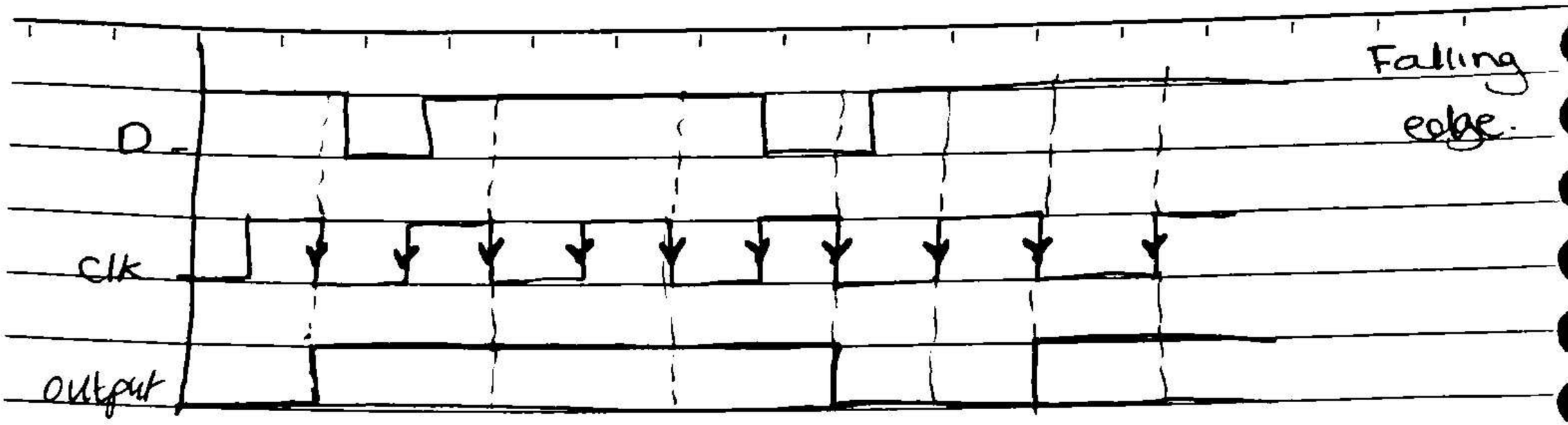
- The output of the D-FF changes in response to the clock input not only in response to a change in D

- D-FF is edge triggered circuits.



D	Q	Q^+
0	0	0
0	1	0
1	0	1
1	1	1

$Q^+ = D$



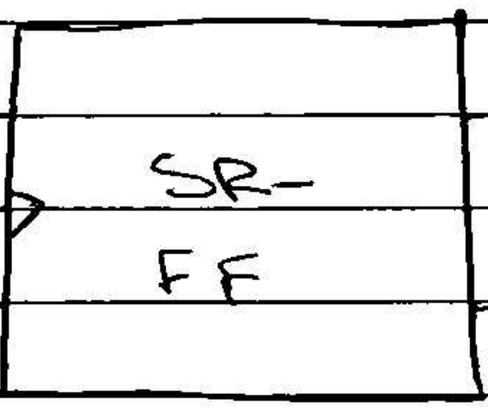
Timing Diagram D- Flip Flop.

* S-R Flip Flop :

- Has 3 inputs :
  1. clock
  2. Set
  3. Reset.
- Similar to SR- latch :

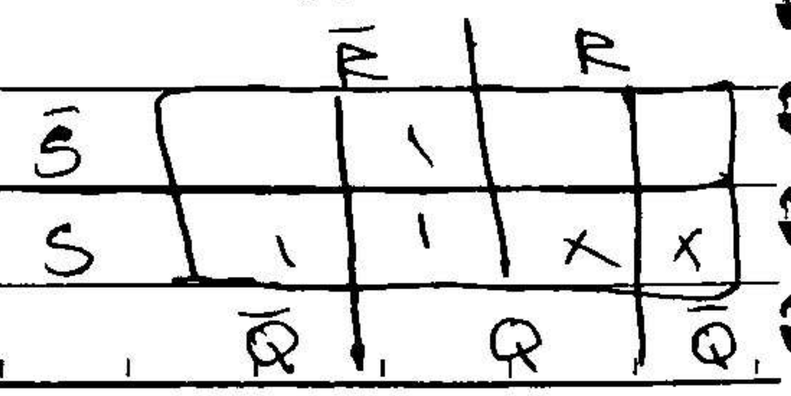
→ S=1 sets the FF rising edge.

→ S=0 Resets the FF



S	R	Q^t	Q^+	
0	0	0	0	No change (store)
0	0	1	1	No change
0	1	0	0	→ R=1 → out = 0
0	1	1	0	
1	0	0	1	→ S=1 → out = 1
1	0	1	1	
1	1	0	X	→ NOT allowed.
1	1	1	X	

$$Q^+ = S + \bar{R}Q$$



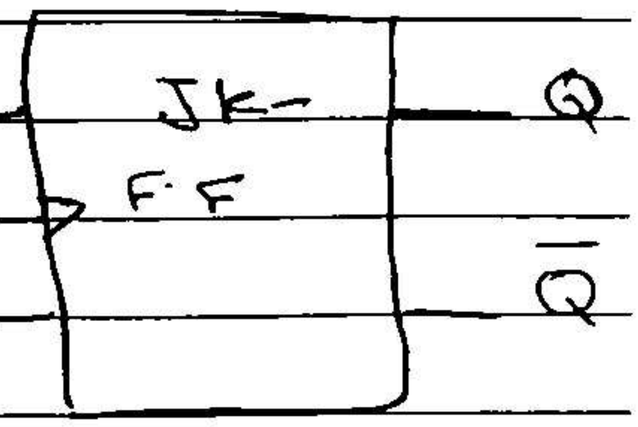
$$Q^+(t+1) = S + \bar{R}Q(t)$$

Five Apple

* JK- Flip-Flop :

- it has 3 inputs : J(set), K(reset), clk.
- it performs three operations :

① set JK = 10
② Reset JK = 01
③ Complement JK = 11
④ Store JK = 00



J	k	Q(t)	Q(t+1)	
0	0	0	0	Store k
0	0	1	1	
0	1	0	0	reset
0	1	1	0	
1	0	0	1	set
1	0	1	1	
1	1	0	1	complement
1	1	1	0	

$$Q(t+1) = J\bar{Q}(t) + \bar{K}Q(t)$$

[redacted]:

J	k	Q(t+1)	
0	0	Q(t)	No change
1	0	1	set
0	1	0	Reset
1	1	$\overline{Q(t)}$	complement.

* T- Flip Flop (Toggle) :

→ Basically Jk FlipFlop.

- Toggle. : has two inputs :
  → But inputs shorted (جميع الـ inputs واصلين على بعض)
  ① T
  ② Clk

when T = 0 , the state does not change
T = 1 , the state changes on active trigger.

T	Q(t)	Q(t+1)	
0	0	0	] → store
0	1	1	]
1	0	1	] → toggle
1	1	0	]

$$Q(t+1) = T \oplus Q(t)$$

↳ XOR.

J	K	Q(t+1)

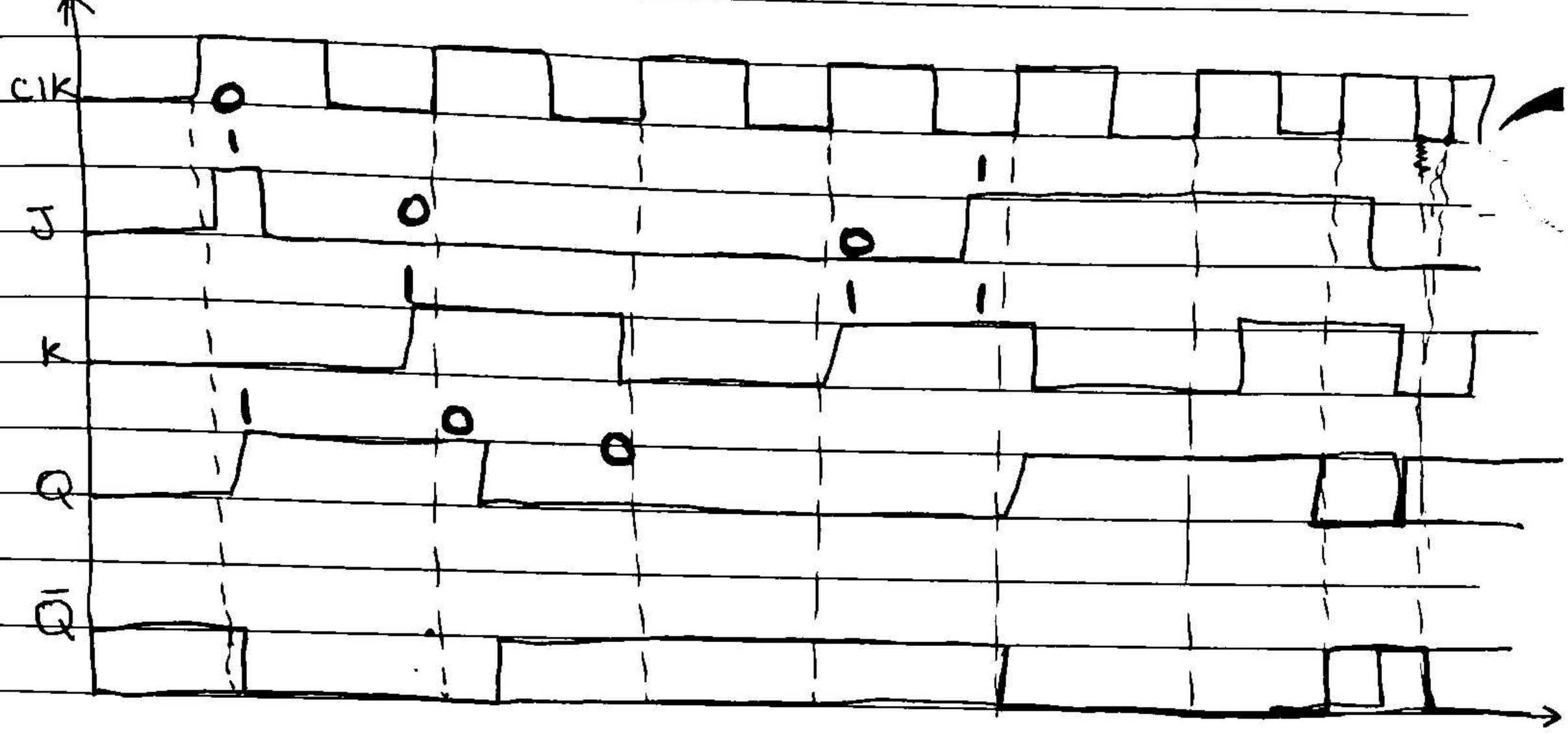
*ex :

الرسمة خطأ ←

ex:

J	k	Q(t+1)
0	0	Q
1	0	1
0	1	0
1	1	$\bar{Q}$

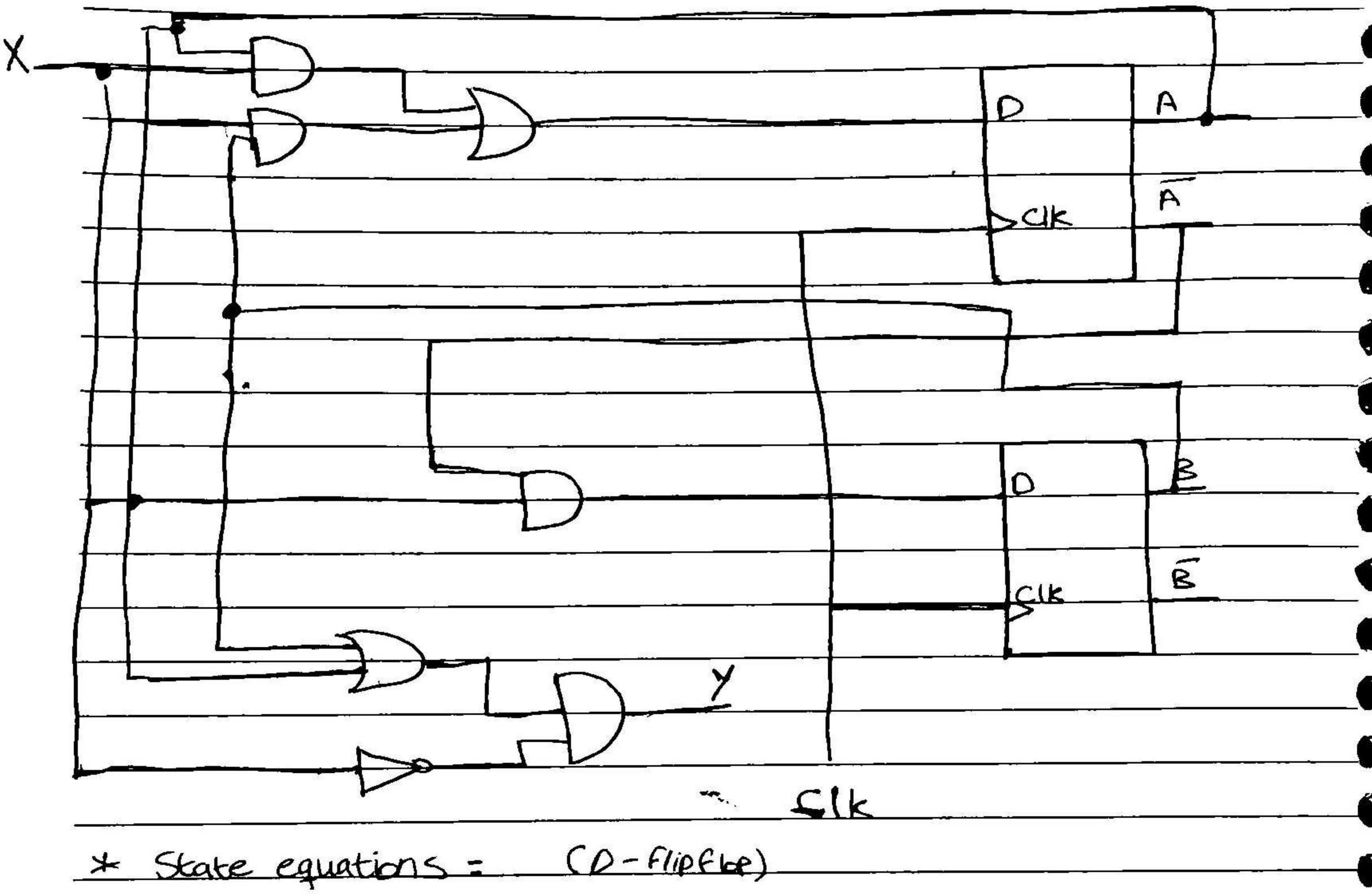
+ve edge.



** Analysis of clocked sequential circuits :

- obtaining a table or diagram for the time sequence of inputs, outputs and internal states.

- The behavior of the circuit can be described algabrically by means of state diagram and state equations that specifies the next state as a function of the present state and inputs



* State equations = (D-FlipFlop)

$A(t+1) = A(t)\,X(t) + B(t)\cdot X(t)$

$B(t+1) = \overline{A}(t)\cdot X(t)$

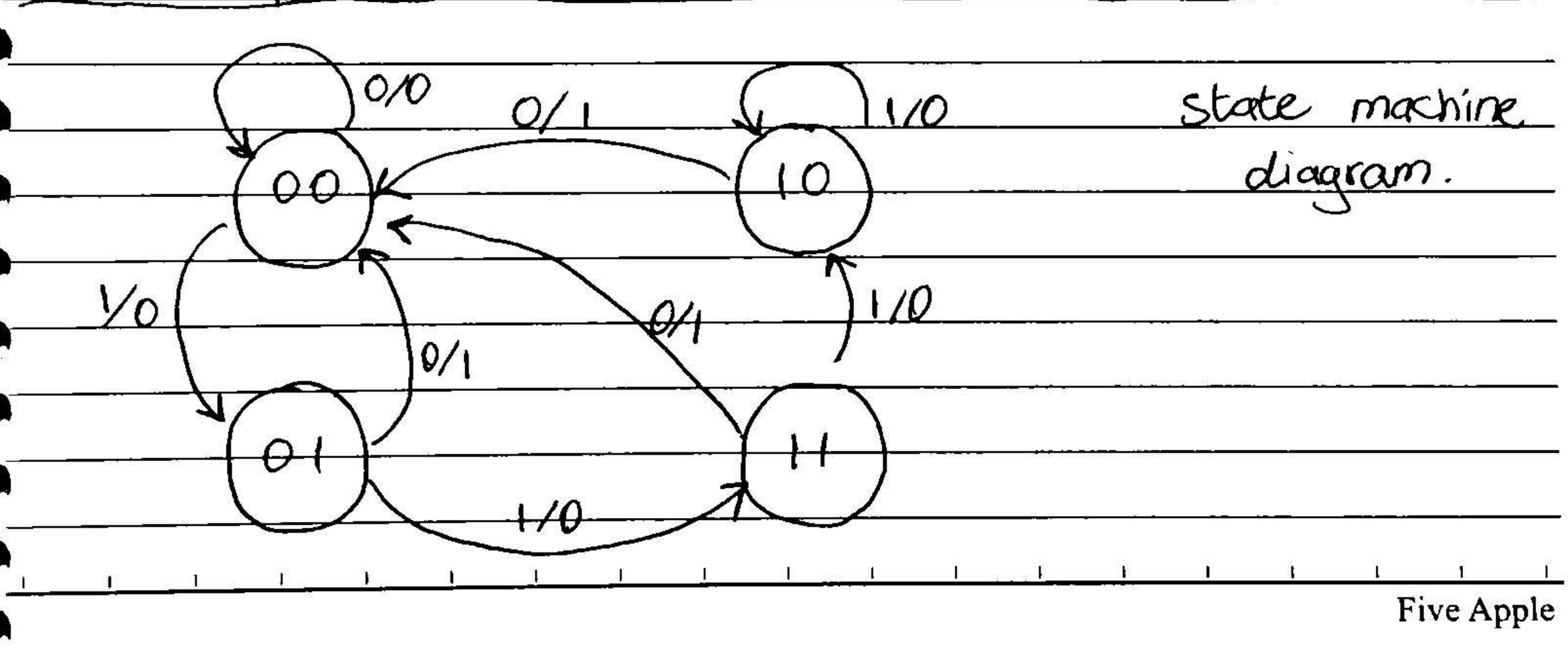
$Y(t) = \overline{X}(t)\,[A(t)+B(t)]$

** Meally Model :

① The output y is presented as a function of both x and the present state

present state A	present state B	input X	Next state A(t+1)	Next state B(t+1)	output Y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	0	1
0	1	1	1	1	0
1	0	0	0	0	1
1	0	1	1	0	0
1	1	0	0	0	1
1	1	1	1	0	0

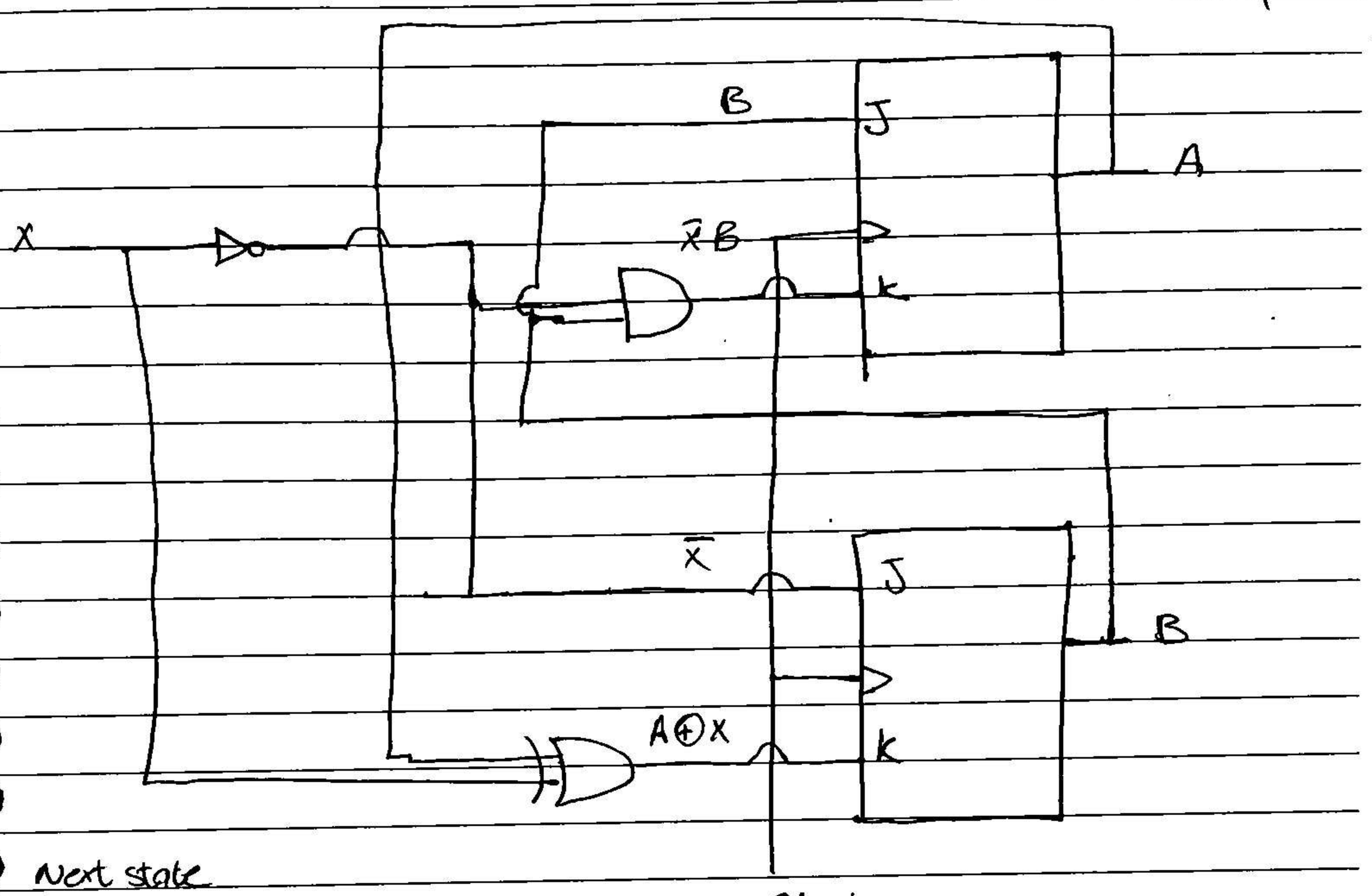
present state A	present state B	Next state x=0 AB	Next state x=1 AB	output x=0 Y	output x=1 Y
0	0	0 0	0 1	0	0
0	1	0 0	1 1	1	0
1	0	0 0	1 0	1	0
1	1	0 0	1 0	1	0



lecture (18)

* Find the state machine diagram for the following circuit :

J	K	Q	$\bar{Q}$
0	0	No change	
0	1	0	1
1	0	1	0
1	1	complement	



Next state

↖ Basic equations * imp

$A(t+1) = J_A \bar{A} + \bar{K}_A A$

$B(t+1) = J_B \bar{B} + \bar{K}_B B$

$Q^+ = J\bar{Q} + \bar{K}Q$

$Q(t+1) = J\bar{Q}(t) + \bar{K}Q(t)$

$J_A = B$, $K_A = \bar{X}B$

$J_B = \bar{X}$, $K_B = A \oplus X$

$A(t+1) = B\bar{A} + (B\bar{x})A$

$= B\bar{A} + (\bar{B}+x)A$

$= B\bar{A} + A\bar{B} + Ax$

$B(t+1) = \bar{x}\bar{A} + (\overline{A \oplus x})B = \bar{B}\bar{x} + ABx + \bar{A}B\bar{x}$

present state		input	Next state	
A	B	x	A	B
0	0	0	0	1
0	0	1	0	0
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	1	0
1	1	0	0	0
1	1	1	1	1

$$A(t+1) = B\bar{A} + (\bar{B}+X)A$$

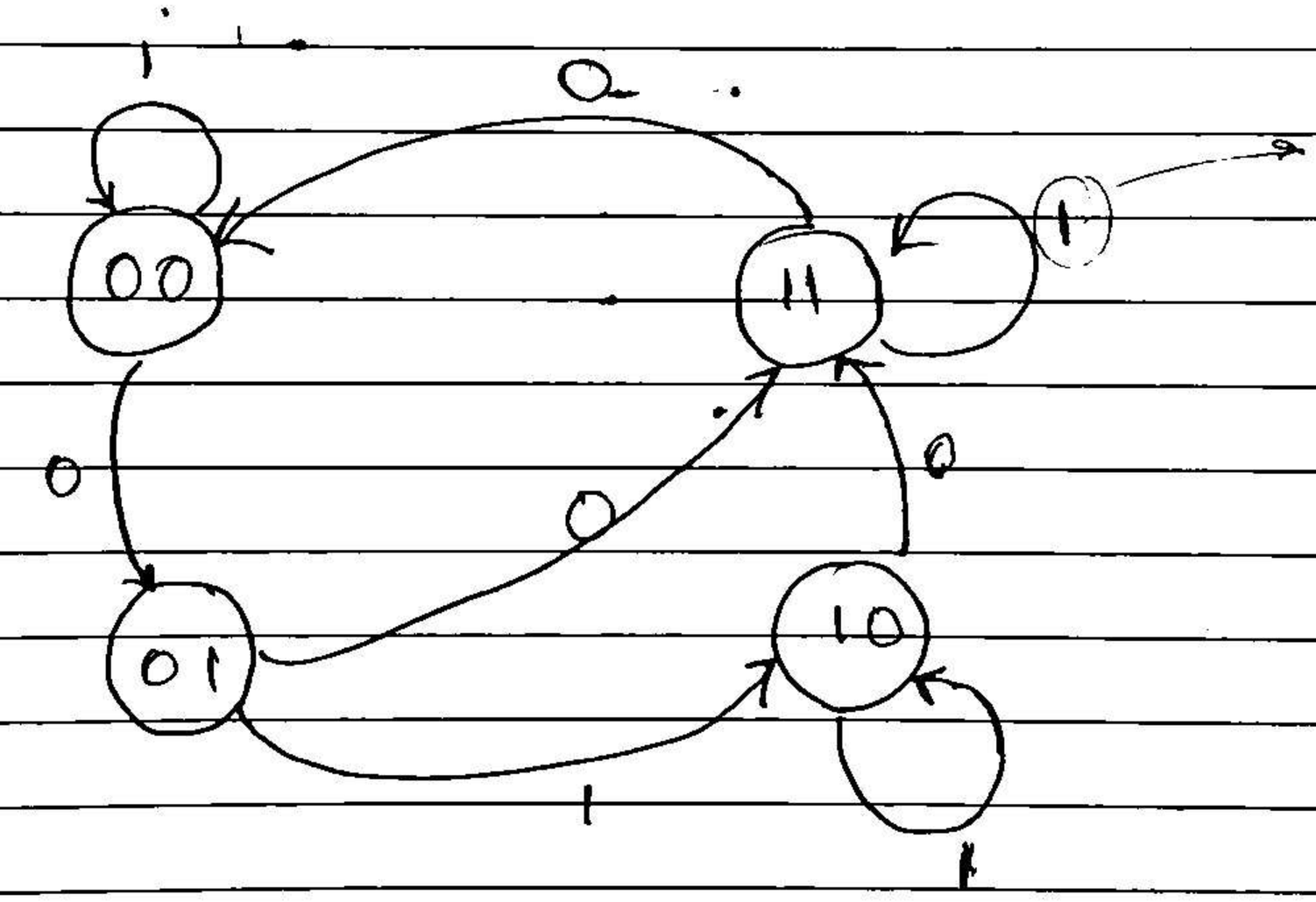
* كل حد متى يساوي واحد؟

B=1 B اما - إذا يساوي 1 ← متى يساوي

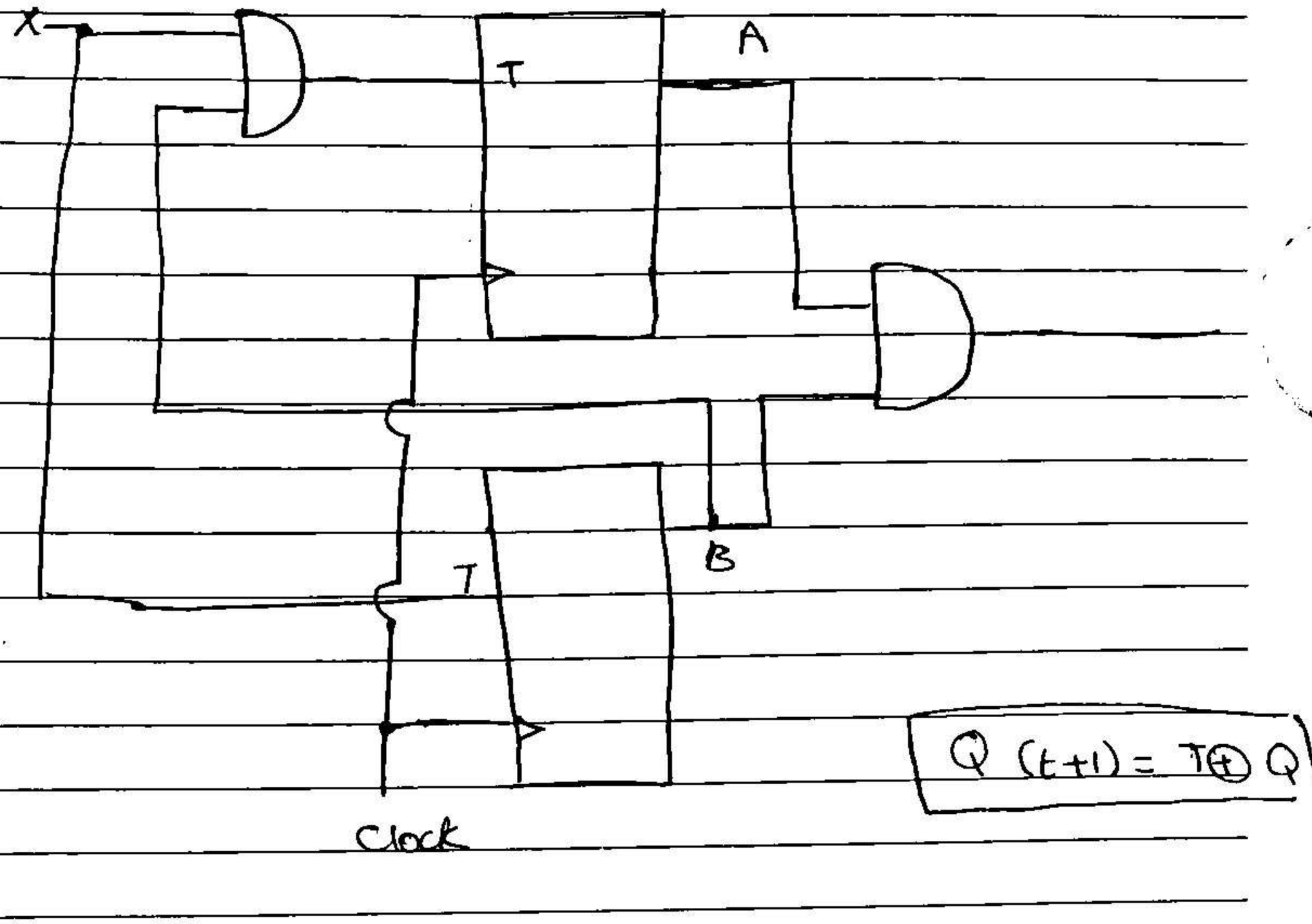
$\bar{A}=0$

وبعبر بالجدول

$$B(t+1) = \bar{B}\bar{X} + AB\bar{X} + \bar{A}B\bar{X}$$



* Find the state machine diagram for the following circuit:



$Q(t+1) = T \oplus Q$

$T_A = BX$ $\quad T_B = X$ $\quad Y = AB$

$$A(t+1) = T_A \oplus A$$
$$= \overline{T_A} A + T_A \overline{A}$$
$$= (\overline{BX}) A + BX \overline{A}$$
$$= (\overline{B} + \overline{X}) A + BX \overline{A}$$
$$= A\overline{B} + \overline{A}\,\overline{X} + \overline{A}BX$$

* ما تنصري

$$B(t+1) = \overline{X} B + X \overline{B}$$

A(t+1)B(t+1)

present state		input	Next state		Y
A	B	X	A(t+1)	B(t+1)	AB
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	1	1
1	1	0	1	1	1
1	1	1	1	0	0

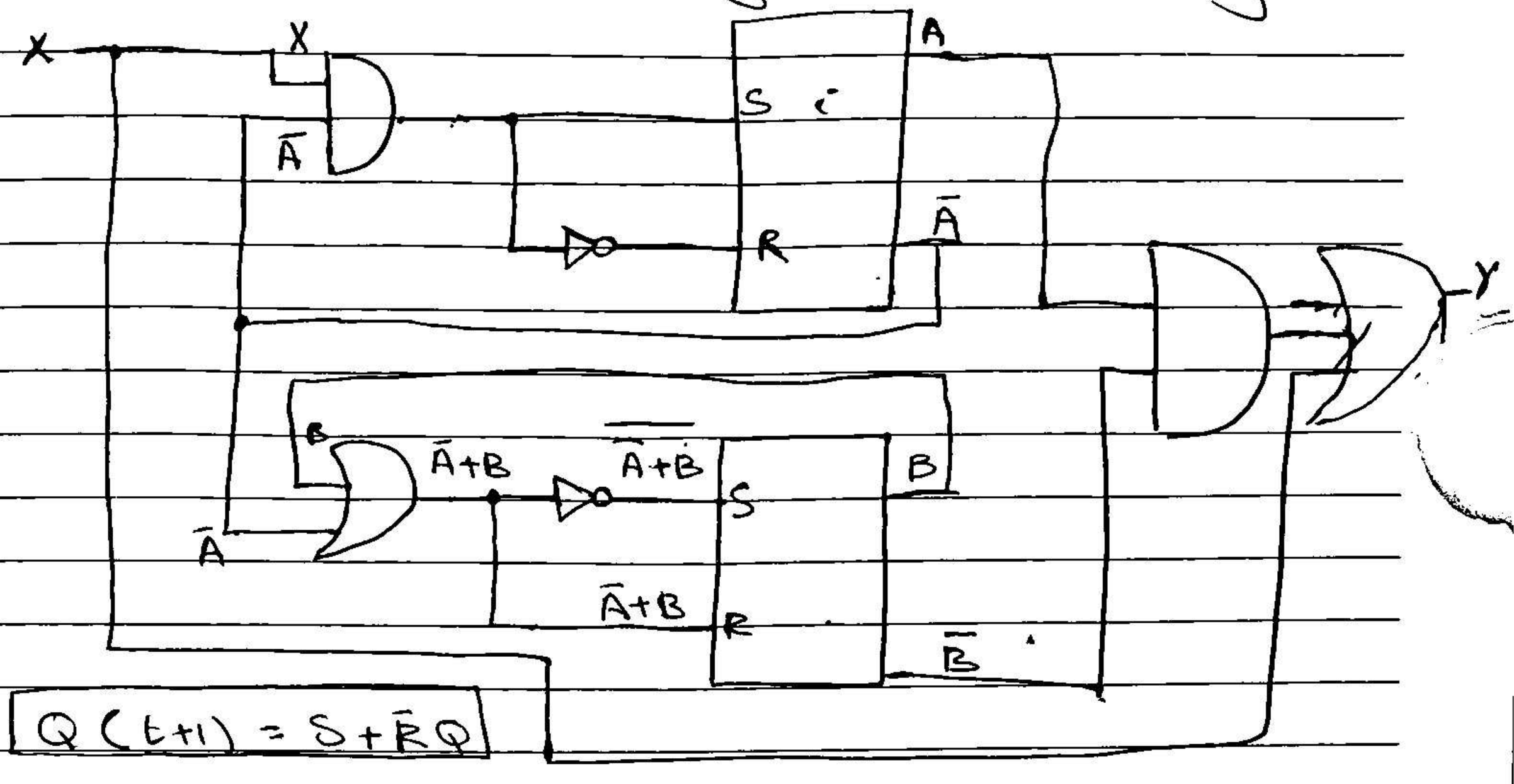
$$A(t+1) = A\bar{B} + A\bar{X} + \bar{A}BX$$

عندي كل حد يساوي واحد ؛ complement = 0

$$B(t+1) = \bar{X}B + X\bar{B}$$

↝ state machine diagram :

* Find the state machine diagram for the following circuit:



$S_A = X\bar{A}$

$R_A = \overline{X\bar{A}} = A + \bar{X}$

$S_B = \overline{\bar{A}+B} = A\bar{B}$

$R_B = \bar{A}+B$

$A(t+1) = S_A + \bar{R}_A A$

$= X\bar{A} + (\overline{A+\bar{X}})A$

$= \bar{A}X + \bar{A}X A = \bar{A}X$

$\boxed{A(t+1) = \bar{A}X}$

$B(t+1) = A\bar{B} + (\overline{\bar{A}+B})B$

$= A\bar{B} + A\bar{B}B$

$\boxed{B(t+1) = A\bar{B}}$

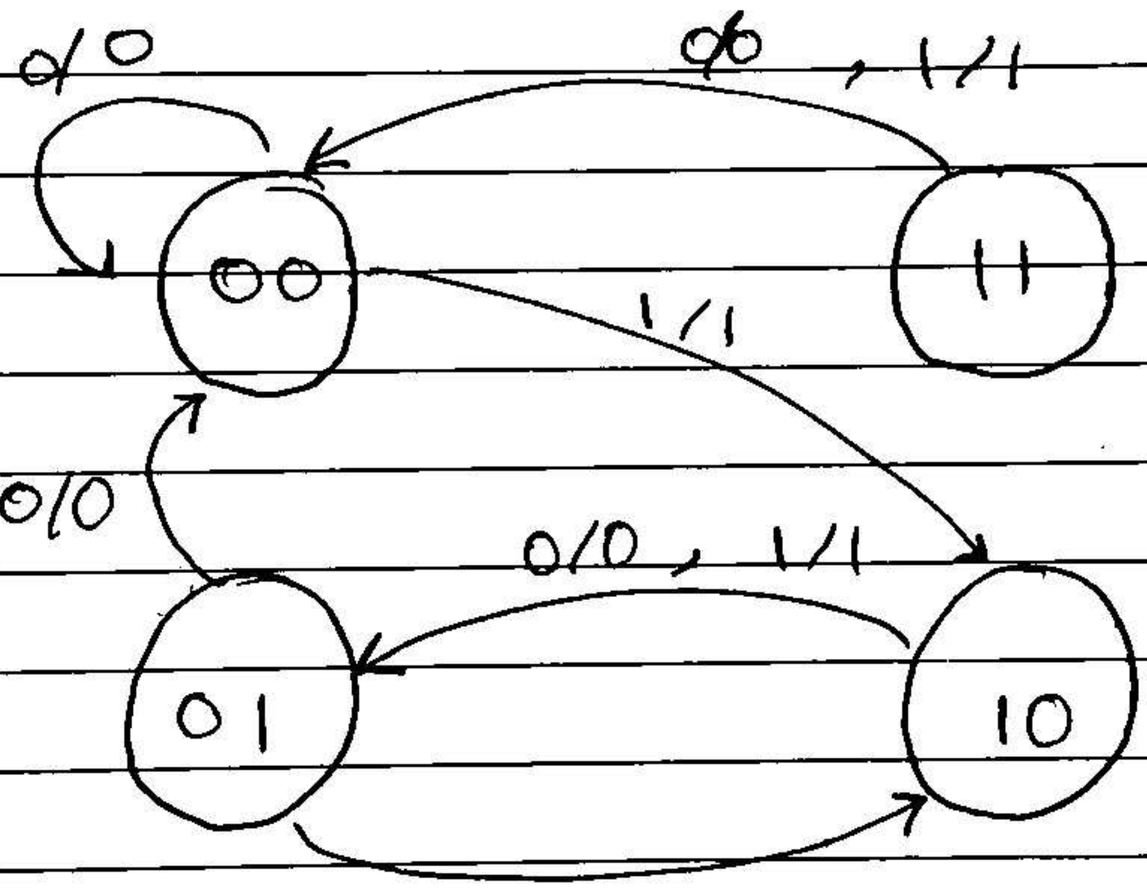
$\boxed{Y = X + A\bar{B}}$

present state		input	Next state		Y
A	B	X	A	B	
0	0	0	0	0	0
0	0	1	1	0	1
0	1	0	0	0	0
0	1	1	1	0	1
1	0	0	0	1	0
1	0	1	0	1	1
1	1	0	0	0	0
1	1	1	0	0	1

$A(t+1) = \bar{A}X$

$B(t+1) = X + A\bar{B}$

متی کل حد برابری ۱



Lecture (19)

**Design of sequential Circuits :

- execution tables:

Q	Q^+	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

S	R	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

D	Q	Q^+
0	0	0
0	1	0
1	0	1
1	1	1

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

XOR

* لما يكونوا مختلفين

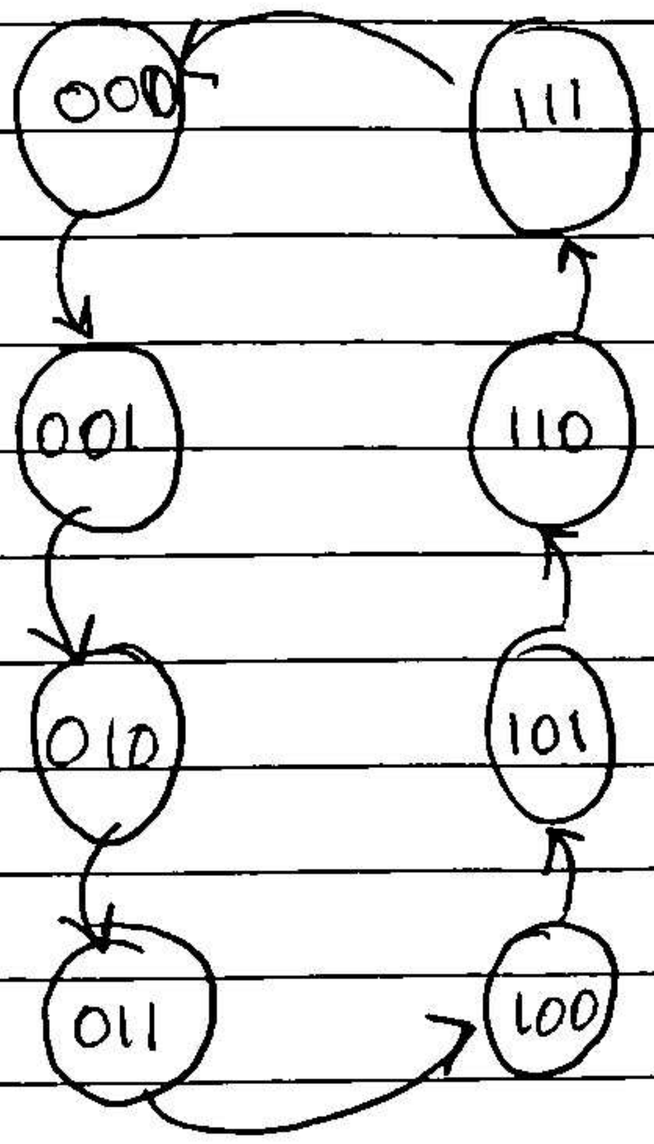
don't care معناها

J	K	Q	Q^+	
0	0	0	0	No change
0	0	1	1	
0	1	0	0	0
0	1	1	0	
1	0	0	1	1
1	0	1	1	
1	1	0	1	Comp
1	1	1	0	

T	Q	Q^+
0	0	0
0	1	1
1	0	1
1	1	0

T	Q	Q^+
0	0	0
0	1	1
1	0	1
1	1	0

* Design a 3-bit up counter using T-FF.

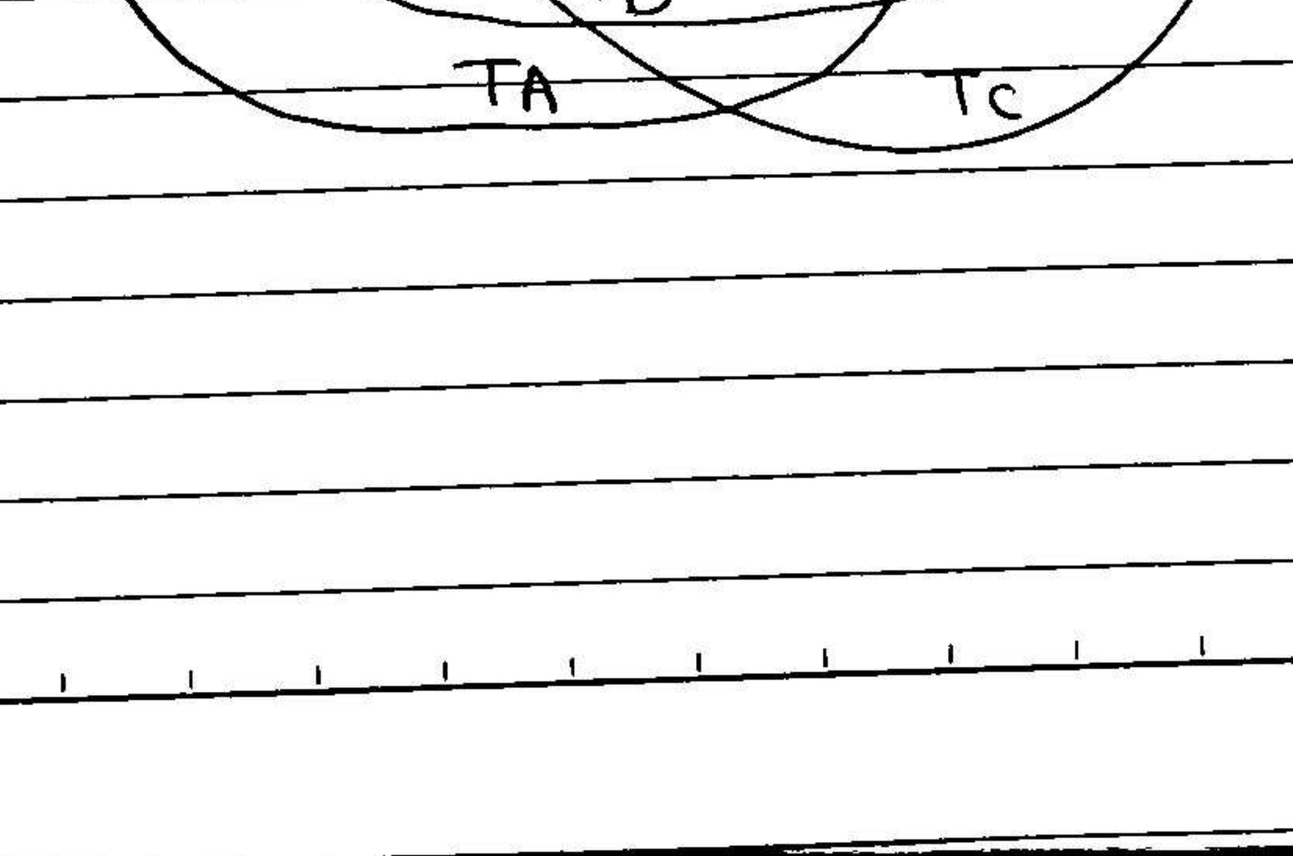


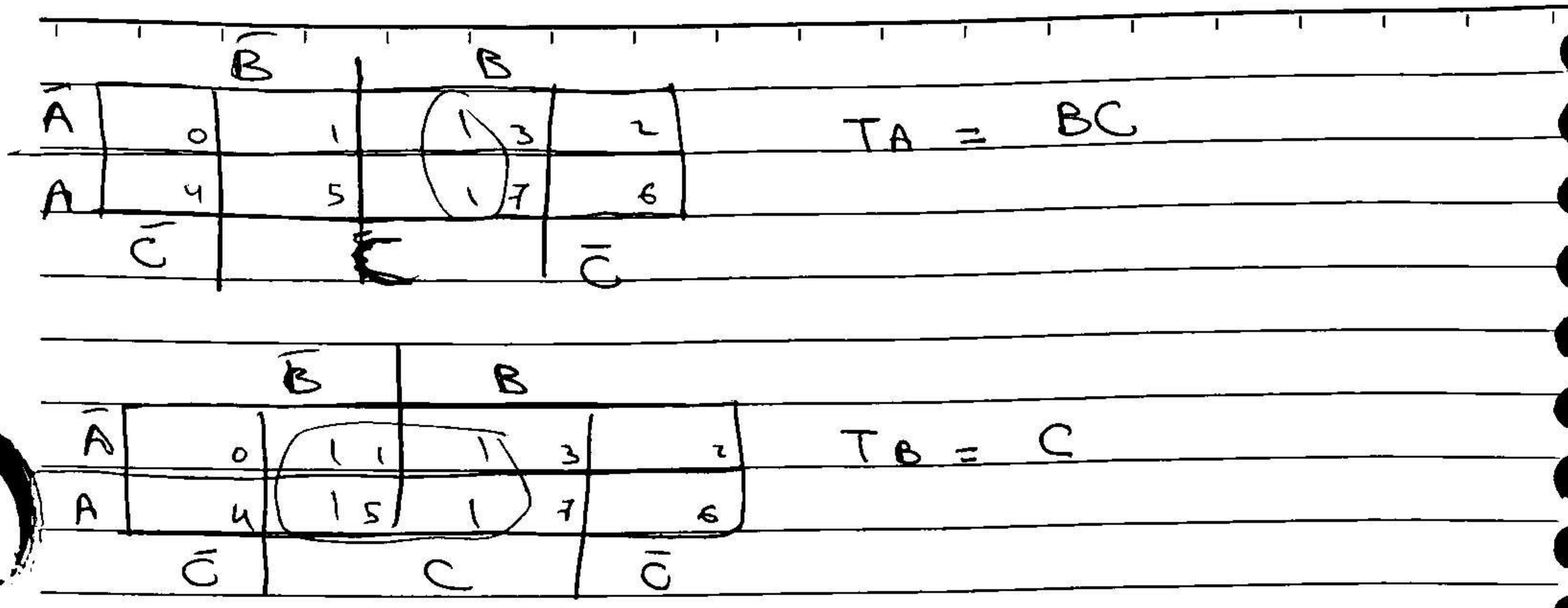
Q	Q^+	t
0	0	0
0	1	1
1	0	1
1	1	0

From state machine diagram.

- State Table:

Current State A B C	Next state A B C	FF inputs T_A T_B T_C
0 0 0	0 0 1	0 0 1
0 0 1	0 1 0	0 1 1
0 1 0	0 1 1	0 0 1
0 1 1	1 0 0	1 1 1
1 0 0	1 0 1	0 0 1
1 0 1	1 1 0	0 1 1
1 1 0	1 1 1	0 0 1
1 1 1	0 0 0	1 1 1

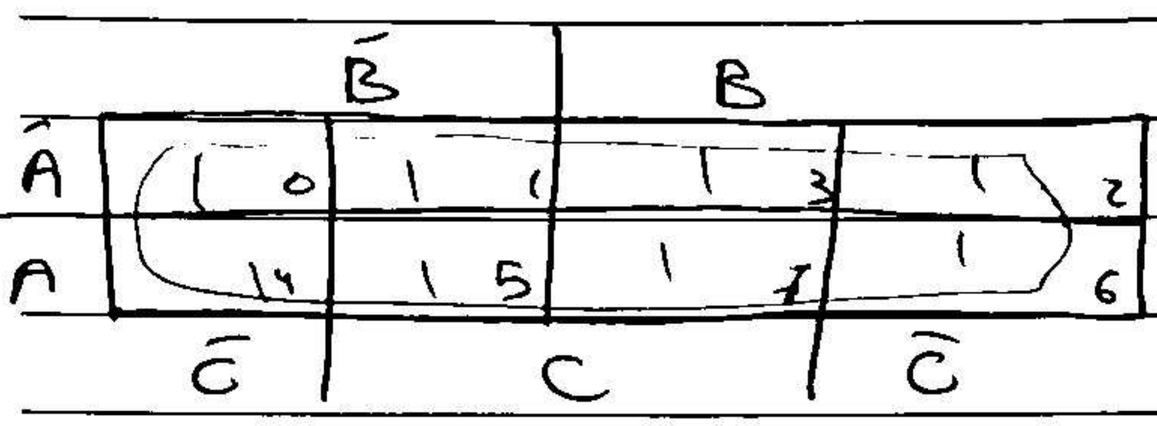



B̄
B
A
0
1
1
3
2
A
4
5
1
7
6
C̄
C
C̄

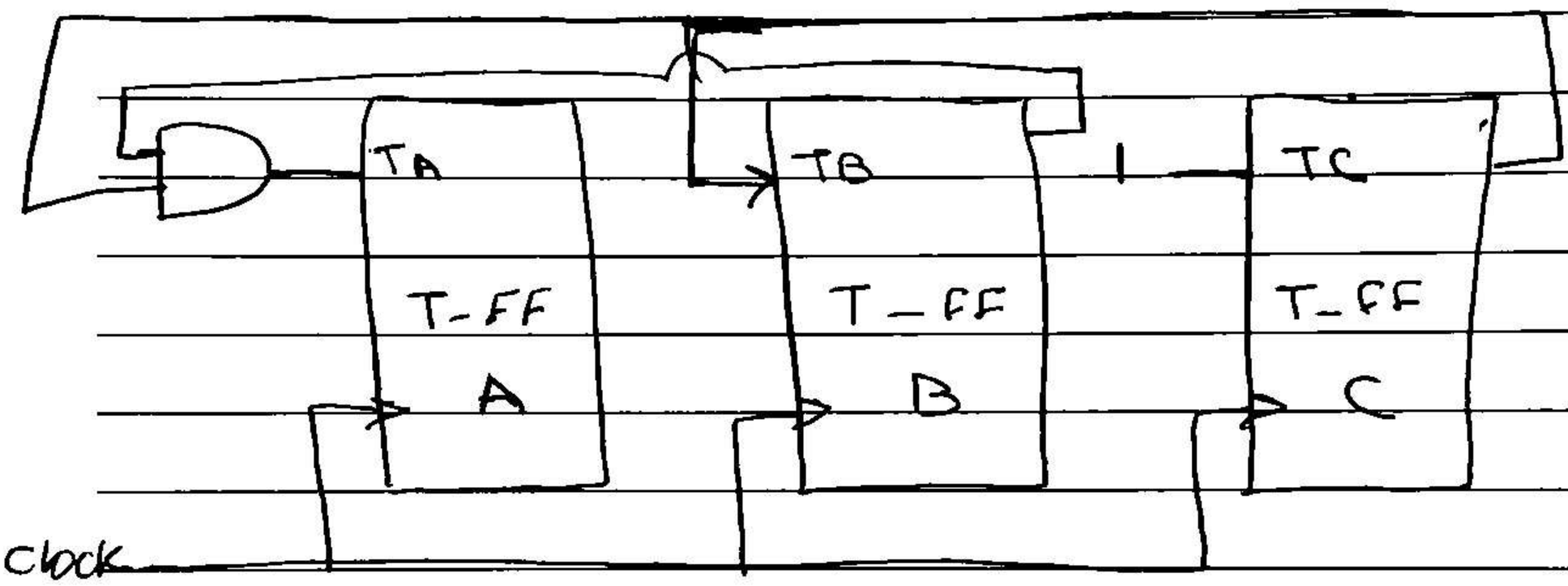
$T_A = BC$

B̄
B
Ā
0
1
1
3
2
A
4
1
5
1
7
6
C̄
C
C̄

$T_B = C$


B̄
B
Ā
0
1
1
3
2
A
4
1
5
1
7
1
6
C̄
C
C̄

$T_C = 1$


TA
TB
1
TC
T-FF
T-FF
T-FF
A
B
C
Clock

* Design a 3-bit up counter using JK-FF

نحتاج السؤال الي قبل state machine diagram ① ←*

present state			Next state.			FF inputs					
Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	0	1	0	X	0	X	1	X
0	0	1	0	1	0	0	X	1	X	X	1
0	1	0	0	1	1	0	X	X	0	1	X
0	1	1	1	0	0	1	X	X	1	X	1
1	0	0	1	0	1	X	0	0	X	1	X
1	0	1	1	1	0	X	0	1	X	X	1
1	1	0	1	1	1	X	0	X	0	1	X
1	1	1	0	0	0	X	1	X	1	X	1

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

$J_1 = Q_0$

Q_2 \ Q_1	$\bar{Q}_1$	Q_1	$\bar{Q}_1$
Q_2	0	1 1 | X 3	X 2
Q_2	4	1 5 | X 7	X 6
	$\bar{Q}_0$	Q_0	$\bar{Q}_0$

$K_1 = Q_0$

Q_2 \ Q_1	$\bar{Q}_1$	Q_1	$\bar{Q}_1$
Q_2	X 0	X 1 | 1 3	2
Q_2	X 4	X 5 | 1 7	6
	$\bar{Q}_0$	Q_0	$\bar{Q}_0$

$J_2 = Q_0 Q_1$

Q_2 \ Q_1	$\bar{Q}_1$	Q_1	$\bar{Q}_1$
Q_2	0	1	3 | 2
Q_2	X 4	X 5	X 7 | X 6
	$\bar{Q}_0$	Q_0	$\bar{Q}_0$

$K_2 = Q_0 Q_1$

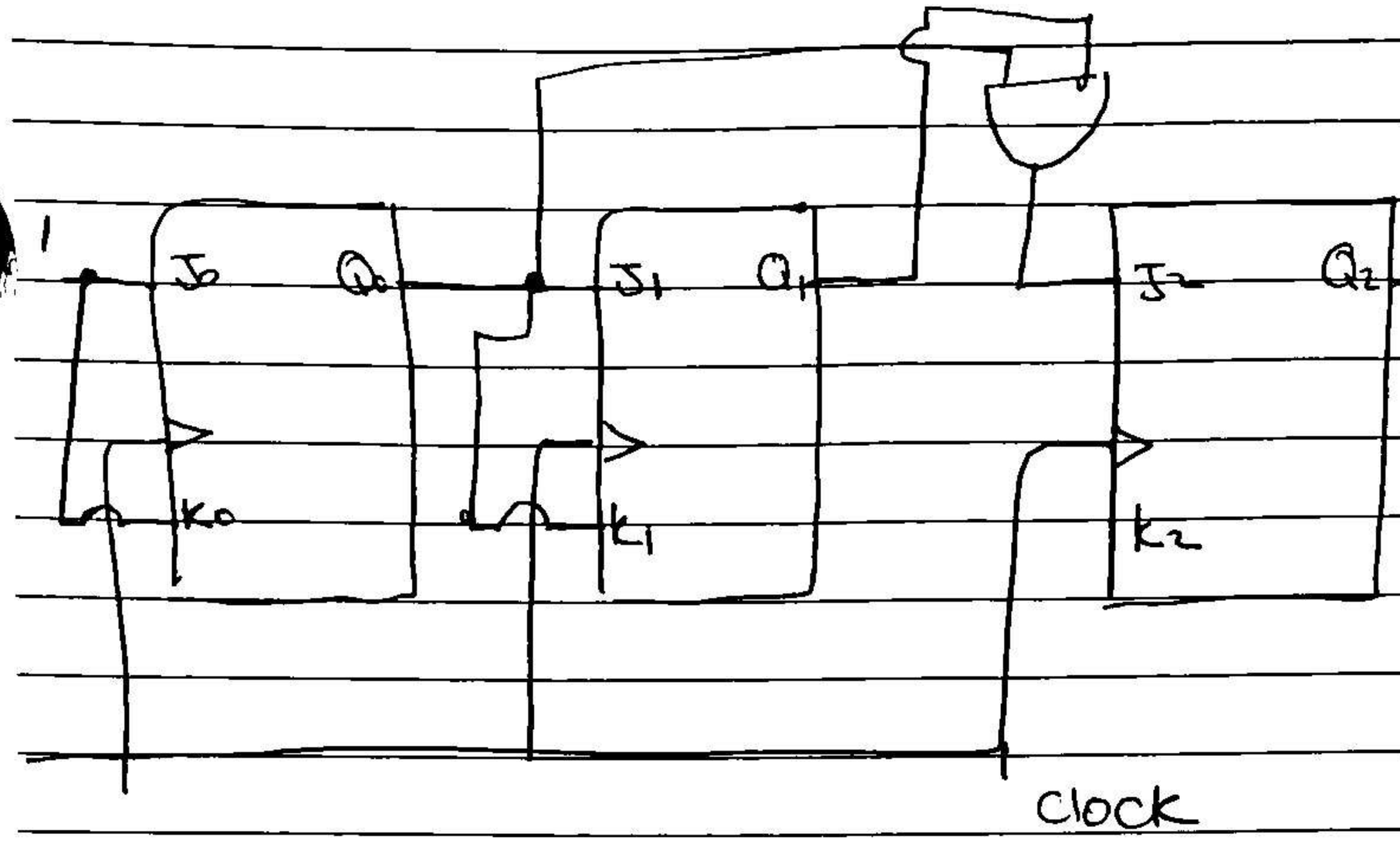
Q_2 \ Q_1	$\bar{Q}_1$	Q_1	$\bar{Q}_1$
Q_2	X 0	X 1	X 3 | X 2
Q_2	7	5	1 7 | 6
	$\bar{Q}_0$	Q_0	$\bar{Q}_0$

$J_0 = 1$

Q_2 \ Q_1	$\bar{Q}_1$	Q_1	$\bar{Q}_1$
Q_2	1 0	X 1	X 3 | 1 2
Q_2	1 4	X 5	X 7 | 1 6
	$\bar{Q}_0$	Q_0	$\bar{Q}_0$

	$\bar{Q}_1$		Q_1	
Q_2	X_0	1 1	1 3	X 2
Q_2	X_1	5	1 7	X 6
	$\bar{Q}_0$	Q_0		$\bar{Q}_0$

$K_0 = 1$



Five Apple

* Design a Counter that goes through the sequence:

0, 3, 2, 4, 1, 5, 7 and repeat, use a D-FF

Current state			Next State			FF outputs		
Q_3	Q_2	Q_1	Q_3	Q_2	Q_1	D_3	D_2	D_1
0	0	0	0	1	1	0	1	1
0	0	1	1	0	1	1	0	1
0	1	0	1	0	0	1	0	0
0	1	1	0	1	0	0	1	0
1	0	0	0	0	1	0	0	1
1	0	1	1	1	1	1	1	1
1	1	0	X	X	X	X	X	X
1	1	1	0	0	0	0	0	0

Q	Q^+	D
0	0	0
0	1	1
1	0	0
1	1	1

	$\bar{Q}_2$		Q_2	
Q_3	0	1	3	2
$\bar{Q}_3$	4	5	7	6
	$\bar{Q}_1$		Q_1	$\bar{Q}_1$

$$D_1 = \bar{Q}_2 Q_3 + Q_2 \bar{Q}_3$$

$$D_2 = \bar{Q}_1 \bar{Q}_2 \bar{Q}_3 + \bar{Q}_1 Q_2 Q_3 + Q_1 \bar{Q}_2 Q_3$$

$$D_3 = \bar{Q}_2$$

* Design a sequential circuit using JK - FF :

present state		input	Next state		output	FF output			
A	B	X	A	B	Y	J_1	k_1	J_0	k_0
0	0	0	0	0	0	0	X	0	X
0	0	1	0	1	1	0	X	1	X
0	1	0	0	1	0	0	X	X	0
0	1	1	1	0	1	1	X	X	1
1	0	0	1	0	1	X	0	0	X
1	0	1	1	1	1	X	0	1	X
1	1	0	1	1	1	X	0	X	0
1	1	1	0	0	1	X	1	X	1

Q	Q^+	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

A \ B	$\bar{B}$	B
A	0	1
A	X 4	X 5
$\bar{A}$	X	X

A \ B	B	
	1 3	2
	X 7	X 6
	X	$\bar{X}$

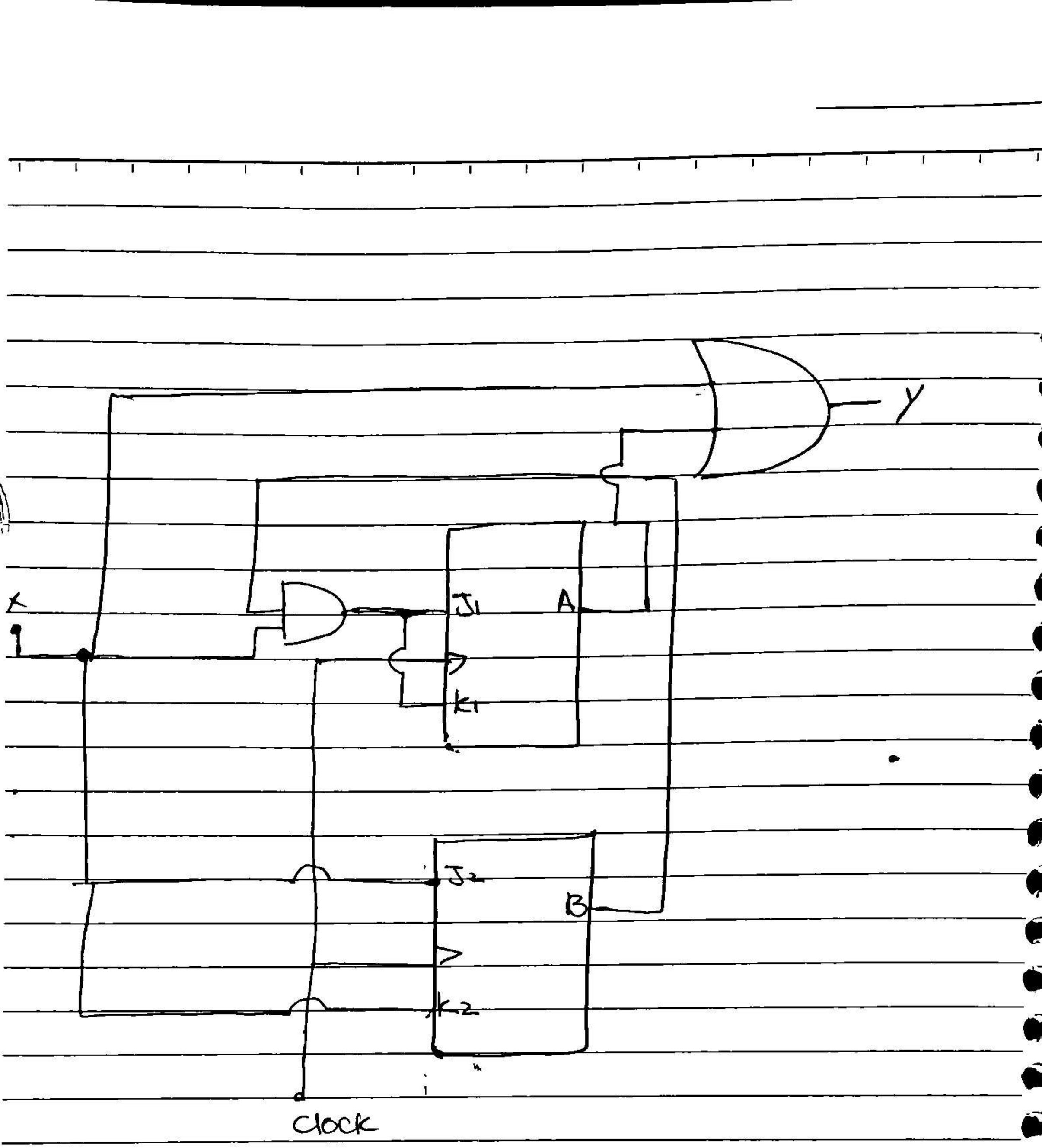
$J_1 = BX$

$k_1 = B\bar{X}$

$J_2 = X$

$k_2 = X$

$Y = X + A$


Y
X
1
J1
A
K1
J2
B
K2
clock